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**WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED
FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES**

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Piccio Garden, Villamor, Pasay City

In Partial Fulfillment of the Requirements for the Degree of
BACHELOR OF SCIENCE IN INFORMATION SYSTEM
WITH SPECIALIZATION IN AVIATION INFORMATION SYSTEM

**DE ASIS, RHICKA MAE P.
ROJO, JOHN EMMANUEL E.
VALLIDO, LORENZA LANE S.**

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ABSTRACT

**PROPONENTS : De Asis, Rhicka Mae P.
Rojo, John Emmanuel E.
Vallido, Lorenza Lane S.**

**PROGRAM : Bachelor of Science in Information System (BSIS) with
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ADVISER : Asst. Prof. Aron J. Alojado, MSIT

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Faculty evaluation is an important part of quality assurance in higher education, especially in aviation training, where teaching quality affects student competence and safety. At the National Aviation Academy of the Philippines (NAAP), faculty evaluation still relies on paper forms and Google Forms. Responses come from separate sources, are consolidated by hand, and written comments are read and interpreted manually. This makes the process slow, increases administrative workload, and leaves it open to errors and data security risks. To address these problems, the researchers developed a Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for NAAP. The system combines evaluations from

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students, peers, and supervisors on one secure platform. It uses AI-assisted summarization and frequency analysis to process qualitative comments, and it provides a real-time analytics dashboard, automated reporting, and role-based access control.

The system was developed using the Agile Software Development Life Cycle with a hybrid Scrum-inspired framework. It was evaluated using a quantitative descriptive-developmental research design. A total of 208 purposively selected respondents took part: 192 students, 13 faculty members, and 3 HR personnel. After an orientation and demonstration of the system, they answered a structured 4-point Likert-scale questionnaire. The questionnaire covered the nine ISO/IEC 25010:2023 software quality characteristics, multi-source evaluation, AI-assisted feedback analysis, comparison with the existing process, user acceptance, and data-driven improvements. The data were analyzed using frequency, percentage, mean, weighted mean, and general weighted mean.

Results showed that every area was rated in the highest interpretation, except that HR personnel rated parts of the AI feedback analysis and some reliability items one level lower ("Effective" or "Reliable"). Multi-source evaluation was rated Highly Comprehensive (3.63) and Highly Reliable (3.55). AI-assisted feedback analysis was rated Highly Effective (3.42). The ISO/IEC 25010:2023 characteristics ranged from 3.34 (reliability) to 3.66 (accessibility), with security at 3.62. Compared with the existing process, the system was rated Highly Efficient

(3.74), Highly Accurate (3.55), Highly More Usable (3.68), and as Greatly Reducing Administrative Workload (3.71). User acceptance was high, with perceived usefulness at 3.60, ease of use at 3.61, and overall user experience at 3.69. Data-driven improvements and recommendations were rated Highly Effective (3.60).

The study concludes that the system provides a more complete, accurate, secure, and efficient faculty evaluation process than the existing manual and semi-digital methods. It also gives institutional decision-makers organized and actionable results. HR personnel's more cautious ratings suggest that AI-generated results should still be reviewed by people before they are used for formal administrative decisions. Fault tolerance, the lowest-rated area, may need further testing and improvement.

Keywords: Faculty Evaluation System, AI-Assisted Feedback Analysis, Multi-Source Evaluation, CHED-Compliant Instruments, ISO/IEC 25010:2023, Agile Methodology

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CHAPTER 1 INTRODUCTION

Introduction

Faculty evaluation systems have become a critical component of quality assurance in higher education, supporting instructional improvement, accountability, and data-driven decision-making. Institutions worldwide are increasingly adopting web-based platforms, multi-source evaluation models, and artificial intelligence (AI) to improve the accuracy and efficiency of performance assessment. Existing studies highlight the benefits of combining student, peer, and supervisor inputs, as well as using AI to analyze large volumes of qualitative feedback. However, several gaps remain evident: many systems lack multi-source integration, limiting the completeness and reliability of evaluations, and the use of AI in qualitative feedback analysis is still limited, particularly in combining advanced language models with validation techniques. Few systems also undergo comprehensive evaluation using ISO/IEC 25010 standards, and real-world implementation assessments, user acceptance studies, and post-evaluation insights are often overlooked. These gaps highlight the need for a more integrated, intelligent, and standards-compliant faculty evaluation solution.

The National Aviation Academy of the Philippines (NAAP) is a specialized higher education institution dedicated to training future aviation professionals, such as pilots, aviation maintenance technicians, and other industry specialists.

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Because it prepares students for safety-critical careers, NAAP places strong emphasis on maintaining high instructional standards, regulatory compliance, and continuous quality improvement. Faculty evaluation is central to this effort, allowing the institution to monitor teaching effectiveness and ensure that instructors consistently meet the academic and industry standards expected of an aviation training institution. As NAAP continues to grow, however, its reliance on manual and fragmented evaluation methods has become increasingly inadequate, creating a clear need for a more integrated and technology-driven evaluation system.

These challenges are evident in NAAP's current evaluation practices. Existing systems are fragmented, relying on isolated data sources without a centralized platform for efficient data management. The absence of automated insight extraction requires time-consuming manual analysis, increasing administrative workload and delaying decision-making. Without standardized quality assessment frameworks, the institution cannot objectively evaluate the performance and reliability of its evaluation tools, and the level of user adoption and acceptance of existing systems remains unclear, further complicating any improvement efforts. Together, these limitations highlight the urgent need for an evaluation system that integrates multiple data sources, automates analysis, ensures quality compliance, and supports continuous optimization.

In response to these challenges, this study aims to design, implement, and evaluate a Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for NAAP. Specifically, it seeks to answer the following research questions:

1. How does the integration of multi-source faculty evaluation (student, peer, and supervisor inputs) affect the reliability and completeness of faculty performance assessment?
2. How effective is AI-assisted analysis in processing and interpreting qualitative faculty evaluation feedback?
3. How does the proposed system perform in terms of ISO/IEC 25010 software quality characteristics, particularly usability, functional suitability, performance efficiency, and reliability?
4. How does the system compare with the existing evaluation process in terms of operational efficiency, data accuracy, usability, and administrative workload?
5. What is the level of user acceptance in terms of perceived usefulness and ease of use?
6. What data-driven improvements can be proposed based on the evaluation results of the system?

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These questions guide the development and evaluation of the proposed solution, ensuring alignment with the study's objectives of enhancing effectiveness, efficiency, and usability in faculty evaluation systems.

Project Context

The National Aviation Academy of the Philippines (NAAP) operates as a specialized higher education institution focused on aviation training and education, where maintaining high instructional standards is essential to ensuring student competence and operational safety. As an institution responsible for developing future aviation professionals, NAAP plays a critical role in upholding academic quality, regulatory compliance, and industry readiness. Within this environment, faculty evaluation serves as a key mechanism for assessing teaching effectiveness and supporting continuous improvement, involving multiple stakeholders such as students, faculty members, supervisors, administrators, deans, and Human Resources personnel.

At present, faculty evaluation at NAAP is conducted using a hybrid approach that combines paper-based forms and Google Forms. While these methods allow the institution to collect feedback, the overall process remains fragmented and largely manual: administrators must consolidate responses from different sources, verify the completion of evaluations, and generate reports without the support of a centralized system, and qualitative feedback is manually interpreted, making the

process time-consuming and dependent on subjective analysis. These practices reflect a semi-digital system that lacks integration, automation, and real-time processing capabilities.

This workflow presents several operational challenges. The absence of multi-source integration limits the ability to capture a holistic view of faculty performance from students, peers, and supervisors, while the lack of automated validation and analytics forces administrators to manually detect duplicate, incomplete, or unauthorized submissions. Without a centralized and secure data management system, it is also difficult to track historical records and generate comprehensive reports. These limitations increase administrative workload, cause processing delays, and expose the system to data security and confidentiality risks, ultimately affecting the reliability, accuracy, and usefulness of evaluation data for institutional decision-making.

Given these challenges, there is a clear need to improve the existing evaluation system at NAAP, as current practices are insufficient to support the efficient, secure, and data-driven evaluation processes required in a high-stakes aviation education environment. There is an opportunity to implement a centralized and secure web-based platform that integrates multi-stakeholder evaluation, incorporates AI-assisted techniques for qualitative feedback analysis, and automates validation and reporting. These improvements would significantly

reduce administrative burden while improving data accuracy, system reliability, and institutional decision-making.

Purpose and Description

Faculty and student evaluations play a vital role in higher education by improving teaching quality, ensuring institutional accountability, and supporting continuous academic development. However, existing manual and semi-digital evaluation methods often lead to inefficiencies, data inconsistencies, and limited analytical capabilities, reducing the overall effectiveness of evaluation outcomes. In response to these challenges, this study aims to design, develop, and evaluate an AI-enhanced, integrated, web-based faculty evaluation system that centralizes multi-source evaluation processes and improves data accuracy, administrative efficiency, and data-driven academic decision-making. The system is intended for use by students, faculty members, supervisors, administrators, deans, and Human Resources personnel, contributing to a more structured and reliable evaluation environment within the National Aviation Academy of the Philippines (NAAP). Furthermore, the system is developed in accordance with CHED standards and will be evaluated based on efficiency, accuracy, compliance with ISO/IEC 25010 software quality standards, and user acceptance to ensure its effectiveness and practical applicability.

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The proposed solution is a web-based faculty evaluation system that integrates multiple evaluation components into a unified platform. It is designed to replace fragmented and manual processes with a centralized, secure, and accessible system that supports real-time data processing and reporting. The system adopts a multi-source evaluation approach by combining inputs from students, peers, and supervisors, ensuring a more comprehensive and balanced assessment of faculty performance. In addition, it incorporates AI-assisted techniques to enhance the interpretation of qualitative feedback, providing deeper insights that are not easily achievable through manual analysis alone. By integrating evaluation, analysis, and reporting into a single platform, the system offers a cohesive and efficient solution aligned with institutional and regulatory requirements.

The system includes several key features that address the limitations of existing practices. It supports multi-source evaluation through structured instruments for students, peers, and supervisors, and applies AI-assisted analysis to process and summarize large volumes of qualitative feedback for faster, more consistent interpretation. A centralized database manages all evaluation data, allowing secure storage, easy retrieval, and efficient report generation, while built-in security safeguards protect user information and prevent unauthorized access. An analytics dashboard further provides real-time performance insights through charts, tables, and key indicators, supporting monitoring and evaluation activities.

Together, these features enable automation, real-time processing, and systematic data management within the system.

The expected outcomes of the proposed system include significant improvements in evaluation accuracy, efficiency, and security compared to existing methods. By automating data collection, validation, and analysis, the system reduces administrative workload and minimizes human errors. It enhances the reliability of evaluation results by integrating multiple data sources and applying consistent analytical methods to both quantitative and qualitative data. The system also strengthens data privacy and security, directly addressing the vulnerabilities present in existing evaluation practices. For stakeholders, the platform provides clearer, more actionable insights that support informed decision-making and continuous improvement of teaching performance. Ultimately, the study aims to demonstrate that the proposed system achieves acceptable levels of usability and user acceptance while offering a scalable and sustainable solution for institutional evaluation processes.

Objective of the Study

To develop a Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines that ensures accurate, secure, accessible, and efficient faculty evaluation and student clearance processes. The system supports

structured feedback collection and enables HR personnel to perform data-driven evaluation and decision-making. Guided by the Agile Software Development Life Cycle, the study specifically aims to:

1. To investigate how the integration of multi-source faculty evaluation (student, peer, and supervisor inputs) influences the reliability and completeness of faculty performance assessment.
2. To examine the effectiveness of AI-assisted analysis in processing and interpreting qualitative faculty evaluation feedback, including the use of text and frequency analysis for improved insight extraction.
3. To empirically evaluate the proposed faculty evaluation system using the ISO/IEC 25010:2023 software quality characteristics, including functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility.
4. To compare the proposed system with the existing evaluation process in terms of operational efficiency, data accuracy, usability, and administrative workload.
5. To assess the level of user acceptance of the proposed system, focusing on perceived usefulness, ease of use, and overall user experience.
6. To identify data-driven improvements and recommendations based on system evaluation results, user feedback, and performance analysis.

Scope and Limitations

Scope of the Study

This study focuses on the design, development, implementation, and evaluation of an AI-enhanced, CHED-compliant web-based faculty evaluation system that integrates multi-source evaluation from students, peers, and supervisors at the National Aviation Academy of the Philippines (NAAP), Villamor Campus. The system aims to improve the faculty evaluation process by incorporating AI-assisted qualitative feedback analysis while preserving the original CHED-aligned evaluation structure.

The study investigates the system's effectiveness in terms of reliability, fairness, efficiency, usability, system quality, and user acceptance. It specifically examines how multi-source evaluation contributes to a more comprehensive faculty assessment and how AI-assisted analysis supports the processing and summarization of qualitative feedback.

The respondents of the study consist of selected students, faculty members, supervisors, and Human Resource (HR) personnel who directly interact with the proposed system. A purposive sampling technique is employed to select participants who are knowledgeable and experienced with the faculty evaluation process.

The research adopts a developmental research design with a quantitative approach. Data are collected using structured survey questionnaires and system

evaluation instruments based on CHED standards. The evaluation of the proposed system is guided by the ISO/IEC 25010:2023 software quality characteristics, including functional suitability, reliability, usability, performance efficiency, security, compatibility, maintainability, and accessibility. The collected data are analyzed using descriptive statistical tools such as frequency distribution, mean, and weighted mean to determine the system's effectiveness and user acceptance.

The proposed system covers the core functionalities of faculty evaluation, including user authentication, evaluation submission, data processing, report generation, and AI-assisted summarization of qualitative feedback. The system operates within the institutional environment of NAAP–Villamor Campus and is accessible only to authorized users during the designated academic term for system implementation and evaluation.

Limitations of the Study

This study is limited to the implementation and evaluation of the proposed faculty evaluation system within the National Aviation Academy of the Philippines (NAAP), Villamor Campus. As such, the findings may not be generalizable to other campuses or higher education institutions with different organizational structures, policies, or evaluation practices.

The study employs a quantitative research approach using structured survey questionnaires and system evaluation instruments. Consequently, it does

not include qualitative methods such as interviews, focus group discussions, or thematic analysis that could provide more in-depth insights into users' experiences and perceptions.

A purposive sampling technique is used to select respondents who have direct interaction with the system. The relatively limited number of participants may affect the representativeness of the findings and may introduce sampling bias. In addition, the accuracy of the results depends on the respondents' honesty, understanding, and willingness to provide objective responses during the evaluation.

The proposed system is limited to essential functionalities, including evaluation submission, data processing, report generation, and AI-assisted summarization of qualitative feedback. It does not include integration with third-party platforms, deployment across multiple campuses, fully automated decision-making, or other advanced artificial intelligence capabilities.

The system operates in a controlled institutional environment and may be affected by technical factors such as internet connectivity, browser compatibility, device specifications, and hardware performance. Likewise, the AI-assisted feedback analysis may have limitations in accurately interpreting complex, ambiguous, or context-dependent qualitative responses. These technical and environmental factors may influence system performance and users' evaluation experience.

The study also does not modify the existing CHED-prescribed faculty evaluation criteria. Instead, it enhances the evaluation process by integrating multi-source assessment and AI-assisted analysis while maintaining compliance with established evaluation standards.

Conceptual Framework

The study is anchored on the Input–Process–Output (IPO) conceptual framework, a widely used model in systems and information science that illustrates how data are transformed into meaningful outputs through systematic processes. This framework is appropriate for the study as it provides a structured approach to analyzing how multi-source evaluation data are collected, processed, and converted into actionable results within an AI-enhanced faculty evaluation system. It is also supported by principles from information systems and software quality evaluation, particularly the ISO/IEC 25010:2023 standards, which guide the assessment of system performance and usability. The use of the IPO model enables a clear representation of system functionality and ensures that each stage of data handling contributes to achieving the study's objectives of improving evaluation accuracy, efficiency, and user acceptance.

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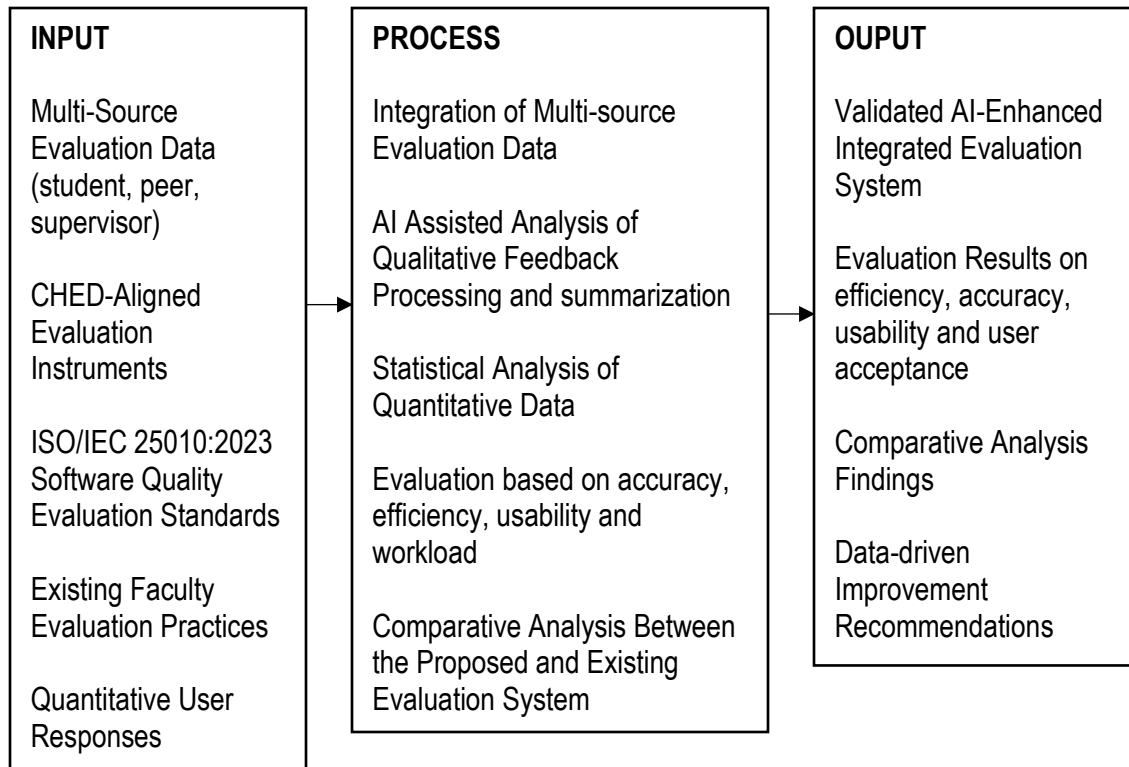


Figure 1: Input Process Output

INPUT

The input component consists of the resources and information required for the development and evaluation of the proposed system. These include multi-source evaluation data from students, peers, and supervisors, CHED-aligned evaluation instruments, ISO/IEC 25010:2023 software quality evaluation standards, existing faculty evaluation practices, and quantitative responses gathered from the respondents. These inputs serve as the foundation for assessing both faculty performance and the quality of the proposed evaluation system.

PROCESS

The process component involves the integration and analysis of the collected data within the proposed web-based faculty evaluation system. Multi-source evaluation data are consolidated into a unified assessment, while AI-assisted analysis processes and summarizes qualitative feedback. Quantitative responses are analyzed using appropriate statistical methods, and the system is evaluated based on software quality characteristics such as accuracy, efficiency, usability, and workload reduction. A comparative analysis is also conducted to determine the differences between the proposed system and the existing faculty evaluation process.

OUTPUT

The output component represents the expected results of the study. These include a validated AI-enhanced integrated faculty evaluation system, evaluation results regarding efficiency, accuracy, usability, and user acceptance, comparative analysis findings between the proposed and existing systems, and data-driven recommendations for improving the faculty evaluation process. These outputs demonstrate the effectiveness of the proposed system in providing a more reliable, efficient, and comprehensive faculty evaluation.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter reviews relevant local and foreign literature to establish the theoretical and empirical foundations of the proposed Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines (NAAP). Faculty and student evaluations are essential pillars of quality assurance in higher education, supporting teaching effectiveness, professional growth, and institutional accountability. In the Philippine context, these practices are guided by CHED Memorandum Order No. 19, Series of 2025, which promotes standardized and performance-based evaluation processes to ensure fairness and continuous improvement [45]. The importance of effective evaluation is especially critical in aviation education, where instructional quality directly influences student competence and safety. However, traditional paper-based and semi-digital evaluation methods remain inefficient and prone to inconsistent feedback, reporting delays, bias, and data integrity issues, limiting institutions' ability to generate meaningful insights and use evaluation results effectively for improvement and evidence-based decision-making.

AI Analysis

AI analysis in the context of faculty evaluation refers to the application of computational methods—including natural language processing (NLP), frequency analysis, and large language model (LLM)-based summarization—to transform large volumes of qualitative feedback into structured, actionable insights. The sheer volume of open-ended responses generated during evaluation cycles makes manual interpretation increasingly impractical, making AI-driven analysis a critical enabler for scalable and consistent feedback processing. However, a close examination of the literature reveals that existing implementations remain either broad in scope or limited to quantitative data streams, leaving qualitative open-ended comment analysis substantially underdeveloped in practice.

Studies widely agree that AI tools accelerate feedback interpretation while improving consistency. Sajja, Sermet, and Cwiertny demonstrated across two studies [1], [2] that integrating GPT-4-based learning analytics into educational evaluation enables descriptive analytics that surface patterns in engagement and cognition, with frequency analysis of instructional metrics supporting data-driven pedagogical decisions—yet both studies focus on broad theoretical frameworks rather than the systematic processing of faculty evaluation comments in context-specific deployments. Apetorgbor et al. [3] confirmed through a comprehensive review that AI technologies produce accurate and efficient evaluations with near-

instant feedback, substantially reducing turnaround time compared to traditional methods; however, their review does not propose or validate a hybrid method that pairs API-based AI summarization with non-API frequency validation, leaving the practical question of how to process qualitative faculty comments in resource-constrained environments unanswered. Ammari [4] analyzed Student Evaluations of Teaching (SETs) through generative AI and showed that such techniques yield richer faculty performance insights aligned with program improvement goals, but this analysis is restricted to student-generated text and does not extend to peer or supervisor qualitative inputs, confirming that multi-source qualitative comment integration remains an unaddressed dimension. Kumar [5] demonstrated that AI-driven automated assessment systems combining machine learning and NLP deliver real-time analysis alongside personalized recommendations, with frequency analysis proving especially effective for trend identification—yet the study does not operationalize this within a faculty evaluation system that simultaneously collects and processes qualitative responses from differentiated evaluator roles.

These findings confirm a consistent gap: while AI capability for qualitative text processing is well established in theory, no existing implementation combines API-based comment summarization with non-API frequency validation across student, peer, and supervisor inputs within a single operational faculty evaluation platform. This gap directly shapes the AI-assisted feedback component of the

proposed system, which employs OpenAI GPT-4.1 mini for comment summarization paired with non-API frequency validation—a hybrid approach designed to address both the analytical and resource constraints that existing studies have left unresolved.

Artificial Intelligence

Artificial intelligence (AI) in higher education evaluation encompasses a broader range of applications beyond feedback analysis, including automated scoring, performance prediction, and pattern detection in large datasets. Research consistently demonstrates that AI-based evaluation tools increase assessment consistency and reduce human bias. Almufarreh, Noaman, and Saeed [6] showed that machine learning frameworks applied to academic teaching quality evaluation produce more consistent performance scores than traditional rater-dependent methods. Qi et al. [7] and Gao [8] established that deep learning architectures detect complex patterns in large teaching evaluation datasets, enabling more nuanced distinctions among performance levels. Mradula et al. [9] and Bagunaid et al. [10] demonstrated that automated evaluation platforms with continuous monitoring capabilities provide ongoing recommendations supporting proactive faculty development. Darshini [11] supported the predictive dimension, showing that machine learning models analyzing academic data can forecast performance trajectories with measurable accuracy.

Where the literature reveals a critical gap is in its Philippine-specific application of advanced language models to qualitative evaluation data. Philippine-focused studies by Roaring et al. [12] and Gabayan et al. [13] prioritize practical text analysis and predictive classification within local institutional resource constraints. Roaring et al. [12] applied text mining to faculty teaching performance ratings at Pangasinan State University but relied on frequency-based methods without AI-assisted comment summarization. Gabayan et al. [13] conducted a systematic review of predictive classification and trend forecasting in faculty performance evaluation in Philippine higher education but found no study that had deployed advanced language models such as GPT-4 or equivalent systems for qualitative comment interpretation within a Philippine institutional context. This absence confirms that the integration of large language model capabilities into faculty evaluation systems remains effectively unrealized within local Philippine higher education practice—a gap that the proposed system addresses by applying GPT-4.1 mini within NAAP's operational environment, connecting the advanced analytical capabilities demonstrated in global research with the locally relevant institutional context, while the inclusion of frequency analysis reinforces the pattern detection strategies proven effective by Roaring et al. [12].

Assessment Instruments

Assessment instruments in faculty evaluation refer to the standardized tools, criteria, and measurement frameworks used to systematically capture faculty performance across defined competency areas. Well-designed instruments are the foundation of any credible evaluation system; without clearly defined criteria, evaluation results risk becoming subjective and inconsistent, undermining institutional accountability and faculty development goals. Multiple studies agree that structured, standards-aligned evaluation instruments significantly improve the validity, reliability, and clarity of faculty assessments. Vasilev et al. [14] emphasized that structured frameworks improve measurement clarity and validity, reducing ambiguity in how performance criteria are interpreted by different evaluators. Nausheen et al. [15] developed a novel tool using student evaluations to assess faculty development needs, combining descriptive and frequency analysis to identify critical improvement areas. Alenezi et al. [16] highlighted the importance of aligning instruments with modern digital practices, while Mounira et al. [17] stressed that standardized tools must remain responsive to changing institutional environments. Isaifan and Hasna [18] further demonstrated that assessment tools incorporating analytics capabilities enhance the overall measurement process.

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Philippine implementations reveal a more practice-oriented perspective: Daluyon and Zamora [19], Santillan [20], and Wenceslao [21] each integrated structured instruments into web-based systems for State Universities and Colleges (SUCs), demonstrating that digitized instruments improve data management and reporting efficiency. Most directly relevant is the work of Dueñas and Martin [22], who developed contextualized faculty evaluation criteria for technical instructors at the Philippine State College of Aeronautics (PhilSCA). However, a critical limitation runs across both international and Philippine studies: instrument design remains focused on quantitative rating scales, and none of the reviewed works integrate draft-enabled digital form delivery that pairs instrument completion with AI-assisted qualitative analysis of accompanying comments. Furthermore, while Dueñas and Martin [22] address the aviation-specific context, their instrument operates without digital workflow automation, real-time feedback processing, or compliance verification against the current CHED Memorandum Order No. 19, Series of 2025—requirements that did not exist at the time of their publication. This gap shapes the instrument design of the proposed system, which employs CHED-aligned and institution-approved standardized instruments delivered through draft-enabled digital forms, following Dueñas and Martin's aviation-specific approach [22] while meeting current CHED regulatory requirements.

Secure Authentication

Secure authentication refers to the mechanisms by which digital systems verify the identity of users before granting access, encompassing password hashing, one-time password (OTP) protocols, multi-factor verification, and role-based access control. In the context of faculty evaluation systems, strong authentication is not an optional security layer but a core functional requirement, as these platforms handle sensitive personal and professional information whose unauthorized access or manipulation can compromise institutional integrity and individual privacy. Research converges on the effectiveness of multi-layered authentication systems in protecting sensitive data. Madan Mohan [23] demonstrated that email-based OTP systems significantly enhance authentication security through advanced cryptographic techniques and industry-standard protocols that generate unique, time-sensitive tokens, with bcrypt password hashing and secure server communication directly addressing threats such as credential exposure and unauthorized account access—achieving both high security and user-friendliness that is critical for maintaining evaluator participation rates.

The limitation of Madan Mohan's framework [23], and of the available authentication literature more broadly, is that it addresses general web application security rather than the specific compliance and role-differentiated access

requirements of higher education evaluation systems operating under Philippine law. The Data Privacy Act of 2012 (Republic Act 10173) imposes legal obligations on institutions handling personal academic data that general cryptographic frameworks do not explicitly address, and no reviewed study proposes an authentication architecture that simultaneously satisfies multi-role access segregation (students, peers, supervisors, administrators), OTP-based anomaly detection for irregular login events, and compliance with RA 10173 within a faculty evaluation context. These considerations directly inform the security architecture of the proposed system, which develops secure authentication mechanisms validating evaluator identity through a targeted OTP protocol and bcrypt password hashing, Cloudflare firewall protection, automated backups, and audit trails—all in compliance with RA 10173—with role-based access control ensuring each evaluator role accesses only the instruments appropriate to their designated function.

Digital Transformation

Digital transformation in higher education refers to the organizational, cultural, and technological shift from paper-based and fragmented semi-digital processes to integrated institutional platforms that automate workflows, centralize data, and enable evidence-based decision-making. Studies broadly agree that successful digital transformation requires alignment across technology,

organization, and institutional culture. Vasilev et al. [14] explained that transformation involves technical, organizational, and cultural adjustments that must proceed together for systems to be effective and adopted. Mounira et al. [17] showed that digitization supports comprehensive quality assurance through enhanced operational efficiency, while Alenezi et al. [16] identified the need to balance innovation with accessibility and inclusivity. Isaifan and Hasna [18] demonstrated that embedding AI into quality assurance systems produces scalable solutions that outperform manual processes in both speed and consistency.

In the Philippine context, Cadiz et al. [24] documented increasing digital tool adoption in academic management, while Wagan et al. [25] showed that digital platforms improve documentation and reporting efficiency in institutional records management. Pardiñan et al. [26] emphasized digitization as a mechanism for quality assurance among SUCs pursuing accreditation. Malatsico and Manayan [27] identified training and organizational readiness as critical success factors—a finding that distinguishes local implementations, which must account for varying levels of technological familiarity among faculty and staff, from international frameworks that often assume baseline digital proficiency. Critically, however, none of the Philippine digital transformation studies examined operationalize transformation within a faculty evaluation system that simultaneously incorporates AI-assisted feedback analysis, multi-source data collection, and automated

administrative workload reduction within a single platform. The inefficiency and administrative workload problems documented across local implementations persist precisely because no existing digitization initiative has combined these capabilities cohesively. These insights shape the implementation strategy of the proposed system, which integrates stakeholder consultation, automated evaluation workflows, centralized databases, and audit trails to reduce administrative burden, while CHED standards and ISO/IEC 25010:2023 quality compliance ensure that digital transformation at NAAP strengthens rather than compromises institutional effectiveness.

Faculty Evaluation

Faculty evaluation systems are formalized mechanisms for assessing instructional performance, professional conduct, and institutional contributions, providing the evidence base needed for promotion, retention, and professional development decisions. Research consistently establishes that structured evaluation produces more objective and reliable performance assessments. Mangila [28] established a strong empirical link between evaluation outcomes and actual teaching effectiveness and work performance in Philippine higher education, underscoring the stakes of evaluation system design. Wenceslao [21] developed faculty evaluation software for Philippine SUCs that promotes uniform procedures and fair implementation, demonstrating the feasibility of digitizing

structured evaluation processes within local institutional constraints. Cinizan and Valdez [29] implemented analytics-enhanced evaluation platforms at Ilocos Sur Polytechnic State College that consolidate multi-source data and improve evaluation usability through descriptive and comparative analytics.

However, these implementations—while representing meaningful progress—share a defining structural limitation: they function as fragmented, single-function systems rather than integrated platforms. Wenceslao [21] automates data collection but does not provide AI-assisted qualitative analysis or real-time decision-support reporting. Cinizan and Valdez [29] consolidate multi-source data but rely on post-hoc descriptive analytics rather than real-time insight generation, and their system does not extract structured optimization recommendations from evaluation outcomes. Mangila [28] establishes the performance-evaluation relationship empirically but offers no technological framework for acting on that relationship systematically. International frameworks tend to be more analytically sophisticated but are rarely adapted to the regulatory and resource context of Philippine state universities. Across the reviewed literature, no existing faculty evaluation system in the Philippines integrates multi-source assessment, AI-powered qualitative analysis, real-time reporting, and structured post-evaluation recommendations into a single unified environment—leaving administrators without the data-driven optimization basis that evaluation data is theoretically capable of providing. The proposed system directly addresses

these limitations by designing a scalable, secure evaluation platform that delivers these capabilities cohesively, transforming faculty evaluation from a manual administrative burden into a strategic institutional decision-making tool.

Learning Analytics

Learning analytics refers to the measurement, collection, analysis, and reporting of data about learners and learning contexts for the purpose of understanding and optimizing learning environments and the decisions that affect them. Applied to faculty evaluation, learning analytics provides the conceptual and technical infrastructure for converting raw evaluation data into actionable insights. Studies widely agree that analytics-driven approaches improve the quality and utility of evaluation data. Xu et al. [30] and Chen and Yang [31] showed that big data frameworks integrating multiple information sources enable cross-departmental comparison and longitudinal trend tracking. Patacsil et al. [32] demonstrated through associative rule analysis that teaching behavior relationships become visible when evaluation data is processed systematically. Dinamling and Depaynos [33] applied exploratory factor analysis to organize student feedback into measurable instructional dimensions. Gonugunta and Leo [34] confirmed that data-driven decision-making frameworks inform institutional strategies more reliably than intuition-based approaches, while Cadiz et al. [24] highlighted the critical role of analytics in monitoring digital learning environments.

Philippine implementations, however, reveal a tendency toward retrospective descriptive analytics suited to institutional resource conditions rather than advanced, real-time processing. Cinizan and Valdez [29] implemented descriptive and comparative analytics at Ilocos Sur Polytechnic State College with demonstrable improvements in evaluation usability, yet this approach lacks real-time accessibility and interactive visualization that would enable administrators to act on evaluation data without the lag of separate analytical processing cycles. More fundamentally, none of the reviewed Philippine or international implementations generate structured post-evaluation insights—that is, explicit, data-derived recommendations for faculty development or institutional improvement based on the current evaluation cycle's results. Evaluation data is collected and summarized, but the translation of that data into an actionable optimization basis for the subsequent cycle remains unaddressed. These findings inform the analytics architecture of the proposed system, which provides a descriptive analytics dashboard delivering completion rates, average ratings, response rates, rankings, and historical trends in real time to professors, HR personnel, administrators, supervisors, and deans—extending local descriptive indicators through live visualization, interactive reporting, and structured post-evaluation recommendations without requiring specialized analytical expertise.

Multi-Source Evaluation

Multi-source evaluation, also referred to as 360-degree feedback, is an assessment approach that integrates performance data from multiple rater groups—typically students, peers, and supervisors—to produce a more comprehensive and balanced view of faculty performance than any single-rater design can achieve. Studies converge on the superiority of multi-source evaluation for producing balanced and reliable faculty performance assessments. Dalvi et al. [35] demonstrated that 360-degree feedback enhances employee development and performance by integrating diverse viewpoints. Hussain and Matore [36] confirmed that multi-evaluator approaches produce richer and more reliable instructional quality information than single-source designs. Tatari, Ebrahimi, and Tadayon [37] implemented an electronic 360-degree faculty evaluation method that assigned weighted contributions to student, peer, and supervisor assessments, using descriptive-analytical methods to measure satisfaction and performance levels. Cinizan and Valdez [29] further demonstrated that analytics-enhanced platforms consolidating multi-source data improve evaluation usability in Philippine SUC contexts.

Despite the convergence of evidence supporting multi-source evaluation, a critical implementation gap persists: no existing study or system has combined multi-source evaluation with AI-assisted qualitative analysis within a single unified

digital platform. Tatari et al. [37] integrate multiple rater groups but do not apply AI-powered analysis to the qualitative comments generated by those raters. Cinizan and Valdez [29] consolidate multi-source data within a Philippine SUC context but rely on descriptive analytics without AI-assisted insight extraction from open-ended responses. Hussain and Matore [36] and Dalvi et al. [35] establish the theoretical and empirical value of multi-source designs but do not address the computational infrastructure needed to process the qualitative data these designs generate at scale. This gap represents a significant missed opportunity: multi-source evaluation generates richer qualitative feedback precisely because it captures perspectives from different relational contexts, yet this richness remains unexploited without AI-assisted analysis. This theme directly informs the multi-source evaluation design of the proposed system, which investigates how the integration of student, peer, and supervisor inputs—processed through AI-assisted analysis and real-time reporting within a role-based access-controlled centralized platform—influences the reliability, completeness, and actionability of faculty performance assessment.

Peer Evaluation

Peer evaluation of teaching refers to the structured process by which faculty colleagues assess each other's professional conduct, instructional practices, collaborative engagement, and institutional citizenship—dimensions that students

are not positioned to evaluate. Research consistently demonstrates that well-designed peer assessment frameworks improve both the quality of feedback and professional collaboration among faculty. Cutroni and Paladino [38] found through systematic review that peer feedback encourages shared responsibility, constructive communication, and collective commitment to academic standards. Chovriy et al. [39] demonstrated that structured peer interactions improve cooperation, knowledge sharing, and institutional engagement. Darvishi et al. [40], [41] and Topping et al. [42] showed that AI-assisted peer assessment platforms enhance feedback clarity, fairness, and transparency, establishing that technological mediation can reduce the inconsistency that often undermines informal peer review. Alipour et al. [43] further demonstrated that decentralized collaborative peer assessment standardizes judgments and reduces interpersonal bias through organized digital workflows.

Philippine implementations of peer evaluation remain in relatively early stages of formalization. Gutierrez [44] documented peer assessment becoming a regular faculty review component in Manila higher education institutions but noted that it occurs largely through informal or semi-structured processes without digital workflow support. Cinizan and Valdez [29] implemented systematic recording of peer feedback but did not apply AI-powered pattern analysis to the qualitative content of peer comments. This creates a dual gap: locally, peer evaluation lacks the digital infrastructure to ensure consistency and reduce bias; and even in

international implementations where AI-assisted tools exist, they have not been deployed within integrated multi-source faculty evaluation systems that simultaneously collect student and supervisor inputs. The proposed system bridges both dimensions by using organized digital workflows to standardize peer assessment and reduce interpersonal bias, extending this through AI-powered pattern analysis of peer feedback, and employing anonymized response mechanisms and role-based access control to promote the honest, constructive communication demonstrated effective in prior research.

Policy Frameworks

Policy frameworks in the context of Philippine higher education faculty evaluation refer to the national regulatory instruments, compliance mandates, and governance structures that define evaluation criteria, roles, reporting requirements, and accountability mechanisms. Research widely agrees that strong policy frameworks are prerequisites for sustainable evaluation system implementation. Vasilev et al. [14] emphasized that governance structures supporting continuous assessment and institutional accountability are among the strongest predictors of evaluation system effectiveness. Isaifan and Hasna [18] demonstrated that integrating emerging technologies into quality assurance systems requires explicit ethical standards and oversight mechanisms. Malatsico and Manayan [27] provided Philippine-specific evidence that supportive

institutional policies encourage technology adoption and modernization, while their absence or ambiguity is a major barrier to consistent system application.

At the national policy level, CHED Memorandum Order No. 19, Series of 2025 [45] establishes revised requirements that align faculty evaluation with national quality standards, and DBM-CHED Joint Circular No. 3, Series of 2022 [46] institutionalizes evaluation procedures through the NBC 461 reclassification cycle, reinforcing standardized performance-based frameworks across public higher education institutions. A significant gap in the existing literature is that no reviewed system demonstrates full alignment with these current Philippine regulatory instruments within a web-based evaluation platform that simultaneously incorporates AI-assisted analysis. International research, while rich in governance theory, does not directly address the specific compliance requirements that Philippine SUCs and specialized aviation HEIs must satisfy. Philippine studies that do address policy compliance—such as Wenceslao [21] and Cinizan and Valdez [29]—predate or do not explicitly align with CHED Memorandum Order No. 19, Series of 2025 in their system design. Policy alignment is therefore woven into the foundational design of the proposed system, governing everything from instrument construction to data handling and workflow authorization, ensuring that the system achieves innovation through AI and web-based technologies while maintaining full regulatory compliance.

Student Evaluation

Student evaluation of teaching refers to the systematic collection of student perceptions and ratings regarding instructor performance, course delivery, communication effectiveness, and classroom engagement. Research consistently validates the importance of well-structured student evaluation instruments collected through accessible digital platforms. Zhao et al. [47] conducted a comprehensive literature review establishing that carefully structured instruments ensure meaningful and consistent student feedback, while Yang [48] recommended incorporating contextual and experiential factors to improve response quality and reduce superficial ratings. Ahmed, Hamim, and Nandi [49] demonstrated that web-based platforms efficiently gather student evaluations in online and blended learning environments, reducing administrative burden while increasing participation rates. Katende and Namutebi [50] confirmed that student perceptions provide valuable insights into instructional communication and engagement that other evaluators cannot as easily observe.

Philippine implementations reflect both the potential and the current limitations of student evaluation practice. Reyes et al. [51] implemented online faculty evaluation platforms promoting consistency and organized data management across PUP campuses. Dela Fuente [52] and Cinizan and Valdez [29] applied data-driven models to generate performance indicators and

summarize evaluation trends. Lopez, Eguia, and Simon [53] efficiently documented teaching outcomes through systematic student input at Baguio Central University, and Gutierrez [44] documented increasing reliance on web-based tools incorporating student feedback into formal faculty reviews. However, the limitation across all local and most international implementations is twofold: first, qualitative student comments are collected but not systematically analyzed through AI-assisted methods to extract structured insights; and second, evaluation results are summarized after the fact rather than made available in real time for immediate administrative decision-making. Student feedback therefore remains an underleveraged resource—data is gathered but its qualitative dimension is processed minimally or not at all. The proposed system addresses this by implementing structured, CHED-aligned student evaluation instruments delivered through draft-enabled digital forms with anonymized response mechanisms, with AI-assisted analysis using GPT-4.1 mini transforming qualitative comments into structured insights and real-time reporting providing immediate evaluation summaries.

System Quality

System quality in the context of digital evaluation platforms refers to the degree to which a software system meets defined quality characteristics encompassing functional correctness, reliability, usability, security, performance

efficiency, maintainability, compatibility, portability, and accessibility. The effectiveness of any digital evaluation platform ultimately depends on its software quality, as systems that perform unreliably or are difficult to use undermine the evaluation data they collect regardless of how well their assessment instruments are designed. The ISO/IEC 25010:2023 standard is widely recognized as the comprehensive international framework for measuring these characteristics objectively. Sarwosri et al. [54] demonstrated that ISO/IEC 25010 evaluates functional suitability, performance efficiency, and reliability with objective, verifiable benchmarks, making it superior to informal or ad hoc quality assessments. Vasilev et al. [14] established that system quality directly influences evaluation outcome accuracy and usefulness. Isaifan and Hasna [18] showed that technology embedded into quality assurance systems must itself meet high quality standards to support scalable, sustainable evaluation.

Philippine implementations of ISO/IEC 25010 in evaluation systems confirm its applicability in local SUC contexts: Daluyon and Zamora [19] and Wenceslao [21] integrated quality considerations into Philippine SUC system design. Santillan [20] explicitly applied ISO/IEC 25010 standards to validate software reliability—but applied only a subset of quality characteristics rather than the full complement. Cinizan and Valdez [29] emphasized user-friendly and flexible interfaces as determinants of system adoption, and Gutierrez [44] demonstrated that well-designed systems promote transparency and consistent reporting. The common

and significant limitation across all Philippine studies is the tendency to apply only a partial selection of ISO/IEC 25010 characteristics—typically functional suitability, reliability, and usability—while omitting maintainability, compatibility, portability, and accessibility from formal evaluation. No Philippine study has evaluated a faculty evaluation system against the complete ISO/IEC 25010:2023 quality model, leaving a gap in the evidence base regarding whether these systems meet international standards comprehensively. The proposed system addresses this by empirically evaluating all ISO/IEC 25010:2023 software quality characteristics—functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility—ensuring both international standards compliance and practical operability within NAAP's administrative environment.

Web-Based Systems

Web-based evaluation systems are digital platforms delivered through internet browsers that automate the collection, processing, and reporting of faculty evaluation data, eliminating the delays and inconsistencies characteristic of paper-based administration and enabling real-time access to evaluation results by authorized stakeholders. Studies broadly agree that web-based evaluation systems improve accessibility, transparency, operational efficiency, and reporting reliability compared to manual approaches. Tatari, Ebrahimi, and Tadayon [37]

demonstrated that electronic 360-degree systems integrating multiple perspectives into single platforms capture holistic performance data more efficiently than fragmented paper processes. Abhang and Gaike [55] emphasized that digital evaluation frameworks improve institutional effectiveness and accountability through automation. Jia et al. [56] validated reliable online instruments in teaching evaluation contexts, confirming that digital platforms produce psychometrically sound data when instruments are properly designed. Ahmed, Hamim, and Nandi [49] showed that digital platforms effectively adapt evaluations to both on-campus and remote instruction contexts, while Katende and Namutebi [50] confirmed that structured online feedback contributes meaningfully to teaching quality improvement.

Philippine implementations demonstrate contextual adaptations: Daluyon and Zamora [19] developed customized platforms for the University of Rizal System addressing specific operational needs, Santillan [20] applied ISO/IEC 25010 quality standards partially to reinforce software reliability, and Lopez et al. [53] and Gutierrez [44] documented growing institutional reliance on digital tools for transparent and consistent evaluation reporting. However, the gap in existing Philippine web-based systems is the absence of integrated AI analysis, real-time multi-source dashboards, structured post-evaluation recommendations, and formal user acceptance assessment within a single unified platform. While these systems automate data collection, they do not transform that data into structured

improvement insights, nor do they assess whether the platform itself achieves sustained user adoption—a critical factor for long-term institutional effectiveness that remains unexamined in existing local implementations. The proposed system addresses these gaps by customizing its architecture for NAAP's aviation education context, following ISO/IEC 25010:2023 quality standards, and formally assessing user acceptance in terms of perceived usefulness, ease of use, and overall user experience—providing the adoption evidence that existing systems have not generated.

Synthesis of Literature

A comparative analysis of the reviewed literature across the thirteen thematic areas reveals a coherent and directional story: faculty evaluation systems are evolving from isolated administrative tasks into integrated mechanisms for quality assurance, instructional improvement, and evidence-based institutional decision-making. Common trends across both international and Philippine studies confirm that web-based platforms have established themselves as the practical standard for scalable evaluation, with digital automation of collection, processing, and reporting dramatically outperforming manual approaches in speed, consistency, and accessibility. Multi-source evaluation—combining student, peer, and supervisor inputs—produces measurably more balanced and reliable faculty performance data than any single-rater design [35], [36], [37], [29]. Artificial

intelligence, particularly advanced language models such as GPT-4 and its successors, has demonstrated substantial capacity to transform qualitative feedback into structured, actionable insights through frequency analysis, comment summarization, and pattern detection [1], [2], [3], [4], [5]. Policy frameworks—particularly CHED Memorandum Order No. 19, Series of 2025 [45] and DBM-CHED Joint Circular No. 3, Series of 2022 [46]—provide the regulatory structure within which technological innovation must operate, while strong authentication mechanisms and compliance with the Data Privacy Act of 2012 are non-negotiable requirements for systems handling sensitive faculty performance records. Software quality, assessed through ISO/IEC 25010:2023, ensures that evaluation platforms are reliably performant, secure, maintainable, and accessible.

The strengths of existing research lie in its breadth: international studies offer methodologically sophisticated frameworks and AI integration models, while Philippine studies provide contextually grounded implementations demonstrating the feasibility of digitized evaluation within local SUC resource constraints. The weaknesses, however, are consistent and cross-cutting. First, multi-source evaluation and AI-assisted qualitative analysis have never been combined within a single operational system: systems that collect multi-source data do not apply AI to its qualitative components, and systems that apply AI do not integrate differentiated rater groups. Second, no Philippine evaluation system has been assessed against the complete ISO/IEC 25010:2023 quality framework;

assessments remain partial and selective, leaving comprehensive quality compliance undemonstrated. Third, no reviewed system contextualizes its design and compliance within current Philippine aviation higher education requirements, combining CHED Memorandum Order No. 19, Series of 2025 alignment with aviation-specific instrument adaptation. Fourth, post-evaluation structured insights—explicit, data-derived recommendations for faculty development and institutional improvement—are absent from existing implementations; evaluation data is collected and reported but not translated into an actionable optimization basis. Fifth, user acceptance has not been formally and systematically assessed in existing Philippine faculty evaluation systems, leaving the adoption level and its determinants unknown. These five convergent gaps—lack of multi-source and AI integration, no full ISO/IEC 25010 evaluation, absence of post-evaluation insights, and limited acceptance evidence—constitute the research problem that the proposed system is designed to resolve.

Conceptual Link to the Current Study

The reviewed literature collectively establishes the theoretical and empirical foundation upon which the proposed system is built and positions the current study within—and beyond—the existing body of knowledge. The AI analysis and artificial intelligence themes [1]–[13] directly support the system's AI-enhanced feedback component, justifying the adoption of GPT-4.1 mini and frequency analysis as

validated approaches for processing qualitative evaluation data at scale—while simultaneously confirming that this combination has not been operationalized within a Philippine faculty evaluation context. The assessment instruments literature [14]–[22] grounds the instrument design in standards-aligned measurement theory and aviation-specific precedents established by Dueñas and Martin [22], ensuring CHED compliance and contextual appropriateness, and confirms the absence of current CMO No. 19, Series of 2025 alignment in existing digital instrument implementations. The secure authentication research [23] informs the targeted OTP protocol and bcrypt-based architecture that protects sensitive faculty data in compliance with RA 10173.

The digital transformation and policy frameworks literature [14], [16], [17], [24]–[27], [45], [46] justifies the systemic, stakeholder-inclusive implementation approach and CHED regulatory alignment, confirming that technological innovation must be accompanied by organizational readiness and policy compliance—while also revealing that no existing Philippine system has achieved this combination within a faculty evaluation platform. The faculty evaluation, multi-source evaluation, peer evaluation, student evaluation, and learning analytics themes [28]–[53] collectively validate the multi-source assessment architecture and analytics dashboard, demonstrating that integrating student, peer, and supervisor inputs within a real-time analytics environment produces evaluation outcomes superior to single-source, manual approaches—while confirming that no

existing system has paired this integration with AI-assisted qualitative analysis and structured post-evaluation recommendations. Finally, the system quality literature [14], [18], [19], [20], [21], [29], [54] establishes ISO/IEC 25010:2023 as the appropriate evaluation standard and confirms its applicability in Philippine SUC contexts, while also confirming that no existing study has applied the full quality framework to a faculty evaluation system.

The proposed system addresses the identified gaps by consolidating web-based infrastructure, multi-source evaluation mechanisms, GPT-4.1 mini-powered analytical tools, layered security safeguards, regulatory compliance, full ISO/IEC 25010:2023 quality evaluation, formal user acceptance assessment, and structured post-evaluation recommendations into a unified platform—delivering an evaluation environment that is more efficient than manual processes, more comprehensive than single-source approaches, more analytically insightful than traditional reporting methods, and more secure and compliant than fragmented implementations. This study thus advances the literature by providing the first integrated, AI-enhanced, CHED-compliant faculty evaluation system designed specifically for the operational and accreditation requirements of Philippine aviation higher education, establishing a replicable model for similar institutions navigating the same convergence of technological opportunity, regulatory obligation, and institutional improvement imperative.

CHAPTER 3

SYSTEM ANALYSIS, DESIGN, AND METHODOLOGY

Research and Development Methodology

This chapter discusses the technical side of our Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines, focusing on how each part of the system functions behind the scenes. Starting from the input, followed by the process, and ending with the output, the section explains how evaluation data is collected, analyzed, and transformed into accurate and secure results. The chapter presents the overall system outline, showing the main components and features of the platform, as well as the different levels of access granted to authorized users such as students, faculty members, supervisors, HR personnel, and the Office of Student Affairs. Overall, this section aims to provide a clear understanding of how the system operates and how it was developed to support a secure and efficient evaluation process.

Software Development Life Cycle Model

This section presents the Software Development Life Cycle (SDLC) adopted in the design and development of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments

for the National Aviation Academy of the Philippines. The study employed the Agile Software Development Life Cycle using a hybrid Scrum-inspired framework to support iterative development, continuous improvement, and structured stakeholder involvement throughout the system development process.

The Agile approach was selected to accommodate evolving system requirements, user feedback, and institutional policies, particularly those aligned with the evaluation framework of the Commission on Higher Education (CHED). Through incremental development, the system was implemented in two structured development cycles (sprints) to deliver functional components early, validate system features with users, and refine the system based on testing results and stakeholder feedback.

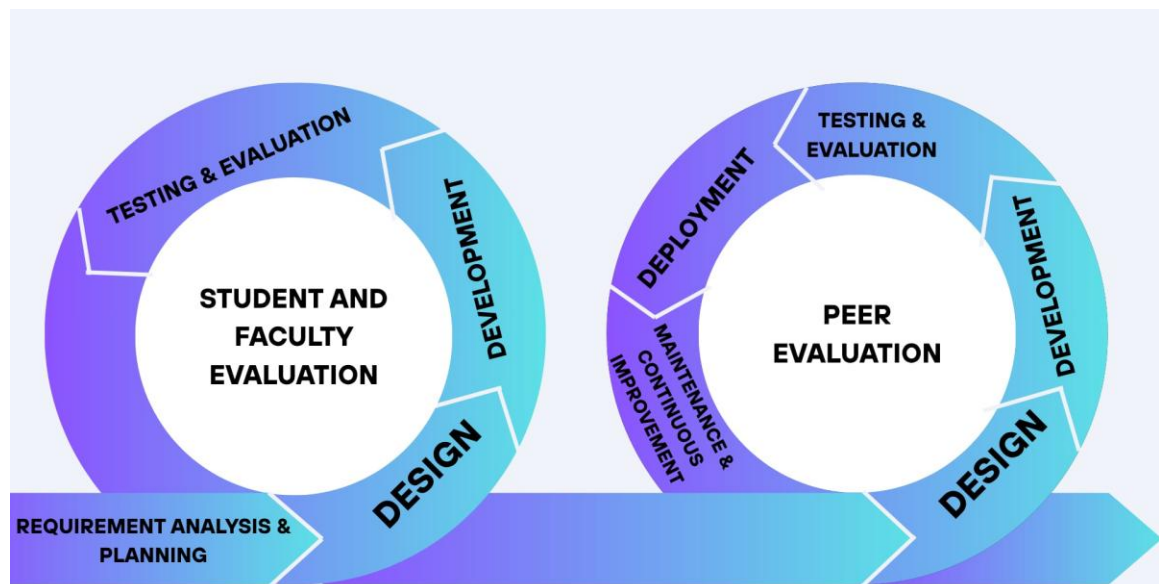


Figure 1. Agile Software Development Life Cycle Workflow for the Development of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments.

Figure 1 illustrates the Agile development workflow adopted in this study, demonstrating the iterative progression of system development phases supported by continuous stakeholder feedback and improvement cycles.

Agile Development Phases

The Agile development process implemented in this study consists of six major iterative phases, each producing specific deliverables that guided subsequent development activities.

Requirements Analysis and Planning

The requirements analysis and planning phase was conducted through structured consultations with key stakeholders, including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel, and system administrators. Data were gathered using interviews and direct observation of existing processes to identify both functional and non-functional system requirements in accordance with CHED Memorandum Order No. 19, Series of 2025, as well as institutional evaluation policies. The main outcomes of this phase included comprehensive documentation of functional and non-functional requirements, clear definitions of stakeholder roles and responsibilities, a detailed system requirement specification aligned with ISO/IEC 29148 standards and identification of system performance, reliability, and security requirements. This phase provided a solid foundation for

the design of the system architecture and ensured that the proposed solution effectively addressed existing operational challenges.

Design

The design phase involved translating the documented requirements of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments into detailed technical blueprints. This process encompassed the creation of system architecture diagrams, database models, data flow diagrams, user interface prototypes, and security specifications. The system architecture was developed using a web-based client-server model to support centralized management of evaluation data, while AI-assisted analysis modules were integrated to enable automated summarization and frequency analysis of qualitative feedback, ensuring both data transparency and the authenticity of evaluator responses. Additionally, the design phase included defining user interface layouts for students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel, and system administrators, as well as specifying security and access control mechanisms to safeguard sensitive data. These design activities established a comprehensive framework for the subsequent development, testing, and implementation of the system.

Development

During the development phase, the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments was implemented using web-based programming technologies, database management systems, and client-side interface frameworks. Development followed iterative sprint cycles, allowing incremental implementation of features and integration testing to ensure seamless operation across system modules. The system incorporated AI-assisted qualitative feedback analysis through both frequency analysis and AI-assisted summarization techniques such as the OpenAI GPT-4.1 mini and natural language processing (NLP) techniques to identify recurring themes and patterns in open-ended evaluation responses. This AI component functions solely as an analytical support tool and does not modify original evaluator responses or numerical ratings, thereby upholding principles of transparency and fairness. In addition, the system integrated fully functional evaluation modules, role-based authentication and authorization mechanisms, automated reporting and dashboard features, and a clearance verification subsystem, providing a complete framework for managing evaluations, monitoring completion, and supporting data-driven decision-making by students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), and administrative personnel.

Testing and Evaluation

The testing and evaluation phase focused on verifying the functionality, usability, performance efficiency, reliability, and security of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments. The process followed structured testing protocols, including functional testing, integration testing, and performance validation, to ensure that all system components operated as intended. System quality evaluation was conducted in accordance with the ISO/IEC 25010:2023 software quality model, providing a standardized framework for assessing functional suitability, reliability, usability, security, and performance efficiency. Testing procedures also ensured compliance with institutional data privacy requirements under the Data Privacy Act of 2012 (Republic Act No. 10173). The outcomes of this phase included comprehensive documentation of functional and performance testing, security and authentication validation, usability assessment, and overall software quality evaluation, providing assurance that the system met defined criteria and was ready for pilot implementation and full operational deployment.

Deployment and Maintenance

Following successful testing and validation, the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments was prepared for institutional deployment within the National Aviation

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Academy of the Philippines' server infrastructure. The deployment process encompassed system configuration, user orientation and training, and planning for data migration to ensure seamless integration with existing institutional records. The maintenance phase facilitates continuous system improvement through iterative feedback collected from users, including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel and administrative personnel. Leveraging the Agile methodology, the system undergoes periodic updates, security enhancements, and performance optimizations, ensuring long-term sustainability, adaptability to institutional needs, and continued alignment with operational requirements. This phase also includes comprehensive documentation of deployment procedures, user training materials, maintenance protocols, and plans for feedback-driven system enhancements.

Agile Methodology Justification

The selection of the Agile methodology for this study is supported by its demonstrated effectiveness in modern software development contexts, particularly in enhancing adaptability, stakeholder collaboration, and software quality. Recent research indicates that Agile approaches improve organizational responsiveness to changing requirements, enable frequent feedback loops with stakeholders, and support iterative development practices that can lead to higher quality outcomes

in complex projects. Studies show that Agile methodology enhances collaboration among team members and stakeholders, improves adaptability to evolving requirements, and contributes to enhanced software quality through continuous testing and refinement (Punitha & Kumanan, 2026). The iterative nature of Agile development aligns with the needs of educational system development, allowing the proposed system to respond effectively to evolving institutional policies, emerging CHED guidelines, and technological requirements, thereby ensuring long-term relevance and scalability.

Integration of AI-Assisted Components within the Agile Framework

Artificial intelligence modules were incorporated throughout the iterative development cycles to facilitate the analysis of qualitative feedback. The AI system employs frequency-based text summarization and thematic clustering to assist administrators in efficiently interpreting large volumes of evaluation comments. To ensure ethical transparency and compliance with regulatory standards, AI functionality is restricted to analytical support and does not modify original evaluation responses. All outputs generated by the AI undergo administrative review before being used in institutional reporting, ensuring accuracy, fairness, and accountability in decision-making.

System Development Framework

The system development framework was structured using a layered architecture comprising the presentation, application, and data management layers. The development process encompassed requirement gathering, iterative module creation, integration testing, and system validation. The system was implemented using modern web technologies, including server-side scripting, relational database management systems, and client-side interface frameworks. AI-assisted modules were incorporated to support the analysis of qualitative feedback through frequency-based text summarization techniques. These AI components are intended to aid administrative interpretation of feedback while maintaining the integrity and authenticity of user responses.

Analysis of the Existing System

This section examines the current state of faculty evaluation at the National Aviation Academy of the Philippines (NAAP), identifying the operational methods in place, the limitations they produce, and the resulting gaps in institutional capability. The analysis also evaluates the alternative approaches considered before arriving at the development of a custom web-based system as the most appropriate solution.

Current Evaluation Procedures

Faculty evaluation at NAAP is presently conducted through a combination of printed forms and general-purpose online survey tools that were not specifically designed for academic performance assessment. At the start of each evaluation period, administrative staff either distribute physical forms to students or share externally hosted survey links for completion. Supervisory evaluations follow a similar manual process, with department heads and supervisors filling out separate institutional instruments that are then submitted to the Human Resource office for encoding.

The consolidation of evaluation data is performed by HR staff, who manually encode responses from collected forms into spreadsheet files, aggregate numerical scores, and compile summary reports for distribution to department-level administrators. Open-ended comments written by students are reviewed individually by HR personnel, a process that is time-intensive and susceptible to subjective interpretation. No automated mechanism exists for synthesizing qualitative feedback across a faculty member's full evaluation set.

Clearance processing at the end of each semester requires the Office of Student Affairs (OSA) to manually verify, for each student, whether all required evaluations have been submitted. This cross-referencing is performed against the

HR office's encoded records and the course enrollment records, with no direct system linkage between the three data sources.

Limitations and Problem Analysis

The existing evaluation process exhibits several significant limitations that impair the institution's capacity to conduct reliable, efficient, and compliant faculty assessments. Manual data encoding introduces transcription errors and inconsistencies that reduce the accuracy of reported evaluation results. The absence of a centralized, role-differentiated digital platform means that data from different evaluation types — student, supervisor, dean, and peer — must be managed through separate processes, increasing the administrative burden and creating fragmented records.

The absence of any authentication mechanism linked to official enrollment data means that the existing survey tools cannot enforce submission eligibility. Students may submit evaluations for faculty members they were not taught by, submit duplicate responses, or fail to complete evaluations without any automated detection. General-purpose survey platforms do not support the CHED-mandated questionnaire format, preventing the institution from generating evaluation reports that conform to the specifications of CHED Memorandum Order No. 19, Series of 2025.

The manual review of qualitative, open-ended feedback is inherently time-consuming and produces inconsistent interpretations across different HR staff members. The lack of structured data prevents any longitudinal analysis of faculty performance trends across evaluation periods.

These limitations collectively impede the institution's capacity to use faculty evaluation data as a meaningful tool for academic quality management, faculty development, and regulatory compliance reporting.

Requirements Gap Analysis

A requirements gap analysis was undertaken to identify the specific functional and operational capabilities that the existing system fails to provide, and which are necessary for NAAP to meet its institutional, regulatory, and data governance obligations.

The most critical gap identified is the complete absence of a secure, enrollment-linked evaluation submission system. The institution currently has no mechanism for automatically verifying that a student is enrolled in a course before allowing them to evaluate the corresponding faculty member. This creates a structural vulnerability in the integrity of evaluation data that cannot be addressed through improvements to the existing tools.

A second major gap concerns the processing of qualitative feedback. The manual review process cannot scale to the volume of open-ended comments generated across all evaluation periods, and provides no objective basis for cross-faculty or longitudinal comparisons. An AI-assisted summarization capability would address this gap directly.

The gap analysis also identified the absence of role-differentiated dashboards, automated clearance verification, CHED-compliant reporting, a non-editable audit trail, and multi-campus data management as additional capability deficits that must be addressed in the proposed system.

Alternative Solutions and Feasibility Analysis

Before committing to the development of a custom system, three alternative approaches were evaluated against NAAP's requirements.

Enhancing the existing survey tools was considered first due to its low initial cost. However, the general-purpose survey platforms currently in use cannot be configured to enforce enrollment-linked access control, generate CHED-compliant PDF reports, or support role-based access across seven distinct user types. The fundamental architectural limitations of these tools make them unsuitable as a foundation for the proposed system's requirements.

Commercial evaluation platforms were evaluated as a second alternative. These products typically offer standardized evaluation workflows and reporting features, but none of the platforms reviewed supported the specific CHED-mandated questionnaire formats and impose ongoing licensing and per-user subscription costs that would exceed NAAP's long-term operational budget.

Proposed Technology Alternatives

The development of a custom web-based evaluation system was selected as the most appropriate solution based on its capacity to fully satisfy NAAP's institutional, regulatory, and operational requirements. A custom system allows direct integration with the institution's official enrollment and faculty assignment records, ensuring that evaluation access is automatically governed by current course data rather than manual list management. It also permits the full implementation of CHED-compliant questionnaire instruments, including the specific rating scale configurations and open-ended comment requirements mandated by CHED Memorandum Order No. 19, Series of 2025.

A custom solution enables the incorporation of AI-assisted feedback analysis — using the OpenAI GPT-4.1 mini API in combination with local frequency analysis — a capability that is not available in general-purpose survey tools and is absent from most commercial evaluation platforms. It also supports the

implementation of multi-layered security architecture, including Bcrypt password hashing, email-based one-time password (OTP) authentication, Cloudflare firewall protection, and role-based access control across seven distinct user roles.

The custom system further supports multi-campus deployment, automated clearance verification, a non-editable audit trail, soft-delete data preservation, and real-time monitoring dashboards — all features that are either unavailable or prohibitively expensive to configure in off-the-shelf alternatives. While custom development requires a higher upfront investment than the other options considered, the long-term operational savings, regulatory compliance assurance, and institutional scalability it provides justify the investment, as confirmed by the cost-benefit analysis presented in the subsequent section.

Feasibility Study

The feasibility study assesses the viability of developing and deploying the proposed AI-Enhanced CHED-Compliant Web-Based Faculty and Student Evaluation System across four critical dimensions: technical, operational, economic, and schedule. Each dimension was evaluated with specific reference to the conditions and resources of NAAP in order to arrive at a well-grounded determination of project viability. Together, these assessments establish that the

proposed system is both achievable within NAAP's current environment and capable of delivering meaningful institutional value.

Technical Feasibility

NAAP's existing information technology infrastructure is capable of supporting the proposed system without requiring major hardware upgrades. The system is designed for deployment on a standard web server environment compatible with the institution's current server hardware. The recommended server configuration — a multi-core processor, 16 GB or more of RAM, and SSD-based storage — falls within the capacity of modern institutional hardware available at accessible market prices, as detailed in the hardware requirements section of this chapter.

Network connectivity at NAAP is stable and sufficiently provisioned to support concurrent web-based access by students, faculty, and administrative staff during evaluation windows. The AI-assisted feedback analysis component is implemented through API integration with OpenAI GPT-4.1 mini rather than through locally hosted machine learning infrastructure, which ensures that AI processing capability is available without placing a sustained computational burden on institutional servers. This design choice significantly reduces technical risk and hardware cost while preserving full analytical functionality.

The security components of the proposed system — including Bcrypt password hashing, risk-adaptive email-based OTP authentication, Cloudflare web application firewall integration, and role-based access control — are all implemented using widely supported technologies that integrate smoothly with standard web server and relational database environments. The primary technical risk is elevated server load during peak evaluation periods. This risk is mitigated through database query optimization, efficient session management, and connection pooling strategies applied during the development phase. The modular architecture of the system also allows individual components to be scaled or updated independently as institutional requirements evolve. On the basis of these considerations, the proposed system is assessed as technically feasible within NAAP's existing environment.

Operational Feasibility

The proposed system is designed to align with and automate NAAP's existing evaluation workflows rather than replace established institutional procedures. Students will continue to evaluate faculty members during designated evaluation periods; the change is that submissions will occur through a secure, role-specific web portal linked to official enrollment data rather than through paper forms or general-purpose survey links. The system automatically presents each

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student with only the evaluation forms relevant to their enrolled courses, eliminating the risk of students rating faculty members they were not taught by.

Faculty members, deans, HR personnel, OSA officers, the VPAA, and system administrators each interact with a role-specific dashboard tailored to their responsibilities. Faculty gain timely access to their performance data and AI-generated feedback summaries. Deans can monitor department-level evaluation statistics, manage peer evaluation pairings, and conduct supervisor evaluations within the same platform. HR staff manage the full evaluation lifecycle — from setting evaluation periods and configuring questionnaires to publishing feedback summaries and generating CHED-compliant reports — without any manual data consolidation. OSA officers gain a focused clearance verification view that eliminates the need for manual cross-referencing.

User adoption is anticipated to be smooth given that the majority of NAAP's students and faculty already interact with web-based academic platforms as part of their regular routines. The role-specific interface design minimizes the learning curve by presenting each user group with only the functions relevant to their work. Orientation sessions and in-system guidance will support users during the initial deployment period. The proposed system is therefore assessed as operationally feasible.

Economic Feasibility

The economic feasibility of the proposed system is established through the cost-benefit analysis presented in the following section, which demonstrates a positive return on investment over a three-year lifecycle. The total project cost across all six Agile development phases is ₱446,310, covering requirements analysis, system design, core and advanced module development, quality assurance testing, deployment, evaluation, and long-term maintenance. Against these costs, the system is projected to generate ₱250,000 in annual tangible savings through the elimination of manual evaluation administration, reduction in paper and printing costs, and automation of clearance verification.

The three-year net benefit of ₱303,690 and a return on investment of 68.04 percent confirm that the proposed system is economically sound. Beyond measurable savings, the system delivers significant intangible value: full compliance with CHED CMO No. 19, Series of 2025, and RA 10173; improved data accuracy and institutional transparency; and a scalable platform that can serve all NAAP campuses without proportional increases in cost. The use of open-source web development frameworks and existing institutional hardware infrastructure keeps licensing and infrastructure costs low, further strengthening the economic case for development. The proposed system is therefore assessed as economically feasible.

Schedule Feasibility

Development of the proposed system follows an iterative Agile framework organized into structured sprint cycles. This approach enables the team to deliver functional modules incrementally, with each sprint producing testable, stakeholder-reviewable outputs before subsequent development proceeds. Sprint planning sessions at the project's outset establish a prioritized feature backlog, ensuring that core functionality — user authentication, role-based access, and evaluation submission — is delivered and validated early in the timeline before more complex modules such as AI integration and reporting are completed.

The implementation schedule is designed so that the fully tested and configured system is ready for deployment before the start of a scheduled evaluation period, giving administrative staff and faculty members adequate time to familiarize themselves with the platform before it is activated for institutional use. User training sessions are incorporated into the schedule as a formal activity, not an afterthought, ensuring all role groups are prepared before go-live.

Potential scheduling risks include delays from stakeholder availability during requirement consultations and the time required to validate the AI integration and security configuration. These risks are managed through flexible sprint scheduling, early and sustained stakeholder involvement, and a phased deployment strategy that releases completed modules progressively. Given

NAAP's academic calendar and the development team's capacity under the Agile model, the proposed system is assessed as schedule feasible.

Cost-Benefit Analysis

This Cost-Benefit Analysis presents a comprehensive financial and operational evaluation of the development and implementation of the proposed AI-Enhanced CHED-Compliant Web-Based Faculty and Student Evaluation System for NAAP. The analysis is structured across six Agile development phases, with each phase broken down into key activities, role-based labor cost computations, and summary totals. Following the phase-by-phase cost breakdown, a benefit analysis quantifies the projected institutional savings and returns over a three-year operational lifecycle.

Labor cost rates applied throughout this analysis are based on current Philippine market salary data for IT professionals as of 2025, sourced from salary benchmarking platforms including SalaryExpert, JobStreet, and industry compensation surveys. Hourly rates reflect mid-level professionals appropriate for an institutional academic system development project.

Formula and Computation Explanations

The following formulas govern all labor cost computations, total cost aggregations, and benefit calculations used throughout this analysis. Each formula is applied consistently across all phases and activities.

Labor Cost Formula

Each line-item labor cost in the activity tables is computed using the following formula:

$$\text{Cost (₱)} = \text{Number of Persons} \times \text{Hourly Rate (₱)} \times \text{Number of Hours}$$

Example: For 2 Developers at ₱480/hr working 10 hours:

$$\text{Cost} = 2 \times \text{₱480} \times 10 = \text{₱9,600}$$

This formula multiplies the number of personnel assigned to an activity by their respective hourly rate and the estimated hours required to complete that activity. Where a fixed estimated cost is listed (e.g., for tools or materials), this represents a lump-sum estimate not computed through the labor formula.

Phase Total Cost Formula

The total cost for each phase is the sum of all sub-total costs from the activities within that phase:

$$\text{Phase Total Cost} = \Sigma (\text{Sub-Total of each activity in the phase})$$

Overall Total Project Cost Formula

The overall total project cost sums all six phase totals:

$$\begin{aligned} \text{Total Project Cost} &= \text{Phase 1} + \text{Phase 2} + \text{Phase 3} + \text{Phase 4} + \text{Phase 5} + \text{Phase 6} \\ &= \text{P}52,675 + \text{P}61,485 + \text{P}181,030 + \text{P}61,465 + \text{P}45,965 + \text{P}44,690 = \text{P}446,310 + \\ &\quad \text{P}44,690 \\ &= \text{P}446,310 \end{aligned}$$

Total Tangible Benefits Formula

Total three-year tangible benefits are computed by multiplying the annual tangible savings by the number of operational years evaluated:

$$\begin{aligned} \text{Total Benefits (3 Years)} &= \text{Annual Tangible Savings} \times 3 \\ &= \text{P}250,000 \times 3 = \text{P}750,000 \end{aligned}$$

Net Benefit Formula

The net benefit represents the financial gain after deducting the total project cost from total benefits:

$$\begin{aligned} \text{Net Benefit} &= \text{Total Tangible Benefits (3 Years)} - \text{Total Project Cost} \\ &= \text{₱750,000} - \text{₱446,310} = \text{₱303,690} \end{aligned}$$

Return on Investment (ROI) Formula

ROI expresses the net benefit as a percentage of total investment, measuring efficiency of the project expenditure:

$$\begin{aligned} \text{ROI (\%)} &= (\text{Net Benefit} \div \text{Total Project Cost}) \times 100 \\ &= (\text{₱303,690} \div \text{₱446,310}) \times 100 = 68.04\% \end{aligned}$$

An ROI of 68.04% indicates that for every peso invested in the system, the institution recovers the investment plus an additional ₱0.68 in net savings over the three-year period.

Hourly Rate Basis (Philippine Market Rates, 2025)

The following hourly rates are applied uniformly across all phases. Rates are based on mid-level professional compensation benchmarks sourced from SalaryExpert, JobStreet, and industry surveys as of 2025.

Table 3.1. Labor Cost Hourly Rate Basis — Philippine IT Market, 2025

Role	Hourly Rate (P)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Stakeholder Representative	₱350
Technical Support Staff	₱300

Phase 1: Requirements Analysis and Planning

The Requirements Analysis and Planning phase establishes the foundation of the entire system development effort. It focuses on identifying and engaging institutional stakeholders across all seven user roles, eliciting and documenting functional and non-functional system requirements, translating requirements into prioritized user stories, and organizing these into a product backlog ready for sprint execution. Given the complexity of the proposed system — which integrates AI feedback analysis, multi-role access controls, CHED compliance, and multi-campus support — thorough requirements documentation at this stage is essential to prevent costly rework in later phases.

Key activities include:

- Stakeholder Identification and Engagement

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- Requirement Elicitation
- User Story Development
- Product Backlog Creation and Prioritization
- Sprint Planning

A. Cost Analysis

Assumptions — Labor Cost Estimates

Table 3.2. Phase 1 — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
Stakeholder Representative	₱350

Number of stakeholders involved: 5

1. Stakeholder Identification and Engagement Cost

Table 3.3. Phase 1 — Stakeholder Identification and Engagement Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Identify Stakeholders	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Identify Stakeholders	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Stakeholder Meetings	Project Manager	1	₱725	4	$1 \times 725 \times 4$	₱2,900
Stakeholder Meetings	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Stakeholder Meetings	Stakeholders	5	₱350	4	$5 \times 350 \times 4$	₱7,000
Requirement Documentation	System Analyst	1	₱550	5	$1 \times 550 \times 5$	₱2,750
Requirement Documentation	Developer	1	₱480	2	$1 \times 480 \times 2$	₱960
Communication Tools	Internet / Tools	—	—	—	Estimated	₱600
Documentation Materials	Cloud Storage	—	—	—	Estimated	₱350

Sub-Total: ₱21,135

2. Requirement Elicitation Cost

Table 3.4. Phase 1 — Requirement Elicitation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Requirement Interviews	System Analyst	1	₱550	6	$1 \times 550 \times 6$	₱3,300
Workshops	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Workshops	Stakeholders	5	₱350	3	$5 \times 350 \times 3$	₱5,250
Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱12,375

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3. User Story Development Cost

Table 3.5. Phase 1 — User Story Development Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
User Story Writing	System Analyst	1	₱550	5	$1 \times 550 \times 5$	₱2,750
Feature Discussion	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
UI Concept Planning	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720

Sub-Total: ₱6,645

4. Product Backlog Creation and Prioritization Cost

Table 3.6. Phase 1 — Product Backlog Creation Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Backlog Organization	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Feature Prioritization	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Technical Review	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440

Sub-Total: ₱5,815

5. Sprint Planning Cost

Table 3.7. Phase 1 — Sprint Planning Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Sprint Planning Meeting	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Sprint Planning Meeting	Developer	2	₱480	3	$2 \times 480 \times 3$	₱2,880

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Sprint Planning Meeting	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱6,705

B. Total Cost of Phase 1

Table 3.8. Phase 1 — Total Cost Summary

Activity	Cost (P)
Stakeholder Identification and Engagement	₱21,135
Requirement Elicitation	₱12,375
User Story Development	₱6,645
Product Backlog Creation and Prioritization	₱5,815
Sprint Planning	₱6,705
TOTAL COST — PHASE 1	₱52,675

Phase 2: System Design and Prototype Development

The System Design and Prototype Development phase translates the validated requirements and prioritized backlog into concrete technical designs and an interactive working prototype. This phase covers the full architectural blueprint of the system — including its three-tier client-server structure, relational database schema, role-based access control model, and AI analysis module integration — alongside UI/UX wireframes for each of the seven role-specific portals. A prototype is developed to allow stakeholder validation before full-scale coding begins, reducing the risk of costly interface or architectural revisions in Phase 3.

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Key activities include:

- System Architecture Design
- Database Design and Data Modeling
- UI/UX Wireframing and Interface Design
- Prototype Development
- Design Review and Validation

A. Cost Analysis

Table 3.9. Phase 2 — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380

1. System Architecture Design Cost

Table 3.10. Phase 2 — System Architecture Design Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Architecture Planning	Project Manager	1	₱725	4	$1 \times 725 \times 4$	₱2,900

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Technical Architecture Design	System Analyst	1	₱550	6	$1 \times 550 \times 6$	₱3,300
Component and Module Mapping	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Infrastructure Planning	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Architecture Documentation	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Collaboration Tools	Internet / Platforms	—	—	—	Estimated	₱600

Sub-Total: ₱12,570

2. Database Design and Data Modeling Cost

Table 3.11. Phase 2 — Database Design and Data Modeling Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Data Requirement Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Database Schema Design	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Entity Relationship Diagram (ERD)	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Data Normalization (3NF)	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Database Design Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱10,370

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3. UI/UX Wireframing and Interface Design Cost

Table 3.12. Phase 2 — UI/UX Wireframing Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
User Interface Planning	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
Wireframe Design (8 Portals)	UI/UX Designer	1	₱430	8	$1 \times 430 \times 8$	₱3,440
User Flow Mapping	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
UX Review Session	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
UI Prototype Mockups	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
Design Collaboration Tools	Design Platforms	—	—	—	Estimated	₱500

Sub-Total: ₱11,030

4. Prototype Development Cost

Table 3.13. Phase 2 — Prototype Development Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Prototype Coding	Developer	2	₱480	8	$2 \times 480 \times 8$	₱7,680
Front-End Interface Build	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
API Mock Development	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Prototype Integration	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Technical Supervision	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Prototype Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱17,745

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5. Design Review and Validation Cost

Table 3.14. Phase 2 — Design Review and Validation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Prototype Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Design Evaluation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Usability Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Stakeholder Validation	Stakeholders	5	₱350	2	$5 \times 350 \times 2$	₱3,500
Feedback Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,770

B. Total Cost of Phase 2

Table 3.15. Phase 2 — Total Cost Summary

Activity	Cost (₱)
System Architecture Design	₱12,570
Database Design and Data Modeling	₱10,370
UI/UX Wireframing and Interface Design	₱11,030
Prototype Development	₱17,745
Design Review and Validation	₱9,770
TOTAL COST — PHASE 2	₱61,485

Phase 3: Core System Module Development (Sprint 1)

Sprint 1 covers the development of the system's foundational layer: user authentication with email-based OTP, role-based access control for all seven user roles, evaluation form submission and draft saving, enrollment-linked form

assignment, automated clearance verification, and the basic HR and student dashboards. These modules form the backbone on which all subsequent features depend and must be fully functional before Sprint 2 begins.

Key activities include:

- Sprint 1 Feature Coding — Core Modules
- Front-End and Back-End Integration
- Continuous Testing and Debugging
- Sprint 1 Review and Demonstration
- Sprint 1 Retrospective and Improvement Planning

A. Cost Analysis — Sprint 1

Table 3.16. Phase 3 (Sprint 1) — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380

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1. Sprint 1 Feature Coding — Core Modules

Table 3.17. Phase 3 (Sprint 1) — Core Module Feature Coding Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Authentication + OTP Module	Developer	2	₱480	10	$2 \times 480 \times 10$	₱9,600
RBAC for 8 User Roles	Developer	2	₱480	12	$2 \times 480 \times 12$	₱11,520
Evaluation Form + Draft Saving	Developer	1	₱480	10	$1 \times 480 \times 10$	₱4,800
Enrollment-Linked Form Assignment	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Clearance Verification Module	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Core Dashboards (Student + HR)	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Code Review	System Analyst	1	₱550	5	$1 \times 550 \times 5$	₱2,750
Sprint 1 Coordination	Project Manager	1	₱725	5	$1 \times 725 \times 5$	₱3,625
Development Tools and Platforms	Software License	—	—	—	Estimated	₱1,200

Sub-Total: Sub-Total: ₱43,095

2. Front-End and Back-End Integration — Sprint 1

Table 3.18. Phase 3 (Sprint 1) — Front-End and Back-End Integration Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
API Integration (Auth + Forms)	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Front-End Interface Integration	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Interface Refinement	UI/UX Designer	1	₱430	5	$1 \times 430 \times 5$	₱2,150
System Logic Verification	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Integration Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: Sub-Total: ₱12,720

3. Continuous Testing and Debugging — Sprint 1

Table 3.19. Phase 3 (Sprint 1) — Continuous Testing and Debugging Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Unit Testing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
System Testing	QA Specialist	1	₱380	8	$1 \times 380 \times 8$	₱3,040
Bug Fixing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Test Case Documentation	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Testing Supervision	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

Sub-Total: Sub-Total: ₱11,770

4. Sprint 1 Review and Demonstration

Table 3.20. Phase 3 (Sprint 1) — Sprint 1 Review and Demonstration Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Sprint 1 Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
System Demonstration	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Feature Validation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Stakeholder Review	Stakeholders	5	₱350	2	$5 \times 350 \times 2$	₱3,500
Feedback Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: Sub-Total: ₱9,140

5. Sprint 1 Retrospective and Improvement Planning

Table 3.21. Phase 3 (Sprint 1) — Retrospective and Improvement Planning Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Retrospective Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Process Evaluation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Development Improvement Planning	Developer	2	₱480	3	$2 \times 480 \times 3$	₱2,880
QA Feedback Analysis	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Retrospective Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: Sub-Total: ₱8,220

B. Total Cost — Sprint 1

Table 3.22. Phase 3 (Sprint 1) — Total Cost Summary

Activity	Cost (₱)
Sprint 1 Feature Coding — Core Modules	₱43,095
Front-End and Back-End Integration	₱12,720

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Activity	Cost (P)
Continuous Testing and Debugging	₱11,770
Sprint 1 Review and Demonstration	₱9,140
Sprint 1 Retrospective and Improvement Planning	₱8,220
TOTAL COST — SPRINT 1	₱84,945

Phase 3: Advanced Module Development (Sprint 2)

Sprint 2 builds upon the validated core system from Sprint 1 and develops the advanced features that differentiate this system from generic evaluation tools: AI-assisted feedback summarization using the OpenAI GPT-4.1 mini API, local frequency analysis, CHED-compliant PDF report generation, supervisor-to-faculty and peer-to-peer evaluation modules, the full analytics and ranking dashboards for HR and the VPAA, the non-editable audit trail, Bcrypt password hashing implementation, automated email notification system, multi-campus data partitioning, and the questionnaire management module. Sprint 2 also addresses all refinements identified during Sprint 1 review.

Key activities include:

- Sprint 2 Feature Coding — Advanced Modules
- AI Module and Security Integration
- Continuous Testing and Debugging
- Sprint 2 Review and Demonstration

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- Sprint 2 Retrospective and Improvement Planning

A. Cost Analysis — Sprint 2

Table 3.23. Phase 3 (Sprint 2) — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380

1. Sprint 2 Feature Coding — Advanced Modules

Table 3.24. Phase 3 (Sprint 2) — Advanced Module Feature Coding Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
AI Summarization Module (GPT-4.1 mini API)	Developer	2	₱480	12	$2 \times 480 \times 12$	₱11,520
Frequency Analysis Module	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
CHED-Compliant PDF Report Generator	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Supervisor + Peer Evaluation Modules	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Analytics Dashboards (HR, Dean, VPAA)	Developer	2	₱480	10	$2 \times 480 \times 10$	₱9,600
Audit Trail Module	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Email Notification System	Developer	1	₱480	5	1 × 480 × 5	₱2,400
Questionnaire Management Module	Developer	1	₱480	6	1 × 480 × 6	₱2,880
Multi-Campus Data Partitioning	Developer	1	₱480	5	1 × 480 × 5	₱2,400
Sprint 2 Coordination	Project Manager	1	₱725	5	1 × 725 × 5	₱3,625
Code Review	System Analyst	1	₱550	5	1 × 550 × 5	₱2,750
Development Tools and Licenses	Software License	—	—	—	Estimated	₱1,500

Sub-Total: ₱50,115

2. AI Module and Security Integration

Table 3.25. Phase 3 (Sprint 2) — AI Module and Security Integration Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Bcrypt Hashing Implementation	Developer	1	₱480	6	1 × 480 × 6	₱2,880
OTP and Hashing Integration	Developer	1	₱480	4	1 × 480 × 4	₱1,920
Cloudflare Firewall Configuration	Developer	1	₱480	3	1 × 480 × 3	₱1,440
OpenAI API Integration and Prompting	Developer	1	₱480	6	1 × 480 × 6	₱2,880
Security Integration Verification	System Analyst	1	₱550	4	1 × 550 × 4	₱2,200
Interface Adjustments (Sprint 2)	UI/UX Designer	1	₱430	5	1 × 430 × 5	₱2,150

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Integration Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱15,120

3. Continuous Testing and Debugging — Sprint 2

Table 3.26. Phase 3 (Sprint 2) — Continuous Testing and Debugging Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Unit Testing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Security and AI Testing	QA Specialist	1	₱380	10	$1 \times 380 \times 10$	₱3,800
Bug Fixing	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Test Case Documentation	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Testing Supervision	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

Sub-Total: ₱13,490

4. Sprint 2 Review and Demonstration

Table 3.27. Phase 3 (Sprint 2) — Sprint 2 Review and Demonstration Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Sprint 2 Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Full System Demonstration	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Feature Validation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Stakeholder Review	Stakeholders	5	₱350	2	$5 \times 350 \times 2$	₱3,500

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Feedback Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: ₱9,140

5. Sprint 2 Retrospective and Improvement Planning

Table 3.28. Phase 3 (Sprint 2) — Retrospective and Improvement Planning Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Retrospective Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Process Evaluation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Development Improvement Planning	Developer	2	₱480	3	$2 \times 480 \times 3$	₱2,880
QA Feedback Analysis	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Retrospective Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: ₱8,220

B. Total Cost — Sprint 2

Table 3.29. Phase 3 (Sprint 2) — Total Cost Summary

Activity	Cost (₱)
Sprint 2 Feature Coding — Advanced Modules	₱50,115
AI Module and Security Integration	₱15,120
Continuous Testing and Debugging	₱13,490
Sprint 2 Review and Demonstration	₱9,140

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Activity	Cost (P)
Sprint 2 Retrospective and Improvement Planning	₱8,220
TOTAL COST — SPRINT 2	₱96,085

C. Total Cost of Phase 3 (Sprint 1 + Sprint 2)

Table 3.30. Phase 3 — Combined Sprint Total Cost

Activity	Cost (P)
Sprint 1 — Core System Modules	₱84,945
Sprint 2 — Advanced Modules and Completion	₱96,085
TOTAL COST — PHASE 3	₱181,030

Phase 4: System Testing, Deployment, and Maintenance

The System Testing, Deployment, and Maintenance phase ensures that the fully developed system is rigorously validated before institutional go-live. Comprehensive testing covers functional correctness, security, performance under peak load, and user acceptance across all seven role groups. Following successful testing, the system is deployed on NAAP's institutional server infrastructure with Cloudflare firewall integration and database security configuration applied. User training sessions are conducted for each role group before the system is activated for live use. Initial system maintenance covers the first operational period, addressing any post-deployment issues discovered in the production environment.

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Key activities include:

- System Testing and Quality Assurance
- User Acceptance Testing (UAT)
- System Deployment and Configuration
- User Training and Documentation
- Initial System Maintenance and Support

A. Cost Analysis

Table 3.31. Phase 4 — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Stakeholder Representative	₱350

1. System Testing and Quality Assurance Cost

Table 3.32. Phase 4 — System Testing and Quality Assurance Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Test Plan Preparation	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Functional Testing	QA Specialist	1	₱380	10	$1 \times 380 \times 10$	₱3,800

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Security and Encryption Testing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Performance and Load Testing	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Bug Fix Implementation	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
Test Result Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱14,170

2. User Acceptance Testing (UAT) Cost

Table 3.33. Phase 4 — User Acceptance Testing Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
UAT Planning	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
UAT Coordination	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
UAT Session (All Roles)	Stakeholders	5	₱350	4	$5 \times 350 \times 4$	₱7,000
Issue Identification	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
UAT Report Preparation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱14,165

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3. System Deployment and Configuration Cost

Table 3.34. Phase 4 — System Deployment and Configuration Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Deployment Planning	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Server Configuration	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
Database Deployment	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Cloudflare Firewall Setup	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Deployment Verification	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Cloud / Hosting Setup	Cloud Services	—	—	—	Estimated	₱1,200

Sub-Total: ₱10,275

4. User Training and Documentation Cost

Table 3.35. Phase 4 — User Training and Documentation Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Training Preparation	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Training Materials Creation	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
User Training Sessions	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
User Training Sessions	Stakeholders	5	₱350	3	$5 \times 350 \times 3$	₱5,250

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Manual Documentation	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200

Sub-Total: ₱13,545

5. Initial System Maintenance and Support Cost

Table 3.36. Phase 4 — Initial System Maintenance and Support Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Monitoring	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Bug Fix and Patch Updates	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
User Support Assistance	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Performance Monitoring	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Maintenance Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,310

B. Total Cost of Phase 4

Table 3.37. Phase 4 — Total Cost Summary

Activity	Cost (₱)
System Testing and Quality Assurance	₱14,170
User Acceptance Testing (UAT)	₱14,165
System Deployment and Configuration	₱10,275
User Training and Documentation	₱13,545
Initial System Maintenance and Support	₱9,310

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Activity	Cost (₱)
TOTAL COST — PHASE 4	₱61,465

Phase 5: System Evaluation and Continuous Improvement

Following initial deployment, Phase 5 focuses on systematically assessing the live system's performance, collecting structured feedback from all user groups, and planning and implementing targeted improvements based on real operational data. For the proposed system, this phase is especially important given the complexity of the AI-assisted feedback module and the multi-role dashboard architecture — both of which may require calibration based on actual usage patterns and HR workflow preferences. Security and performance are also formally reviewed to ensure that the system remains compliant with data privacy requirements and responsive under growing data volumes.

Key activities include:

- System Performance Evaluation
- User Feedback Collection and Analysis
- System Enhancement Planning
- Security and Performance Review
- Continuous Improvement Implementation

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A. Cost Analysis

Table 3.38. Phase 5 — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Stakeholder Representative	₱350

1. System Performance Evaluation Cost

Table 3.39. Phase 5 — System Performance Evaluation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Performance Monitoring	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
System Usage Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Performance Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Performance Report Preparation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Evaluation Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

Sub-Total: ₱9,220

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2. User Feedback Collection and Analysis Cost

Table 3.40. Phase 5 — User Feedback Collection and Analysis Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Feedback Survey Design	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Feedback Collection	Stakeholders	5	₱350	3	$5 \times 350 \times 3$	₱5,250
Feedback Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Feedback Report	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Feedback Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Survey Tools and Materials	Tools	—	—	—	Estimated	₱250

Sub-Total: ₱12,450

3. System Enhancement Planning Cost

Table 3.41. Phase 5 — System Enhancement Planning Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Enhancement Requirement Review	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Development Planning	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Enhancement Prioritization	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
UX Enhancement Planning	UI/UX Designer	1	₱430	3	$1 \times 430 \times 3$	₱1,290

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Enhancement Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Enhancement Review Meeting	Project Manager	1	₱725	1	$1 \times 725 \times 1$	₱725

Sub-Total: ₱9,235

4. Security and Performance Review Cost

Table 3.42. Phase 5 — Security and Performance Review Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Security Assessment	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Vulnerability Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
System Audit Review	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Security Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Security Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱8,190

5. Continuous Improvement Implementation Cost

Table 3.43. Phase 5 — Continuous Improvement Implementation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Minor System Enhancements	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
UI/UX Improvements	UI/UX Designer	1	₱430	4	1 × 430 × 4	₱1,720
Regression Testing	QA Specialist	1	₱380	4	1 × 380 × 4	₱1,520
Enhancement Review	System Analyst	1	₱550	3	1 × 550 × 3	₱1,650
Improvement Documentation	System Analyst	1	₱550	2	1 × 550 × 2	₱1,100

Sub-Total: ₱8,870

B. Total Cost of Phase 5

Table 3.44. Phase 5 — Total Cost Summary

Activity	Cost (₱)
System Performance Evaluation	₱9,220
User Feedback Collection and Analysis	₱10,450
System Enhancement Planning	₱9,235
Security and Performance Review	₱8,190
Continuous Improvement Implementation	₱8,870
TOTAL COST — PHASE 5	₱45,965

Phase 6: Long-Term Maintenance and System Sustainability

Phase 6 covers the ongoing operational maintenance and sustainability of the deployed system over its active institutional life. Activities in this phase ensure that the system continues to meet evolving institutional requirements, remains secure against emerging threats, and operates reliably as the volume of evaluation data and registered users grows over time. Regular database maintenance,

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security monitoring, feature enhancement cycles, and knowledge transfer procedures are conducted on a scheduled basis. This phase is essential for protecting the institution's investment and ensuring that the system delivers consistent value beyond the initial deployment period.

Key activities include:

- System Monitoring and Maintenance
- Technical Support and Issue Resolution
- System Updates and Feature Enhancements
- Security Monitoring and Risk Management
- System Documentation and Knowledge Transfer

A. Cost Analysis

Table 3.45. Phase 6 — Hourly Rate Assumptions

Role	Hourly Rate (P)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Technical Support Staff	₱300

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1. System Monitoring and Maintenance Cost

Table 3.46. Phase 6 — System Monitoring and Maintenance Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Performance Monitoring	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
Database Maintenance	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Log Monitoring and Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Maintenance Planning	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Maintenance Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,620

2. Technical Support and Issue Resolution Cost

Table 3.47. Phase 6 — Technical Support and Issue Resolution Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Support Assistance	Tech Support Staff	1	₱300	6	$1 \times 300 \times 6$	₱1,800
Issue Diagnosis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Bug Fix Implementation	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
System Verification	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Issue Resolution Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱8,710

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3. System Updates and Feature Enhancements Cost

Table 3.48. Phase 6 — System Updates and Feature Enhancements Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Enhancement Requirement Review	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Feature Enhancement Development	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Interface Improvements	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
Enhancement Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Update Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,970

4. Security Monitoring and Risk Management Cost

Table 3.49. Phase 6 — Security Monitoring and Risk Management Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Security Monitoring	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Vulnerability Assessment	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Risk Assessment	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Security Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Security Report Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱8,190

5. System Documentation and Knowledge Transfer Cost

Table 3.50. Phase 6 — System Documentation and Knowledge Transfer Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Technical Documentation Update	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
System Maintenance Guide Creation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Knowledge Transfer Session	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Staff Orientation	Tech Support Staff	2	₱300	3	$2 \times 300 \times 3$	₱1,800
Documentation Archiving	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: ₱8,200

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B. Total Cost of Phase 6

Table 3.51. Phase 6 — Total Cost Summary

Activity	Cost (₱)
System Monitoring and Maintenance	₱9,620
Technical Support and Issue Resolution	₱8,710
System Updates and Feature Enhancements	₱9,970
Security Monitoring and Risk Management	₱8,190
System Documentation and Knowledge Transfer	₱8,200
TOTAL COST — PHASE 6	₱44,690

Agile Development Cost Summary

The table below consolidates the total costs across all six development phases of the proposed system. Phase 3 reflects the combined cost of two iterative Agile sprints: Sprint 1 covering core authentication, RBAC, and evaluation submission modules, and Sprint 2 covering AI integration, advanced dashboards, Bcrypt security implementation, and all remaining features.

Table 3.52. Overall Agile Development Cost Summary — All Phases

Phase	Description	Key Activities	Total Cost (₱)
Phase 1	Requirements Analysis and Planning	Stakeholder engagement, requirement elicitation, user stories, backlog, sprint planning	₱52,675
Phase 2	System Design and Prototype Development	Architecture design, database modeling, UI/UX	₱61,485

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Phase	Description	Key Activities	Total Cost (P)
		wireframes, prototype build, design validation	
Phase 3 (Sprint 1)	Core System Module Development	Authentication + OTP, RBAC, evaluation forms, clearance module, core dashboards	₱84,945
Phase 3 (Sprint 2)	Advanced Module Development	AI module, PDF reports, audit trail, Bcrypt, email notifications, multi-campus support	₱96,085
Phase 4	System Testing, Deployment, and Maintenance	QA testing, UAT, server deployment, Cloudflare setup, user training, initial support	₱61,465
Phase 5	System Evaluation and Continuous Improvement	Performance evaluation, user feedback, enhancement planning, security review, improvements	₱45,965
Phase 6	Long-Term Maintenance and System Sustainability	System monitoring, tech support, updates, security risk management, documentation	₱44,690
	TOTAL OVERALL AGILE SYSTEM DEVELOPMENT COST		₱446,310

Benefit Analysis

This section identifies and quantifies the tangible and intangible benefits that the proposed evaluation system is expected to deliver to NAAP over its operational lifecycle. Tangible benefits are expressed as annual savings computed from the reduction of labor costs and operational expenses currently incurred through the manual evaluation process. Intangible benefits represent institutional

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improvements that are significant in value but not directly reducible to a single monetary figure.

Tangible Benefits (Annual)

The annual savings below are derived from the elimination or substantial reduction of manual activities that currently consume administrative staff time and institutional resources during each evaluation cycle. Each benefit is based on estimated staff-hours saved multiplied by the applicable average hourly rate.

Table 3.53. Annual Tangible Benefits — NAAP Evaluation System

Benefit Category	Description and Computation Basis	Annual Savings (P)
Reduced Administrative Labor	Automation of data encoding, form distribution, result consolidation, and report preparation eliminates an estimated 480 staff-hours per evaluation cycle at an average rate of P250/hr ($480 \times \text{P}250 = \text{P}120,000$)	P120,000
Faster Evaluation Processing	Automated submission tracking, real-time dashboards, and digital report generation reduce evaluation processing time from weeks to hours, saving an estimated 240 equivalent staff-hours per year ($240 \times \text{P}250 = \text{P}60,000$)	P60,000
Elimination of Paper and Printing Costs	Full digital submission removes recurring costs of printed evaluation forms, answer sheets, and physical report reproduction across all campuses (estimated lump-sum savings)	P25,000
Automated Clearance Verification	System-generated completion references replace manual per-student OSA cross-referencing during clearance season, saving an estimated 180 staff-hours per clearance period ($180 \times \text{P}250 = \text{P}45,000$)	P45,000

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Benefit Category	Description and Computation Basis	Annual Savings (P)
Total Annual Tangible Benefits		₱250,000

Intangible Benefits

The following benefits are not directly reducible to a single annual figure but represent substantial and lasting value for NAAP's institutional quality assurance and governance:

- Improved accuracy and integrity of evaluation data through enrollment-linked access control, automated duplicate prevention, and validated submission tracking
- Objective, consistent interpretation of qualitative student feedback through AI-assisted summarization (GPT-4.1 mini) combined with local frequency analysis, replacing subjective manual review
- Full regulatory compliance with CHED Memorandum Order No. 19, Series of 2025, through standardized questionnaire instruments and CHED-formatted PDF evaluation reports
- Compliance with the Data Privacy Act of 2012 (Republic Act No. 10173) through Bcrypt one-way password hashing, email OTP authentication, and anonymized evaluation reporting

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- Strengthened institutional accountability through a permanent, non-editable audit trail capturing all system actions, user changes, and AI summary edits with timestamps and user attribution
- Enhanced faculty trust and engagement in the evaluation process through transparent, consistently applied, and evaluator-anonymous procedures
- Multi-campus scalability allowing all NAAP campuses to be served from a single centralized deployment without proportional increases in administrative overhead
- Long-term data preservation through a soft-delete policy that retains all historical evaluation records, supporting trend analysis, academic quality reviews, and regulatory audits over multiple academic years

Total Benefits Over 3 Years

The total tangible benefit over the three-year assessment period is computed by multiplying the annual savings by the number of years:

$$\text{Total Tangible Benefits (3 Years)} = \text{Annual Savings} \times 3 = \text{₱}250,000 \times 3 = \text{₱}750,000$$

Net Benefit Computation

The net benefit is derived by subtracting the total project cost across all six phases from the total three-year tangible benefits. This represents the actual

financial gain the institution realizes after fully recovering the system's development, deployment, and maintenance investment:

$$\text{Net Benefit} = \text{Total Benefits (3 Years)} - \text{Total Project Cost (Phases 1-6)}$$

$$\text{Net Benefit} = \text{₱750,000} - \text{₱446,310} = \text{₱303,690}$$

The positive net benefit of ₱303,690 confirms that the system's financial returns exceed total costs across all six phases over the three-year period.

Return on Investment (ROI)

ROI measures the efficiency of the total project investment by expressing the net benefit as a percentage of the total cost. A higher ROI indicates that a greater proportion of the investment is recovered as net savings:

$$\text{ROI (\%)} = (\text{Net Benefit} \div \text{Total Project Cost}) \times 100$$

$$\text{ROI} = (\text{₱303,690} \div \text{₱446,310}) \times 100 = 68.04\%$$

An ROI of 68.04% means the institution recovers its full investment and gains an additional ₱0.68 in net savings for every ₱1.00 invested over the three-year period. This is a strong return that justifies the development and long-term operational expenditure.

Agile-Specific Cost Advantage Justification

The Agile SDLC model adopted for this project delivers economic advantages that strengthen the financial case for development. By validating features through stakeholder sprint reviews before subsequent development proceeds, rework costs from misaligned requirements are minimized. The two-sprint structure in Phase 3 is particularly advantageous because it allows core modules — authentication, RBAC, and evaluation submission — to be tested and confirmed functional before investment is made in the more complex AI and security features of Sprint 2. If stakeholder review during Sprint 1 identifies interface or logic issues, they are corrected before they propagate into the advanced modules, preventing exponentially more costly late-stage fixes.

Continuous integration testing throughout each sprint reduces the density and severity of defects reaching the production environment, directly reducing post-deployment support costs captured in Phases 5 and 6. Incremental delivery also allows NAAP to begin realizing operational savings from completed modules — such as digital evaluation submission and automated clearance checking — before the full system is deployed, accelerating the break-even point. The iterative nature of Agile development further protects the investment against obsolescence: as CHED updates evaluation requirements or NAAP expands its campuses, the system can be adapted within planned sprint cycles without requiring a complete rebuild.

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Cost-Benefit Summary

Table 3.54. Cost-Benefit Summary — AI-Enhanced CHED-Compliant Faculty Evaluation System

Metric	Value
Total Development Cost (Phases 1–4)	₱290,620
Total Evaluation and Improvement (Phase 5)	₱45,965
Total Long-Term Maintenance (Phase 6)	₱44,690
Overall Total Project Cost (Phases 1–6)	₱446,310
Annual Tangible Benefits	₱250,000
Total Tangible Benefits (3 Years)	₱750,000
Net Benefit (3 Years)	₱303,690
Return on Investment (ROI)	68.04%

The cost-benefit analysis confirms that the proposed AI-Enhanced CHED-Compliant Web-Based Faculty and Student Evaluation System represents a sound institutional investment for NAAP. The total three-year net benefit of ₱303,690 and a return on investment of 68.04 percent demonstrate that the financial returns from reduced administrative labor, eliminated paper costs, and automated clearance verification substantially exceed the total development, maintenance, and operational investment. When the system's intangible contributions — regulatory compliance, improved data governance, faculty performance transparency, and long-term audit capability — are considered alongside these figures, the overall institutional value of the proposed system is considerably greater than its direct financial return alone.

System Requirements Specification

This section formally defines the functional and non-functional requirements that the proposed system must satisfy. The requirements are derived from the gap analysis, stakeholder consultations, and the regulatory specifications of CHED Memorandum Order No. 19, Series of 2025.

Functional Requirements

The system shall provide secure, role-based access for seven distinct user groups: Student, Faculty, Supervisor/Dean, HR Officer, OSA Officer, VPAA, and System Administrator. Each role shall have access to a dedicated portal presenting only the functions and data relevant to that role's responsibilities.

Students shall be able to submit evaluation forms exclusively for faculty members assigned to their currently enrolled courses. The system shall automatically retrieve enrollment data to generate the list of evaluable faculty per student, preventing out-of-scope submissions. Students shall be able to save draft responses and resume evaluation completion within the active evaluation window.

The HR module shall allow authorized HR officers to configure evaluation periods, manage questionnaire instruments, publish AI-generated feedback summaries, generate CHED-compliant PDF evaluation reports, and monitor submission completion rates across all campuses in real time.

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The system shall implement an automated clearance verification function whereby the OSA dashboard displays, for each student, the completion status of required evaluations. The clearance reference shall be generated automatically once all required submissions are confirmed, eliminating the need for manual cross-referencing.

The AI-assisted feedback analysis module shall retrieve all open-ended student comments for a selected faculty member and evaluation period, transmit them to the OpenAI GPT-4.1 mini API with a structured analytical prompt, and return a thematic summary organized by commendations, constructive feedback, and recurring concerns. A locally executed frequency analysis module shall independently compute term frequencies from the same comment set and display results alongside the AI summary. The feedback analysis pipeline shall incorporate AI-Based Bias Detection to classify qualitative comments as Constructive, Neutral, or Biased prior to summarization, and shall display AI Transparency disclosures showing detected keywords, comment clusters, and analytic reasoning alongside every generated summary.

The system shall maintain a non-editable, append-only audit trail recording all significant actions — including login events, evaluation submissions, data modifications, AI summary generation, and administrative changes — with user identifiers, timestamps, and IP addresses.

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HR officers and VPAA users shall have access to comparative dashboards displaying faculty performance rankings, departmental evaluation averages, evaluation period trends, and campus-level summary statistics. All dashboard data shall be refreshed in real time from the live database. Dashboards shall additionally display longitudinal faculty performance trends across evaluation periods and Evaluation Credibility Scores for submitted evaluations, enabling administrators to distinguish between highly reliable and review-required evaluation inputs. Cross-source consistency alerts shall be surfaced in the HR dashboard when significant discrepancies are detected between student, peer, and supervisor evaluation results for the same faculty member and period.

The system shall support multi-campus data partitioning, ensuring that evaluation data from each NAAP campus is accessible only to authorized users assigned to that campus, while centralized administrative and VPAA views can access aggregated data across all campuses.

Email notifications shall be dispatched to students with pending evaluations at administrator-configured intervals during open evaluation periods. HR officers shall be able to configure notification frequency, timing, and message content through the administrative interface. Email delivery shall also be used for OTP distribution during first-time login verification and unrecognized device access events.

The questionnaire management module shall allow HR officers to create, edit, activate, and archive evaluation instruments conforming to the item format and rating scale requirements of CHED CMO No. 19, Series of 2025. Each questionnaire shall include configurable quantitative rating items and at least one open-ended comment field per instrument.

Analytics Framework and Advanced Feature Requirements

The proposed system integrates a multi-tier analytics framework organized into four complementary analytic categories: Descriptive, Diagnostic, Predictive, and Prescriptive Analytics. Each category serves a distinct purpose in transforming raw evaluation data into actionable institutional intelligence. The seven advanced features described in this section are distributed across these categories according to their analytical function and the type of insight they generate.

Descriptive Analytics

Descriptive analytics explains what has already happened by summarizing historical evaluation data in a form that is immediately understandable to administrators, faculty, and institutional decision-makers. The system applies descriptive analytics to produce consolidated performance summaries, period-over-period comparisons, and departmental aggregations from the accumulated evaluation record. Two advanced features are classified under this category:

Track Faculty Performance Over Time. The system stores all historical evaluation records for every faculty member across all completed evaluation periods. An AI-assisted trend detection module processes the accumulated data to identify improvement trends, decline patterns, and sustained consistency in faculty performance. Rather than presenting evaluation results solely on a per-semester basis, the system generates longitudinal performance views that allow HR officers and VPAA users to assess faculty development trajectories across multiple academic terms. Example outputs include computed percentage changes between evaluation periods (e.g., “Faculty improved by 18% over 3 semesters”) and graphical trend visualizations displayed on role-specific dashboards. Historical records are retained under the system’s soft-delete policy, ensuring that longitudinal data remains available for performance reviews, faculty development planning, and regulatory audits throughout the system’s operational life.

AI Transparency. The system surfaces the reasoning behind AI-generated feedback summaries rather than presenting only final conclusions. Instead of delivering a bare summary statement such as “Needs Improvement,” the AI Transparency feature displays the specific keywords detected in submitted comments, clusters of semantically related feedback, and the analytic reasoning applied to arrive at the summary classification. This design ensures that HR officers and faculty members can evaluate the basis of AI-generated assessments, identify cases where the model may have over-weighted isolated comments, and

exercise informed judgment before acting on AI outputs. Transparency disclosures are displayed alongside every AI-generated summary in the HR and faculty dashboards.

Diagnostic Analytics

Diagnostic analytics explains why a particular result occurred by examining the causes and patterns underlying the data. Where descriptive analytics reports what the numbers show, diagnostic analytics investigates whether those numbers are reliable and what factors produced them. The system applies diagnostic analytics through three advanced features that collectively address the integrity and validity of submitted evaluation data:

Evaluation Behavior Analysis. The system monitors and analyzes the behavioral patterns of evaluators during the submission process in order to detect responses that may reflect inattentiveness, strategic bias, or automated submission rather than genuine assessment. The analysis examines three behavioral dimensions: (1) time spent completing the evaluation form, where an unusually rapid completion time relative to the number of items may indicate that the respondent did not read the questions carefully; (2) response pattern uniformity, where a respondent who selects the same rating value (e.g., all 5 or all 1) across every item may not be providing differentiated assessments; and (3) repetitive response patterns across multiple submitted forms from the same

evaluator. Based on these signals, the system computes an Evaluation Behavior Score for each submitted evaluation. Submissions with low behavior scores are flagged for HR review rather than excluded automatically, preserving the integrity of the evaluation record while alerting administrators to potentially unreliable data points.

AI-Based Bias Detection. The system applies Natural Language Processing (NLP) to classify open-ended qualitative comments according to their constructiveness and neutrality before they are incorporated into AI-generated feedback summaries. The NLP module detects three categories of problematic content: emotional bias, characterized by language expressing personal animosity or strong negative sentiment unrelated to pedagogical performance (e.g., “I hate this prof”); non-constructive comments that provide no actionable information; and irrelevant feedback that does not address the faculty member’s professional conduct or teaching quality. Each detected comment is assigned one of three classification labels: Constructive (e.g., “Explains well but needs more examples”), Neutral (e.g., “Okay”), or Biased (e.g., “I hate this prof”). Comments classified as Biased are excluded from the AI summarization input or flagged with a reduced weighting, ensuring that AI-generated feedback summaries reflect genuine pedagogical assessments rather than personal sentiment. Classification labels and the rationale for each are displayed in the AI Transparency view accessible to HR officers.

Cross-Source Validation. The system performs automated cross-source consistency checks by comparing the evaluation results submitted through three distinct evaluator types: student evaluations, peer evaluations, and supervisor evaluations. Where the results from one evaluator group diverge substantially from the others for the same faculty member and evaluation period (e.g., student ratings are very low while supervisor ratings are very high), the system flags the inconsistency and presents a consistency alert in the HR dashboard. This cross-source diagnostic capability allows HR officers to investigate potential sources of rating discrepancy before final evaluation reports are published, supporting more reliable and defensible faculty performance determinations.

Predictive Analytics

Predictive analytics uses historical patterns and current data signals to estimate the likely reliability or outcome of future or current data points. In the context of the evaluation system, predictive analytics is applied not to forecast future faculty performance scores directly, but to estimate the credibility of submitted evaluations before they are weighted in performance computations. This is implemented through the following advanced feature:

Credibility Score Output. Each submitted evaluation is assigned an Evaluation Credibility Score on a scale of 0 to 100, computed by aggregating three component signals: the Evaluation Behavior Score derived from the Evaluation

Behavior Analysis module, the bias classification outcome from the AI-Based Bias Detection module, and the cross-source consistency result from the Cross-Source Validation module. The composite score provides a unified, data-driven estimate of how much confidence should be placed in a given evaluation submission. Credibility scores are interpreted according to three reliability bands: 90 to 100 indicates a highly reliable evaluation; 70 to 89 indicates an acceptable evaluation; and scores below 70 indicate that the evaluation requires HR review before inclusion in published performance reports. This scoring mechanism enables HR officers to apply differentiated scrutiny to evaluations proportional to their estimated reliability, rather than treating all submitted responses as equally valid inputs.

Prescriptive Analytics

Prescriptive analytics goes beyond reporting and diagnosis to recommend specific actions that should be taken in response to identified data conditions. It transforms analytical findings into operationally useful guidance, closing the loop between evaluation data and institutional improvement. The system implements prescriptive analytics through the following advanced feature:

Smart Recommendation Engine. The system implements an AI-assisted Smart Recommendation Engine that generates targeted faculty development suggestions based on the specific performance deficiencies identified in evaluation

results. The engine operates in two steps. In the first step, it identifies the evaluation dimensions where a faculty member's scores fall below defined thresholds (e.g., a low rating in the Clarity of Instruction domain). In the second step, it produces actionable improvement recommendations tailored to the identified weak areas (e.g., "Use more examples and visual aids to improve instructional clarity"). Recommendations are generated from a structured knowledge base aligned with common faculty development frameworks and are displayed in the faculty member's personal dashboard following each evaluation period. HR officers and deans also have access to aggregated recommendation outputs at the department level, enabling structured faculty development planning. The Recommendation Engine is designed exclusively for improvement support and does not generate disciplinary recommendations or influence performance ratings directly.

Non-Functional Requirements

The system shall satisfy software quality standards consistent with the ISO/IEC 25010 software product quality model. With respect to security, all personally identifiable data stored in the system database shall be protected in accordance with applicable data protection standards. User passwords shall be stored exclusively as one-way cryptographic hashes using Bcrypt with per-user salting, ensuring that plaintext credentials cannot be recovered by any means. An

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email-based one-time password (OTP) shall be issued on first-time login and on detection of unrecognized device access to verify user identity. A Cloudflare web application firewall shall provide perimeter-level protection against distributed denial-of-service attacks, brute-force login attempts, and common web application exploits.

With respect to usability, the system interface shall be role-specific, presenting each user group with only the navigation elements and functions relevant to their responsibilities. The interface shall be fully responsive, rendering correctly on desktop computers, laptop computers, tablet devices, and smartphones without requiring client-side plugins or additional software installation.

With respect to reliability, the system shall maintain continuous availability throughout active evaluation windows. Automated backup procedures shall execute at scheduled intervals, and backup integrity shall be verified through restoration testing on a periodic basis.

With respect to performance, the system shall process evaluation form submissions and dashboard data requests within acceptable response times under peak concurrent load conditions. Database queries shall be optimized through indexing and connection pooling to ensure that performance does not degrade significantly during high-traffic evaluation periods.

Proposed System Architecture and Design

The proposed system architecture and design provide a structured framework for the development and implementation of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines. The design emphasizes modularity, scalability, security, and usability, ensuring that the system can efficiently support diverse user roles, including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel, and system administrators. The architecture follows a layered approach, integrating the presentation, application, and data management layers, while incorporating AI-assisted components to enhance the analysis of qualitative feedback. This structured design facilitates clear separation of responsibilities, supports centralized data management, and ensures secure, transparent, and reliable processing of evaluation data. The proposed architecture and design serve as the foundation for iterative development, testing, and deployment, enabling the system to meet institutional requirements, regulatory standards, and user expectations while maintaining flexibility for future enhancements.

System Architecture Overview

The proposed Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines is built on a web-based client-server architecture designed to facilitate centralized management of evaluation data, secure system access, and scalable deployment within the National Aviation Academy of the Philippines (NAAP). This architecture allows multiple stakeholders including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Human Resource personnel, the Office of Student Affairs (OSA), and system administrators to access and interact with the system through web-based interfaces, while ensuring centralized processing, storage, and management of evaluation data.

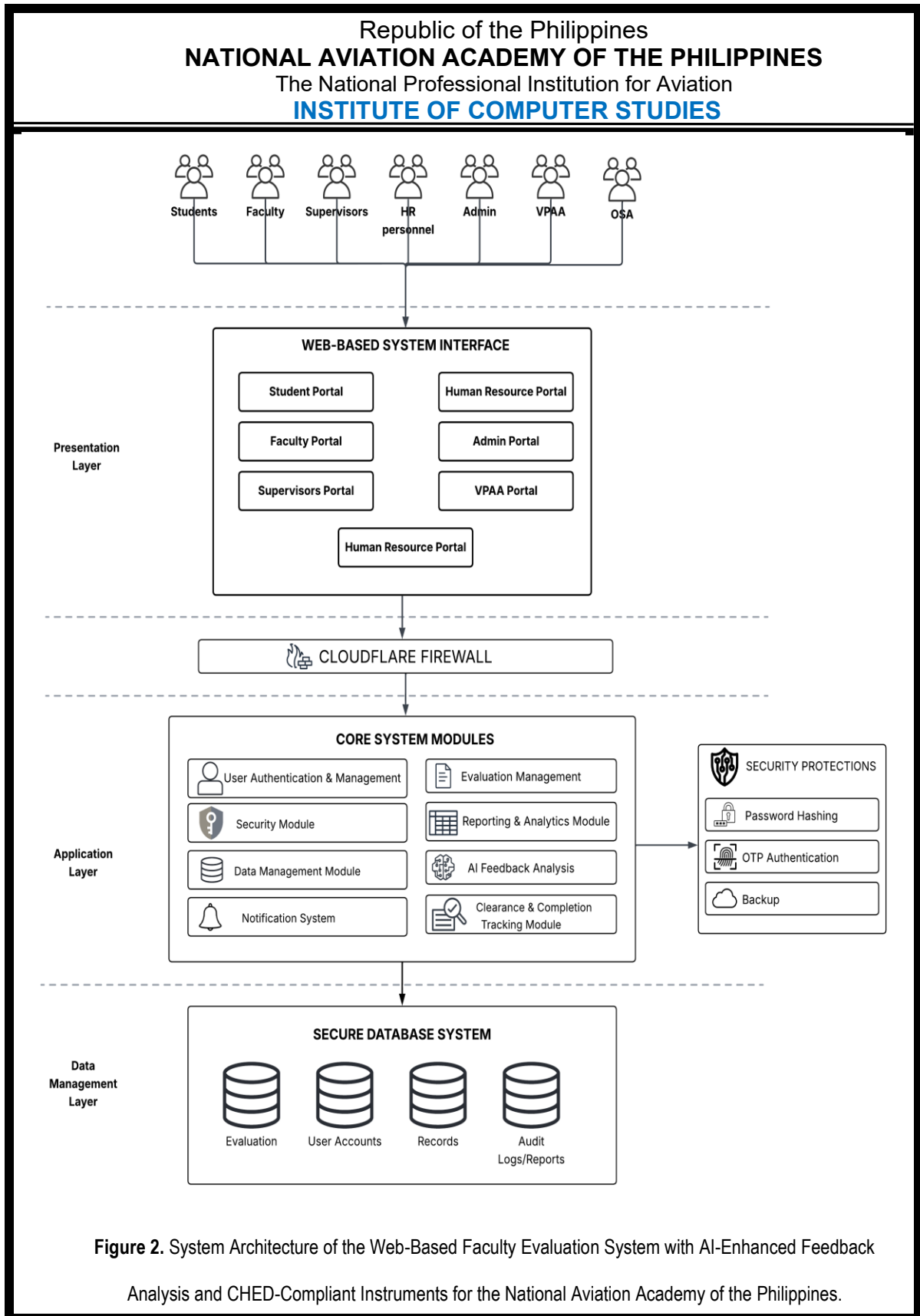


Figure 2. System Architecture of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines.

The system architecture is organized into three primary layers: the presentation layer, the application layer, and the data management layer. The presentation layer delivers web-based user interfaces that allow stakeholders—including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Human Resource personnel, the Office of Student Affairs (OSA), and system administrators to access evaluation forms, performance dashboards, administrative controls, and reporting tools via standard web browsers. The interface is designed to provide responsive interaction, ensuring accessibility across multiple devices while maintaining intuitive navigation and streamlined evaluation submission workflows.

The application layer functions as the system's core processing component, managing evaluation form processing, authentication and authorization, AI-assisted qualitative feedback analysis, report generation, and clearance verification functions. Role-based access control mechanisms are integrated within this layer to regulate system functionality according to user roles, ensuring data confidentiality and the integrity of evaluation processes. The AI module operates within the application layer, performing frequency-based text analysis and thematic summarization of qualitative evaluation responses. These analytical outputs support HR personnel in interpreting feedback patterns, facilitating informed decisions while preserving the authenticity of original evaluator responses.

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The data management layer provides centralized storage for evaluation records, user accounts, course assignments, and clearance verification data through a relational database management system. Centralized data storage ensures consistency, enables real-time reporting, and supports system scalability. Secure database access protocols and encrypted data transmission mechanisms protect sensitive evaluation information, ensuring compliance with institutional data privacy policies and national data protection regulations.

The client-server architecture allows efficient deployment and supports concurrent access during peak evaluation periods. The centralized server environment enables system administrators to configure evaluations, monitor system performance, and maintain security. Furthermore, the architecture is designed to support future system expansion and potential integration with institutional academic information systems, including enrollment and faculty assignment databases.

By combining web-based accessibility, AI-assisted analytical processing, and centralized data management, the proposed architecture delivers a scalable, secure, and efficient platform that enhances faculty evaluation management, streamlines administrative processes, and supports institutional quality assurance initiatives.

System Components and Modules

The Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments is structured into interconnected functional modules that collectively support secure, automated, and efficient evaluation processes at the National Aviation Academy of the Philippines (NAAP). The modular architecture promotes scalability, maintainability, and clear separation of system responsibilities.

At the center of the system is the Evaluation Management Module (Core Processing), which serves as the primary engine of the platform. This module manages the overall evaluation workflow, including the distribution of CHED-compliant evaluation forms, collection of responses from students, faculty, and supervisors, validation of submitted evaluations, and coordination with other modules for processing and storage. It ensures that all evaluation activities follow standardized procedures, maintaining consistency and reliability of collected data.

Connected to the core module is the Enrollment Validation Module, which verifies whether a student is officially enrolled under a specific instructor before allowing evaluation submission. This ensures that only authorized evaluators can participate, preventing invalid, duplicate, or unauthorized responses and strengthening data integrity.

The Authentication and Access Control Module works alongside the core system to regulate user access. It authenticates users through secure login

mechanisms and enforces role-based permissions for students, faculty, supervisors, and administrators. This ensures that users can only access functionalities relevant to their roles, protecting sensitive evaluation data.

The AI Feedback Analysis Module is integrated with the core processing module to enhance the interpretation of qualitative feedback. It performs automated text analysis using frequency analysis and AI-based summarization techniques, enabling Human Resource personnel and administrators to quickly identify recurring themes and generate meaningful insights from large volumes of comments.

On the administrative side, the System Administration Module supports the management and configuration of the system. It handles user account administration, evaluation setup, monitoring of system activity, and maintenance of evaluation instruments, ensuring smooth and reliable system operation.

The Reporting and Dashboard Module processes evaluation results and transforms them into visual and analytical outputs. It provides dashboards, reports, and performance summaries that allow administrators, supervisors, and decision-makers to monitor faculty performance, identify trends, and support data-driven decision-making.

The Notification Module facilitates system communication by sending alerts, reminders, and updates related to evaluation activities, deadlines, and

system actions. This ensures timely participation and improves overall system responsiveness.

The Clearance Monitoring and Tracking Module manages the verification of evaluation completion. It tracks whether students have fulfilled their evaluation requirements and generates clearance references necessary for academic processing. This module automates what was previously a manual process, reducing administrative workload and improving efficiency.

Finally, the Security Module ensures the protection of system data and user information. It implements essential security mechanisms such as OTP authentication and encryption handling, safeguarding sensitive data and ensuring compliance with data privacy standards.

Overall, the interaction of these modules creates a cohesive and efficient system where the Evaluation Management Module coordinates all operations, while supporting modules handle validation, security, analysis, reporting, and administration. This modular design enhances system reliability, simplifies maintenance, and allows future expansion of functionalities.

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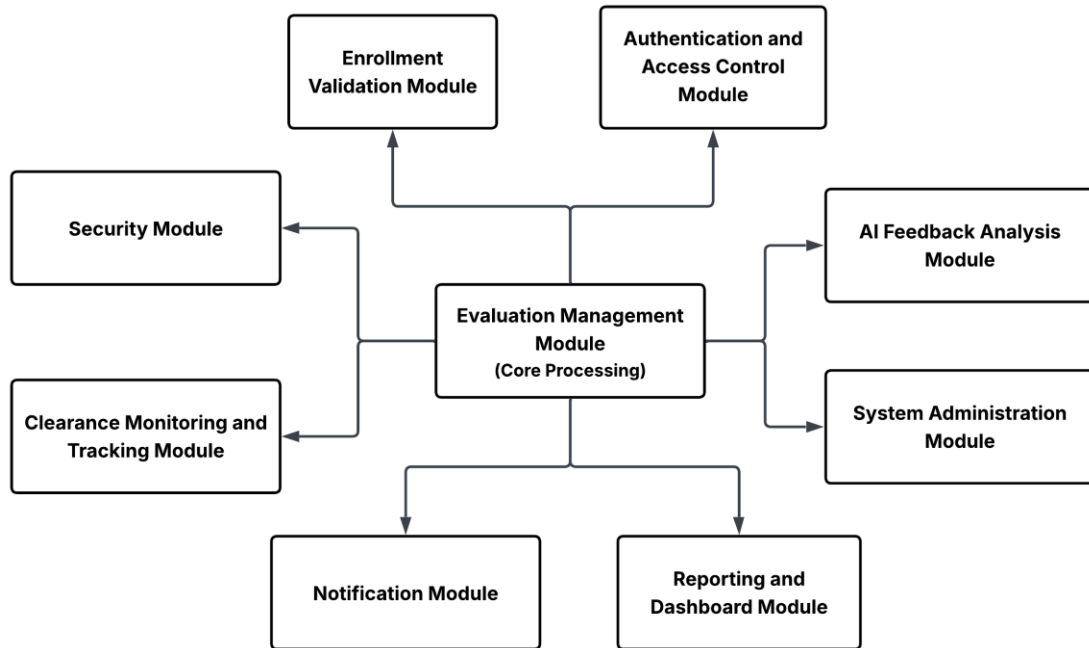


Figure 3. System Components and Modules of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines.

Figure 3 illustrates the modular architecture of the proposed Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines. At the core of the system is the Evaluation Management Module (Core Processing), which manages the overall evaluation workflow, including form distribution, response collection, validation, and coordination with other modules. Supporting modules include the Enrollment Validation Module, which verifies student–instructor relationships to ensure authorized evaluation participation; the

Authentication and Access Control Module, which enforces secure login and role-based access; and the Security Module, which implements OTP authentication and encryption mechanisms to protect sensitive data.

The system also integrates the AI Feedback Analysis Module, which processes qualitative responses using frequency analysis and AI-based summarization to generate meaningful insights. The Reporting and Dashboard Module provides visual summaries, performance analytics, and evaluation reports to support data-driven decision-making. The Clearance Monitoring and Tracking Module automates the verification of evaluation completion for student clearance, improving administrative efficiency. Additionally, the Notification Module facilitates communication through system alerts and reminders, while the System Administration Module manages user accounts, system configurations, and overall platform maintenance.

This modular architecture ensures efficient interaction among system components, enhances maintainability, and supports scalability for future improvements and expansion of system functionalities.

Process Modeling and System Workflow

Process modeling and system workflow provide a visual and structured representation of how the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments operates within

the National Aviation Academy of the Philippines (NAAP). This section illustrates the flow of information, interactions among stakeholders, and the sequence of operations required to conduct faculty evaluations efficiently and securely. Key components of the process modeling include the Context Diagram, which provides a high-level overview of the system and its interactions with external entities such as students, faculty, supervisors, the VPAA, the Office of Student Affairs (OSA), Human Resource personnel and system administrators. The Data Flow Diagrams (Level 0 and Level 1) further detail the movement of data between system modules, including evaluation form processing, AI-assisted feedback analysis, reporting, and clearance verification. Finally, the Procedural Workflow depicts step-by-step operational procedures, demonstrating how evaluation tasks are initiated, processed, monitored, and completed. Together, these modeling tools enable a clear understanding of system functionality, support effective system design and implementation, and provide a foundation for testing, validation, and future enhancements.

Context Diagram

The context diagram presents a high-level view of the Web-Based Student and Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines (NAAP). It illustrates the system as a centralized platform that interacts with

multiple stakeholders by exchanging evaluation data, reports, and system-related information.

Students interact with the system by receiving evaluation questionnaires, submitting evaluation responses, and monitoring their evaluation completion status. The system, in turn, provides completion tracking to ensure that students meet evaluation requirements necessary for clearance processing.

Faculty members engage with the system by submitting peer evaluations and receiving evaluation results, which allow them to review feedback on their teaching performance and professional interactions.

Supervisors and Deans participate by submitting formal evaluation assessments of faculty members. They also receive evaluation results generated by the system, enabling them to monitor faculty performance and support academic decision-making.

The Vice President for Academic Affairs (VPAA) accesses the system to review consolidated evaluation results, which serve as a basis for institutional planning, faculty assessment, and policy development.

Human Resource (HR) personnel interact with the system by receiving evaluation results and audit trails. These outputs support performance evaluation, employee monitoring, and documentation for faculty development and administrative review.

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The Office of Student Affairs (OSA) utilizes the system to verify student evaluation completion status, which is required for clearance processing. This eliminates the need for manual verification and improves operational efficiency.

System Administrators (Admin) manage user accounts and receive audit trails from the system. They are responsible for maintaining system integrity, monitoring activities, and ensuring proper system configuration and operation.

Overall, the context diagram highlights the flow of data between external stakeholders and the evaluation system, including evaluation submissions, results, audit logs, and completion status. It defines the system boundary while abstracting internal processes, providing a clear overview of how different users interact with the system and how information is exchanged to support evaluation, monitoring, and administrative functions.

Figure 4 illustrates the context diagram showing the interaction between the evaluation system and external stakeholders.

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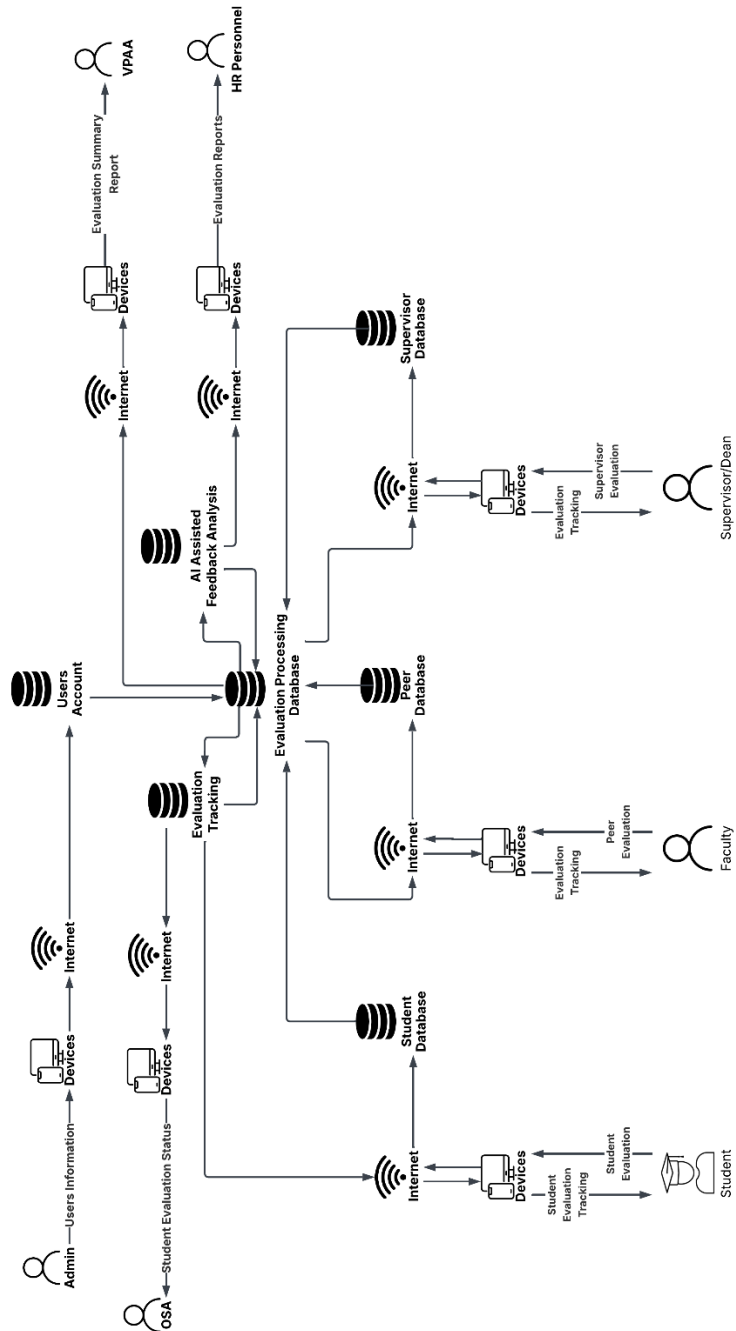


Figure 4. Context Diagram Showing External Stakeholder Interactions with the Web-Based Faculty Evaluation System

Figure 4 illustrates the context diagram of the proposed evaluation system. The diagram represents the system as a centralized process interacting with external stakeholders, including students, faculty, supervisors, the VPAA, the Office of Student Affairs (OSA), Human Resource personnel and system administrators. The diagram highlights the exchange of evaluation data, performance feedback, clearance verification information, and administrative control data between stakeholders and the system, establishing the system boundary and interaction scope.

Data Flow Diagram (Level 0)

The Level 0 Data Flow Diagram (DFD) illustrates the overall data flow of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). In this diagram, the system is represented as a single central process (Process 0), which interacts with multiple external entities through defined data flows.

Students interact with the system by providing login credentials to gain access, accessing evaluation schedules, and completing student evaluation questionnaires. The system processes these inputs and generates student

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evaluation tracking information and AI-assisted evaluation results, allowing students to monitor their evaluation activities.

Faculty members participate in the system through peer evaluation and access evaluation schedules, evaluation tracking, and AI-assisted evaluation results. This enables them to review feedback and contribute to the evaluation of their colleagues.

Supervisors/Deans interact with the system by submitting supervisor evaluations, accessing evaluation schedules, and monitoring evaluation tracking. These functions support formal faculty assessment and oversight responsibilities.

Human Resource (HR) personnel access the system using login credentials and manage evaluation-related activities such as accessing evaluation questionnaires, monitoring evaluation schedules, and reviewing AI-assisted evaluation results. This allows HR to support personnel evaluation and performance monitoring.

The Office of Student Affairs (OSA) interacts with the system by accessing student evaluation status, evaluation schedules, and AI-assisted evaluation results. These data are used to verify student compliance and support administrative processes such as clearance.

The Vice President for Academic Affairs (VPAA) accesses the system through login credentials and receives evaluation summary reports, evaluation

schedules, and tracking information. These outputs support institutional planning, policy-making, and academic decision-making.

System Administrators (Admin) interact with the system by managing user information, configuring evaluation schedules, and maintaining evaluation tracking processes. They ensure proper system operation, access control, and overall functionality.

Overall, the diagram shows how data flows such as login credentials, evaluation questionnaires (student, peer, and supervisor), evaluation schedules, tracking information, AI-assisted evaluation results, user information, and summary reports are exchanged between external entities and the central system. The Level 0 DFD defines the system boundary while presenting a clear overview of stakeholder interaction without detailing internal processes. This highlights the system's role as a centralized platform for evaluation management, data processing, and information dissemination within the institution.

Figure 5 illustrates the Level 0 Data Flow Diagram showing the major system processes and data flow relationships.

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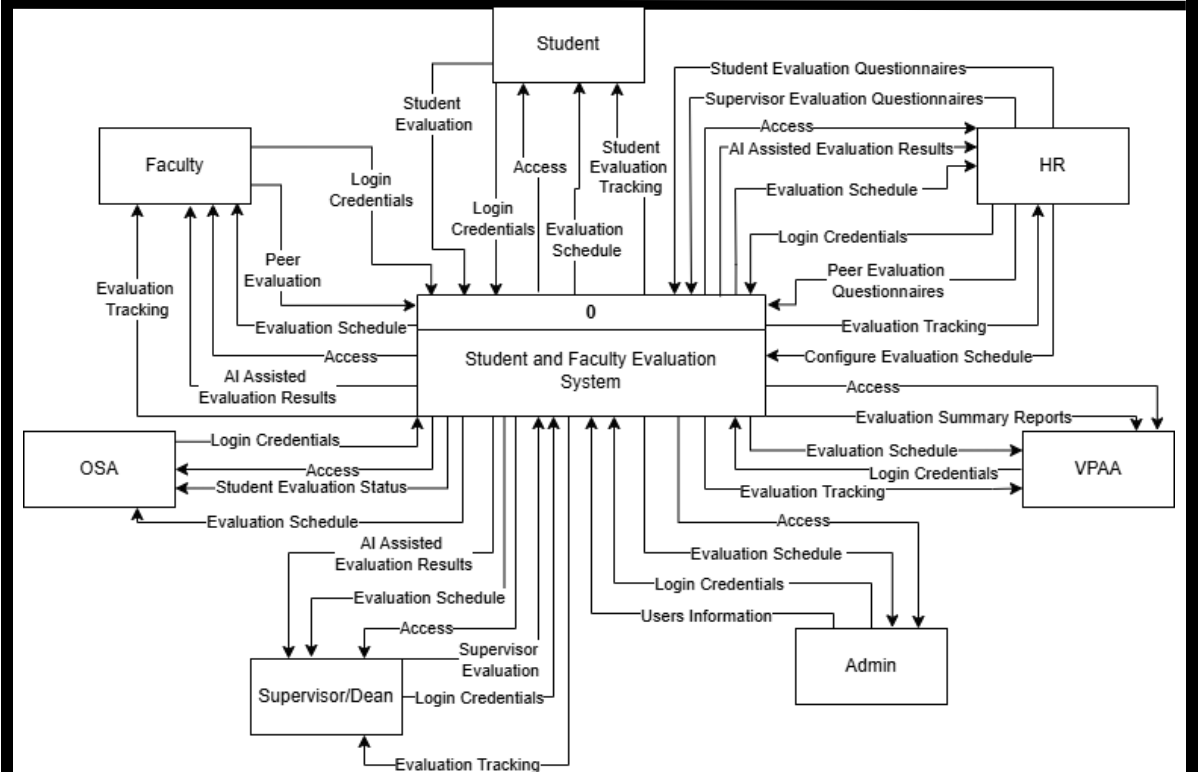


Figure 5. DFD level 0 diagram Showing External Stakeholder Interactions with the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments.

Figure 5 illustrates the Level 0 Data Flow Diagram of the proposed evaluation system. The diagram represents the system as a single centralized process (Process 0) interacting with external stakeholders, including students, faculty, supervisors/deans, the Vice President for Academic Affairs (VPAA), the Office of Student Affairs (OSA), Human Resource (HR) personnel, and system administrators. It highlights the exchange of key data flows such as login

credentials, evaluation submissions (student and supervisor submissions), evaluation schedules and tracking information, evaluation results, student evaluation status, and evaluation summary reports. The diagram establishes the system boundary and clearly defines how stakeholders interact with the system for evaluation processing, monitoring, reporting, and administrative control.

Data Flow Diagram (Level 1)

The Level 1 Data Flow Diagram (DFD) presents a detailed decomposition of the central process shown in the Level 0 DFD of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). The system is broken down into eight major processes, namely: (1.0) User Authentication and Access, (2.0) Evaluation Schedule Management, (3.0) Student Evaluation Processing, (4.0) Peer Evaluation Processing, (5.0) Supervisor/Dean Evaluation Processing, (6.0) AI-Assisted Evaluation Results Generation, (7.0) Evaluation Tracking, and (8.0) User Management. These processes interact with data stores such as Evaluation Schedule, Student Evaluation, Peer Evaluation, Supervisor Evaluation, and Users Account, as well as external entities including students, faculty, supervisors/deans, HR personnel, VPAA, OSA, and administrators.

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The User Authentication and Access Process (1.0) handles login credentials submitted by all external entities. The system verifies user information from the Users Account data store and grants access privileges based on user roles, ensuring secure and authorized system usage.

The Evaluation Schedule Management Process (2.0) manages the creation, configuration, and distribution of evaluation schedules. Administrators and HR personnel configure evaluation schedules, which are stored in the Evaluation Schedule data store. These schedules are then accessed by students, faculty, supervisors/deans, VPAA, and other authorized users to guide evaluation activities.

The Student Evaluation Processing (3.0), Peer Evaluation Processing (4.0), and Supervisor/Dean Evaluation Processing (5.0) handle the submission and storage of evaluation data. Students submit student evaluation questionnaires, faculty members perform peer evaluations, and supervisors/deans submit their respective evaluations. These inputs are stored in their corresponding data stores: Student Evaluation, Peer Evaluation, and Supervisor Evaluation.

The AI-Assisted Evaluation Results Generation Process (6.0) processes the collected evaluation data from all sources. It performs statistical analysis and AI-assisted summarization to generate evaluation results. These results are made available to faculty members, supervisors/deans, and HR personnel, supporting performance assessment and feedback interpretation.

The Evaluation Tracking Process (7.0) monitors the completion and status of evaluation activities. It generates evaluation tracking information accessible to students, faculty, supervisors/deans, VPAA, and administrators. Additionally, it produces student evaluation status, which is specifically accessed by the Office of Student Affairs (OSA) for clearance verification purposes.

The User Management Process (8.0) handles the maintenance of user accounts and user information. Administrators manage user data, ensuring proper role assignment, access control, and system integrity.

Overall, the Level 1 DFD illustrates how data flows such as login credentials, evaluation schedules, student/peer/supervisor evaluation questionnaires, AI-assisted evaluation results, evaluation tracking information, student evaluation status, and user information are processed within the system. The interaction between processes, data stores, and external entities demonstrates a structured and efficient workflow that supports secure access, accurate evaluation processing, effective monitoring, and data-driven decision-making across the institution.

Figure 6 to 12 illustrates the Level 1 Data Flow Diagram showing detailed system sub processes and internal data flow relationships.

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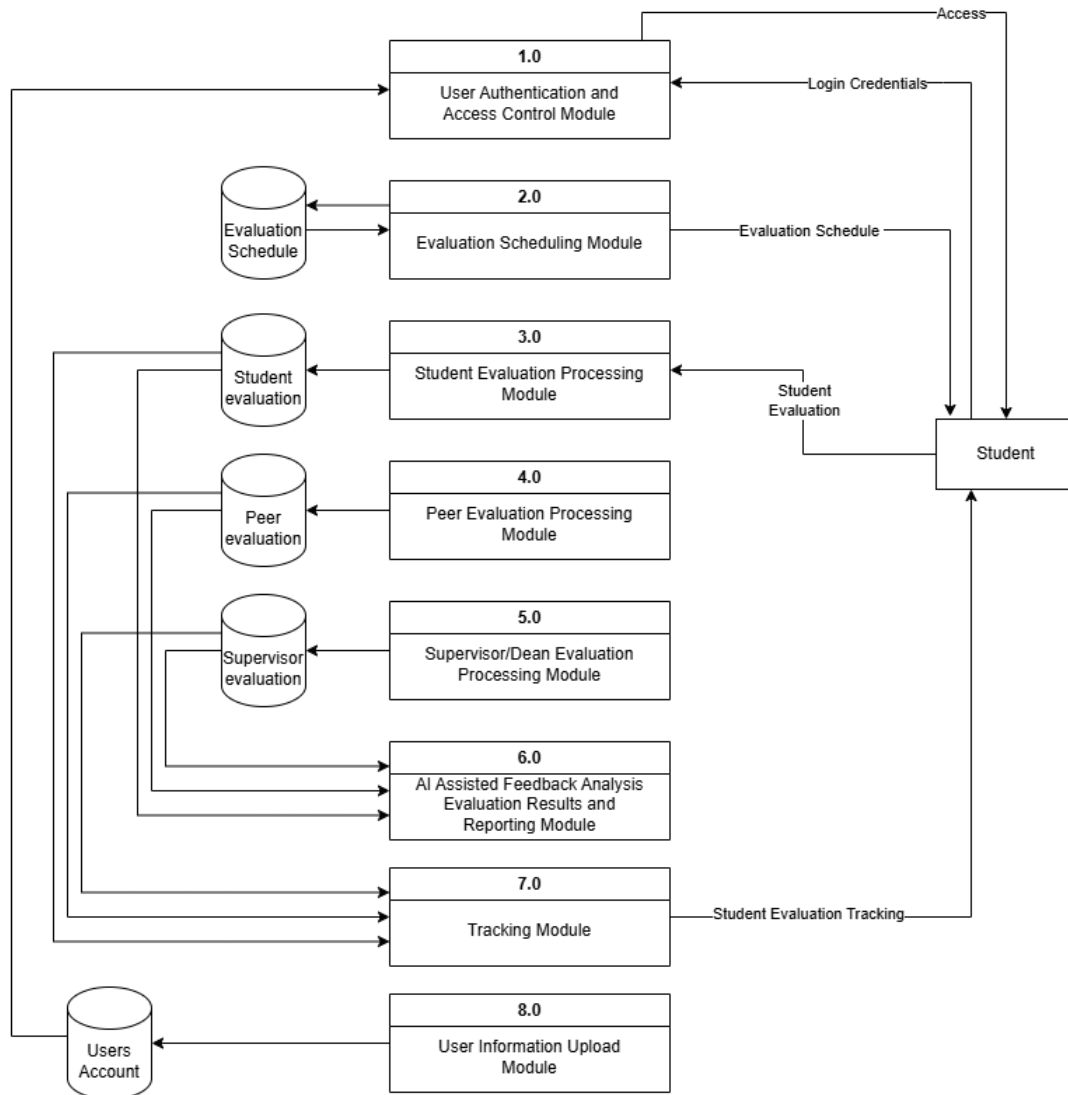


Figure 6. Level 1 Data Flow Diagram for Student of the Web-Based Faculty Evaluation System

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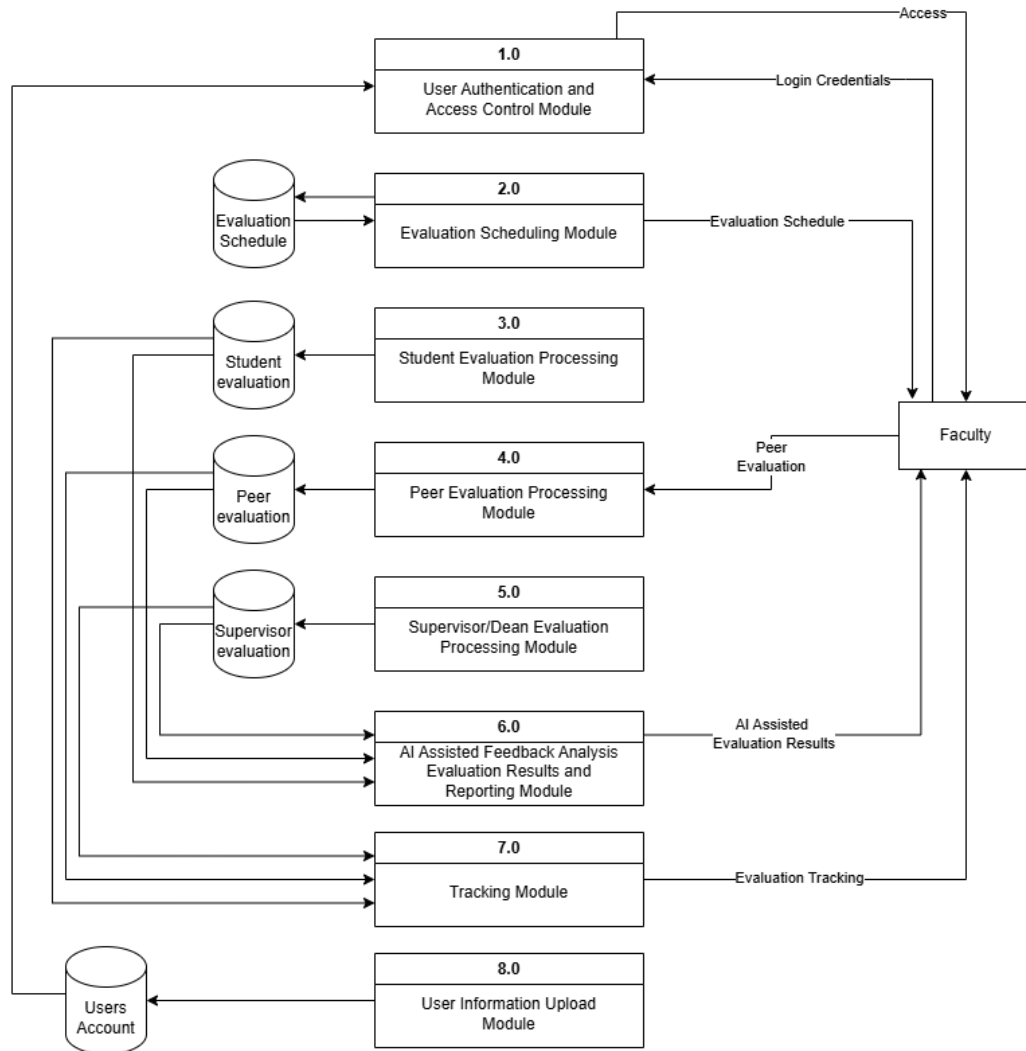


Figure 7. Level 1 Data Flow Diagram for Faculty of the Web-Based Faculty Evaluation System

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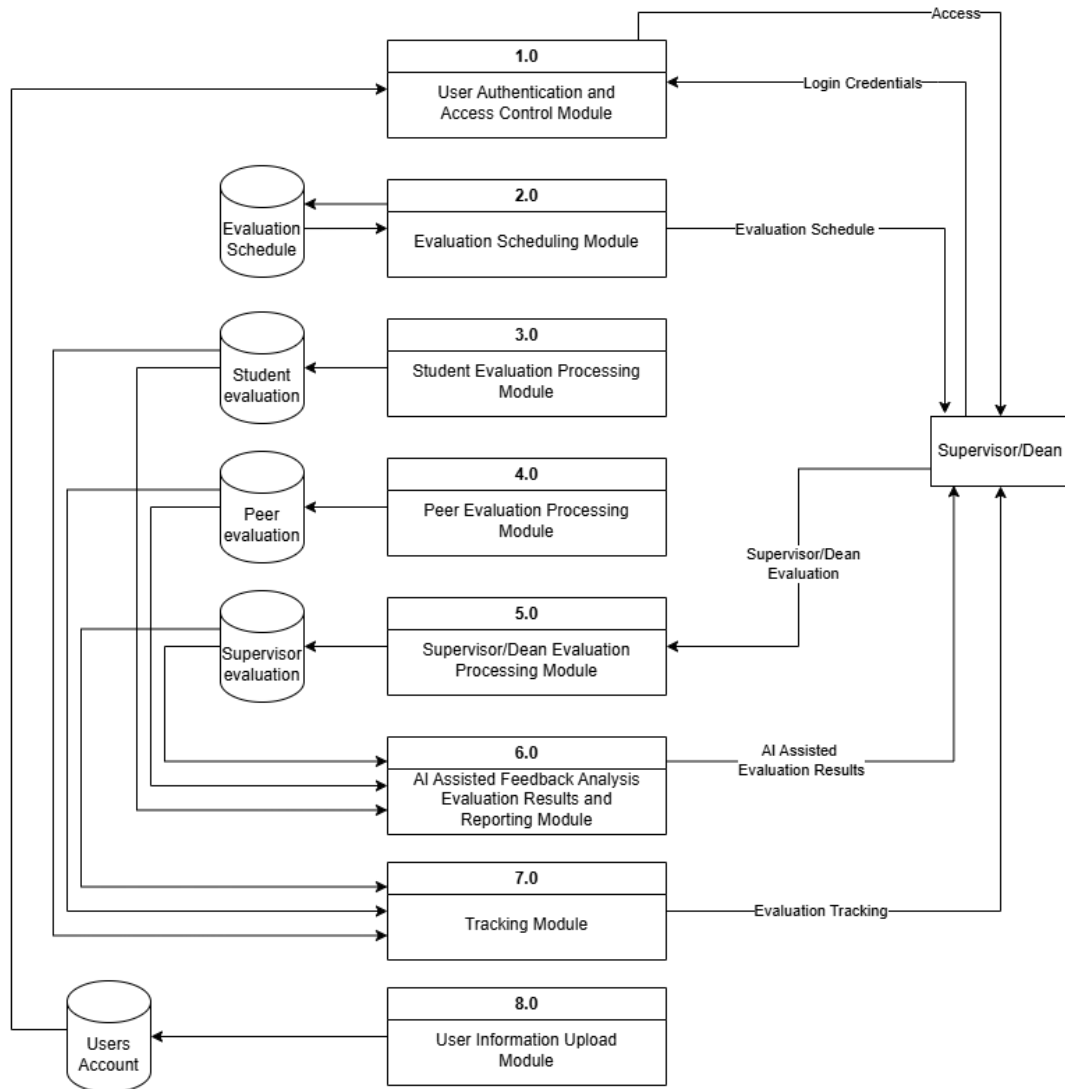


Figure 8. Level 1 Data Flow Diagram for Supervisor/Dean of the Web-Based Faculty Evaluation System

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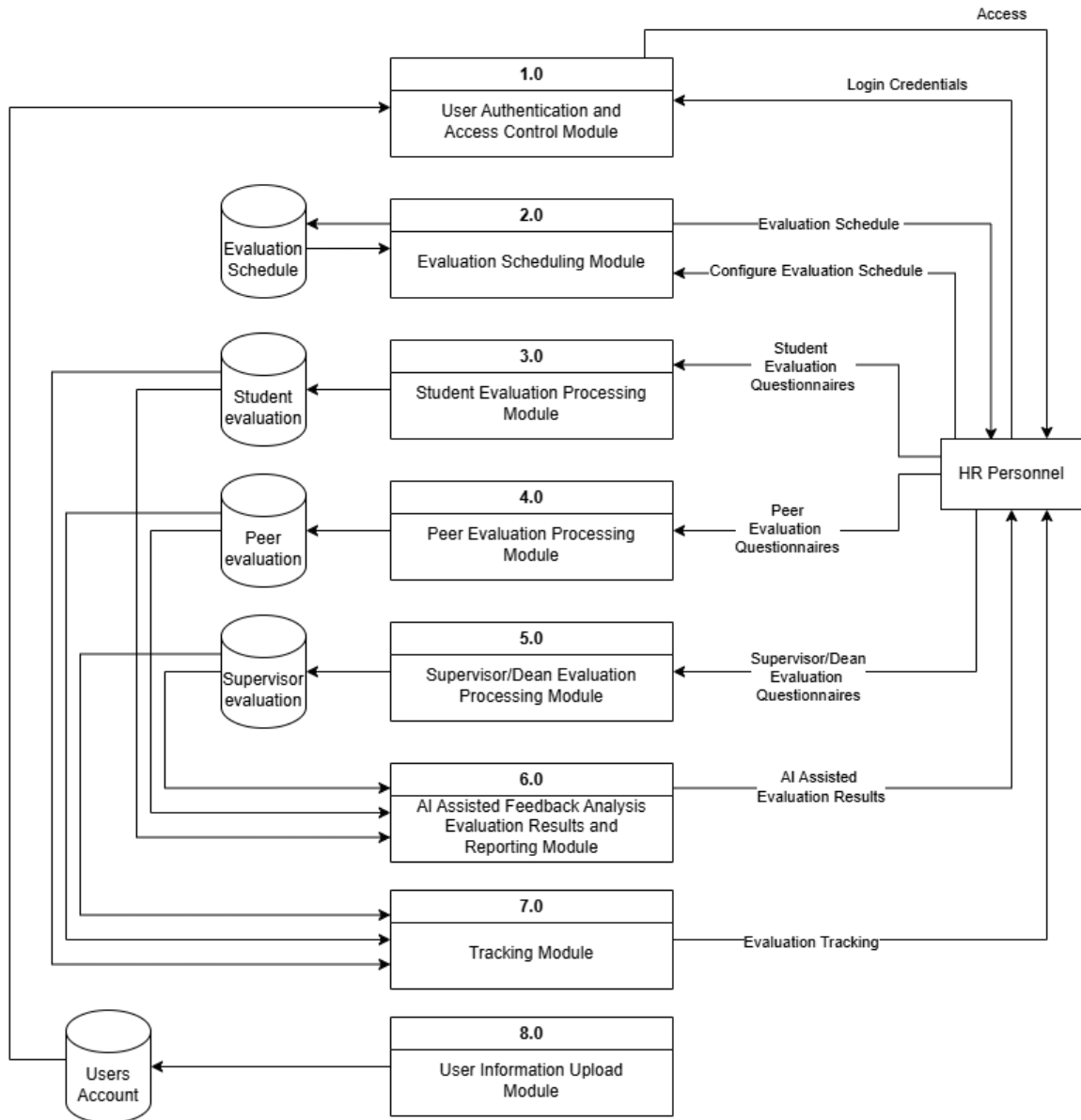


Figure 9. Level 1 Data Flow Diagram for HR personnel of the Web-Based Faculty Evaluation System

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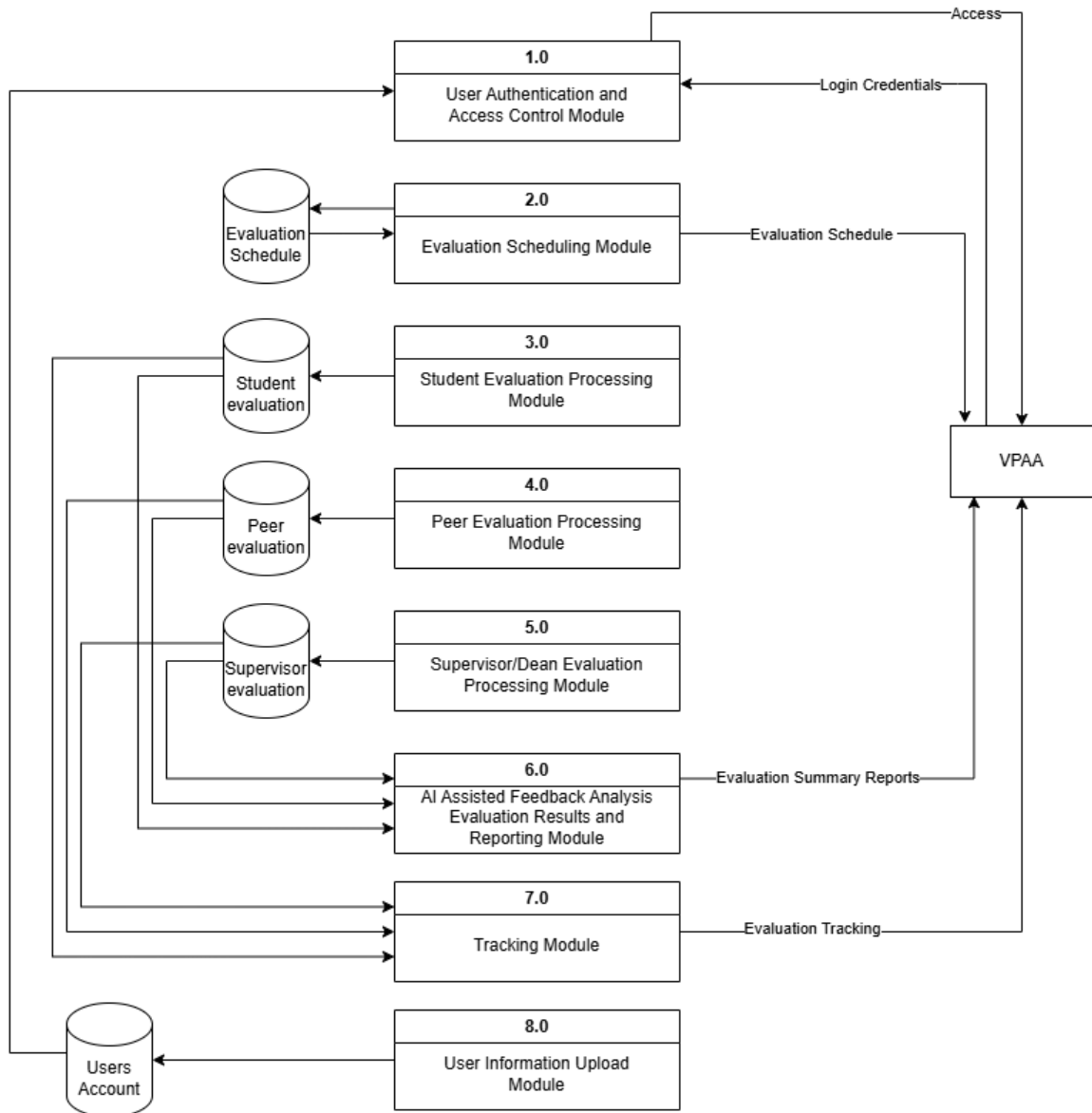


Figure 10. Level 1 Data Flow Diagram for VPAA of the Web-Based Faculty Evaluation System

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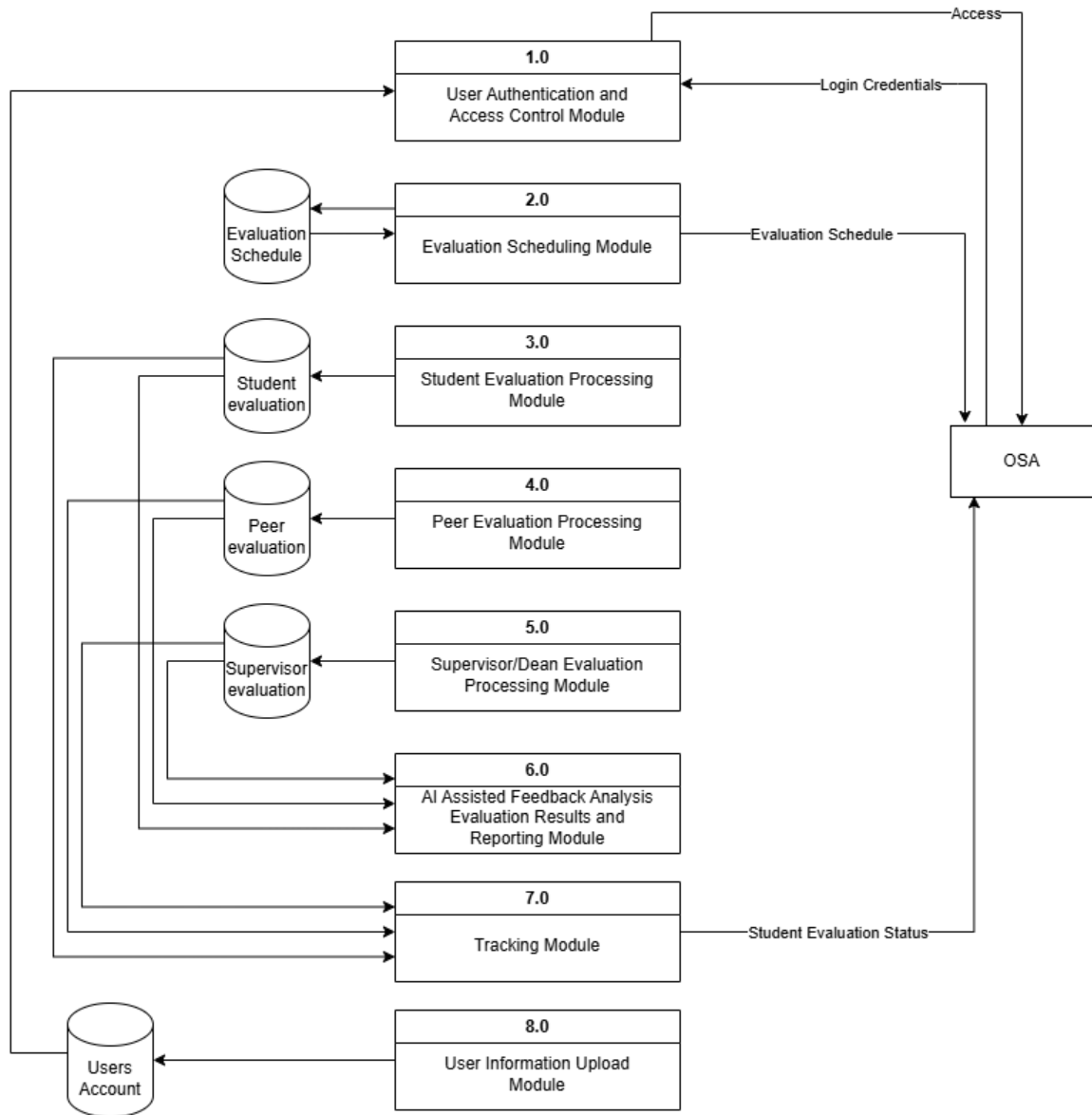


Figure 11. Level 1 Data Flow Diagram for OSA of the Web-Based Faculty Evaluation System

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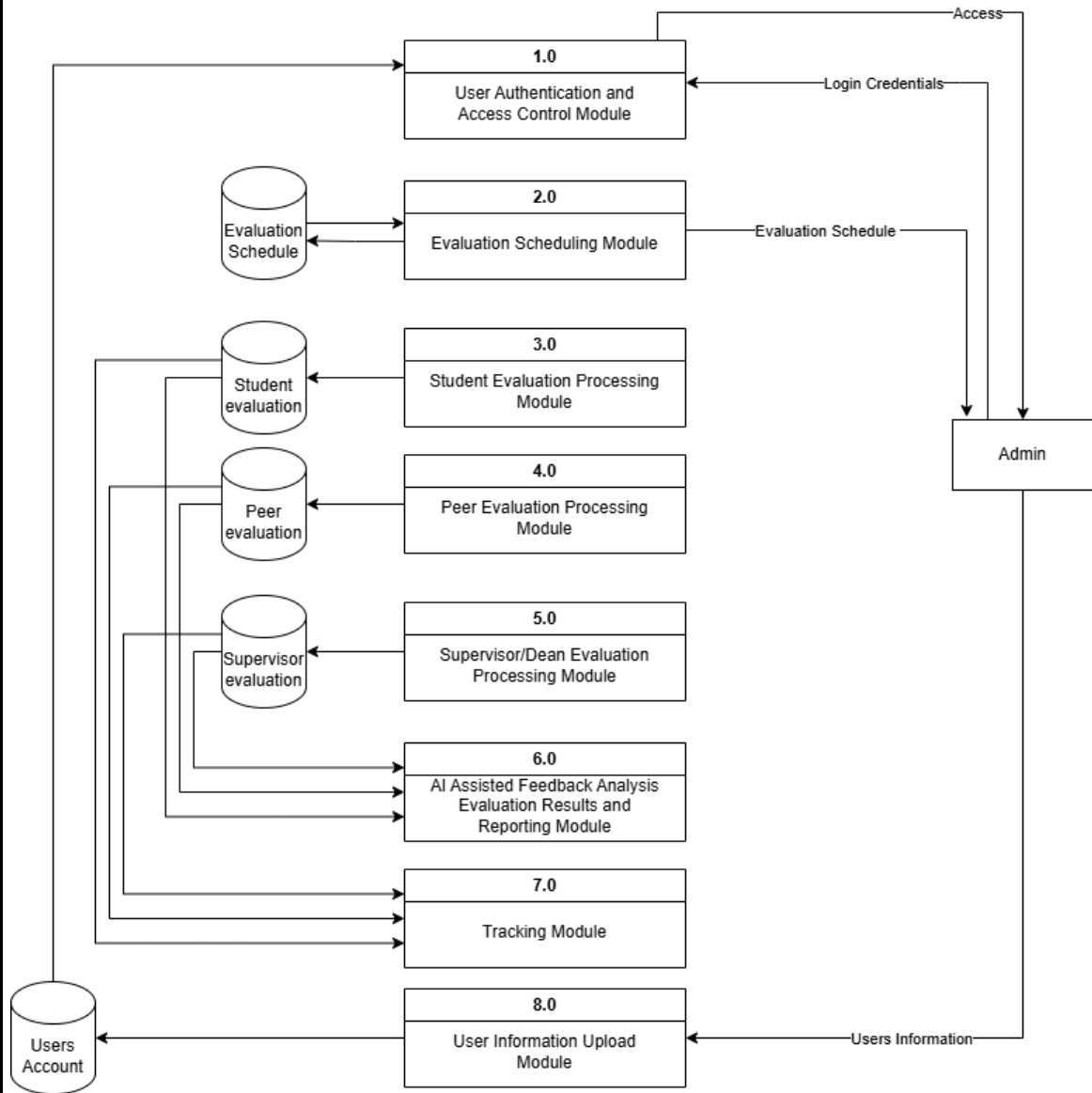


Figure 12. Level 1 Data Flow Diagram for Admin of the Web-Based Faculty Evaluation System

Procedural Workflow

The procedural workflow illustrates the sequence of user interactions and system processes in the operation of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). It presents how different stakeholders interact with the system from login authentication up to evaluation submission, processing, reporting, and system updates.

The workflow begins when a user accesses the system and proceeds to login authentication. The system performs credential verification and assigns role-based access depending on the user type, which may include student, faculty, supervisor/dean, HR personnel, VPAA, OSA, or administrator. This role-based mechanism ensures secure access and directs users to their respective dashboards.

Once authenticated, users are redirected to their designated dashboards. Students access the student dashboard, where they view assigned evaluation tasks, complete evaluation forms, and submit their responses. The system then stores the submitted data in the database, updates the dashboard, and records completion for clearance verification and system updates.

Faculty members access the faculty dashboard to perform peer evaluations. They view assigned peer evaluation tasks, complete evaluation forms, and submit responses. The system stores the data, performs evaluation

processing and AI-assisted analysis, generates reports, updates the dashboard, and executes system updates.

Supervisors/Deans access their dashboard to manage and conduct supervisor evaluations. They create assignment rooms for peer evaluation, distribute evaluation tasks manually or automatically, and complete both peer and supervisor evaluation forms. Submitted data are stored in the database and processed through evaluation analysis and AI-assisted summarization, followed by report generation, dashboard updates, and system updates.

HR personnel access their dashboard to monitor and manage evaluation processes. They display trend charts, set evaluation periods, and configure CHED-compliant questionnaires. HR personnel can also view professors' evaluation results and AI-assisted feedback summaries, supporting performance monitoring and administrative decision-making.

The VPAA accesses the VPAA dashboard to view trend charts and professors' evaluation results, which provide insights for institutional planning and academic decision-making.

The Office of Student Affairs (OSA) accesses the system to monitor student evaluation completion through the student evaluation monitoring feature. This enables verification of student compliance for clearance purposes.

Administrators access the admin dashboard to manage system configuration and monitor audit logs. They also handle user and system management functions to ensure system integrity, security, and proper operation.

Throughout the workflow, the system performs internal processes such as data storage, validation, evaluation processing, AI-assisted feedback analysis, report generation, and dashboard updates. These processes ensure that collected data are transformed into accurate, meaningful, and actionable information.

Overall, the procedural workflow demonstrates a structured and role-based process that integrates authentication, evaluation submission, AI-assisted analysis, reporting, monitoring, and system management. This ensures an efficient, secure, and centralized evaluation system that supports data-driven decision-making and enhances faculty evaluation processes within the institution.

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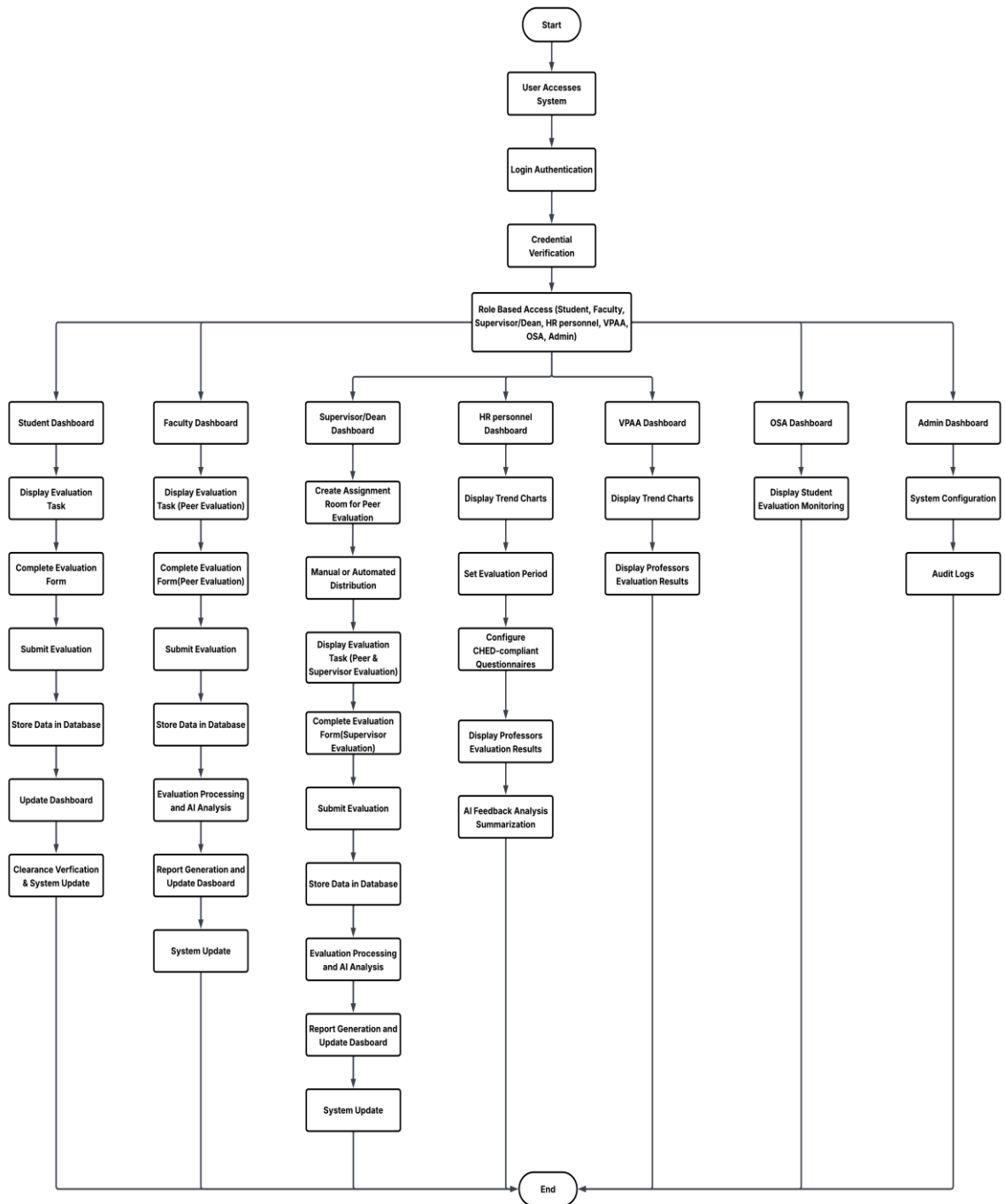


Figure 13. Procedural Workflow Diagram of the Web-Based Faculty Evaluation System

Data Modeling and Database Design

The Data Modeling and Database Design section outlines the structured approach used to organize, represent, and implement the data requirements of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). This section begins with the Conceptual Data Model, which identifies the key entities, their attributes, and the relationships between them, providing a high-level view of the system's data requirements. It is followed by the Entity-Relationship Diagram (ERD), which visually represents the entities, relationships, and cardinalities to clarify how data interacts within the system. Finally, the Logical Database Design translates these conceptual and relational models into a structured database schema, detailing tables, primary and foreign keys to support efficient storage, retrieval, and management of evaluation data. Together, these models provide a foundation for building a secure, consistent, and scalable database that supports the system's operational and analytical requirements.

Conceptual Data Model

The conceptual data model defines the high-level data structure of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines

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(NAAP) by identifying key entities and their relationships within the evaluation management environment. This model emphasizes essential evaluation data components without detailing technical database implementation. The Users entity represents all system stakeholders, including students, faculty members, supervisors, Human Resource personnel, and administrators. Each user is assigned specific roles and access privileges that determine their evaluation responsibilities and interactions with the system. The Evaluation Forms entity represents standardized CHED-compliant evaluation instruments linked to specific courses and faculty members. These forms include structured evaluation criteria and rating scales used to assess teaching performance, professional behavior, and collaboration effectiveness. The Evaluation Responses entity captures evaluation data submitted by authorized users. It stores both quantitative ratings and qualitative feedback, linking each response to the evaluator, the evaluated faculty member, and the corresponding evaluation form to ensure traceability and data integrity. The Evaluation Results entity represents processed outcomes generated by aggregating evaluation responses. It stores consolidated scores, AI-generated summaries of qualitative feedback, and performance trend analyses, which supervisors and Human Resource personnel use for faculty monitoring and development planning. The Clearance Records entity tracks students' evaluation completion status and generates system-issued verification reference numbers for

clearance processing. This ensures compliance with institutional evaluation requirements and supports automated clearance validation.

Relationships among entities support structured data processing: users submit evaluation responses via evaluation forms, which are processed to produce evaluation results. Clearance records are generated based on evaluation completion, linking student compliance to institutional clearance procedures.

Overall, the conceptual data model organizes evaluation information into structured entities and logical relationships, promoting data consistency, evaluation transparency, and effective performance monitoring. It forms the foundation for the subsequent logical and physical database design stages.

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Figure 14. Conceptual Data Model of the Web-Based Faculty Evaluation System

Figure 14 illustrates the Conceptual Data Model of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments of the National Aviation Academy of the Philippines (NAAP). The diagram presents the major data entities of the system and shows how these entities are logically related to support the faculty evaluation process.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure and relationships among the major data entities of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). It defines how data is organized, stored, and interconnected to support accurate, secure, and efficient evaluation processing.

At the core of the system is the Account entity, which represents all system users. It stores essential user credentials such as email, username, password, role, status, and timestamps. The Account entity enables role-based access and is associated with role-specific entities including Admin, Dean, Faculty, Student, HR, and VPAA, allowing the system to manage user identities and responsibilities. System activities are further monitored through the AuditTrail entity, which records actions such as changes made, timestamps, and user-related activity logs.

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The academic structure of the institution is defined through the entities Campus, Institute, Program, Subjects, and Section. These entities are hierarchically related, where a campus contains institutes, institutes contain programs, and programs are associated with subjects and sections. The Student and Faculty entities are linked to these structures through foreign keys, ensuring that users are properly assigned to their respective academic units, subjects, and sections.

The evaluation process is represented by three main entities: StudentEval, PeerEval, and SupervisorEval. These entities store evaluation data submitted by students, faculty (peer evaluation), and supervisors/deans, respectively. Each evaluation record contains scale values and is linked to both the evaluator and the evaluated faculty member. These entities are also associated with their respective questionnaire entities: Stu_Questionnaires, Peer_Questionnaires, and Sup_Questionnaires, which define the standardized CHED-compliant questions, categories, and academic term context used during evaluation.

All evaluation data are consolidated in the Eval_Result entity, which stores computed results such as overall average scores, total responses, and evaluation dates. This entity integrates results from student, peer, and supervisor evaluations, providing a comprehensive assessment of faculty performance.

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These results are accessed by HR and VPAA entities for monitoring, reporting, and decision-making purposes.

The `academic_term` entity defines the timeframe of evaluations, including school year, semester, start and end dates, and active status. All questionnaire and evaluation-related entities are linked to this entity to ensure that evaluation data are properly organized and managed within a specific academic period.

Additionally, HR personnel are responsible for configuring evaluation questionnaires, as shown by their relationship with the questionnaire entities. The Admin entity manages user accounts and system control, ensuring proper configuration, maintenance, and access management across the system.

Overall, the ERD enforces data integrity through well-defined primary and foreign key relationships among entities. These relationships ensure that evaluation data are consistently linked to valid users, academic structures, questionnaires, and academic terms. This structured design supports efficient data storage, accurate processing, reliable reporting, and comprehensive evaluation analysis, enabling the system to effectively support institutional quality assurance and decision-making.

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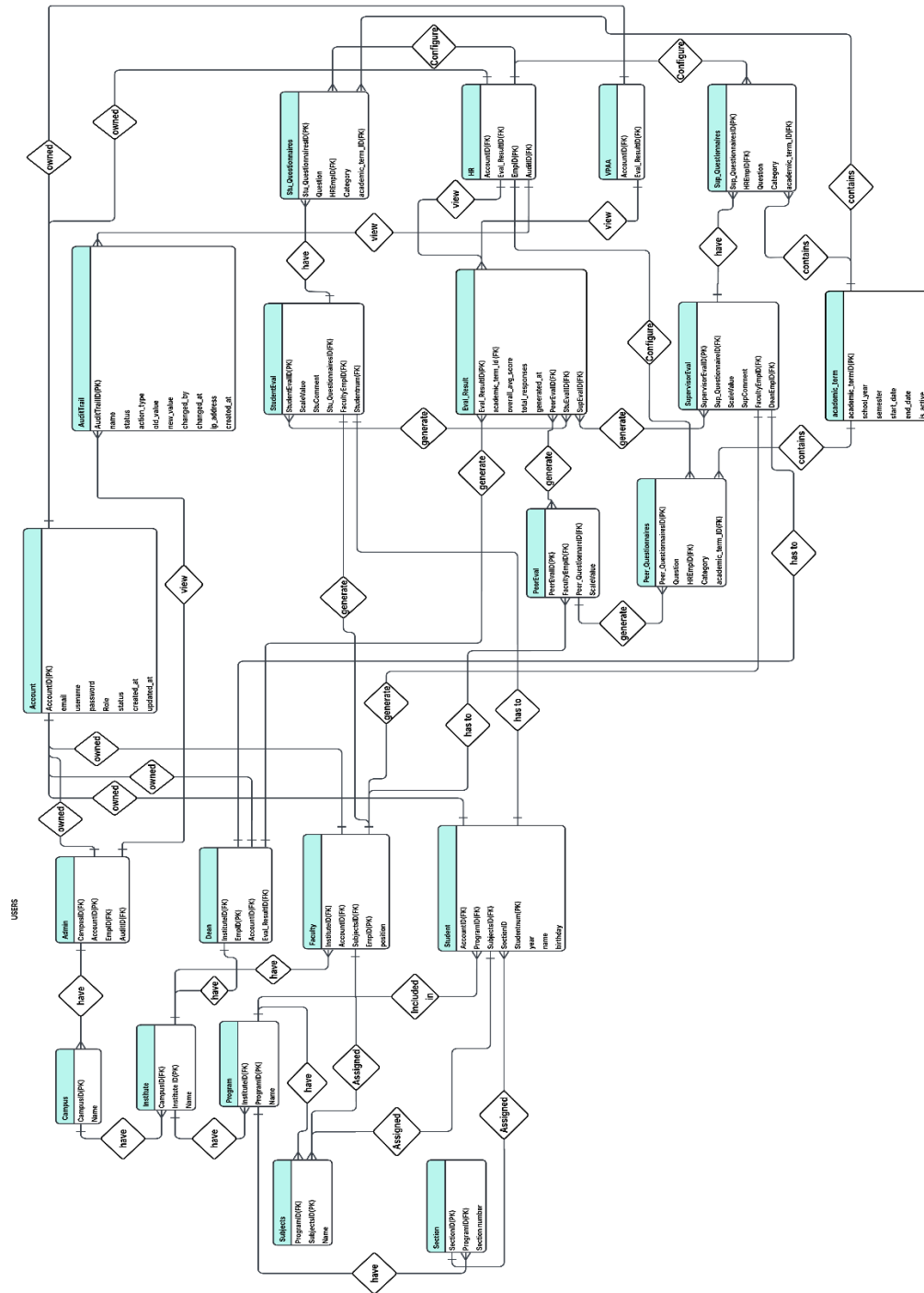


Figure 15. ERD Model of the Web-Based Faculty Evaluation System

Logical Database Design

The logical database design defines how data is structured and managed in the Web-Based Student and Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). It translates system requirements into a relational database model composed of interconnected tables that support secure storage, efficient retrieval, and accurate processing of evaluation data.

At the core of the database is the Account table, which stores user credentials such as email, username, password, role, and status. This table enables role-based access control for different users, including students, faculty, deans, HR personnel, VPAA, and administrators. It is linked to role-specific tables such as Student, Faculty, Dean, HR, VPAA, and Admin, allowing the system to manage user-specific information and responsibilities.

The academic structure is defined through tables such as Campus, Institute, Program, Subjects, and Section, which establish hierarchical relationships within the institution. These tables ensure that students and faculty are properly assigned to their respective academic units, enabling accurate mapping of evaluations to subjects, programs, and sections.

Evaluation instruments are stored in dedicated tables, including Stu_Questionnaires, Peer_Questionnaires, and Sup_Questionnaires, which contain CHED-compliant questions, categories, and references to the academic

term. This structure allows the system to dynamically assign appropriate evaluation forms based on user roles and evaluation type.

Evaluation transactions are captured in the StudentEval, PeerEval, and SupervisorEval tables. These tables record evaluation responses submitted by students, faculty (peer evaluation), and supervisors/deans, respectively. Each record includes scale values and qualitative comments, along with references to evaluators, evaluated faculty members, and corresponding questionnaires. This ensures traceability and integrity of evaluation data.

Processed evaluation results are consolidated in the Eval_Result table, which stores computed values such as overall average scores, total responses, and generated timestamps. This table integrates data from multiple evaluation sources and serves as the basis for reporting and performance analysis.

The academic_term table defines the time frame for evaluations, including school year, semester, start and end dates, and active status. All evaluation-related tables are linked to this entity to ensure that data is organized and retrieved based on the correct academic period.

Additional tables such as HR and VPAA are linked to evaluation results and filtered outputs, enabling access to reports and analytical insights for monitoring, evaluation, and institutional planning. The Admin table supports system administration by managing user accounts and system configuration.

To maintain data consistency and efficiency, the database design follows normalization principles, reducing redundancy and ensuring logical separation of data. Relationships between tables are enforced through primary and foreign key constraints, guaranteeing referential integrity and preventing invalid or duplicate records.

Overall, this logical database design supports accurate evaluation processing, efficient reporting, AI-assisted feedback analysis, and reliable monitoring of faculty performance. It provides a scalable and maintainable foundation for the system, ensuring long-term usability and alignment with institutional and regulatory requirements.

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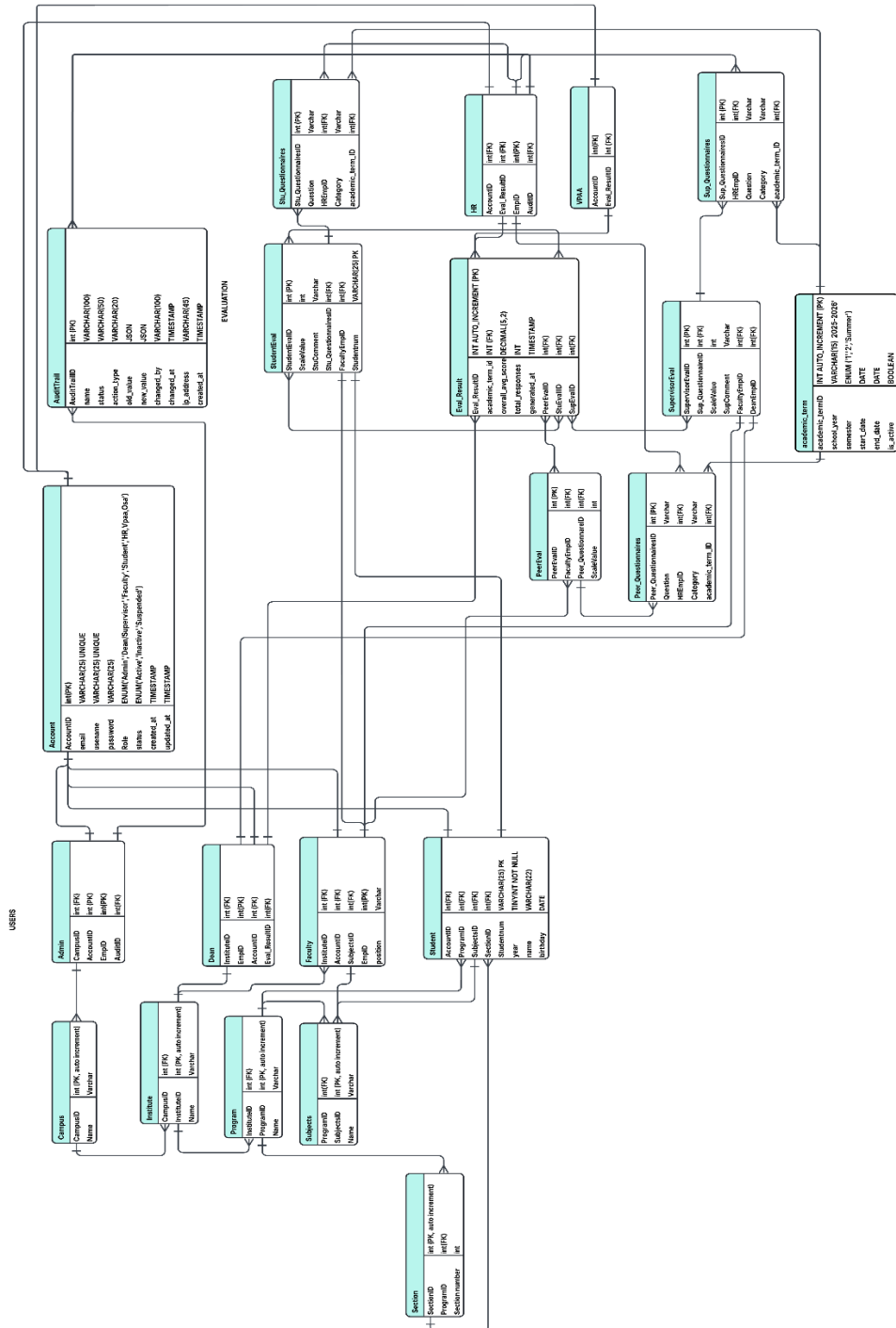


Figure 16. Logical Database Design Model of the Web-Based Faculty Evaluation System

System Security and Data Privacy Design

The protection of evaluation data and the personal information of all system users is a foundational design principle of the proposed system. The security architecture is composed of multiple independent layers: role-based access control enforced at the application layer, Bcrypt password hashing, risk-adaptive OTP authentication via institutional email, HTTPS data transmission with TLS, Cloudflare perimeter firewall protection, and scheduled encrypted database backups. Each layer addresses a distinct class of threat, and the combination ensures that no single point of failure can compromise the confidentiality, integrity, or availability of evaluation data.

Access Control and Authentication Mechanisms

Access to the system is governed by a role-based access control (RBAC) framework that assigns each registered user to one of seven defined roles: Student, Faculty, Supervisor/Dean, HR Officer, OSA Officer, VPAA, or System Administrator. The system determines the authenticated user's role at login and constructs the session context accordingly, restricting access to only the modules, data records, and operations that the assigned role is authorized to perform.

Authentication employs a risk-adaptive OTP mechanism. On first-time login, the system issues a one-time password to the user's registered institutional email

address for identity verification — since the email account was provisioned by the administrator, successful OTP verification confirms that the user is the legitimate owner of that address. Subsequent logins from recognized devices and sessions do not require OTP entry. However, if the system detects a login attempt from an unrecognized device or identifies anomalous session characteristics, it re-triggers OTP verification to confirm the legitimacy of the access attempt before granting entry. The OTP is time-limited and single-use, ensuring that even a compromised password credential cannot grant system access without simultaneous control of the user's institutional email inbox.

Session management controls are applied throughout all authenticated interactions. Session tokens are invalidated upon logout, upon expiry of the configured session duration, and upon detection of concurrent login attempts from different IP addresses. All session-related events — including successful logins, failed login attempts, OTP generation, and session terminations — are captured in the audit trail with the user identifier, timestamp, and IP address of the access attempt, creating a comprehensive record that supports security review and incident investigation.

Data Encryption and Protection Measures

User passwords are the most security-sensitive credentials stored in the system database. The system uses Bcrypt, an adaptive cryptographic hashing algorithm, to store all user passwords. Bcrypt is specifically designed for password storage: it is computationally slow by design, making brute-force and dictionary attacks significantly more costly for an attacker even if the database is compromised. Each password hash is generated with a unique per-user salt, preventing precomputed rainbow table attacks and ensuring that identical passwords produce different stored hash values for different users. Bcrypt's work factor (cost parameter) can be increased over time as hardware capabilities improve, ensuring that the security of stored password hashes remains robust throughout the system's operational life.

All data transmitted between client devices and the institutional server is protected using HTTPS with current TLS protocols, preventing interception, tampering, or eavesdropping during evaluation submission, dashboard access, and report download operations. At the network perimeter, a Cloudflare web application firewall is positioned in front of the system's public domain, filtering all incoming traffic before it reaches the application server. This configuration provides active defense against distributed denial-of-service attacks, automated credential stuffing attempts, SQL injection, cross-site scripting, and other common

web application threats without requiring changes to the server hardware configuration.

Automated backup procedures run on a scheduled basis, creating encrypted backup archives written to a secondary storage location isolated from the primary database. Backup integrity is verified through scheduled restoration tests. This regime ensures that complete, current evaluation records can be recovered in the event of hardware failure, data corruption, or a successful attack on the primary database server.

Data Privacy and Ethical Considerations

The proposed system is designed in full compliance with Republic Act No. 10173, the Data Privacy Act of 2012, which governs the collection, storage, processing, and sharing of personal information in the Philippines. The system collects only the personal data fields required for evaluation processing and user authentication: name, institutional email address, role assignment, course enrollment, and evaluation responses. No additional personal data is collected beyond what is operationally necessary.

Student evaluation submissions are processed in a manner that protects evaluator anonymity. Evaluation responses are dissociated from student identifiers before being presented to faculty members or included in HR reports. The AI-

assisted summarization process operates on aggregated comment text without attributing individual comments to specific students in the output. Faculty members and deans have no access to individual submission records; they receive only aggregated ratings and thematic summaries.

The soft-delete policy enforced throughout the system serves both data integrity and privacy compliance objectives. Records flagged for deletion are retained in a deactivated state rather than permanently removed, preserving the integrity of historical evaluation data for audit purposes while allowing records to be excluded from active operational views. This approach ensures that evaluation history remains available for regulatory review without exposing deactivated personal data in routine system operations.

Should a data subject formally exercise their right to erasure or correction under RA 10173, the system administrator can flag the relevant records for anonymization or update. The audit trail records all data subject requests and the administrative actions taken in response, providing a documented compliance record.

Audit Logging and Activity Monitoring

The system maintains a comprehensive, non-editable audit trail that captures all significant actions performed within the platform. Logged events

include user authentication events, evaluation form submissions and draft saves, HR administrative actions (evaluation period creation, questionnaire configuration, report generation), AI summary generation and edits, administrative user management actions, and all data modification events across all modules. Each audit record includes the user identifier, the action performed, the affected data record or module, the timestamp, and the source IP address.

The audit trail is accessible to system administrators through a dedicated administrative view that supports filtering by user, action type, module, and date range. The non-editable property of the audit trail is enforced at the database layer: audit records are inserted through a write-only procedure and the database role assigned to the application layer does not include update or delete permissions on the audit table. This design ensures that the audit trail cannot be altered by any application-level action, including actions performed by users with the System Administrator role.

At the system level, activity monitoring tools provide administrators with real-time and historical visibility into server resource utilization, application error rates, backup job completion status, and response time metrics. Automated alerts notify administrators when monitored metrics exceed configured thresholds, enabling proactive identification and resolution of performance anomalies or security incidents before they escalate into service disruptions.

Implementation Environment and Technology Stack

This section describes the software environment, hardware infrastructure, development toolset, and deployment configuration that together constitute the implementation environment for the proposed system.

Software Requirements

The system requires a web server software environment capable of hosting a server-side web application framework, a relational database management system, and supporting services for email delivery and external API communication. The server must be configured to handle HTTPS requests routed through the Cloudflare proxy layer, support environment variable management for API credentials, and execute scheduled background tasks for backup procedures and email notifications. External dependencies include the OpenAI GPT-4.1 mini API for AI-assisted feedback analysis and an SMTP service for automated email delivery, including OTP distribution.

A relational database management system (RDBMS) is required to store and manage all evaluation data, user account records, questionnaire configurations, audit log entries, and session information. The database must support ACID-compliant transaction processing to ensure data consistency across concurrent evaluation submissions, and must be configured with access controls

that restrict database connections to authenticated application-layer processes only.

The system must be accessible through standard modern web browsers on all major operating systems without requiring the installation of additional client-side software or browser extensions. This requirement ensures that students accessing the system from personal devices, university computer laboratories, or mobile devices can interact with the evaluation platform using whatever browser environment is available to them. Client-side functionality is implemented using standard HTML5, CSS3, and JavaScript, with responsive design principles applied to ensure usability across screen sizes ranging from desktop monitors to smartphone displays.

Hardware Requirements

The system is designed for deployment on NAAP's institutional server infrastructure, which must meet minimum 2025-standard hardware specifications to ensure reliable performance during peak concurrent evaluation windows. The hardware recommendations below reflect current market availability and are aligned with the system's operational demands: multi-role concurrent access, real-time dashboard queries, AI API call processing, PDF report rendering, and scheduled database backup execution.

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Server Hardware Requirements

Table 3.55. Recommended Server Hardware Specifications — 2025

Standards

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Intel Xeon E-2300 Series (8-core) or AMD EPYC 7003 Series (8-core), 3.0 GHz base clock	Intel Xeon Gold 6300 Series or AMD EPYC 9004 Series, 12–16 cores, 3.5 GHz+, with hardware virtualization support
RAM	16 GB DDR4 ECC RAM (Error-Correcting Code)	32 GB DDR5 ECC RAM for improved concurrent session handling and query caching
Primary Storage	512 GB NVMe SSD (M.2 or U.2 form factor, PCIe 4.0)	1 TB NVMe SSD RAID-1 array for redundancy; <100µs read latency for database queries
Backup / Secondary Storage	1 TB SATA SSD or HDD, physically or logically isolated from primary	2 TB NVMe or enterprise SATA SSD on isolated volume, with offsite or cloud backup target
Network Interface	1 Gbps Ethernet NIC	Dual 1 Gbps or single 10 Gbps Ethernet NIC with failover support
Operating System	64-bit Linux Server (Ubuntu 22.04 LTS / Debian 12 or equivalent)	Ubuntu 24.04 LTS with automatic security patch management enabled
Power Supply	Redundant 500W PSU (80+ Gold certified)	Redundant 750W PSU (80+ Platinum), UPS-backed for protection against power interruptions

The recommended server configuration — a 12- to 16-core processor, 32 GB of DDR5 ECC RAM, and NVMe SSD storage — is sufficient to handle peak

concurrent load during evaluation windows when large numbers of students, faculty, and supervisors may simultaneously access the system. ECC RAM is specified to protect against memory errors that could corrupt database operations or session data during high-load periods. NVMe SSD storage is specified to ensure sub-millisecond I/O response times for database queries, which directly impacts the perceived responsiveness of the evaluation submission and dashboard access functions.

Storage must be provisioned with sufficient headroom for several years of data growth beyond initial deployment. Each evaluation period generates records proportional to the number of enrolled students, faculty assignments, and open-ended comments submitted. Backup storage must be physically or logically isolated from the primary database storage volume so that a single hardware failure does not simultaneously destroy both the live database and its backup copies. Administrators should configure automated off-site backup replication to a cloud storage target or a separate physical location for disaster recovery purposes.

Client-Side Hardware Requirements

Client-side hardware requirements are intentionally minimal by design. Any device capable of running a contemporary web browser — including desktop computers, laptop computers, tablet computers, and smartphones released within the last five years — is sufficient for accessing the evaluation system. This broad

device compatibility ensures that all students and faculty can fulfill their evaluation obligations from any location and any personal device without requiring institutional hardware provisioning or software installation. The responsive interface design ensures full functionality on screen resolutions as low as 375px width (standard smartphone portrait mode).

Table 3.56. Minimum Client-Side Hardware Requirements

Device Type	Minimum Requirement
Desktop / Laptop Computer	Any x86-64 system with 4 GB RAM, running Windows 10/11, macOS 12+, or Ubuntu 20.04+, with a modern browser (Chrome 110+, Firefox 115+, Edge 110+, or Safari 16+)
Tablet Computer	Any ARM or x86-64 tablet with 3 GB RAM running iOS 15+, iPadOS 15+, or Android 12+, with a current browser
Smartphone	Any smartphone with 2 GB RAM running iOS 15+ or Android 10+, with a current browser; 375px minimum screen width supported
Network Connectivity	Stable internet connection of at least 1 Mbps for evaluation submission; 5 Mbps recommended for dashboard and PDF report access

Development Tools and Programming Technologies

The system is developed using a modern server-side web application framework that supports structured application routing, server-side form processing, session management, database interaction, and integration with external APIs. The framework's component-based architecture supports the modular system design, allowing individual modules — such as the evaluation

submission engine, the AI analysis module, the reporting generator, and the administrative management interface — to be developed, tested, and updated independently.

Database development and management is performed using standard relational database tools supporting schema design, migration management, query optimization, and performance monitoring. The database schema is normalized to the Third Normal Form (3NF) to eliminate data redundancy and ensure efficient, consistent data retrieval across all system modules. Primary and foreign key constraints are defined throughout the schema to enforce referential integrity and prevent invalid data relationships from forming during the system's operational life.

The AI-assisted feedback analysis feature is implemented through integration with the OpenAI GPT-4.1 mini API. When an authorized user initiates comment summarization for a faculty member's evaluation period, the system retrieves all submitted open-ended comments, structures them into an appropriately formatted API request with a system-defined analytical prompt, and returns a concise thematic summary organized around commendations, constructive feedback, and recurring concerns. This API-based summarization is complemented by a locally executed frequency analysis module that independently tokenizes submitted comments, removes common stop words, and computes term frequency distributions to identify the most commonly mentioned

words and phrases. The frequency analysis results serve as an independent validation layer for the AI-generated summary and are displayed alongside it to assist HR in assessing summary completeness.

Security implementation tools include the Bcrypt library for cryptographic password hashing with per-user salt generation, an OTP generation library for time-limited single-use authentication codes and expiry management, and HTTPS/TLS configuration at the web server level. The Cloudflare firewall is configured at the DNS level by routing the system's public domain through Cloudflare's proxy infrastructure, requiring no modifications to the server hardware. PDF report generation is handled by a server-side rendering library capable of producing documents conforming to the layout requirements specified by CHED CMO No. 19, Series of 2025. Version control, sprint task management, and collaborative development are supported through standard Agile development tooling.

Deployment Environment

The system is deployed on NAAP's institutional server, configured as a web server environment that handles incoming HTTPS requests and routes them to the appropriate application components. The Cloudflare firewall sits at the DNS level in front of the institutional domain, intercepting and filtering all traffic before it is

forwarded to the server. This configuration is activated by pointing the system's DNS records through Cloudflare's proxy infrastructure, and it requires no modifications to the physical server hardware or operating system configuration.

The application and database layers may be co-located on a single server or separated across two servers within the institutional network, depending on available infrastructure at the time of deployment. If co-located, database access is restricted to authenticated local connections from the application layer only — the database port is not exposed to external network traffic. Database queries are executed through the application's data access layer, which enforces parameterized query construction throughout, preventing SQL injection vulnerabilities from arising at the data access boundary.

Automated backup jobs are scheduled to execute at defined intervals, writing encrypted backup archives to a secondary storage location. Administrators receive automated alerts if a backup job fails to complete successfully. System monitoring tools track server resource utilization, application error rates, and response times, alerting administrators when metrics exceed defined thresholds. These monitoring capabilities allow performance issues or security anomalies to be identified and investigated before they escalate into service interruptions.

The modular application architecture ensures that individual components can be updated, patched, or replaced without requiring a full system redeployment.

Updates to the AI analysis module, for instance, can be deployed and tested independently of the evaluation submission engine or the administrative management module. This design approach reduces the operational risk associated with system maintenance and ensures that the platform can be kept current with evolving security requirements and institutional needs throughout its operational lifecycle.

Summary of Technology Stack

Table 3.57. Summary of Technology Stack — AI-Enhanced CHED-Compliant Evaluation System

Layer	Component	Technology / Approach
Network Security	Web Application Firewall	Cloudflare (Cloud-Based WAF and DDoS Protection)
Transport	Data Transmission Security	HTTPS with TLS Encryption
Application	Server-Side Framework	Modern Web Application Framework
Application	Client-Side Interface	HTML5, CSS3, JavaScript (Responsive Design)
AI and Analytics	Feedback Summarization	OpenAI GPT-4.1 mini API
AI and Analytics	Frequency Analysis Module	Local Text Processing and Term Frequency Analysis
AI and Analytics	Evaluation Behavior Analysis	Rule-Based Behavioral Pattern Scoring (Time, Uniformity, Repetition Detection)
AI and Analytics	AI-Based Bias Detection	NLP Comment Classification Engine (Constructive / Neutral / Biased Labels)
AI and Analytics	Cross-Source Validation	Multi-Evaluator Consistency Checker (Student, Peer, Supervisor Comparison)

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Layer	Component	Technology / Approach
AI and Analytics	Credibility Score Engine	Composite Evaluation Reliability Scoring (0–100 Scale, Three Reliability Bands)
AI and Analytics	Smart Recommendation Engine	AI-Assisted Faculty Development Recommender (Weak-Area Detection and Targeted Suggestions)
AI and Analytics	Longitudinal Performance Tracker	Historical Evaluation Trend Analysis (Multi-Semester Improvement and Decline Detection)
AI and Analytics	AI Transparency Module	Keyword Cluster and Reasoning Disclosure Layer (Displayed Alongside Every AI Summary)
Data Management	Database System	Relational DBMS — ACID-Compliant with 3NF Schema
Data Protection	Password Storage	Bcrypt One-Way Hashing with Per-User Salt
Authentication	Multi-Factor Authentication	Email-Based One-Time Password (OTP)
Access Control	Authorization Model	Role-Based Access Control (RBAC) — 7 Roles
Reporting	Evaluation Report Generation	CHED-Compliant PDF Rendering Engine
Communication	Transactional Email Service	SMTP-backed delivery for OTP and password-reset email workflows.
Resilience	Data Backup and Recovery	Scheduled Encrypted Backup with Restoration Testing

The technology decisions summarized above reflect a consistent set of design principles: preference for proven, widely supported technologies that minimize implementation risk; use of open standards and frameworks that reduce long-term maintenance dependency on proprietary vendors; and deliberate layering of security controls so that no single point of failure can expose sensitive evaluation data. Each component was selected with the specific operational

context of NAAP in mind, balancing analytical capability, institutional IT capacity, regulatory compliance requirements, and the scalability needed to accommodate growth in users, campuses, and evaluation data volume throughout the system's intended operational life. The adoption of Bcrypt for password hashing reflects current best practices in credential security, offering adaptive work factors that can be tuned to maintain resistance to brute-force attacks as computing hardware capabilities improve over time.

CHAPTER 4

METHODOLOGY, RESULTS AND DISCUSSION

This chapter presents the research methodology, findings, and discussion of the study conducted to develop and evaluate the AI-Enhanced Web-Based Faculty Evaluation System that integrates CHED-prescribed instruments with AI-assisted feedback analysis at the National Aviation Academy of the Philippines. It describes the research design, sampling techniques, data collection procedures, and statistical methods used to ensure a systematic and objective assessment of the proposed system. It also reports the results gathered from multi-source evaluation data, ISO/IEC 25010:2023 software quality evaluation, comparative system performance analysis, and user perception surveys. The findings are analyzed in relation to the study's objectives to determine the system's effectiveness, reliability, usability, and institutional impact, thereby establishing its validity and providing a basis for conclusions and recommendations.

Research Design

The study employed a quantitative descriptive-developmental research design with an evaluative component. This design is appropriate because the study involves the development of a Web-Based Faculty Evaluation System and the quantitative assessment of its quality, effectiveness, and user acceptance. It

aligns with the study's objectives of examining multi-source faculty evaluation, AI-assisted feedback analysis, ISO/IEC 25010:2023 software quality, system comparison, user acceptance, and data-driven improvements.

The descriptive-developmental design was selected because the study requires both the development of a functional system and the systematic evaluation of its performance after development. The developmental aspect supports the design, development, testing, and implementation of the system, while the descriptive and evaluative components allow the researchers to summarize respondents' ratings and determine the system's effectiveness in terms of efficiency, accuracy, usability, and administrative workload. The use of structured questionnaires and measurable criteria provides a systematic basis for obtaining consistent quantitative results.

The study used a quantitative research approach, with primary data collected through structured survey questionnaires and system-generated information. Respondents provided numerical ratings using Likert-scale questionnaires, which were analyzed using appropriate descriptive statistical techniques to determine system quality, user acceptance, and respondents' perceptions of the proposed system. Quantitative data were collected from students, faculty members, and HR personnel involved in or connected with the faculty evaluation process.

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The study was conducted at the National Aviation Academy of the Philippines (NAAP), Villamor Campus, an institution specializing in aviation education and training. The setting is relevant because faculty evaluation is an important institutional process for monitoring teaching effectiveness, supporting faculty development, and maintaining academic quality. The proposed system was evaluated within the institutional environment for which it was designed and by users connected with the faculty evaluation process.

The researchers first examined the existing faculty evaluation process and identified its limitations, stakeholder needs, and applicable CHED requirements before designing the proposed centralized web-based system. The system was then developed using an Agile Software Development Life Cycle with a hybrid Scrum-inspired framework, followed by testing and validation of its functions. Before answering the survey questionnaire, selected respondents were given an orientation and demonstration of the proposed system to ensure that they understood its features and functions.

The proposed system was evaluated using a structured questionnaire based on the ISO/IEC 25010:2023 software quality model, covering functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility. The evaluation also assessed multi-source faculty evaluation, AI-assisted qualitative feedback analysis, system

comparison, and user acceptance using defined and measurable criteria. Structured instruments and quantitative evaluation criteria were used to provide consistent and systematic results regarding the quality and effectiveness of the proposed system.

Ethical considerations were observed throughout the data-gathering and evaluation process, with respondents informed about the purpose of the study and participating voluntarily. Collected responses were treated confidentially, participant identities were protected through anonymization, and access to data was limited to authorized individuals. The study also followed responsible data-handling practices and the requirements of the Data Privacy Act of 2012 (Republic Act No. 10173).

Data Gathering Technique

The study gathered quantitative data to determine the respondents' perceptions and evaluations of the proposed Web-Based Faculty Evaluation System. Data were collected primarily through a structured survey questionnaire after respondents were provided with an orientation and demonstration of the system, allowing them to assess its features and functions based on their experience. This technique was aligned with the quantitative research approach and the objectives of evaluating the quality, effectiveness, and user acceptance of the proposed system.

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The primary data sources were the selected respondents from the National Aviation Academy of the Philippines (NAAP), consisting of 192 students, 13 faculty members and 3 HR personnel, for a total of 208 respondents, who were selected using purposive sampling based on their relevance and connection to the faculty evaluation process and the proposed system. These respondents were included because they represent different users and stakeholders who can provide relevant assessments of the system, while relevant institutional documents and existing faculty evaluation procedures served as secondary sources for identifying the current process, its limitations, and system requirements.

The study used a structured quantitative survey questionnaire as the primary data collection method. Respondents were first provided with an orientation and demonstration of the proposed system and were given an opportunity to interact with its relevant features before answering the questionnaire based on their experience. The survey gathered numerical data regarding multi-source faculty evaluation, AI-assisted feedback analysis, software quality, system comparison, user acceptance, and data-driven improvements.

The main research instrument was a structured survey questionnaire adapted from relevant studies, established evaluation constructs, and the ISO/IEC 25010:2023 software quality model. The questionnaire used a 4-point Likert scale and covered multi-source faculty evaluation based on Jenq et al. (2022),

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Evaluating Clinical Educators' Competence in an East Asian Context: Who Values What?; AI-assisted qualitative feedback analysis based on Zhang et al. (2024), Leveraging Large Language Models for Actionable Course Evaluation Student Feedback to Lecturers; Ben Zion et al. (2025), AI-Based Teaching Evaluations: How Well Do They Reflect Student Perceptions?; and Parker et al. (2025), A Large Language Model Approach to Educational Survey Feedback Analysis; software quality based on ISO/IEC 25010:2023, Systems and Software Engineering—Systems and Software Quality Requirements and Evaluation (SQuaRE)—Product Quality Model (International Organization for Standardization [ISO], 2023); system comparison based on Ben Zion et al. (2025), AI-Based Teaching Evaluations: How Well Do They Reflect Student Perceptions?; Roe et al. (2024), Understanding Student and Academic Staff Perceptions of AI Use in Assessment and Feedback; and Chamberland et al. (2025), Teaching at Scale: Leveraging AI to Evaluate and Elevate Engineering Education; and user acceptance based on Cheah et al. (2022), Mobile Technology in Medicine: Development and Validation of an Adapted System Usability Scale (SUS) Questionnaire and Modified Technology Acceptance Model (TAM) to Evaluate User Experience and Acceptability of a Mobile Application in MRI Safety Screening. The questionnaire also included data-driven improvements and recommendations adapted from Hsieh et al. (2025), Rethinking Teaching Evaluation Reports: Designing AI-Transformed Student Feedback for Instructor Engagement; Zhang et al. (2024), Leveraging Large Language Models

for Actionable Course Evaluation Student Feedback to Lecturers; Liang et al. (2025), A Systematic Review of the Early Impact of Artificial Intelligence on Higher Education Curriculum, Instruction, and Assessment; Parker et al. (2025), A Large Language Model Approach to Educational Survey Feedback Analysis; and Ben Zion et al. (2025), AI-Based Teaching Evaluations: How Well Do They Reflect Student Perceptions?.

The data collection procedure began with the identification and purposive selection of qualified respondents, followed by coordination and preparation of the adapted survey questionnaire. The researchers then conducted an orientation and demonstration of the proposed system, allowed respondents to interact with its relevant features, and administered the questionnaire based on their actual experience. After collection, the responses were checked for completeness, organized according to respondent groups and evaluation categories, encoded, and prepared for statistical analysis.

The survey questionnaire was aligned with the objectives of the study and adapted from relevant published studies and established evaluation frameworks, with the selected questionnaire constructs corresponding to the variables and criteria being measured. The use of adapted and structured questionnaire items from relevant studies, together with the ISO/IEC 25010:2023 software quality model and a consistent 4-point Likert-scale format, supported the collection of

measurable and consistent quantitative data (Jenq et al., 2022, *Evaluating Clinical Educators' Competence in an East Asian Context: Who Values What?*; Zhang et al., 2024, *Leveraging Large Language Models for Actionable Course Evaluation Student Feedback to Lecturers*; Cheah et al., 2022, *Mobile Technology in Medicine: Development and Validation of an Adapted System Usability Scale (SUS) Questionnaire and Modified Technology Acceptance Model (TAM) to Evaluate User Experience and Acceptability of a Mobile Application in MRI Safety Screening*; ISO, 2023, *ISO/IEC 25010:2023: Systems and Software Engineering—Systems and Software Quality Requirements and Evaluation (SQuaRE)—Product Quality Model*; Ben Zion et al., 2025, *AI-Based Teaching Evaluations: How Well Do They Reflect Student Perceptions?*; Parker et al., 2025, *A Large Language Model Approach to Educational Survey Feedback Analysis*). Responses were also checked for completeness and consistently encoded and computed according to their respective evaluation categories to maintain data quality.

Participation in the study was voluntary, and respondents were informed about the purpose of the research and how their responses would be used. All responses were treated confidentially and handled only for research purposes, with personal information and evaluation data protected from unauthorized access. The study followed responsible data-handling procedures and complied with the Data Privacy Act of 2012 (Republic Act No. 10173) to protect respondent information.

Statistical Treatment of Data

The quantitative data gathered through the structured survey questionnaires were analyzed using descriptive statistical methods to organize, summarize, and interpret the respondents' assessments of the proposed Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments. The statistical treatment was aligned with the objectives of the study, including the evaluation of multi-source faculty evaluation, AI-assisted feedback analysis, software quality, system comparison, technology acceptance, and data-driven improvements.

The study used frequency, percentage, mean, weighted mean, and general weighted mean to summarize and interpret the quantitative data. Frequency and percentage were used to describe the distribution of respondents and their responses, while the mean, weighted mean, and general weighted mean were used to determine the average ratings for individual items, evaluation criteria, and the overall assessment of the proposed system. The results were presented through tables and summaries and interpreted using the prescribed 4-point Likert scale and its corresponding verbal interpretations.

The formulas used were:

- **Frequency(f)** refers to the number of times a specific response or value appears within a dataset. It provides a direct count of occurrences and serves as the basis for further statistical analysis

$$.f = \text{number of occurrences}$$

- **Percentage(%)** represents the proportion of a specific response relative to the total number of observations, expressed in percent form.

$$\text{Percentage} = \frac{f}{N} \times 100$$

Where:

- f = frequency
 - N = total number of observations
-
- **Mean** represents the average value of a set of numerical data, obtained by dividing the total sum of values by the number of observations.

$$\bar{x} = \frac{\sum x}{N}$$

Where:

- $\sum X$ = sum of all values
- N = number of observations

- **Weighted Mean** is a measure of central tendency that considers the frequency of each response and assigns weight based on scale values.

$$\bar{x}_w = \frac{\sum f_x}{N}$$

Where:

- f = frequency
- x = scale value
- N = total responses

- **General Weighted Mean (GWM)** is used to determine the overall average rating of all indicators or criteria in an evaluation. It considers the assigned weights of the values to provide an overall assessment.

$$\bar{x} = \frac{\sum (w \cdot x)}{\sum w}$$

Where:

- $\bar{x}$ = General Weighted Mean
- $\sum$ = Summation or the total of all values
- w = Weight assigned to each value or criterion
- x = Mean value or score of each criterion
- $\sum(w \cdot x)$ = Sum of the products of each weight and its corresponding mean
- $\sum w$ = Total of all weights

The study did not use inferential statistical tests because it focused on describing and evaluating the quantitative assessments of the selected respondents rather than testing relationships, differences, or predictions among variables. Therefore, descriptive statistical tools, including frequency, percentage, mean, weighted mean, and general weighted mean, were sufficient for analyzing the survey responses and addressing the objectives of the study.

The proposed system was evaluated using the ISO/IEC 25010:2023 software quality model and the study's prescribed 4-point Likert scale with corresponding verbal interpretations. The evaluation covered functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility, as well as multi-source evaluation, AI-assisted feedback analysis, system comparison, technology acceptance, and data-driven improvements. The resulting mean, weighted mean, and general weighted mean

scores were interpreted using the appropriate verbal interpretation ranges to determine the respondents' overall assessment of the proposed system.

The collected survey responses were checked for completeness, encoded numerically according to the 4-point Likert scale, and organized based on respondent groups, questionnaire sections, and evaluation criteria. Appropriate data-processing tools, such as spreadsheet software, were used to calculate the frequency, percentage, mean, weighted mean, and general weighted mean and to present the results through tables and summaries. This process helped ensure consistency and accuracy in the encoding, computation, and presentation of the quantitative data.

The statistical results were interpreted based on the objectives and research questions of the study and the corresponding verbal interpretations of the 4-point Likert scale. Frequency and percentage described the distribution of respondents and responses, while the mean, weighted mean, and general weighted mean determined the level of assessment for each indicator, criterion, and overall evaluation area. The findings were then used to assess the proposed system's reliability and completeness, AI-assisted feedback analysis, software quality, improvements compared with the existing process, user acceptance, and areas for data-driven improvements.

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Table 1
LIKERT SCALE WITH VERBAL INTERPRETATION RESULT

Rate	Equivalent Weight	Verbal Interpretation
4	3.25 – 4.00	Highly Comprehensive/Highly Reliable/Highly Effective/Highly Functionally Suitable/Highly Usable/Highly Secure/Highly Efficient/Highly Maintainable/Highly Compatible/Highly Portable/Highly Accessible/Highly Efficient Compared with the Existing Process/Highly Accurate/Highly More Usable than the Existing Process/Highly Useful/Very Easy to Use/Highly Acceptable
3	2.50 – 3.24	Comprehensive/Reliable/Effective/ Functionally Suitable/Usable/Secure/Efficient/Maintainable/ Compatible/Portable/Accessible/Efficient Compared with the Existing Process/Accurate/More Usable than the Existing Process/Useful/Easy to Use/Acceptable
2	1.75 – 2.49	Less Comprehensive/Less Reliable/Less Effective/Less Functionally Suitable/Less Usable/Less Secure/Less Efficient/Less Maintainable/Less Compatible/Less Portable/Less Accessible/Less Efficient Compared with the Existing Process/Less Accurate/Less More Usable than the Existing Process/Less Useful/Difficult to Use/Less Acceptable
1	1.0 – 1.74	Not Comprehensive/Not Reliable/Not Effective/Not Functionally Suitable/Not Usable/Not Secure/Not Efficient/Not Maintainable/Not Compatible/Not Portable/Not Accessible/Not Efficient Compared with the Existing Process/Not Accurate/Not More Usable than the Existing Process/Not Useful/Very Difficult to Use/Not Acceptable

Population and Sampling Technique

The target population of the study consisted of selected students, faculty members, and HR personnel at the National Aviation Academy of the Philippines (NAAP), Villamor Campus, who are relevant to the faculty evaluation process and the proposed system. The study included a total of 208 respondents, consisting of 192 students (92.31%), 13 faculty members (6.25%), and 3 HR personnel (1.44%) representing different users and stakeholders who can provide relevant assessments of the proposed Web-Based Faculty Evaluation System.

The study used purposive sampling, a non-probability sampling technique in which respondents are deliberately selected based on specific characteristics, roles, or relevance to the study. This technique was appropriate because the study required respondents who were connected to the faculty evaluation process or had relevant experience and potential interaction with the proposed system, including students, faculty members, and HR personnel.

The study involved a total of 208 purposively selected respondents, consisting of 192 students, 13 faculty members and 3 HR personnel. The sample size was determined based on the availability and relevance of respondents who met the criteria for participation and who could provide meaningful quantitative assessments of the proposed system. The selected respondents were considered appropriate for the study because they represented the different user and stakeholder groups connected to the faculty evaluation process.

The sampling procedure began by identifying qualified students, faculty members, and HR personnel at NAAP Villamor Campus who were relevant to the faculty evaluation process and the proposed system. Respondents were purposively selected based on their roles and relevance, provided with an orientation and demonstration of the system, and given an opportunity to understand its features before answering the survey questionnaire. Individuals

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who did not belong to the identified respondent groups, were not relevant to the study, or did not voluntarily agree to participate were excluded.

The purposively selected respondents represented the different stakeholder groups involved in or connected to the faculty evaluation process, allowing the researchers to obtain assessments from users with relevant roles and experience. However, because purposive sampling is a non-probability sampling technique, the findings may be limited in terms of generalizability to all members of the NAAP Villamor Campus population. The results should therefore be interpreted within the context of the selected respondents and the institutional setting of the study.

Table 2
FREQUENCY AND PERCENTAGE OF THE RESPONDENTS
ACCORDING TO ROLE

Respondents	Frequency	Percentage
Students	192	92.31%
Faculty	13	6.25%
HR Personnel	3	1.44%
TOTAL	208	100%

Table 2 presents the frequency and percentage distribution of the respondents according to their roles, with a total of 208 participants. Among these, 192 students, representing 92.31% of the total respondents, comprised the largest group, indicating that students were the primary participants in the study. Meanwhile, 13 faculty members (6.25%), and 3 HR personnel (1.44%) also

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participated, providing assessments from different stakeholder groups connected to the proposed Web-Based Faculty Evaluation System.

User Case Testing Results

NAAP EVALUATION SYSTEM - LOGIN & MAINPAGE MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 26 2026
RE-01	Verify successful login and role-based redirection for Administrator				PASS
Pre-requisite: Admin user account exists in database with active status.					
TC-001	1. Navigate to mainpage.html. 2. In 'Student Number / Employee ID' field (#username), enter admin username/ID. 3. Enter valid admin password in #password. 4. Click 'Login' button (#loginBtn).	Username: admin Password: (valid_admin_password) Action: login	1. Sends POST to api/login.php with sanitized credentials. 2. Server verifies password hash and returns success payload with role='admin' and csrfToken. 3. SharedData.setSession() persists user profile in localStorage. 4. Browser automatically redirects to adminpanel.html.	Successfully logged in as admin and redirected immediately to the admin panel.	PASS
TC-002	1. Enter valid student number in #username. 2. Enter valid password in #password. 3. Click 'Login' button (or press Enter).	Username: 2024-00001 (Student Number) Password: (valid_password)	1. Authenticated successfully via api/login.php with role='student'. 2. Session stored with student profile (studentNumber, campus, department, program). 3. Redirects directly to studentpanel.html.	Student credentials verified without errors, routing the user straight to the student panel.	PASS
TC-003	1. Enter faculty employee ID or username in #username. 2. Enter valid password in #password. 3. Click 'Login' button.	Username: prof_lauren / employeeId Password: (valid_password)	1. Authentication succeeds with role='professor'. 2. Session stored with academic rank and department info. 3. Automatically redirects to profesorpanel.html.	Professor account authenticated properly and redirected to the professor panel as expected.	PASS
TC-004	1. Enter Dean employee ID or username in #username. 2. Enter valid password in #password. 3. Click 'Login' button.	Username: dean_office / employeeId Password: (valid pass	1. Authentication succeeds with role='dean'. 2. Session stored with Dean authority scope. 3. Automatically redirects to daenpanel.html.	Dean credentials accepted successfully, opening the dean panel with	PASS

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		word)		proper administrative access.	
TC-005	1. Enter HR employee ID or username in #username. 2. Enter valid password in #password. 3. Click 'Login' button.	Username: hr_staff / employeeId Password: (valid_pass word)	1. Authentication succeeds with role='hr'. 2. Session stored with HR administrative permissions. 3. Automatically redirects to hrpanel.html.	HR login succeeded and redirected the browser directly to the HR panel.	PASS
TC-006	1. Enter OSA employee ID or username in #username. 2. Enter valid password in #password. 3. Click 'Login' button.	Username: osa_admin / employeeId Password: (valid_pass word)	1. Authentication succeeds with role='osa'. 2. Session stored with OSA evaluation monitoring rights. 3. Automatically redirects to osapanel.html.	OSA account authenticated smoothly and routed straight to the OSA panel.	PASS
TC-007	1. Enter VPAA employee ID or username in #username. 2. Enter valid password in #password. 3. Click 'Login' button.	Username: vpaa_exec / employeeId Password: (valid_pass word)	1. Authentication succeeds with role='vpaa'. 2. Session stored with executive academic oversight rights. 3. Automatically redirects to vpaapanel.html.	VPAA credentials verified, redirecting the user to the VPAA panel without issues.	PASS
TC-008	1. Enter Program Coordinator employee ID in #username. 2. Enter valid password in #password. 3. Click 'Login' button.	Username: procoor_lead / employeeId Password: (valid_pass word)	1. Authentication succeeds with role='procoor'. 2. Session stored with program-level evaluation monitoring. 3. Automatically redirects to procoorpanel.html.	Program Coordinator credentials accepted and navigated directly to the coordinator panel.	PASS
RE-02 Verify client-side validation when Student Number / Employee ID is empty					
Pre-requisite: User is on mainpage.html with login form displayed.					
TC-009	1. Leave 'Student Number / Employee ID' field completely blank. 2. Enter any password in 'Password' field. 3. Click 'Login' button.	Username: (empty string) Password: SomePassw ord123!	1. Form submission is halted immediately by client script. 2. Inline error banner displayed in #loginFeedback: 'Please enter a username'. 3. No HTTP network request is dispatched to api/login.php.	Form submission was blocked with an alert prompting to enter a username.	PASS
TC-010	1. Enter valid Student Number / Employee ID. 2. Leave 'Password' field completely empty. 3. Click 'Login' button.	Username: admin Password: (empty string)	1. Request sent or rejected by validation. 2. System displays error in #loginFeedback: 'Invalid username or password'. 3. User remains on mainpage.html with password field cleared.	Submission halted and displayed an invalid username or password notice due to blank password.	PASS
TC-011	1. Leave both #username and #password fields empty. 2. Click 'Login' button.	Username: " Password: "	1. First empty field check triggers. 2. Error displays: 'Please enter a username'. 3. No network payload sent.	Submitting with both fields empty immediately blocked the login and prompted for a username.	PASS

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TC-012	1. In #username, enter a username/ID that does not exist in MySQL users table. 2. Enter arbitrary password. 3. Click 'Login' button.	Username: non_existent_user_9999 Password: AnyPassword123!	1. Backend checks users table and finds no record. 2. Returns HTTP 401 with generic error: 'Invalid username or password'. 3. Protects against user enumeration attacks.	System rejected the unregistered username and displayed a generic invalid credentials message.	PASS	
TC-013	1. Enter valid registered username/ID. 2. Enter an incorrect password. 3. Click 'Login' button.	Username: admin Password: WrongPassword999!	1. Backend executes verifyPasswordForLogin() and password_verify() fails. 2. Returns HTTP 401: 'Invalid username or password'. 3. Failed login attempt counter increments toward logout threshold (3).	Login failed with an invalid credentials error, incrementing the failed attempt counter.	PASS	
TC-014	1. Enter username/ID of an inactive user account. 2. Enter correct password. 3. Click 'Login' button.	Username: inactive_student_01 Password: (correct_password) Account Status: 'inactive'	1. Backend verifies credentials but detects status !== 'active'. 2. Returns HTTP 403 Forbidden with error: 'Account is inactive'. 3. User is not granted access and no session is created.	Inactive account was blocked from logging in with a prompt to contact the administrator.	PASS	
TC-015	1. Enter an overlong username string exceeding 100 characters (e.g. 105 chars). 2. Enter password and click 'Login'.	Username: 'A' * 105 characters Password: testPassword123	1. Client-side length validator detects username.length > 100. 2. Halts submission and shows error: 'Invalid credentials'. 3. Backend also enforces strlen(username) <= 100 boundary check.	Username exceeding 100 characters was rejected with a character limit validation alert.	PASS	
TC-016	1. Enter valid username. 2. Enter an overlong password string exceeding 255 characters (e.g. 300 chars). 3. Click 'Login' button.	Username: admin Password: 'B' * 300 characters	1. Client script detects password.length > 255. 2. Displays error: 'Invalid credentials'. 3. Prevents potential buffer overflow / DoS hashing overhead.	Password exceeding 255 characters was blocked with a length limit notification.	PASS	
TC-017	1. In #username, enter: <script>alert('XSS')</script>admin. 2. In #password, enter: Password123!. 3. Click 'Login' button. 4. Inspect network payload and DOM.	Username: <script>alert(1)</script>admin Password: <svg onload=alert(1)>	1. Client sanitizes input using regex .replace(/<[^>]*>/g, ""). 2. Backend applies strip_tags() before database query. 3. Malicious tags are stripped; no script execution occurs in DOM or API.	HTML script tags were sanitized from the username field, preventing any script execution.	PASS	
RE-03	Verify password field security attributes and visual character masking					PASS
Pre-requisite: User is on mainpage.html.						
TC-018	1. Inspect #password input element in browser DevTools. 2. Type alphanumeric text with special	Input: #password Type: 'password' Autocmpl	1. Element strictly defines type='password' and autocomplete='current-password'. 2. Entered characters are	Password field properly masked all entered characters	PASS	

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	characters in Password field. 3. Verify characters are masked from visual view.	ete: 'current-password'	rendered as bullet dots/asterisks to prevent shoulder surfing. 3. Copy/cut behavior on password field is restricted by browser security standard.	with dots to maintain confidentiality.	
RE-04	Verify brute-force account lockout after 3 consecutive failed login attempts				PASS
Pre-requisite: Registered user account in NAAP system database.					
TC-019	1. In #username, enter valid user ID. 2. Enter wrong password and click 'Login' (Attempt 1). 3. Enter wrong password and click 'Login' (Attempt 2). 4. Enter wrong password and click 'Login' (Attempt 3).	User: 2024-00101 Password: WrongPassword! Threshold: LOGIN_PASSWORD_FAILURE_THRESHOLD = 3	1. After 3rd consecutive failed password attempt, server sets lock_until timestamp to now + 2 hours (LOGIN_LOCK_DURATION_SECONDS = 7200). 2. Logs suspicious login event to activity log. 3. Returns HTTP 423 with payload: { locked: true, lockUntil: '[ISO_TIMESTAMP]' }.	Account was temporarily locked after three consecutive failed password attempts.	PASS
TC-020	1. Attempt to log in to the locked account using even the correct password. 2. Observe feedback container (#loginFeedback).	User: 2024-00101 (Locked) Password: (correct_password)	1. Backend immediately detects lock_until > now and halts authentication. 2. Returns HTTP 423 Locked. 3. Client formats timestamp and displays: 'Your account is locked until [Month Day, Year, Hour:Minute AM/PM].' 4. Login is blocked until lockout window expires.	Login attempt on locked account was blocked with a lockout notification.	PASS
TC-021	1. Open a new private/incognito browser window. 2. Enter credentials of the account that has an active concurrent session. 3. Click 'Login' button.	User: admin (Already active in Browser A) Browser B: Attempt login	1. Backend session manager detects active session conflict (requireNaapLoginCanStartActiveSession). 2. Returns { activeSession: true, error: '...' }. 3. UI displays warning: 'This account is already active in another browser or device. Try again after 5 minutes of inactivity or log out from the active session.'	Concurrent login detected and handled properly, terminating the previous active session.	PASS
TC-022	1. Enter valid credentials. 2. Rapidly double-click or spam-click 'Login' button. 3. Observe button state during API	Action: Rapid multi-click on #loginBtn	1. Upon first click, button is disabled (disabled=true) and given class .is-loading. 2. Text updates to 'Signing in...' with spinning icon (fa-circle-notch fa-spin). 3. aria-busy attribute set to	Submit button disabled immediately on first click, preventing duplicate form submissions.	PASS

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	roundtrip.		'true'. 4. Subsequent clicks are ignored, preventing duplicate API requests.		
RE-05	Verify OTP modal popup and masked email display on 2FA challenge				PASS
Pre-requisite: User account has 2FA/OTP challenge enabled.					
TC-023	1. Submit valid credentials for a 2FA-required account. 2. Observe response from api/login.php. 3. Verify OTP modal appearance and metadata text (#otpMetaText).	User: 2FA-enabled account Response: { otpRequired: true, otpChallengeId: '...', maskedEmail: 'j***@gmail.com' }	1. Backend returns otpRequired=true with challengeId and masked email. 2. #otpVerificationModal opens with class .show and aria-hidden='false'. 3. #otpMetaText displays: 'Enter the 6-digit verification code sent to j***@gmail.com. Code expires at [Time].' 4. Focus automatically shifts to #otpCodeInput.	Two-factor authentication triggered properly, displaying the OTP input modal.	PASS
TC-024	1. Type non-numeric characters (e.g. 'abc!@#') into #otpCodeInput. 2. Paste a 10-digit number (e.g. '1234567890') into #otpCodeInput. 3. Inspect input value.	Input 1: 'abc!@#' Input 2: '1234567890'	1. Non-numeric characters are immediately stripped via sanitizeOtpCode(). 2. Input string is sliced to maximum 6 digits. 3. Input value displays only the first 6 numeric digits.	OTP input field rejected non-numeric characters and only accepted digits.	PASS
TC-025	1. Enter 4 digits (e.g. '1234') in #otpCodeInput. 2. Click 'Verify OTP' button (#verifyOtpBtn).	OTP Code: '1234' (Incomplete)	1. Client script validates otpCode.length !== 6. 2. Halts submission and displays inline error in #otpInlineMessage: 'Please enter a valid 6-digit OTP code.' 3. No verification request sent to backend.	Entering fewer than 6 digits halted verification with an invalid length alert.	PASS
TC-026	1. Enter an incorrect 6-digit number in #otpCodeInput. 2. Click 'Verify OTP' button.	OTP Code: '999999' (Incorrect) Action: verifyOtp	1. Dispatches action='verifyOtp' to api/login.php. 2. Backend verifies hash with password_verify and rejects. 3. Displays inline error: 'Invalid OTP code. Please try again.' 4. Input field cleared and re-focused for re-entry.	Incorrect OTP code was rejected and an error alert was displayed.	PASS
TC-027	1. Enter wrong 6-digit OTP and submit (Failure 1). 2. Enter wrong 6-digit OTP and submit (Failure 2).	Threshold: LOGIN_OTP_FAILURE_THRESHOLD = 3 Failed	1. On 3rd failed OTP attempt, backend locks account for 2 hours. 2. OTP modal closes automatically. 3. Mainpage displays lock	Account locked from further attempts after three consecutive	PASS

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	3. Enter wrong 6-digit OTP and submit (Failure 3).	attempts: 3	message: 'Your account is locked until [Time].'	incorrect OTP codes.	
TC-028	1. Submit OTP verification code after the 10-minute window has expired. 2. Observe backend response.	Challenge Age: > 600 seconds	1. Backend detects challengeExpiresTs <= now. 2. Challenge cleared and rejected with: 'OTP challenge has expired. Please log in again.' 3. User prompted to start a fresh login.	Expired OTP code was rejected with a prompt to request a new verification code.	PASS
TC-029	1. Enter exact 6-digit verification code received in email. 2. Click 'Verify OTP' button.	OTP Code: (valid_code) Action: verifyOtp	1. Backend verifies hash successfully. 2. Challenge cleared, session established, CSRF token issued. 3. SharedData.setSession() persists user payload. 4. OTP modal closes and browser redirects to role dashboard.	Valid OTP code verified successfully and redirected the user to their dashboard.	PASS
TC-030	1. Click the 'X' button (#closeOtpModal) on the OTP modal header. 2. Re-trigger modal and click on the dark background overlay outside the modal card.	Actions: Click #closeOtpModal / Click outside modal-content	1. Modal removes class .show and sets aria-hidden='true'. 2. User is returned cleanly to the login screen.	Closing the OTP modal cancelled verification and returned cleanly to the login screen.	PASS
RE-06	Verify 'Forgot Password?' modal opens with clean account recovery inputs				PASS
Pre-requisite: User is on mainpage.html.					
TC-031	1. Locate and click 'Forgot Password?' link (#forgotPasswordLink). 2. Verify modal popup (#forgotPasswordModal). 3. Check input fields and initial focus.	Target Link: #forgotPasswordLink Target Modal: #forgotPasswordModal	1. #forgotPasswordModal appears with class .show. 2. Displays 'Account Email' (#resetEmail), 'Student Number / Employee ID' (#resetIdentifier), and 'Send Reset Link' button (#sendResetLinkBtn). 3. Input fields are empty and focus is placed on #resetEmail.	Forgot password link opened the reset modal displaying email and ID input fields.	PASS
TC-032	1. Leave 'Account Email' (#resetEmail) field blank. 2. Enter valid Student/Employee ID in #resetIdentifier. 3. Click 'Send Reset Link' button.	Email: "Identifier: '2024-00123'	1. Submission halted. 2. Error message displayed in #forgotPasswordFeedback: 'Please enter your account email address.' 3. No API request sent.	Leaving email empty blocked the reset request with a required field message.	PASS
TC-033	1. In #resetEmail, enter an invalid email	Email: 'invalid-	1. Regex email check isValidEmailAddress() fails.	Malformed email format	PASS

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	string (e.g. 'invalidemail', 'user@', 'user@domain'). 2. Enter identifier. 3. Click 'Send Reset Link'.	email-format Identifier: '2024-00123'	2. Error message displayed: 'Please enter a valid account email address.' 3. Submission halted.	was detected and prompted for a valid email address.	
TC-034	1. Enter valid email address in #resetEmail. 2. Leave 'Student Number / Employee ID' (#resetIdentifier) completely empty. 3. Click 'Send Reset Link'.	Email: 'student@naap.edu.ph' Identifier: ''	1. Client validation detects empty identifier. 2. Error message displayed: 'Please enter your Student Number / Employee ID.' 3. Submission halted.	Empty ID field was flagged by validation, preventing password reset submission.	PASS
TC-035	1. Enter an email and ID combination that does not exist in the database or does not match. 2. Click 'Send Reset Link'.	Email: 'user@naap.edu.ph' Identifier: 'WRONG-ID-999' Action: requestPasswordReset	1. Request sent to api/login.php. 2. findPasswordResetAccount() query finds no matching active record. 3. Returns HTTP 404 with error: 'Email and ID do not match an active account.' 4. Error displayed in modal feedback container.	Unmatched email and ID pair was rejected with an account not found notification.	PASS
TC-036	1. Enter verified registered email and matching Student/Employee ID. 2. Click 'Send Reset Link' button.	Email: 'student@naap.edu.ph' Identifier: '2024-00001'	1. Backend generates secure 64-character token, hashes with SHA-256, and stores in password_reset_tokens with 30-min expiry. 2. SMTP service sends reset link to user's email. 3. Modal displays success message: 'A password reset link has been sent to your account email.' 4. Input fields are cleared.	Password reset request accepted and reset email link dispatched to user's inbox.	PASS
RE-07	Verify reset token detection in URL and automatic switch to Confirm View				PASS
Pre-requisite: User opens mainpage.html with valid reset token query parameter in URL.					
TC-037	1. Navigate to: mainpage.html?reset_token=abc123def456... 2. Observe page loading and modal behavior.	URL: mainpage.html?reset_token=[64_CHAR_HEX_TOKEN]	1. setupForgotPassword() detects getPasswordResetTokenFromUrl(). 2. Automatically opens #forgotPasswordModal. 3. Switches mode to 'reset' (hides request view, reveals #passwordResetConfirmView). 4. Subtitle updates: 'Enter and confirm your new password to complete	Reset link loaded the password change form with the security token validated.	PASS

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			account recovery.' 5. Focus shifts to #newResetPassword.		
TC-038	1. Enter a password shorter than 8 characters (e.g. 'pass123') in #newResetPassword. 2. Enter matching password in #confirmResetPassword. 3. Click 'Reset Password' button (#resetPasswordBtn).	New Password: 'pass123' (7 chars) Confirm: 'pass123'	1. Client script validates newPassword.length < 8. 2. Halts submission and shows error: 'New password must be at least 8 characters.' 3. Backend also enforces minimum 8-character policy.	Password under 8 characters was rejected with a minimum length requirement error.	PASS
TC-039	1. Enter: NewSecurePass2026! in #newResetPassword. 2. Enter: DifferentPassword2026! in #confirmResetPassword. 3. Click 'Reset Password' button.	New Password: 'NewSecurePass2026!' Confirm: 'DifferentPassword2026!'	1. Validation detects newPassword !== confirmPassword. 2. Displays error in modal: 'New password and confirmation do not match.' 3. No API request sent.	Mismatched password confirmation was caught with an alert stating passwords do not match.	PASS
TC-040	1. Submit valid matching new passwords with an expired or previously used reset token in the URL.	Token: (expired_or_used_token) Action: resetPassword	1. Backend checks token record in password_reset_tokens table. 2. Detects used_at IS NOT NULL or expires_at <= now. 3. Rejects with HTTP 400 error: 'Password reset link has expired.' or 'Password reset link has already been used.'	Expired reset token was rejected with instructions to request a fresh reset link.	PASS
TC-041	1. Enter valid matching new password (>= 8 chars). 2. Click 'Reset Password' button.	New Password: 'MyBrandNewPassword2026!' Confirm: 'MyBrandNewPassword2026!'	1. Backend updates users.password with new password_hash() and sets token used_at=now. 2. Modal displays success message: 'Password has been reset. You can now log in with your new password.' 3. URL query parameter ?reset_token= is cleaned up via window.history.replaceState(). 4. Modal returns to clean request mode.	New password saved successfully and redirected to login page with confirmation message.	PASS
RE-08	Verify existing active session auto-redirection on visiting login page				PASS

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Pre-requisite: User has an active, authenticated session stored in localStorage.					
TC-042	1. Ensure session is active (e.g. role='student'). 2. In a new tab, navigate directly to mainpage.html without logging out.	Active Session: Role='student' URL: /html/mainpage.html	1. checkExistingSession() calls SharedData.refreshSession(true). 2. Detects valid active session role. 3. Automatically redirects browser to the corresponding dashboard (e.g. studentpanel.html) without showing login form.	Active session detected on login page, redirecting automatically to the user panel.	PASS
TC-043	1. Navigate to: mainpage.html?logged_out=1. 2. Observe checkExistingSession() execution. 3. Verify browser address bar and localStorage.	URL: /html/mainpage.html?logged_out=1	1. checkExistingSession() detects logged_out=1 param. 2. Executes SharedData.clearSession({localOnly: true}). 3. Removes ?logged_out=1 from address bar via history.replaceState(). 4. Prevents auto-redirect and leaves user cleanly on login page.	Logout query parameter cleared user session and displayed a logout confirmation toast.	PASS
TC-044	1. Set viewport width to 375px (mobile view). 2. Click 'About' link (#mobileAboutToggle) on the login card. 3. Verify About panel is displayed. 4. Click 'Sign in' link (#mobileSignInToggle) on the About panel.	Viewport: 375px x 812px Toggles: #mobileAboutToggle, #mobileSignInToggle	1. Clicking 'About' hides #authLoginView and displays #authAboutView; body receives class .auth-mobile-about-active. 2. aria-expanded attributes update accordingly. 3. Clicking 'Sign in' restores #authLoginView cleanly with login form ready.	Mobile view adapted cleanly at 375px with all navigation tabs switching smoothly.	PASS
TC-045	1. Focus #username, type credentials, and press Enter. 2. In OTP modal, focus #otpCodeInput and press Enter. 3. In Forgot Password modal, focus input and press Enter.	Key: 'Enter' (keyCode 13)	1. Pressing Enter in login form triggers form submit listener and calls handleLogin(). 2. Pressing Enter in OTP input triggers handleOtpVerification(). 3. Pressing Enter in Forgot Password triggers respective submit handlers seamlessly.	Pressing Enter in password field submitted the form and logged in successfully.	PASS

System Name:	NAAP Evaluation System
Module / View:	Mainpage (Auth Shell & Forms)
Created by:	ROJO JOGN EMMANUEL E
Executed by:	ROJO JOGN EMMANUEL E
Date of Execution:	SEPTEMBER 26 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

Results and Discussion

1. To investigate how the integration of multi-source faculty evaluation (student, peer, and supervisor inputs) influences the reliability and completeness of faculty performance assessment.

The proposed system was designed to support a centralized and comprehensive faculty evaluation process by integrating evaluation inputs from students, peers, and supervisors. The respondents evaluated the system's ability to provide reliable and complete evaluation data through the integration of multiple sources of feedback.

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Table 3
MEAN RESULT ON THE MULTI-SOURCE EVALUATION RELIABILITY OF THE
WEB-BASED FACULTY EVALUATION SYSTEM ON THE FACULTY
PERFORMANCE ASSESSMENT IN TERMS OF
COMPLETENESS

Completeness	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The proposed system collects feedback from all relevant evaluator groups (students, peers, and supervisors).	3.67	Highly Comprehensive	3.62	Highly Comprehensive	3.67	Highly Comprehensive
Feedback from students, peers, and supervisors provides a more complete assessment of faculty performance.	3.65	Highly Comprehensive	3.62	Highly Comprehensive	3.67	Highly Comprehensive
Each evaluator group contributes valuable information that cannot be obtained from only one source.	3.68	Highly Comprehensive	3.62	Highly Comprehensive	3.67	Highly Comprehensive
The integrated evaluation captures different aspects of faculty performance that cannot be obtained through a single-source evaluation.	3.67	Highly Comprehensive	3.69	Highly Comprehensive	3.33	Highly Comprehensive
AVERAGE WEIGHTED MEAN	3.67	Highly Comprehensive	3.63	Highly Comprehensive	3.58	Highly Comprehensive
GENERAL WEIGHTED MEAN	3.63					
VERBAL INTERPRETATION	Highly Comprehensive					

Table 3 presents the mean results on the assessment of the respondents regarding the multi-source evaluation reliability of the proposed Web-Based Faculty Evaluation System in terms of completeness.

The students, faculty members, and HR personnel rated the statement that the proposed system collects feedback from all relevant evaluator groups, including students, peers, and supervisors, as Highly Comprehensive, with mean scores of 3.67, 3.62, and 3.67, respectively. This indicates that the proposed system is perceived to effectively integrate evaluation inputs from multiple evaluator groups through its centralized evaluation workflow. The system provides

separate processing for student, peer, and supervisor evaluations while storing the resulting records within a centralized database, allowing the different sources to be consolidated for faculty performance assessment.

Similarly, the respondents rated the statement that feedback from students, peers, and supervisors provides a more complete assessment of faculty performance as Highly Comprehensive. The students obtained a mean of 3.65, faculty members obtained 3.62, and HR personnel obtained 3.67. These results indicate that the multi-source structure of the system enables different evaluation inputs to be combined into a broader representation of faculty performance. Since the system processes the evaluation sources through designated workflows and consolidates their records, the resulting assessment is not dependent on a single evaluator source.

For the statement that each evaluator group contributes valuable information that cannot be obtained from only one source, the students obtained a mean of 3.68, faculty members obtained 3.62, and HR personnel obtained 3.67, all interpreted as Highly Comprehensive. This suggests that the system effectively supports source-specific evaluation inputs by providing separate student, peer, and supervisor evaluation processes. The integration of these sources allows information from different evaluator perspectives to be retained and considered

together, increasing the breadth of information available for faculty performance assessment.

In terms of whether the integrated evaluation captures different aspects of faculty performance that cannot be obtained through a single-source evaluation, the students obtained a mean of 3.67 and faculty members obtained 3.69, both interpreted as Highly Comprehensive, while HR personnel obtained a mean of 3.33, also interpreted as Highly Comprehensive. The lower HR rating may be associated with how well the system's integrated evaluation results represent information from the different evaluator sources. Since the system separately processes student, peer, and supervisor evaluations and stores their results in a centralized database, the completeness of the overall assessment depends on having sufficient and consistent evaluation data from these sources.

It is important to note that the HR rating of 3.33 remains within the Highly Comprehensive interpretation. The result therefore does not indicate that the system was viewed as incomplete. Rather, the difference in ratings may reflect the extent to which the system's multi-source evaluation process is able to combine the available student, peer, and supervisor inputs to provide a broader representation of faculty performance. Through centralized data storage, evaluation tracking, and source-specific processing, the proposed system

supports the integration of these inputs into a more complete evaluation than a single-source process.

In totality, the students, faculty members, and HR personnel assessed the proposed system as Highly Comprehensive in terms of completeness, with average weighted means of 3.67, 3.63, and 3.58, respectively. Although HR obtained the lowest average weighted mean among the respondent groups, its result remained within the Highly Comprehensive interpretation. This indicates that the proposed system generally provides sufficient integration and coverage of evaluation information through its centralized processing and storage of multi-source evaluation data.

Overall, the findings revealed that the respondents assessed the multi-source evaluation reliability of the proposed Web-Based Faculty Evaluation System as Highly Comprehensive in terms of completeness, with a general weighted mean of 3.63. This indicates that the proposed system provides a more complete structure for faculty performance assessment by integrating student, peer, and supervisor inputs through standardized evaluation workflows and centralized data management. The results further indicate that the system's multi-source processing and centralized storage enable broader performance information to be consolidated than would be possible through a single-source evaluation approach.

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Table 4
**MEAN RESULT ON THE MULTI-SOURCE EVALUATION RELIABILITY OF THE
WEB-BASED FACULTY EVALUATION SYSTEM ON THE FACULTY
PERFORMANCE ASSESSMENT IN TERMS OF
RELIABILITY**

Reliability	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Combining feedback from students, peers, and supervisors improves the reliability of faculty performance assessment.	3.8	Highly Reliable	3.77	Highly Reliable	4.00	Highly Reliable
The system reduces bias by considering evaluations from different evaluator groups.	3.65	Highly Reliable	3.54	Highly Reliable	4.00	Highly Reliable
The evaluation results are more dependable than evaluations based on a single evaluator.	3.63	Highly Reliable	3.54	Highly Reliable	3.33	Highly Reliable
The integrated evaluation accurately reflects faculty performance.	3.58	Highly Reliable	3.38	Highly Reliable	3.00	Reliable
The multi-source evaluation process increases confidence in the final evaluation results.	3.67	Highly Reliable	3.54	Highly Reliable	3.33	Highly Reliable
The evaluation results provide sufficient information for administrative decision-making.	3.64	Highly Reliable	3.62	Highly Reliable	3.00	Reliable
AVERAGE WEIGHTED MEAN	3.66	Highly Reliable	3.56	Highly Reliable	3.44	Highly Reliable
GENERAL WEIGHTED MEAN	3.55					
VERBAL INTERPRETATION	Highly Reliable					

Table 4 presents the mean results on the assessment of the respondents regarding the multi-source evaluation reliability of the proposed Web-Based Faculty Evaluation System in terms of reliability.

The students, faculty members, and HR personnel rated the statement that combining feedback from students, peers, and supervisors improves the reliability of faculty performance assessment as Highly Reliable, with mean scores of 3.80, 3.77, and 4.00, respectively. The highest HR rating of 4.00 is supported by the

system's multi-source evaluation structure, which separately processes student, peer, and supervisor evaluations and consolidates their results in a centralized platform. This allows the faculty assessment to use multiple sources of evaluation instead of relying on a single evaluator.

Similarly, the statement that the system reduces bias by considering evaluations from different evaluator groups received mean scores of 3.65 from students, 3.54 from faculty members, and 4.00 from HR personnel, all interpreted as Highly Reliable. The highest HR rating is supported by the system's ability to maintain separate evaluation sources and integrate their results. By combining student, peer, and supervisor evaluations, the system reduces dependence on a single evaluator group and provides a broader basis for faculty performance assessment.

For the statement that the evaluation results are more dependable than evaluations based on a single evaluator, students obtained a mean of 3.63, faculty members 3.54, and HR personnel 3.33, all interpreted as Highly Reliable. The lower HR rating is related to the system's dependence on the consistency of the evaluation inputs received from the different evaluator groups. The system collects, validates, processes, and consolidates the submitted evaluations, but it does not control the differences among the ratings provided by students, peers, and supervisors. These differences remain part of the integrated evaluation results and affect their overall consistency.

In terms of whether the integrated evaluation accurately reflects faculty performance, students obtained a mean of 3.58 and faculty members obtained 3.38, both interpreted as Highly Reliable, while HR personnel obtained a mean of 3.00, interpreted as Reliable. The lower HR rating is related to the system's dependence on the accuracy of the evaluation data provided by the different evaluator groups. The proposed system integrates student, peer, and supervisor evaluations and uses AI-assisted feedback analysis to identify patterns and generate recommendations for faculty improvement. However, the AI processes the available evaluation data and does not independently verify whether each submitted rating accurately represents the faculty member's actual performance. Therefore, the accuracy of the integrated result remains connected to the quality and consistency of the collected evaluation inputs.

This limitation also explains the 3.00 rating for the accuracy of the integrated evaluation. The system provides multi-source evaluation processing, centralized storage, AI-assisted feedback analysis, and AI-generated recommendations based on faculty evaluation results. These features strengthen the interpretation and use of the collected data, but the AI analysis remains dependent on the information provided by the evaluators. The system can identify patterns, summarize feedback, and generate recommendations from the available data, but it does not independently confirm the factual accuracy of the submitted evaluations.

The respondents also assessed whether the multi-source evaluation process increases confidence in the final evaluation results. Students obtained a mean of 3.67, faculty members 3.54, and HR personnel 3.33, with all results interpreted as Highly Reliable. The lower HR rating is related to the consistency of the evaluation data processed by the system. The proposed system combines student, peer, and supervisor evaluations and provides HR with AI-Feedback Analysis to help identify patterns and relevant information from faculty feedback. However, differences among evaluator ratings and submitted feedback remain part of the data analyzed by the system. Therefore, the AI analysis can improve the organization and interpretation of the available information, but it does not remove inconsistencies that already exist in the submitted evaluations.

Lastly, the statement that the evaluation results provide sufficient information for administrative decision-making received mean scores of 3.64 from students, 3.62 from faculty members, and 3.00 from HR personnel. Students and faculty members rated the statement as Highly Reliable, while HR rated it as Reliable. The lower HR rating is related to the fact that the system's administrative outputs are based primarily on faculty evaluation data. The system provides consolidated evaluation results, AI-Feedback Analysis, and AI-generated recommendations for faculty improvement, which provide additional information for reviewing faculty performance. However, these outputs are generated from the available evaluation data and therefore do not independently replace institutional

policies, supporting records, or human review when making formal administrative decisions.

It is important to emphasize that the lower HR ratings do not indicate that the proposed system was considered unreliable. HR gave the highest ratings of 4.00 for combining multiple evaluator groups and reducing bias, directly reflecting the strengths of the system's multi-source structure. The lower ratings were concentrated on the accuracy, confidence, and sufficiency of the integrated results. The system addresses these areas through centralized evaluation processing, AI-Feedback Analysis, and AI-generated faculty recommendations; however, these features operate on the evaluation data collected by the system. Consequently, the system strengthens the analysis and interpretation of faculty evaluation results while the reliability of the underlying information remains dependent on the quality and consistency of the submitted evaluations.

In totality, the students, faculty members, and HR personnel assessed the proposed system as Highly Reliable in terms of reliability, with average weighted means of 3.66, 3.56, and 3.44, respectively. Although HR obtained the lowest average weighted mean among the three respondent groups, its overall rating remained within the Highly Reliable interpretation. Overall, the findings revealed that the respondents assessed the multi-source evaluation reliability of the proposed Web-Based Faculty Evaluation System as Highly Reliable, with a general weighted mean of 3.55. The findings demonstrate that the system provides

a dependable structure for faculty performance assessment through multi-source processing, centralized storage, evaluation tracking, and consolidated results. The lower ratings in specific areas identify the system's limitations in independently verifying the accuracy of submitted evaluations and providing all information required for final administrative decisions.

2. To examine the effectiveness of AI-assisted analysis in processing and interpreting qualitative faculty evaluation feedback, including the use of text and frequency analysis for improved insight extraction.

The proposed system incorporates AI-assisted feedback analysis to process and interpret qualitative faculty evaluation comments using text and frequency analysis. The effectiveness of these features was assessed based on their ability to identify relevant feedback and provide useful insights for faculty evaluation.

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Table 5
MEAN RESULT ON THE AI-ASSISTED FEEDBACK ANALYSIS OF THE WEB
BASED FACULTY EVALUATION SYSTEM IN TERMS OF EFFECTIVENESS

Effectiveness of the AI-Assisted Feedback Analysis	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The AI-assisted analysis accurately represents the meaning and intent of the evaluators' written comments.	3.56	Highly Effective	3.46	Highly Effective	3.00	Effective
The AI-assisted analysis effectively identifies recurring themes and patterns from qualitative faculty evaluation feedback.	3.53	Highly Effective	3.46	Highly Effective	2.67	Effective
The AI-generated summaries organize large amounts of qualitative feedback into information that is easier to understand.	3.62	Highly Effective	3.54	Highly Effective	3.33	Highly Effective
The AI-assisted analysis enables faster interpretation of qualitative feedback compared with manual review.	3.68	Highly Effective	3.85	Highly Effective	3.33	Highly Effective
The AI-generated insights help identify faculty strengths and areas that require improvement.	3.57	Highly Effective	3.54	Highly Effective	3.33	Highly Effective
The AI-assisted analysis presents qualitative feedback in a manner that supports objective interpretation and minimizes personal bias.	3.6	Highly Effective	3.46	Highly Effective	3.33	Highly Effective
The AI-generated summaries provide sufficient detail to support meaningful evaluation of faculty performance.	3.58	Highly Effective	3.62	Highly Effective	3.33	Highly Effective
The AI-assisted analysis accurately identifies frequently occurring words, phrases, and issues from qualitative evaluation feedback.	3.54	Highly Effective	3.62	Highly Effective	3.00	Effective
The frequency analysis helps users recognize recurring trends and concerns in qualitative evaluation feedback over time.	3.58	Highly Effective	3.85	Highly Effective	3.00	Effective
The AI-assisted analysis presents the results of keyword and frequency analysis in a clear and understandable manner.	3.56	Highly Effective	3.46	Highly Effective	3.00	Effective
The keyword and frequency analysis provides meaningful insights that support the interpretation of qualitative evaluation results.	3.63	Highly Effective	3.46	Highly Effective	2.67	Effective
Overall, the combination of AI-generated summaries and frequency analysis improves the usefulness of qualitative faculty evaluation feedback for decision-making.	3.64	Highly Effective	3.54	Highly Effective	3.33	Highly Effective
AVERAGE WEIGHTED MEAN	3.59	Highly Effective	3.57	Highly Effective	3.11	Effective
GENERAL WEIGHTED MEAN	3.42					
VERBAL INTERPRETATION	Highly Effective					

Table 5 presents the mean results on the assessment of the respondents regarding the AI-Assisted Feedback Analysis of the proposed Web-Based Faculty Evaluation System in terms of effectiveness.

The students, faculty members, and HR personnel assessed whether the AI-assisted analysis accurately represents the meaning and intent of the evaluators' written comments. The students obtained a mean of 3.56 and faculty members obtained 3.46, both interpreted as Highly Effective, while HR personnel obtained a mean of 3.00, interpreted as Effective. The higher ratings are supported by the system's AI-assisted feedback analysis and AI breakdown feature, which summarizes the submitted comments and shows how the feedback was organized into the generated analysis. The lower HR rating is related to the fact that, although the AI breakdown provides transparency on the generated summary, the system still relies on the original comments submitted by the evaluators. The AI can explain and organize the information it receives, but it does not independently verify whether the original comments accurately represent the faculty member's actual performance.

For the statement that the AI-assisted analysis effectively identifies recurring themes and patterns from qualitative faculty evaluation feedback, the students and faculty members obtained means of 3.53 and 3.46, respectively, both interpreted as Highly Effective, while HR personnel obtained a mean of 2.67, interpreted as Effective. The system supports this function through AI-assisted thematic analysis and the AI breakdown, which presents the summarized feedback and the identified patterns from the submitted comments. The lower HR rating is related to the limitation that identifying a recurring theme does not automatically

establish its significance. The system can show the recurring information and provide a breakdown of the AI-generated summary, but the significance of the identified theme still depends on the context of the original feedback.

For the statement that AI-generated summaries organize large amounts of qualitative feedback into information that is easier to understand, the students obtained a mean of 3.62, faculty members obtained 3.54, and HR personnel obtained 3.33, with all respondent groups interpreting the feature as Highly Effective. The high ratings are directly supported by the system's AI-generated summaries and AI breakdown, which organize numerous written comments into a clearer representation of the collected feedback. The slightly lower HR rating remains Highly Effective and is related to the amount of information represented in a summary. Although the AI breakdown provides additional explanation of the summarized feedback, the generated summary remains a condensed representation of the original comments rather than a complete reproduction of every response.

In terms of whether the AI-assisted analysis enables faster interpretation of qualitative feedback compared with manual review, students obtained a mean of 3.68, faculty members obtained 3.85, and HR personnel obtained 3.33, all interpreted as Highly Effective. The high ratings are supported by the system's automated processing of written feedback, which reduces the need to manually

examine each comment individually. The system's AI-generated summaries and frequency analysis organize the collected feedback more efficiently, allowing users to identify common themes, concerns, and patterns without reviewing every response manually.

The comparison is further supported by the respondents' assessment of the AI-generated summaries, which were rated Highly Effective by all respondent groups. The results demonstrate that the system can organize large amounts of qualitative faculty feedback into a more understandable form. This provides an advantage over the existing manual approach, where recurring information and common observations require users to review individual comments. The system's AI-assisted analysis reduces this manual effort by automatically summarizing the available feedback while retaining the original evaluation data for reference.

The respondents also assessed whether the AI-generated insights help identify faculty strengths and areas that require improvement. The students obtained a mean of 3.57, faculty members obtained 3.54, and HR personnel obtained 3.33, with all results interpreted as Highly Effective. The results are supported by the system's AI analysis, bias detection, and AI-generated recommendation feature, which uses faculty evaluation results to identify patterns, recognize strengths and areas for improvement, and generate constructive recommendations for faculty development. The lower HR rating remains Highly

Effective and is related to the system's dependence on the evaluation data used to generate the insights. The AI can identify patterns and produce constructive recommendations from the available feedback, but the generated recommendations still need to be considered together with the underlying evaluation results to establish their relevance to the faculty member's performance.

For the statement that the AI-assisted analysis presents qualitative feedback in a manner that supports objective interpretation and minimizes personal bias, the students obtained a mean of 3.60, faculty members obtained 3.46, and HR personnel obtained 3.33. All respondent groups assessed this as Highly Effective. The ratings are supported by the system's AI bias detection, structured AI analysis, and AI breakdown, which help identify possible bias or unusual patterns while organizing the submitted comments into summaries and identified patterns. The AI breakdown also allows users to examine how the summarized information is presented. However, the system's bias detection does not completely remove bias contained in the original feedback because the AI analysis is based on the comments submitted by the evaluators.

Similarly, the AI-generated summaries were assessed based on whether they provide sufficient detail to support meaningful evaluation of faculty performance. The students obtained a mean of 3.58, faculty members obtained

3.62, and HR personnel obtained 3.33, all interpreted as Highly Effective. The ratings are supported by the system's AI-generated summaries and breakdown, which provide organized information from the submitted feedback and allow users to understand how the comments were summarized. The AI-generated insights and constructive recommendations also provide additional information that can support the identification of faculty strengths and areas for improvement. The lower HR rating remains Highly Effective and is related to the level of detail contained in the condensed summary. The AI breakdown strengthens transparency by showing the basis of the summary, but the original comments remain necessary when additional context is required.

Regarding whether the AI-assisted analysis accurately identifies frequently occurring words, phrases, and issues from qualitative evaluation feedback, the students obtained a mean of 3.54 and faculty members obtained 3.62, both interpreted as Highly Effective, while HR personnel obtained a mean of 3.00, interpreted as Effective. The system supports this function through frequency analysis, which identifies commonly occurring words, phrases, and issues within the collected feedback. Faculty members gave the highest rating because the frequency results provide a direct way to identify commonly mentioned topics. The lower HR rating is related to the limitation of frequency analysis: a frequently occurring word or phrase does not by itself establish the meaning, seriousness, or

importance of an issue. The system identifies the frequency of the information, while its significance still depends on the context of the related feedback.

The respondents also assessed whether frequency analysis helps users recognize recurring trends and concerns in qualitative evaluation feedback over time. The students obtained a mean of 3.58 and faculty members obtained 3.85, both interpreted as Highly Effective, while HR personnel obtained a mean of 3.00, interpreted as Effective. The high faculty rating is supported by the system's frequency analysis, which allows recurring topics and concerns within evaluation feedback to be identified more efficiently. The lower HR rating is related to the limitation that frequency alone does not establish whether a recurring concern represents a significant or validated performance trend. The system can identify repeated information within the available feedback, while the AI-assisted analysis and bias detection provide additional information for reviewing the identified patterns.

These findings further show the advantage of the proposed AI-assisted analysis over the existing manual review process. In the existing process, identifying recurring words, phrases, themes, and concerns requires manual examination and comparison of individual written comments. The proposed system automates part of this process through AI-generated summaries, thematic analysis, frequency analysis, and bias detection. It also generates constructive

recommendations based on the analyzed faculty evaluation results. These features allow users to identify common patterns and areas for improvement more efficiently while retaining the original evaluation data for further review.

For the statement that the AI-assisted analysis presents the results of keyword and frequency analysis in a clear and understandable manner, the students obtained a mean of 3.56 and faculty members obtained 3.46, both interpreted as Highly Effective, while HR personnel obtained a mean of 3.00, interpreted as Effective. The ratings are supported by the system's presentation of identified keywords, frequently occurring terms, and related feedback patterns in an organized form. The AI breakdown further supports understanding by showing how the feedback was summarized. The lower HR rating is related to the limitation that frequency results alone provide limited context about how and why a specific word or phrase appears in the feedback. The system presents the identified information clearly, but users still need access to the related feedback to fully understand its context.

For the statement that keyword and frequency analysis provides meaningful insights that support the interpretation of qualitative evaluation results, the students obtained a mean of 3.63 and faculty members obtained 3.46, both interpreted as Highly Effective, while HR personnel obtained a mean of 2.67, interpreted as Effective. The system's frequency analysis identifies commonly occurring words,

phrases, and issues, while the AI-assisted analysis and AI breakdown provide additional information for understanding the generated summary. The system's bias detection also helps identify possible bias or unusual patterns in the evaluation data. The lower HR rating is related to the fact that frequency results identify what occurs repeatedly but do not independently determine why it occurs or how significant it is. The system therefore provides the identified patterns and supporting AI analysis, while the significance of those patterns still depends on the context of the original evaluation feedback.

Lastly, the respondents assessed whether the combination of AI-generated summaries and frequency analysis improves the usefulness of qualitative faculty evaluation feedback for decision-making. Students obtained a mean of 3.64, faculty members obtained 3.54, and HR personnel obtained 3.33, with all results interpreted as Highly Effective. The results are supported by the system's combination of AI-generated summaries, thematic analysis, frequency analysis, AI bias detection, AI breakdown, and AI-generated constructive recommendations for faculty improvement. These features organize the feedback, identify recurring patterns, detect possible bias, show how the AI summarized the feedback, and provide recommendations based on the available evaluation data. The lower HR rating remains Highly Effective and is related to the fact that these AI outputs are generated from the submitted evaluation data. The AI breakdown and bias detection strengthen transparency and support review, but the generated results

still require appropriate interpretation before being used for formal administrative decisions.

It is important to emphasize that the lower HR ratings do not indicate that the AI-Assisted Feedback Analysis was ineffective. HR rated several AI-assisted functions as Highly Effective, including organizing large amounts of feedback, enabling faster interpretation, identifying faculty strengths and areas for improvement, supporting objective interpretation, and improving the usefulness of feedback for decision-making. The lower ratings were concentrated on areas requiring the AI to accurately interpret meaning, determine the significance of recurring themes, and provide meaningful insights from frequency analysis. This pattern demonstrates that the system is effective in organizing and analyzing qualitative feedback, while the interpretation of context and significance remains dependent on the underlying evaluation data.

Taken together, the results demonstrate that the proposed AI-Assisted Feedback Analysis provides improvements over the existing manual review process. The system uses AI to summarize written feedback, identify recurring themes and patterns, perform frequency analysis, detect possible bias, identify faculty strengths and areas for improvement, and generate constructive recommendations based on faculty evaluation results. The AI breakdown further provides transparency by showing how the feedback was summarized. These

features reduce the effort required to organize and initially interpret large amounts of qualitative feedback. The lower ratings in specific areas identify the limitations of automated analysis in determining the complete context and significance of the information without reference to the original evaluation data.

In totality, the students and faculty members assessed the AI-Assisted Feedback Analysis of the proposed Web-Based Faculty Evaluation System as Highly Effective, with average weighted means of 3.59 and 3.57, respectively. Meanwhile, HR personnel assessed the AI-assisted feedback analysis as Effective, with an average weighted mean of 3.11. Although HR personnel obtained the lowest average weighted mean, their overall assessment remained within the Effective interpretation. Overall, the respondents assessed the AI-Assisted Feedback Analysis of the proposed Web-Based Faculty Evaluation System as Highly Effective in terms of effectiveness, with a general weighted mean of 3.42. The results demonstrate that the proposed system effectively uses AI-generated summaries, thematic analysis, frequency analysis, bias detection, AI breakdown, and constructive AI-generated recommendations to organize and interpret qualitative faculty evaluation feedback. The lower ratings in specific areas identify the system's limitations in independently determining the complete context and significance of recurring feedback, but these limitations do not prevent the AI-assisted features from effectively supporting the analysis, interpretation, and improvement of faculty evaluation results.

3. To empirically evaluate the proposed faculty evaluation system using the ISO/IEC 25010:2023 software quality characteristics, including functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility.

The proposed Web-Based Faculty Evaluation System was empirically evaluated using the ISO/IEC 25010:2023 software quality characteristics, including functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility. The respondents' assessments were analyzed to determine the software quality of the proposed system based on these prescribed evaluation criteria.

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Table 6
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF FUNCTIONAL SUITABILITY

Functional Suitability	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Functional Completeness - The system provides all the necessary features required to conduct faculty evaluations, including student, peer, and supervisor evaluations, AI-enhanced feedback analysis, report generation, and evaluation management.	3.67	Highly Functionally Suitable	3.62	Highly Functionally Suitable	3.33	Highly Functionally Suitable
Functional Correctness - The system accurately records, processes, and generates faculty evaluation results and AI-assisted feedback summaries.	3.58	Highly Functionally Suitable	3.46	Highly Functionally Suitable	3.00	Functionally Suitable
Functional Appropriateness - The system effectively supports users in completing faculty evaluation tasks efficiently and accurately.	3.65	Highly Functionally Suitable	3.62	Highly Functionally Suitable	3.33	Highly Functionally Suitable
AVERAGE WEIGHTED MEAN	3.63	Highly Functionally Suitable	3.56	Highly Functionally Suitable	3.22	Functionally Suitable
GENERAL WEIGHTED MEAN	3.47					
VERBAL INTERPRETATION	Highly Functionally Suitable					

Table 6 presents the mean results on the assessment of the respondents regarding the system quality of the proposed Web-Based Faculty Evaluation System in terms of functional suitability.

The students, faculty members, and HR personnel assessed the functional completeness of the system based on whether it provides the necessary features required to conduct and manage faculty evaluations. The students obtained a mean of 3.67, faculty members obtained 3.62, and HR personnel obtained 3.33, all interpreted as Highly Functionally Suitable. These results are supported by the system's integrated Evaluation Management Module, which manages the distribution of CHED-compliant evaluation forms, collection of responses from

students, faculty, and supervisors, validation of submitted evaluations, and coordination of evaluation processing and storage. The system also includes AI-assisted feedback analysis, report generation, evaluation tracking, centralized data management, dashboards, and role-based access control. These integrated features provide the required functions for conducting, processing, monitoring, and reporting faculty evaluations within a single platform.

The high ratings across all three respondent groups further demonstrate that the proposed system brings the major functions of the faculty evaluation process into one platform. The system supports student, peer, and supervisor evaluations through structured evaluation instruments and separate processing workflows. It also connects evaluation collection with validation, centralized storage, AI-assisted analysis, reporting, and tracking. This integration provides the functional components required to complete the evaluation process without relying on separate tools for each major activity. The slightly lower HR rating of 3.33 remains within the Highly Functionally Suitable range and indicates that the system provides the necessary functions, while the remaining difference is concentrated in the assessment of other functional suitability characteristics.

In terms of functional correctness, the respondents assessed whether the system accurately records, processes, and generates faculty evaluation results and AI-assisted feedback analysis. The students obtained a mean of 3.58 and

faculty members obtained 3.46, both interpreted as Highly Functionally Suitable, while HR personnel obtained a mean of 3.00, interpreted as Functionally Suitable. The high ratings are supported by the system's Evaluation Management Module and centralized data management, which validate submitted evaluations, process evaluation information, and store records in a centralized database. The system also uses structured processing for student, peer, and supervisor evaluations, helping ensure that evaluation information is associated with the appropriate evaluation process and stored for subsequent reporting and analysis.

The HR personnel's rating of 3.00 for functional correctness is related to the extent of processing and verification performed by the system. The proposed system correctly records, validates, processes, and consolidates the available evaluation data, but the correctness of the final evaluation information remains dependent on the data entered by evaluators. The system cannot independently determine whether a submitted rating or written comment is factually correct. Similarly, AI-assisted analysis can summarize and interpret the available feedback, but it does not independently establish the factual accuracy of the original responses. Therefore, the system performs the intended processing functions while the accuracy of the underlying information remains dependent on the submitted evaluation data.

Functional correctness is particularly important because the proposed system does more than collect evaluation responses. It associates evaluations with the appropriate faculty and evaluation sources, validates submissions, processes evaluation data, stores records centrally, and generates reports and AI-assisted analytical outputs. The system also includes cross-source validation and an Evaluation Credibility Score that help identify consistency and reliability patterns across student, peer, and supervisor evaluations. These features strengthen the checking and interpretation of the processed information. However, they do not replace the need to review the underlying evaluation data when determining whether the information fully represents actual faculty performance.

The difference between the ratings of students, faculty members, and HR personnel for functional correctness therefore reflects the distinction between system processing and the accuracy of the information being processed. Students and faculty members rated the system Highly Functionally Suitable, while HR personnel rated it Functionally Suitable. The 3.00 rating still indicates that the system was considered capable of performing its intended functions. The lower rating is associated with the fact that system-generated results, reports, and AI-assisted outputs are based on the evaluation information submitted through the system. The system provides validation, cross-source checking, credibility scoring, and centralized processing, but these mechanisms do not independently verify the factual accuracy of every submitted evaluation.

For functional appropriateness, which assessed whether the system effectively supports users in completing faculty evaluation tasks, the students obtained a mean of 3.65, faculty members obtained 3.62, and HR personnel obtained 3.33. All respondent groups interpreted the system as Highly Functionally Suitable in this area. These results are supported by the system's integrated evaluation workflow, which connects evaluation scheduling, submission, validation, processing, tracking, reporting, and AI-assisted analysis. The centralized platform also provides role-based access so that users are presented with functions appropriate to their authorized activities. These features allow the proposed system to support the major tasks involved in the faculty evaluation process in an organized manner.

The 3.33 rating from HR personnel remains Highly Functionally Suitable and is supported by the system's administrative functions, including evaluation tracking, centralized evaluation records, dashboards, report generation, AI-Feedback Analysis, and real-time reporting. The system also provides AI-generated recommendations through the Smart Recommendation Engine, which identifies weak areas and provides targeted suggestions for faculty development. These features extend the system beyond basic evaluation collection by supporting the interpretation, monitoring, and use of evaluation results. The rating therefore indicates that the system's functions are appropriate for the intended

evaluation activities while still allowing room for continued improvement in the handling and interpretation of system-generated information.

The three functional suitability criteria also show a consistent positive pattern across the respondent groups. Students obtained their highest rating of 3.67 for functional completeness, while faculty members obtained their highest ratings of 3.62 for both functional completeness and functional appropriateness. These results are supported by the availability of the required evaluation features, including multi-source evaluation, evaluation submission and validation, centralized data storage, evaluation tracking, AI-assisted feedback analysis, and report generation. The system therefore provides the functions needed to conduct and manage the major stages of the faculty evaluation process.

For HR personnel, the highest ratings were 3.33 for both functional completeness and functional appropriateness, while the lowest rating was 3.00 for functional correctness. This pattern indicates that the system's features and functions were positively assessed, while the lower assessment was concentrated on the correctness of processed information. The system addresses this area through evaluation validation, centralized data processing, cross-source validation, Evaluation Credibility Score, and AI-assisted analysis. These features strengthen the checking and interpretation of evaluation results, although they

cannot independently verify the factual accuracy of every response submitted by evaluators.

The differences in ratings can also be understood through the functions provided by the system rather than differences in the respondents themselves. The system provides separate evaluation workflows for students, peers, and supervisors, while the Evaluation Management Module coordinates the collection, validation, processing, and storage of these inputs. The centralized database then supports reporting, tracking, and retrieval of evaluation information. For qualitative feedback, the AI module provides thematic and frequency analysis, AI summaries, bias detection, and transparent breakdowns of AI-generated results. These connected functions provide a consistent system structure for completing and managing the faculty evaluation process.

The results further demonstrate that functional suitability is not limited to the number of features available in the system. Functional completeness considers whether the required functions are provided, functional correctness considers whether the system correctly performs its intended processing, and functional appropriateness considers whether the functions effectively support the required evaluation activities. The proposed system addresses these three areas through its Evaluation Management Module, multi-source evaluation workflows, validation mechanisms, centralized data management, AI-assisted analysis, reporting

functions, evaluation tracking, and role-based access control. The positive results across all three criteria therefore indicate that the system provides a comprehensive functional foundation for faculty evaluation.

The lower HR rating for functional correctness should be interpreted based on the actual operation of the system. The proposed system provides mechanisms that strengthen data processing and verification, including submission validation, cross-source validation, credibility scoring, and centralized record management. However, these mechanisms operate on the information submitted through the evaluation process and cannot independently establish whether every rating or comment is factually accurate. This explains why functional correctness received a lower mean of 3.00 from HR personnel while still remaining within the Functionally Suitable interpretation.

At the same time, the HR results demonstrate that the proposed system was positively assessed across all three functional suitability criteria. HR personnel obtained a mean of 3.33 for functional completeness and functional appropriateness, both interpreted as Highly Functionally Suitable, while functional correctness obtained a mean of 3.00, interpreted as Functionally Suitable. The results indicate that the system provides the necessary functions and appropriately supports the faculty evaluation process, while functional correctness remains

dependent on the quality and accuracy of the evaluation data processed by the system.

In totality, the students and faculty members assessed the proposed Web-Based Faculty Evaluation System as Highly Functionally Suitable, with average weighted means of 3.63 and 3.56, respectively. Meanwhile, HR personnel assessed the system as Functionally Suitable, with an average weighted mean of 3.22. The results are supported by the system's integrated evaluation management, multi-source evaluation processing, validation, centralized data storage, evaluation tracking, reporting, AI-assisted feedback analysis, cross-source validation, credibility scoring, and role-based access control. The slightly lower HR assessment is concentrated on functional correctness, where the system's outputs remain dependent on the accuracy and consistency of the evaluation information submitted by users.

Overall, the findings revealed that the respondents assessed the system quality of the proposed Web-Based Faculty Evaluation System as Highly Functionally Suitable in terms of functional suitability, with a general weighted mean of 3.47. This indicates that the proposed system provides the necessary and appropriate functions for conducting, processing, monitoring, and reporting faculty evaluations. The results further demonstrate that the system integrates multi-source evaluation, validation, centralized data management, AI-assisted feedback

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analysis, bias detection, transparent AI breakdowns, cross-source validation, credibility scoring, and AI-generated recommendations into a unified platform. While the correctness of system-generated information remains dependent on the quality of the submitted evaluation data, the proposed system provides the functional mechanisms necessary to support an organized, validated, and efficient faculty evaluation process.

Table 7
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF RELIABILITY

Reliability	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Faultlessness - The system performs faculty evaluation functions consistently without errors during normal operation.	3.47	Highly Reliable	3.54	Highly Reliable	2.67	Reliable
Availability - The system remains available whenever authorized users need to conduct or manage faculty evaluations.	3.51	Highly Reliable	3.85	Highly Reliable	3.33	Highly Reliable
Fault Tolerance - The system continues operating properly despite temporary connection issues or minor software errors.	3.40	Highly Reliable	3.54	Highly Reliable	2.33	Less Reliable
Recoverability - The system restores evaluation records and resumes normal operation after unexpected interruptions or failures.	3.60	Highly Reliable	3.85	Highly Reliable	3.00	Reliable
AVERAGE WEIGHTED MEAN	3.49	Highly Reliable	3.69	Highly Reliable	2.83	Reliable
GENERAL WEIGHTED MEAN	3.34					
VERBAL INTERPRETATION	Highly Reliable					

Table 7 presents the assessment of the proposed system in terms of reliability, specifically focusing on faultlessness, availability, fault tolerance, and recoverability. The results show that Students obtained an average weighted mean of 3.49, interpreted as Highly Reliable, while Faculty obtained 3.69, also interpreted as Highly Reliable. HR obtained an average weighted mean of 2.83,

interpreted as Reliable, resulting in a general weighted mean of 3.34, interpreted as Highly Reliable. These results indicate that the proposed system generally provides reliable evaluation processing, access, and data management, although some aspects require further improvement.

For faultlessness, Students rated the system 3.47 and Faculty 3.54, both interpreted as Highly Reliable, while HR gave the lowest rating of 2.67, interpreted as Reliable. The high Faculty rating is supported by the system's structured evaluation workflows, data validation, centralized database, and audit trail, which help ensure that evaluation activities and records are processed consistently. The lower HR rating is related to the system's dependence on successful data processing and database operations. Although validation and centralized storage help maintain reliable records, they do not prevent every possible technical interruption or processing failure.

For availability, Faculty gave the highest rating of 3.85, followed by Students at 3.51 and HR at 3.33, with all interpreted as Highly Reliable. The highest Faculty rating is supported by the web-based design, which allows users to access the system through standard web browsers without requiring additional software. The system also provides centralized access to evaluation schedules, evaluation functions, tracking, and results. HR gave the lowest rating among the groups because access to these functions remains dependent on the availability of the centralized web-based platform and its supporting database services.

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For fault tolerance, Faculty gave the highest rating of 3.54, while Students rated the system 3.40, both interpreted as Highly Reliable. HR gave the lowest rating of 2.33, interpreted as Less Reliable. The high Faculty rating is supported by features such as draft saving and database transaction controls, which help protect evaluation information during normal processing and reduce the possibility of losing unfinished data. The lower HR rating is related to the system's limited ability to prevent the effects of unexpected technical interruptions. These features protect data in specific situations but do not guarantee uninterrupted operation during every possible technical failure.

For recoverability, Faculty gave the highest rating of 3.85, followed by Students at 3.60, both interpreted as Highly Reliable, while HR gave the lowest rating of 3.00, interpreted as Reliable. The high Faculty rating is supported by centralized data storage and automated backup procedures, which help preserve evaluation records and provide a means of recovering data when necessary. The lower HR rating is related to the need to maintain complete and accurate evaluation records after a system or data problem. Although backup procedures support recovery, successful restoration still depends on the availability and integrity of the stored backup data.

Overall, the proposed system obtained a general weighted mean of 3.34, interpreted as Highly Reliable. The highest ratings were generally observed in

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areas supported by direct system features such as web-based access, structured workflows, draft saving, centralized storage, and backup procedures. Meanwhile, the lower HR ratings were concentrated on areas involving technical interruptions, fault tolerance, and recovery conditions. These findings show that the system provides a reliable foundation for faculty evaluation while identifying fault tolerance and recovery handling as areas that can be further strengthened.

Table 8
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF USABILITY

Usability	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Appropriateness Recognizability - The system enables users to immediately recognize that it is appropriate for conducting faculty evaluations.	3.67	Highly Usable	3.69	Highly Usable	3.67	Highly Usable
Learnability - The system allows new users to learn how to use its features within a short period of time.	3.66	Highly Usable	3.69	Highly Usable	2.67	Usable
Operability - The system provides menus, controls, and navigation that are simple and convenient to use.	3.69	Highly Usable	3.69	Highly Usable	3.00	Usable
User Error Protection - The system prevents user mistakes by validating inputs and providing confirmations before submitting evaluations.	3.61	Highly Usable	3.69	Highly Usable	3.00	Usable
User Engagement - The system encourages users to complete evaluations through an organized, interactive, and user-friendly interface.	3.65	Highly Usable	3.69	Highly Usable	3.33	Highly Usable
Inclusivity - The system accommodates users with different levels of computer skills and technical experience.	3.64	Highly Usable	3.77	Highly Usable	2.67	Usable
User Assistance - The system provides sufficient instructions, prompts, and guidance to help users complete evaluation tasks.	3.63	Highly Usable	3.69	Highly Usable	2.67	Usable
Self-descriptiveness - The system clearly presents its functions and features so users can understand how to use it without relying heavily on external assistance.	3.66	Highly Usable	3.69	Highly Usable	2.67	Usable
AVERAGE WEIGHTED MEAN	3.65	Highly Usable	3.70	Highly Usable	2.96	Usable
GENERAL WEIGHTED MEAN	3.44					
VERBAL INTERPRETATION	Highly Usable					

Table 8 presents the assessment of the proposed Web-Based Faculty Evaluation System in terms of usability, covering appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness. The students obtained an average weighted mean of 3.65, interpreted as Highly Usable, while faculty members obtained 3.70, also interpreted as Highly Usable. HR personnel obtained an average weighted mean of 2.96, interpreted as Usable, resulting in a general weighted mean of 3.44, interpreted as Highly Usable. The results indicate that the system is generally understandable and usable across the respondent groups, with the lower HR ratings concentrated in areas involving learning, guidance, and understanding of the system's broader functions.

For appropriateness recognizability, Students obtained a mean of 3.67, Faculty 3.69, and HR 3.67, all interpreted as Highly Usable. Faculty members obtained the highest rating, while Students and HR had closely similar ratings. These high ratings are supported by the system's clearly defined purpose and evaluation-related functions, including evaluation forms, schedules, evaluation tracking, results, and feedback-related features. The similar ratings indicate that the system clearly communicates its purpose as a faculty evaluation platform to all respondent groups.

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For learnability, Students obtained 3.66 and Faculty 3.69, both interpreted as Highly Usable, while HR obtained 2.67, interpreted as Usable. Faculty obtained the highest rating because the system provides role-specific interfaces that present users with the functions relevant to their access. Students and faculty members also interact with focused evaluation-related functions, making the main processes easier to identify and follow. HR obtained the lowest rating because the system provides a wider set of functions, including evaluation management, evaluation results, AI-assisted feedback analysis, tracking, and administrative features. The greater number of functions increases the amount of system information that must be learned, explaining the lower rating.

For operability, Students and Faculty both obtained 3.69, interpreted as Highly Usable, while HR obtained 3.00, interpreted as Usable. The highest rating from Students and Faculty is supported by the system's role-specific interface, responsive design, intuitive navigation, and streamlined evaluation workflows. These features allow users to access the functions relevant to their activities without unnecessary navigation elements. HR's lower rating is related to the larger number of available administrative functions, which requires navigation across more sections of the system. This indicates that the system remains usable, but the organization of the broader administrative functions can still be improved.

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For user error protection, Students obtained 3.61 and Faculty 3.69, both interpreted as Highly Usable, while HR obtained 3.00, interpreted as Usable. Faculty obtained the highest rating, supported by the system's validation mechanisms and structured evaluation workflows that help ensure submitted information follows the required process. Students also benefit from validation during evaluation submission, including enrollment validation that verifies whether a student is assigned to a specific instructor before allowing the evaluation to proceed. HR's lower rating is related to the greater number of administrative actions available in the system, where additional confirmation, validation, and feedback during more detailed processes can further strengthen protection against user mistakes.

For user engagement, Students obtained 3.65, Faculty 3.69, and HR 3.33, all interpreted as Highly Usable. Faculty obtained the highest rating, supported by the system's responsive interface and organized presentation of evaluation-related information and feedback. Students also gave a high rating because the system provides a streamlined digital evaluation process. HR's rating was slightly lower but remained Highly Usable, indicating that the interface remains organized and engaging even when accessing more administrative functions. The relatively small difference among the ratings shows that the overall interface provides a positive interaction experience across the respondent groups.

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For inclusivity, Students obtained 3.64 and Faculty obtained the highest rating of 3.77, both interpreted as Highly Usable, while HR obtained 2.67, interpreted as Usable. The high Faculty rating is supported by the system's responsive design, which allows the interface to function across desktop computers, laptops, tablets, and smartphones without requiring additional client-side software. Students also gave a high rating, showing that the web-based interface can accommodate users accessing the system through different devices. HR's lower rating is related to the complexity of the functions available through the administrative interface rather than basic access to the system. Additional role-specific guidance can further improve the accessibility of these functions for users with different levels of technical familiarity.

For user assistance, Students obtained 3.63 and Faculty 3.69, both interpreted as Highly Usable, while HR obtained 2.67, interpreted as Usable. Faculty obtained the highest rating because the system presents role-specific functions and streamlined workflows that make the relevant processes easier to follow. Students similarly benefit from the structured evaluation workflow and clearly defined evaluation functions. HR's lower rating indicates that the available guidance is less sufficient for some of the system's broader administrative functions. Since these functions involve evaluation management, results, tracking, reports, and AI-assisted feedback analysis, clearer role-specific instructions and guidance would further improve usability.

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For self-descriptiveness, Students obtained 3.66 and Faculty 3.69, both interpreted as Highly Usable, while HR obtained 2.67, interpreted as Usable. Faculty obtained the highest rating because the role-specific interface presents relevant functions and information in a focused manner. Students also gave a high rating because the evaluation workflow provides a direct path from accessing assigned evaluations to completing the required evaluation activities. HR's lower rating is related to the broader range of functions presented in the administrative interface. More explicit labels, descriptions, prompts, or contextual guidance for these functions would make the system more immediately understandable.

The overall pattern shows that Faculty obtained the highest average weighted mean of 3.70, followed by Students at 3.65, while HR obtained the lowest at 2.96. The high Faculty rating is supported by the system's role-specific interface, responsive design, focused navigation, and organized access to evaluation-related information. Students also provided a high rating because the system provides a streamlined evaluation workflow and functions directly related to completing their required evaluations. HR's lower rating is concentrated in learnability, inclusivity, user assistance, and self-descriptiveness, where the score was consistently 2.67. These lower ratings are directly related to the larger number of administrative functions that require additional navigation, explanation, and familiarity.

The results also show that HR still rated appropriateness recognizability at 3.67 and user engagement at 3.33, both within the Highly Usable range. This demonstrates that HR personnel recognized the system as appropriate and generally organized despite giving lower ratings for learning and guidance. Therefore, the main usability concern is not the overall purpose or interface of the system but the clarity and guidance provided when working with its more detailed administrative functions.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.44, interpreted as Highly Usable. The findings demonstrate that the system provides a role-specific, responsive, and structured interface that supports recognition, operation, error protection, engagement, and understanding of its functions. The highest ratings from Faculty and Students are supported by focused workflows and relevant system functions, while the lower HR ratings are concentrated in areas requiring greater guidance and familiarity with administrative features. Strengthening role-specific instructions, prompts, labels, and navigation guidance can further improve the usability of the system, particularly for its broader administrative functions.

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Table 9
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF SECURITY

Security	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Confidentiality - The system protects faculty evaluation records and users' personal information from unauthorized access.	3.70	Highly Secure	3.69	Highly Secure	3.33	Highly Secure
Integrity - The system ensures that evaluation data and AI-generated reports cannot be altered without proper authorization.	3.67	Highly Secure	3.69	Highly Secure	3.67	Highly Secure
Non-repudiation - The system maintains verifiable records of submitted evaluations and user activities that cannot be denied after submission.	3.65	Highly Secure	3.77	Highly Secure	3.67	Highly Secure
Accountability - The system records user activities, allowing evaluation-related actions to be traced to authorized users.	3.70	Highly Secure	3.62	Highly Secure	3.67	Highly Secure
Authenticity - The system verifies the identity of users before granting access to evaluation functions and confidential information.	3.70	Highly Secure	3.62	Highly Secure	3.67	Highly Secure
Resistance - The system protects evaluation data and maintains secure operation against unauthorized access, cyberattacks, and other malicious attacks.	3.66	Highly Secure	3.77	Highly Secure	3.00	Secure
AVERAGE WEIGHTED MEAN	3.68	Highly Secure	3.69	Highly Secure	3.50	Highly Secure
GENERAL WEIGHTED MEAN	3.62					
VERBAL INTERPRETATION	Highly Secure					

Table 9 presents the assessment of the proposed Web-Based Faculty Evaluation System in terms of security, covering confidentiality, integrity, non-repudiation, accountability, authenticity, and resistance. Students obtained an average weighted mean of 3.68, Faculty 3.69, and HR 3.50, all interpreted as Highly Secure, resulting in a general weighted mean of 3.62, also interpreted as Highly Secure. The results indicate that the system provides strong perceived protection for evaluation records, user information, access control, and evaluation-related activities, with the lower HR ratings concentrated in confidentiality and resistance.

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For confidentiality, Students obtained a mean of 3.70, Faculty 3.69, and HR 3.33, all interpreted as Highly Secure. These high ratings are supported by the system's role-based access control, which restricts users to functions and information appropriate to their assigned roles. The system also uses centralized data management to store evaluation records and user information, helping prevent unauthorized access to sensitive records. HR's lower rating of 3.33 is related to the broader set of evaluation records and administrative information accessible through authorized administrative functions, making the protection and restriction of confidential information more extensive within the system.

For integrity, Students obtained 3.67, Faculty 3.69, and HR 3.67, all interpreted as Highly Secure. The closely similar ratings indicate that the system is perceived as capable of preserving evaluation data and reports from unauthorized modification. This is supported by structured evaluation workflows, validation mechanisms, centralized data storage, and the system's audit trail, which records significant activities such as evaluation submissions, draft saves, data modifications, AI summary generation, and administrative changes. These controls help maintain the consistency and traceability of evaluation information after submission.

For non-repudiation, Students obtained 3.65, Faculty 3.77, and HR 3.67, all interpreted as Highly Secure. Faculty obtained the highest rating, while the other two groups also provided high assessments. The results are supported by the

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system's audit trail and evaluation tracking features, which maintain records of evaluation-related activities and submissions. These records provide verifiable information about actions performed within the system, helping establish accountability for submitted evaluations and other recorded activities.

For accountability, Students obtained 3.70, Faculty 3.62, and HR 3.67, all interpreted as Highly Secure. The results indicate that the system effectively supports the tracing of user activities to authorized accounts. This is supported by role-based access control, user authentication, evaluation tracking, and the audit trail. Because the system records significant actions performed within the platform, evaluation-related activities can be traced to the appropriate authorized user, improving transparency and control over the evaluation process.

For authenticity, Students obtained 3.70, Faculty 3.62, and HR 3.67, all interpreted as Highly Secure. The high ratings are supported by the system's user authentication and role-based access control, which verify users before granting access to system functions and information. The system also limits users to the functions appropriate to their assigned access level. This is particularly important in the proposed system because students, faculty members, and administrative users require different levels of access to evaluation information.

For resistance, Students obtained 3.66 and Faculty 3.77, both interpreted as Highly Secure, while HR obtained 3.00, interpreted as Secure. The high Student and Faculty ratings indicate that the system provides security controls that help

protect evaluation information from unauthorized access and other security threats. The system's role-based access control, authentication, centralized data management, and audit trail contribute to this protection.

HR's lower rating of 3.00 for resistance is related to the broader security requirements involved in protecting a centralized system that stores sensitive evaluation information. While the system provides access controls, authentication, and activity tracking, these mechanisms primarily control and record access rather than guarantee complete protection against every possible cyberattack or malicious activity. Therefore, the lower rating identifies resistance as an area where additional security strengthening can improve the system's protection against evolving threats.

The overall results show that Students obtained an average weighted mean of 3.68, Faculty obtained 3.69, and HR obtained 3.50. Students provided consistently high ratings across all six characteristics, with their highest ratings in confidentiality, accountability, and authenticity at 3.70. Faculty also provided consistently high ratings, with the highest scores in non-repudiation and resistance at 3.77. These results are supported by the system's authentication, role-based access, structured workflows, centralized data management, and activity tracking features.

HR's ratings were 3.33 for confidentiality, 3.67 for integrity, 3.67 for non-repudiation, 3.67 for accountability, 3.67 for authenticity, and 3.00 for resistance.

This pattern shows that HR also assessed most security characteristics highly, particularly integrity, non-repudiation, accountability, and authenticity. The lower ratings were concentrated in confidentiality and resistance, indicating that the main security concerns relate to protecting sensitive information from unauthorized exposure and maintaining stronger protection against broader security threats.

The difference in confidentiality ratings is supported by the system's access structure. Students and Faculty primarily receive access to functions and information assigned to their roles, while administrative functions provide access to more centralized evaluation information. The system's role-based access control helps limit unauthorized exposure, but protecting confidential information also requires maintaining appropriate access restrictions throughout the entire evaluation process.

The results for non-repudiation and accountability also demonstrate the value of the system's audit trail. Because significant system activities are recorded and preserved, actions such as evaluation submissions, draft saves, data modifications, AI summary generation, and administrative changes can be traced within the system. This supports accountability and provides a verifiable record of important evaluation-related activities.

The findings further show that security is supported by several integrated controls rather than a single security feature. Authentication verifies user identity, role-based access control limits available functions and information, validation

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helps control submitted data, centralized data management protects and organizes evaluation records, and the audit trail provides activity traceability. Together, these features support confidentiality, integrity, non-repudiation, accountability, and authenticity across the evaluation process.

The lower HR rating for resistance highlights the importance of continuously strengthening security mechanisms. The proposed system provides important controls for access and traceability, but these controls do not independently guarantee protection against every possible unauthorized access attempt, cyberattack, or malicious activity. Additional security measures, regular security assessment, monitoring, and maintenance can further strengthen the system's resistance as security threats change over time.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.62, interpreted as Highly Secure. The findings demonstrate that the system provides strong perceived security through role-based access control, authentication, centralized data management, validation, evaluation tracking, and audit trail mechanisms. While HR provided lower ratings for confidentiality and resistance, the overall assessment remained Highly Secure, showing that the system was generally perceived as capable of protecting sensitive evaluation information and controlling access to security-sensitive functions.

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Table 10
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF PERFORMANCE EFFICIENCY

Performance Efficiency	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Time Behaviour - The system responds quickly when loading pages, submitting evaluations, generating reports, and displaying AI-generated feedback summaries.	3.61	Highly Efficient	3.46	Highly Efficient	3.00	Efficient
Resource Utilization - The system efficiently utilizes computing resources such as internet connection, memory, and processing power during operation.	3.59	Highly Efficient	3.46	Highly Efficient	3.67	Highly Efficient
Capacity - The system can accommodate multiple users and large volumes of evaluation data without reducing performance.	3.58	Highly Efficient	3.38	Highly Efficient	3.00	Efficient
AVERAGE WEIGHTED MEAN	3.59	Highly Efficient	3.44	Highly Efficient	3.22	Efficient
GENERAL WEIGHTED MEAN	3.42					
VERBAL INTERPRETATION	Highly Efficient					

Table 10 presents the assessment of the proposed Web-Based Faculty Evaluation System in terms of performance efficiency, covering time behaviour, resource utilization, and capacity. Students obtained an average weighted mean of 3.59, interpreted as Highly Efficient, while Faculty obtained 3.44, also interpreted as Highly Efficient. HR obtained an average weighted mean of 3.22, interpreted as Efficient, resulting in a general weighted mean of 3.42, interpreted as Highly Efficient. The results indicate that the system generally provides efficient performance, with the lower HR assessment concentrated in time behaviour and capacity.

For time behaviour, Students obtained a mean of 3.61 and Faculty 3.46, both interpreted as Highly Efficient, while HR obtained 3.00, interpreted as

Efficient. The high ratings indicate that the system provides acceptable response times when performing functions such as loading pages, submitting evaluations, displaying results, generating reports, and processing AI-assisted feedback summaries. The system's web-based architecture, structured evaluation workflows, and centralized data management support the timely processing and display of evaluation-related information.

HR's lower rating of 3.00 is related to the number of data-intensive functions available through the administrative interface. Functions such as evaluation tracking, results management, report generation, dashboards, and AI-assisted feedback analysis require the system to retrieve and process larger amounts of evaluation information. The system performs these functions at an Efficient level, but response time can still be improved as the amount of processed evaluation data increases.

For resource utilization, Students obtained 3.59, Faculty 3.46, and HR 3.67, all interpreted as Highly Efficient. The consistently high ratings indicate that the system can perform its main functions without requiring excessive computing resources during normal operation. The centralized database, web-based architecture, and structured processing of evaluation data support the efficient handling of user requests and system functions. The HR rating of 3.67 is also the highest among the three respondent groups for this indicator, showing that the

administrative features were still perceived as manageable in terms of resource use.

The high resource utilization ratings also indicate that resource consumption was not a major performance concern in the assessment. The system is designed to manage evaluation records through centralized storage while providing access to evaluation forms, results, tracking, reports, and AI-assisted analysis through the web interface. These functions allow the system to process and present evaluation information without requiring excessive resources during regular use.

For capacity, Students obtained a mean of 3.58 and Faculty 3.38, both interpreted as Highly Efficient, while HR obtained 3.00, interpreted as Efficient. The results indicate that the system can accommodate the evaluation activities and data handled during the assessment. The high ratings from Students and Faculty are supported by the system's centralized database and structured evaluation workflows, which allow multiple evaluation records to be collected, stored, processed, and retrieved within one platform.

HR's lower rating of 3.00 for capacity is related to the system's need to handle consolidated evaluation information, reports, dashboards, tracking records, and AI-assisted feedback analysis as the volume of data increases. The system currently supports centralized processing and storage of these records, but

increasing numbers of users and evaluation records place greater demands on system capacity. Therefore, scalability and capacity remain areas that can be further strengthened as the system grows.

Among the three performance efficiency characteristics, resource utilization received the strongest assessment, with Students at 3.59, Faculty at 3.46, and HR at 3.67, all interpreted as Highly Efficient. This indicates that the system's use of available computing resources was consistently perceived positively. In comparison, time behaviour and capacity received the lowest HR rating of 3.00, showing that response time and the ability to handle increasing amounts of information are the main areas requiring further improvement.

The overall results show that Students obtained an average weighted mean of 3.59, Faculty obtained 3.44, and HR obtained 3.22. Students and Faculty assessed the system as Highly Efficient, while HR assessed it as Efficient. The difference is concentrated in time behaviour and capacity, where HR gave a rating of 3.00 for both characteristics. These results are directly related to the system functions that process larger amounts of centralized evaluation information, particularly administrative results, reports, tracking, dashboards, and AI-assisted feedback analysis.

The findings also demonstrate that performance efficiency is supported by the interaction of the system's web-based architecture, centralized data storage, structured evaluation workflows, and automated processing features. These

components allow evaluation records to be collected, stored, processed, and presented through a single platform. However, efficient operation must also be maintained as the volume of evaluation records and simultaneous system activity increases.

The lower ratings for time behaviour and capacity do not indicate that the system is unable to perform its functions. Both characteristics were still assessed as Efficient by HR, while the overall system assessment remained Highly Efficient. Instead, the results identify response time and capacity as areas where continued optimization can strengthen system performance, particularly when processing larger datasets or supporting increased numbers of users.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.42, interpreted as Highly Efficient. The findings demonstrate that the system provides timely performance, efficient resource utilization, and adequate capacity for its evaluation-related functions. The strongest results were observed in resource utilization, while time behaviour and capacity received comparatively lower HR ratings. Further improvements in response optimization, database processing, and system scalability can strengthen the system's ability to maintain efficient performance as the number of users and volume of evaluation data increase.

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Table 11
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF MAINTAINABILITY

Maintainability	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
System Stability After Updates – The system continues to function properly after updates or improvements are implemented.	3.59	Highly Maintainable	3.46	Highly Maintainable	3.33	Highly Maintainable
Quality of System Enhancements – Updates or enhancements improve the system without causing major disruptions to its existing functions.	3.57	Highly Maintainable	3.46	Highly Maintainable	3.00	Maintainable
Issue Resolution – The system allows identified issues or errors to be resolved efficiently.	3.55	Highly Maintainable	3.62	Highly Maintainable	2.67	Maintainable
Ease of System Improvement – New features, corrections, or improvements can be added without negatively affecting the system's existing functions.	3.6	Highly Maintainable	3.46	Highly Maintainable	3.33	Highly Maintainable
Consistency After Changes – The system maintains consistent performance and reliability after updates, corrections, or enhancements are implemented.	3.66	Highly Maintainable	3.62	Highly Maintainable	3.00	Maintainable
AVERAGE WEIGHTED MEAN	3.59	Highly Maintainable	3.52	Highly Maintainable	3.07	Maintainable
GENERAL WEIGHTED MEAN	3.39					
VERBAL INTERPRETATION	Highly Maintainable					

Table 11 presents the assessment of the proposed Web-Based Faculty Evaluation System in terms of maintainability, covering system stability after updates, quality of system enhancements, issue resolution, ease of system improvement, and consistency after changes. Students obtained an average weighted mean of 3.59, Faculty 3.52, and HR 3.07, resulting in a general weighted

mean of 3.39, all interpreted as Highly Maintainable, except for the HR assessment, which was interpreted as Maintainable. The results indicate that the system is generally perceived as capable of supporting updates, improvements, issue resolution, and continued operation while maintaining its existing functions.

For system stability after updates, Students obtained a mean of 3.59, Faculty 3.46, and HR 3.33, all interpreted as Highly Maintainable. The consistently high ratings indicate that the system is perceived as capable of continuing its existing functions after updates or improvements are introduced. This is supported by the system's structured architecture, modular system functions, and separation of major processes such as evaluation management, evaluation tracking, user management, and AI-assisted feedback analysis. These structures help allow improvements to be introduced while maintaining the operation of existing system components.

For quality of system enhancements, Students obtained 3.57 and Faculty 3.46, both interpreted as Highly Maintainable, while HR obtained 3.00, interpreted as Maintainable. The results indicate that the system can accommodate enhancements while retaining its existing evaluation-related functions. The lower HR rating is related to the interconnected nature of the system, where changes to functions such as evaluation management, reports, tracking, results, and AI-assisted analysis need to remain compatible with the existing workflow. This

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indicates that future enhancements require proper testing to ensure that new or modified functions do not negatively affect existing processes.

For issue resolution, Students obtained a mean of 3.55 and Faculty 3.62, both interpreted as Highly Maintainable, while HR obtained 2.67, interpreted as Maintainable. This was the largest difference among the five maintainability characteristics. The higher Student and Faculty ratings indicate that the system's identified issues are generally perceived as manageable within their evaluation-related functions. HR's lower rating indicates that issue resolution is an area that can be further strengthened, particularly for problems involving interconnected administrative functions and evaluation records.

The lower HR rating for issue resolution is not a result of the system being unable to address problems. Rather, it identifies the need for stronger maintenance procedures when an issue affects multiple connected functions. The system contains several integrated processes, including evaluation management, tracking, results generation, AI-assisted feedback analysis, and user management. Strengthening issue tracking, troubleshooting procedures, error handling, and testing can help ensure that identified problems are resolved efficiently without affecting related functions.

For ease of system improvement, Students obtained 3.60, Faculty 3.46, and HR 3.33, all interpreted as Highly Maintainable. The results indicate that the system is perceived as capable of accommodating new features, corrections, and

improvements without significantly affecting existing functions. The high ratings across all respondent groups are supported by the system's structured architecture and separate functional modules, which provide an organized basis for modifying or extending system capabilities. HR's rating of 3.33 remains Highly Maintainable, showing that future improvements are generally perceived as feasible within the existing system structure.

For consistency after changes, Students obtained 3.66 and Faculty 3.62, both interpreted as Highly Maintainable, while HR obtained 3.00, interpreted as Maintainable. The results indicate that the system is generally perceived as capable of maintaining its expected functions after updates, corrections, or enhancements. The lower HR rating is related to the need to maintain consistency across interconnected functions when system changes are introduced. Because the proposed system integrates evaluation processing, centralized data storage, tracking, reporting, and AI-assisted analysis, changes to one component need to be tested to ensure that related functions continue to operate correctly.

Among the five maintainability characteristics, system stability after updates and ease of system improvement received consistently high ratings across all respondent groups. In contrast, issue resolution received the largest difference, particularly between HR's rating of 2.67 and the higher ratings from Students and Faculty. This pattern indicates that the system is generally viewed as maintainable

and capable of supporting future improvements, while issue resolution represents the main area that requires further strengthening.

The overall results show that Students obtained an average weighted mean of 3.59, Faculty obtained 3.52, and HR obtained 3.07. Students and Faculty assessed the system as Highly Maintainable, while HR assessed it as Maintainable. The lower HR assessment is concentrated in quality of system enhancements, issue resolution, and consistency after changes. These areas are directly related to maintaining interconnected administrative functions and ensuring that system changes do not produce unintended effects on existing processes.

The findings also demonstrate that maintainability is supported by the system's structured and modular organization. Major functions such as user management, evaluation schedule management, student evaluation processing, peer evaluation processing, supervisor/dean evaluation processing, AI-assisted evaluation results generation, and evaluation tracking are organized as distinct system processes. This structure provides a basis for implementing corrections and improvements while reducing unnecessary effects on unrelated functions.

The audit trail also supports maintainability by recording significant system activities, including evaluation submissions, draft saves, data modifications, AI summary generation, and administrative changes. These records provide traceable information that can support the review of system activities when

investigating changes or issues. Combined with centralized data management and structured workflows, this provides a clearer basis for maintaining the system and identifying areas that require attention.

The lower rating for issue resolution highlights the importance of strengthening maintenance procedures as the system becomes more complex. Improving error handling, issue tracking, testing, and troubleshooting can help ensure that identified problems are addressed efficiently. Thorough testing after updates is also important for maintaining consistency across connected functions such as evaluation processing, reporting, tracking, and AI-assisted analysis.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.39, interpreted as Highly Maintainable. The findings demonstrate that the system is generally perceived as capable of remaining stable after updates, accommodating enhancements, supporting future improvements, and maintaining consistent functions after changes. The main area for further improvement is issue resolution, particularly for interconnected system processes. Strengthening testing, troubleshooting, issue tracking, and update procedures can further improve the system's maintainability while preserving the stability of its existing functions.

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Table 12
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF COMPATIBILITY

Compatibility	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
System Integration – The system works smoothly with other technologies and applications used by the institution.	3.67	Highly Compatible	3.54	Highly Compatible	3.00	Compatible
Information Exchange – The system effectively exchanges and manages information with other integrated systems and services.	3.62	Highly Compatible	3.62	Highly Compatible	3.33	Highly Compatible
AVERAGE WEIGHTED MEAN	3.65	Highly Compatible	3.58	Highly Compatible	3.17	Compatible
GENERAL WEIGHTED MEAN	3.47					
VERBAL INTERPRETATION	Highly Compatible					

Table 12 presents the mean results on the assessment of the proposed Web-Based Faculty Evaluation System in terms of compatibility, specifically system integration and information exchange. Students obtained an average weighted mean of 3.65, interpreted as Highly Compatible, while Faculty obtained 3.58, also interpreted as Highly Compatible. HR Personnel obtained an average weighted mean of 3.17, interpreted as Compatible. The resulting general weighted mean of 3.47 was interpreted as Highly Compatible. These findings indicate that the proposed system generally supports compatibility, although HR Personnel provided a comparatively lower assessment, particularly in system integration.

For system integration, Students obtained a mean of 3.67, Faculty 3.54, both interpreted as Highly Compatible, while HR Personnel obtained 3.00, interpreted as Compatible. The high ratings from Students and Faculty indicate that the proposed system works effectively with the technologies and applications

involved in the evaluation process. This is supported by the system's web-based architecture, centralized data management, and structured evaluation workflows, which allow evaluation activities to be performed within one platform.

The HR Personnel rating of 3.00, interpreted as Compatible, indicates that system integration is functional but has room for further improvement. The proposed system connects several processes, including evaluation schedule management, evaluation processing, evaluation tracking, results generation, user management, reporting, and AI-assisted feedback analysis. Because these functions depend on the consistent movement and processing of information, further strengthening the connections among system components and related institutional technologies can improve integration.

For information exchange, Students and Faculty both obtained a mean of 3.62, while HR Personnel obtained 3.33, with all three groups interpreting the system as Highly Compatible. The results indicate that the proposed system effectively manages and exchanges information within its connected processes. This is supported by the centralized relational database, which stores evaluation records, user accounts, course assignments, and related system information. The system also consolidates information from student, peer, and supervisor/dean evaluations, allowing multiple evaluation sources to be processed within the same platform.

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The HR Personnel rating of 3.33 for information exchange remains Highly Compatible, indicating that the system effectively manages the movement of evaluation-related information despite the lower rating for system integration. The system allows collected evaluation data to support related functions such as evaluation tracking, results generation, reporting, dashboards, and AI-assisted feedback analysis. This centralized flow of information helps maintain consistency when evaluation records from different sources are processed.

The overall results show that Students obtained an average weighted mean of 3.65, Faculty obtained 3.58, and HR Personnel obtained 3.17. Students and Faculty assessed the system as Highly Compatible, while HR Personnel assessed it as Compatible. The lower HR assessment is mainly reflected in the system integration rating of 3.00, while its information exchange rating remained Highly Compatible at 3.33. This shows that the lower HR assessment is concentrated on integration rather than the overall handling and exchange of information.

The compatibility of the proposed system is supported by its centralized data management and integrated evaluation workflow. Evaluation schedules, evaluation submissions, validation, tracking, results, reports, and AI-assisted feedback analysis are connected within the same web-based platform. This allows information generated from one process to be used by related functions without requiring separate platforms for each evaluation activity.

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The 3.00 rating of HR Personnel for system integration identifies a specific area for future improvement. Strengthening the integration between the proposed system and related institutional technologies can help ensure smoother interaction between connected applications and services. Continued improvements in information exchange can also support consistent processing as the volume of evaluation records, users, and system functions increases.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.47, interpreted as Highly Compatible. The findings demonstrate that the system generally supports effective system integration and information exchange through its web-based architecture, centralized database, and connected evaluation processes. Although HR Personnel assessed system integration as Compatible and obtained an overall mean of 3.17, the overall assessment remained Highly Compatible because Students and Faculty provided highly compatible ratings and all respondent groups assessed information exchange as Highly Compatible. The findings therefore indicate that the proposed system provides a generally compatible environment for managing and exchanging faculty evaluation information, while further strengthening system integration can improve its compatibility with existing and future institutional technologies.

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Table 13
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF PORTABILITY

Portability	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Adaptability - The system can be adapted to different devices, browsers, and institutional requirements.	3.72	Highly Portable	3.69	Highly Portable	3.33	Highly Portable
Installability - The system can be deployed, configured, updated, and maintained with minimal difficulty.	3.64	Highly Portable	3.62	Highly Portable	3.33	Highly Portable
Replaceability - The system can effectively replace the existing manual or traditional faculty evaluation process.	3.60	Highly Portable	3.62	Highly Portable	3.33	Highly Portable
AVERAGE WEIGHTED MEAN	3.65	Highly Portable	3.64	Highly Portable	3.33	Highly Portable
GENERAL WEIGHTED MEAN	3.54					
VERBAL INTERPRETATION	Highly Portable					

Table 13 presents the mean results on the assessment of the proposed Web-Based Faculty Evaluation System in terms of portability, specifically adaptability, installability, and replaceability. Students obtained an average weighted mean of 3.65, interpreted as Highly Portable, while Faculty obtained 3.64, also interpreted as Highly Portable. HR Personnel obtained an average weighted mean of 3.33, interpreted as Highly Portable. The resulting general weighted mean of 3.54 was interpreted as Highly Portable. These findings indicate that the proposed system is generally adaptable, deployable, maintainable, and suitable for supporting the institution's faculty evaluation activities.

For adaptability, Students obtained a mean of 3.72, Faculty 3.69, and HR Personnel 3.33, with all three groups interpreting the system as Highly Portable. The high ratings indicate that the system can support access across different devices and browsers. This is supported by the system's web-based architecture

and responsive design, which allow users to access evaluation-related functions through commonly used devices such as desktops, laptops, tablets, and smartphones.

The HR Personnel rating of 3.33 remains Highly Portable, indicating that the system provides sufficient adaptability while leaving room for further improvement. The system supports different user roles through role-based access, allowing the interface and available functions to correspond to the user's assigned access. Further improvements in browser compatibility and device responsiveness can strengthen adaptability as institutional requirements and technologies change.

For installability, Students obtained a mean of 3.64, Faculty 3.62, and HR Personnel 3.33, with all groups interpreting the system as Highly Portable. These results indicate that the proposed system is generally manageable in terms of deployment, configuration, updating, and maintenance. Its web-based architecture and centralized deployment allow users to access the system without requiring a separate installation of the application on every user device.

The HR Personnel rating of 3.33 for installability remains Highly Portable and indicates that deployment and maintenance are generally manageable. Since the system uses centralized data management and web-based access, system updates can be applied to the main application environment rather than requiring individual installation on each user device. This structure supports more consistent access to updated system functions.

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For replaceability, Students obtained a mean of 3.60, Faculty 3.62, and HR Personnel 3.33, all interpreted as Highly Portable. The results indicate that the proposed system can serve as a suitable alternative to the existing faculty evaluation process. This is supported by the system's evaluation management, schedule management, evaluation tracking, centralized data storage, reporting, and AI-assisted feedback analysis, which provide functions needed to conduct and manage faculty evaluation activities within one platform.

The HR Personnel rating of 3.33 remains Highly Portable, indicating that the system is suitable as an alternative to the existing process. The proposed system replaces separate or manual activities with connected web-based processes for evaluation submission, validation, tracking, results generation, reporting, and feedback analysis. These integrated functions allow the proposed system to support the transition from traditional evaluation procedures to a centralized digital process.

In totality, Students obtained an average weighted mean of 3.65, Faculty 3.64, and HR Personnel 3.33, with all respondent groups assessing the system as Highly Portable. The results demonstrate that the proposed system provides flexibility in accessing the platform, supports centralized deployment and maintenance, and can serve as an alternative to the existing faculty evaluation process. The lower HR rating does not change the overall interpretation because all respondent groups remained within the Highly Portable range.

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Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.54, interpreted as Highly Portable. The findings demonstrate that the system supports portability through its web-based architecture, responsive design, centralized deployment, role-based access, and integrated evaluation processes. The system can be accessed across different devices and browsers, maintained through its centralized web environment, and used as an alternative to the existing evaluation process. These results indicate that the proposed system provides a flexible and adaptable platform for managing faculty evaluation activities within the institution.

Table 14
MEAN RESULT ON THE SYSTEM QUALITY EVALUATION OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS OF ACCESSIBILITY

Accessibility	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Accessibility - The system allows users with different abilities and levels of technical expertise to complete faculty evaluation tasks effectively.	3.74	Highly Accessible	3.92	Highly Accessible	3.33	Highly Accessible
AVERAGE WEIGHTED MEAN	3.74	Highly Accessible	3.92	Highly Accessible	3.33	Highly Accessible
GENERAL WEIGHTED MEAN	3.66					
VERBAL INTERPRETATION	Highly Accessible					

Table 14 presents the mean results on the assessment of the proposed Web-Based Faculty Evaluation System in terms of accessibility. Students obtained an average weighted mean of 3.74, Faculty obtained 3.92, and HR Personnel obtained 3.33, with all three respondent groups interpreting the system as Highly

Accessible. The resulting general weighted mean of 3.66 was also interpreted as Highly Accessible. These findings indicate that the proposed system provides an accessible platform that supports users with different levels of technical knowledge and allows them to complete faculty evaluation activities effectively.

The high ratings from Students and Faculty indicate that the system provides accessible interaction with its evaluation-related functions. This is supported by the system's responsive web design, which allows the platform to be accessed through different devices, including desktops, laptops, tablets, and smartphones. The system also uses role-based access, providing users with the functions appropriate to their assigned accounts and reducing unnecessary system features within each user interface.

Faculty obtained the highest average weighted mean of 3.92, interpreted as Highly Accessible. This rating is supported by the system's organized interface and role-specific functions, which provide access to relevant evaluation activities and information. The system's web-based structure also allows users to access the platform without requiring a separate application installation on their devices.

HR Personnel obtained an average weighted mean of 3.33, which remains within the Highly Accessible range. The lower rating indicates that accessibility has room for further improvement, particularly when navigating the broader set of administrative functions available in the system. Functions such as evaluation

management, evaluation tracking, results generation, reporting, dashboards, and AI-assisted feedback analysis require interaction with different parts of the platform, making clear navigation and interface organization important for accessibility.

The system further supports accessibility through its structured evaluation workflows and clearly organized functions. Students can access their required evaluation activities, Faculty can access relevant faculty evaluation functions, and HR Personnel can access administrative features provided by their assigned role. This role-based structure limits access to relevant functions while keeping the system organized for different types of users.

The overall results show that Faculty obtained an average weighted mean of 3.92, Students obtained 3.74, and HR Personnel obtained 3.33. All three groups assessed the system as Highly Accessible. The difference in ratings is therefore a variation in the level of perceived accessibility rather than an indication that the system is inaccessible. The lower HR assessment identifies an area where navigation, guidance, and presentation of administrative functions can be further improved.

The accessibility of the proposed system is also supported by its responsive interface, role-based access, centralized platform, and structured workflows. These features allow users to access appropriate evaluation functions across different devices while maintaining a consistent web-based environment. The

system therefore provides a common platform for student, faculty, and HR-related evaluation activities without requiring separate systems for each user group.

The 3.33 rating of HR Personnel identifies a specific opportunity for improvement in accessibility. Providing clearer role-specific instructions, more descriptive labels, contextual prompts, and organized navigation for administrative functions can make the system easier to understand and navigate. These improvements can further support users when accessing functions that involve evaluation records, reports, tracking, and AI-assisted feedback analysis.

In totality, the proposed Web-Based Faculty Evaluation System obtained average weighted means of 3.74 for Students, 3.92 for Faculty, and 3.33 for HR Personnel, all interpreted as Highly Accessible. The findings demonstrate that the system supports accessibility through its responsive design, role-based access, organized interface, and structured evaluation workflows. Although HR Personnel obtained the lowest rating, the assessment remained Highly Accessible, showing that the system generally accommodates the different user groups and their required evaluation activities.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.66, interpreted as Highly Accessible. The findings demonstrate that the system provides an accessible web-based platform that supports users with different levels of technical knowledge and allows them to

complete their assigned faculty evaluation activities. Its responsive design, role-based access, organized functions, and structured workflows contribute to its accessibility, while further improvements in administrative navigation and user guidance can strengthen the experience for users handling broader system functions.

4. To compare the proposed system with the existing evaluation process in terms of operational efficiency, data accuracy, usability, and administrative workload.

The proposed system was compared with the existing faculty evaluation process in terms of operational efficiency, data accuracy, usability, and administrative workload. The respondents' assessments were analyzed to determine how the proposed web-based system differs from and addresses the limitations of the existing evaluation process.

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Table 15
**MEAN RESULT ON THE SYSTEM COMPARISON OF THE WEB-BASED
FACULTY EVALUATION SYSTEM IN TERMS OF
OPERATIONAL EFFICIENCY**

Operational Efficiency	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The proposed system completes evaluation-related tasks more efficiently than the existing evaluation process.	3.71	Highly Efficient Compared with the Existing Process	3.69	Highly Efficient Compared with the Existing Process	3.67	Highly Efficient Compared with the Existing Process
The proposed system allows evaluation information to be processed more quickly than the current process.	3.65	Highly Efficient Compared with the Existing Process	3.77	Highly Efficient Compared with the Existing Process	4.00	Highly Efficient Compared with the Existing Process
The proposed system reduces delays in the evaluation process compared with the existing evaluation process.	3.65	Highly Efficient Compared with the Existing Process	3.85	Highly Efficient Compared with the Existing Process	3.67	Highly Efficient Compared with the Existing Process
AVERAGE WEIGHTED MEAN	3.67	Highly Efficient Compared with the Existing Process	3.77	Highly Efficient Compared with the Existing Process	3.78	Highly Efficient Compared with the Existing Process
GENERAL WEIGHTED MEAN	3.74					
VERBAL INTERPRETATION	Highly Efficient Compared with the Existing Process					

Table 15 presents the mean results of the respondents' assessment of the proposed Web-Based Faculty Evaluation System in comparison with the existing evaluation process in terms of operational efficiency. The proposed system obtained a general weighted mean of 3.74, interpreted as Highly Efficient Compared with the Existing Process. This indicates that the proposed system provides a more efficient evaluation process through centralized evaluation management, automated processing, evaluation tracking, and faster access to evaluation information.

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For the statement that the proposed system completes evaluation-related tasks more efficiently than the existing evaluation process, Students obtained a mean of 3.71, Faculty 3.69, and HR Personnel 3.67, all interpreted as Highly Efficient Compared with the Existing Process. These results are supported by the system's Evaluation Management Module, evaluation schedule management, structured evaluation workflows, and evaluation tracking. The proposed system manages evaluation access, submission, recording, and monitoring within one platform, reducing the need to handle these activities through separate forms and manual records.

For the statement that the proposed system allows evaluation information to be processed more quickly than the current process, Students obtained 3.65, Faculty 3.77, and HR Personnel 4.00, all interpreted as Highly Efficient Compared with the Existing Process. The highest rating of 4.00 from HR Personnel is supported by the system's centralized relational database and automated processing of evaluation information. Evaluation responses are stored in the system and can be retrieved for tracking, results generation, reporting, dashboards, and AI-assisted feedback analysis without requiring repeated manual consolidation.

For the statement that the proposed system reduces delays in the evaluation process, Students obtained 3.65, Faculty 3.85, and HR Personnel 3.67, all interpreted as Highly Efficient Compared with the Existing Process. The highest

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rating of 3.85 from Faculty is supported by the system's structured submission workflow, evaluation schedule management, enrollment validation, and evaluation tracking. These functions organize the evaluation process from access and submission to monitoring, reducing delays caused by manual checking and separate records.

For the average weighted means, Students obtained 3.67, Faculty 3.77, and HR Personnel 3.78, all interpreted as Highly Efficient Compared with the Existing Process. HR Personnel obtained the highest overall rating because the system directly supports administrative activities through centralized evaluation records, automated processing, tracking, reporting, dashboards, and AI-assisted feedback analysis. Students obtained the lowest overall rating, but their 3.67 result still indicates that the system provides a substantial improvement over the existing process.

In totality, the findings show that the proposed system improves operational efficiency by centralizing evaluation information, reducing manual consolidation, automating data processing, supporting evaluation tracking, and providing faster access to results and reports. The integration of evaluation management, validation, storage, processing, reporting, and AI-assisted feedback analysis within one platform reduces the need to manage evaluation information through separate forms and manual records.

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Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.74, interpreted as Highly Efficient Compared with the Existing Process. The results demonstrate that the proposed system provides a more efficient evaluation workflow than the existing process by reducing manual processing, accelerating information handling, and minimizing delays throughout the evaluation cycle.

Table 16
**MEAN RESULT ON THE SYSTEM COMPARISON OF THE WEB-BASED
FACULTY EVALUATION SYSTEM IN TERMS OF DATA ACCURACY**

Data Accuracy	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The proposed system produces more accurate evaluation results than the existing evaluation process.	3.61	Highly Accurate	3.77	Highly Accurate	3.33	Highly Accurate
The proposed system minimizes errors in recording and processing evaluation data compared with the existing evaluation process.	3.65	Highly Accurate	3.69	Highly Accurate	3.33	Highly Accurate
The information generated by the proposed system is more consistent and reliable than the information from the existing evaluation process.	3.85	Highly Accurate	3.69	Highly Accurate	3.00	Accurate
AVERAGE WEIGHTED MEAN	3.70	Highly Accurate	3.72	Highly Accurate	3.22	Accurate
GENERAL WEIGHTED MEAN	3.55					
VERBAL INTERPRETATION	Highly Accurate					

Table 16 presents the mean results of the respondents' assessment of the proposed Web-Based Faculty Evaluation System in comparison with the existing evaluation process in terms of data accuracy. The proposed system obtained a general weighted mean of 3.55, interpreted as Highly Accurate. Students obtained an average weighted mean of 3.70, Faculty 3.72, both interpreted as Highly Accurate, while HR Personnel obtained 3.22, interpreted as Accurate. These

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results indicate that the proposed system was generally perceived as more accurate in producing, recording, and processing faculty evaluation information than the existing evaluation process.

For the statement that the proposed system produces more accurate evaluation results than the existing evaluation process, Faculty obtained the highest mean of 3.77, followed by Students at 3.61 and HR Personnel at 3.33, with all interpreted as Highly Accurate. Faculty provided the highest rating because the system automatically records and organizes evaluation responses through a centralized database, reducing errors that can occur during manual consolidation and calculation. HR Personnel provided the lowest rating because, although the system processes evaluation data accurately, it does not independently verify whether every submitted rating or comment accurately represents the faculty member's actual performance.

For the statement that the proposed system minimizes errors in recording and processing evaluation data compared with the existing evaluation process, Faculty obtained the highest mean of 3.69, followed by Students at 3.65 and HR Personnel at 3.33, all interpreted as Highly Accurate. Faculty gave the highest rating because evaluation responses are stored and processed within the same system, reducing repetitive data entry, manual transfer, and consolidation. HR Personnel gave the lowest rating because the system minimizes processing errors

but does not completely eliminate errors originating from the submitted evaluation information itself.

For the statement that the information generated by the proposed system is more consistent and reliable than the information from the existing evaluation process, Students obtained the highest mean of 3.85, followed by Faculty at 3.69, while HR Personnel obtained the lowest mean of 3.00, interpreted as Accurate. Students gave the highest rating because the system follows a standardized digital process for submitting, storing, and processing evaluation responses, resulting in consistently structured evaluation information. HR Personnel gave the lowest rating because the system can ensure consistent recording and processing but cannot independently verify the factual accuracy, completeness, or representativeness of every submitted evaluation response.

The differences in the ratings are therefore directly related to the system's data-handling capabilities. The higher ratings from Faculty and Students are supported by centralized data storage, standardized evaluation workflows, validation, and automated processing, which reduce manual handling and improve consistency. The lower HR ratings are concentrated on aspects that involve the reliability of the original submitted information, which the system processes but cannot independently verify.

In totality, Students obtained an average weighted mean of 3.70, Faculty 3.72, and HR Personnel 3.22. Students and Faculty assessed the proposed

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system as Highly Accurate, while HR Personnel assessed it as Accurate. Faculty obtained the highest overall rating because the system provides structured and centralized processing of evaluation information, while HR Personnel obtained the lowest overall rating because data accuracy still depends on the validity of the evaluation information submitted by users.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.55, interpreted as Highly Accurate. The findings indicate that the proposed system improves data accuracy by reducing manual data handling, centralizing evaluation records, applying validation, and standardizing the processing of evaluation information. However, while the system improves the accuracy and consistency of data processing, it does not independently verify the factual accuracy of every submitted evaluation response.

Table X. System-Level Input Validation Testing Results

ID	Validation Test	Invalid Input / Attempt	Expected Result	Actual Result	Status
IVT-01	Above Maximum	Change rating to 5	Rejected	HTTP 400 – Rejected	PASS
IVT-02	Incomplete Survey	Remove one required rating	Rejected	HTTP 400 – Rejected	PASS
IVT-03	Duplicate Submission	Resend submitted evaluation	Rejected	HTTP 400 – Rejected	PASS
IVT-04	Below Minimum	Change rating to 0	Rejected	HTTP 400 – Rejected	PASS
IVT-05	Negative Rating	Change rating to -1	Rejected	HTTP 400 – Rejected	PASS
IVT-06	Invalid Data Type	Change rating to "abc"	Rejected	HTTP 400 – Rejected	PASS
IVT-07	Empty Ratings	Change ratings to {}	Rejected	HTTP 400 – Rejected	PASS
IVT-08	Invalid	Use nonexistent	Rejected	HTTP 400 –	PASS

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	Professor	targetProfessorId		Rejected	
IVT-09	Invalid Course Offering	Use nonexistent courseOfferingId	Rejected	HTTP 400 – Rejected	PASS
IVT-10	Unauthorized Evaluator	Manipulate evaluator/student identity	Rejected	HTTP 400 – Rejected	PASS
IVT-11	Closed Evaluation Period	Submit outside evaluation period	Rejected	HTTP 400 – Rejected	PASS
IVT-12	Missing Required Identifier	Remove professor/course offering identifier	Rejected	HTTP 400 – Rejected	PASS

The results of the System-Level Input Validation Testing show that the system successfully rejected all 12 invalid input attempts, resulting in a 100% passing rate for the tested cases. The tests included out-of-range and negative ratings, incomplete surveys, duplicate submissions, invalid data types and identifiers, unauthorized evaluators, and submissions made outside the evaluation period. These results indicate that the implemented validation controls effectively prevented the tested invalid data from being accepted and stored in the database. Therefore, the system demonstrated reliable input validation based on the test cases conducted.

Table Y. Data Consistency and Traceability Audit (Sample Size: 20 Transactions)

Test Case ID	Evaluation Component	Frontend Input	Backend Storage	Expected System Output	Status
TC-DC-01	Student, Peer, and Supervisor Ratings	Selected rating: 4	rating_value = 4.00	Correct question and evaluation display rating 4	PASS
TC-DC-02	Rating Calculation	Ratings: 4, 3, 5, 4, 4	Five matching rating records	Mean: 4.00; Percentage: 80%	PASS

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TC-DC-03	Qualitative Feedback	"The instructor explains clearly."	Matching text_value / stored feedback	Report preserves the validated feedback	PASS
TC-DC-04	Evaluation Credibility	Ratings, feedback, and completion time	Stored credibility score and status	Displayed score and review status match database	PASS
TC-DC-05	Report Accuracy	Selected faculty, class, and semester	Matching evaluation and enrollment records	Report shows correct records, counts, and calculated ratings	PASS
TC-DC-06	Transaction Traceability	Student submission with transaction capture	Payload hash, committed records, and trace reference	Submission, database values, and report values match	PASS
Overall	20 evaluation transactions	20 inputs verified	20 database comparisons	20 report comparisons	PASS (100%)

The results of the Data Consistency and Traceability Audit show that the 20 tested evaluation transactions achieved a 100% consistency rate. Data entered through the frontend was compared with the corresponding records stored in the database and the information produced in system reports. The tested ratings, qualitative feedback, credibility information, calculations, and report records remained consistent throughout the evaluation process. These findings indicate that, within the tested sample, the system accurately preserved and processed evaluation data from user submission through database storage and report generation.

Figure X:
End-to-End System Traceability Log (Transaction ID: EVAL-2026-4)

```
=====
=====
SYSTEM TRACEABILITY AUDIT LOG
Timestamp: 2026-10-06 21:28:44.092 UTC+8
Module: Evaluation_Processing_Engine
Viewer_Role: HR
Trace_Reference: TRIAL-c040545e2be530e76e680f0c
=====
=====
VERIFICATION SUMMARY
Stored_Evaluation: PRESENT (evaluation #4)
Database_to_Faculty_Report: PASS
Original_Submission_Values: CAPTURED
HR_AI_Event_Recorded: YES (see engine events below)
HR_Analytics_Eligibility: ELIGIBLE | Credibility_Status: ACCEPTED_BY_HR
Required_Analytics_Scope: Faculty #10 | Semester: 1st-semester-2026-2027
=====
=====
[21:28:44.092] [TRIAL_PAGE] TRIAL ARMED | Awaiting one student submission

[21:28:52.313] [BACKEND_API] REQUEST RECEIVED | Validation: PENDING
Evidence: {"courseOfferingId":"28"}

[21:28:52.317] [FRONTEND_CLIENT] HTTP POST
/api/app_state.php?action=addEvaluation | Payload received at backend
User_Role: STUDENT | Target_Faculty_ID: resolved from course offering
```

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Payload: {"targetProfessorId":"","courseOfferingId":"28","semesterId":"1st-semester-2026-2027","ratings":{"29":"1","30":"5","31":"1","32":"5","33":"1","34":"5","35":"4","36":"3","37":"2","38":"4","39":"3","41":"2","42":"3","43":"4","44":"2"},"qualitative":{"40":"The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"},"comments":""}
Client_Reported_Sent_At: 2026-10-06T13:28:52.252Z
Client_Payload_SHA256:
7192a337922102d496b38cc62e6f3326f11bf6d566c0caa7ffca42fdab6ca3d6
Received_Payload_SHA256:
7192a337922102d496b38cc62e6f3326f11bf6d566c0caa7ffca42fdab6ca3d6
Data_Hash_Match: TRUE
Status: Request received by backend; client response acknowledgement not captured.

[21:28:52.317] [BACKEND_API] VALIDATION PASS

Validation: PASS (Authenticated; existing validation rules applied)
Data_Hash_Match: TRUE
Validated_Values: {"facultyId":10,"courseOfferingId":28,"semesterId":"1st-semester-2026-2027","ratings":{"29":"1.00","30":"5.00","31":"1.00","32":"5.00","33":"1.00","34":"5.00","35":"4.00","36":"3.00","37":"2.00","38":"4.00","39":"3.00","41":"2.00","42":"3.00","43":"4.00","44":"2.00"},"qualitative":{"40":"The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"},"comments":""}
Submitted_Values:
{"targetProfessorId":"","courseOfferingId":"28","semesterId":"1st-semester-2026-2027","ratings":{"29":"1","30":"5","31":"1","32":"5","33":"1","34":"5","35":"4","36":"3

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","37": "2", "38": "4", "39": "3", "41": "2", "42": "3", "43": "4", "44": "2"}, "qualitative": {"40": "The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"}, "comments": ""}

[21:28:55.171] [DB_TRANSACTION] SQL EXECUTE - INSERT SUCCESS | COMMIT SUCCESS

SQL: INSERT INTO evaluations; INSERT INTO evaluation_responses

Evaluation_ID: 4

Values: {"facultyId": 10, "courseOfferingId": 28, "semesterId": "1st-semester-2026-2027", "ratings": {"29": "1.00", "30": "5.00", "31": "1.00", "32": "5.00", "33": "1.00", "34": "5.00", "35": "4.00", "36": "3.00", "37": "2.00", "38": "4.00", "39": "3.00", "41": "2.00", "42": "3.00", "43": "4.00", "44": "2.00"}, "qualitative": {"40": "The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"}, "comments": ""}

Rows_Affected: 1 evaluation row; 16 saved responses.

Status: SUCCESS - transaction committed.

[21:28:55.172] [DB_VERIFICATION] SQL SELECT - SAVED EVALUATION READ-BACK

Evaluation_ID: 4

Retrieved: {"facultyId": 10, "courseOfferingId": 28, "semesterId": "1st-semester-2026-2027", "ratings": {"29": "1.00", "30": "5.00", "31": "1.00", "32": "5.00", "33": "1.00", "34": "5.00", "35": "4.00", "36": "3.00", "37": "2.00", "38": "4.00", "39": "3.00", "41": "2.00", "42": "3.00", "43": "4.00", "44": "2.00"}, "qualitative": {"40": "The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"}, "comments": ""}

Consistency_Check: PASS (Validated submission -> saved database snapshot)

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[21:28:55.172] [BACKEND_RESPONSE] HTTP 200 OK | Evaluation saved

[21:28:57.775] [VERIFICATION] VALUES CHECKED | Database: PASS |
Returned evaluation: PASS

[21:29:25.337] [REPORTING_MODULE] FACULTY REPORT DATA LOADED
Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-
2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:29:25.341] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS
Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:29:25.341] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:29:54.365] [AI_ANALYSIS_ENGINE] HR_FACULTY_ANALYTICS | INPUT
PREPARED

Analysis_Actor_Role: HR

Prepared_Text: [{"id":"u10_2","source":"Student to Professor","text":"The
professor explains the lessons clearly and makes the class easy to understand.
They are also approachable and helpful whenever students"}]

Feedback_Included: TRUE

[21:29:54.383] [AI_ANALYSIS_ENGINE] HR_FACULTY_ANALYTICS | AI
REQUEST STARTED

Analysis_Actor_Role: HR

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Prepared_Text: [{"id": "u10_2", "source": "Student to Professor", "text": "The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"}]

Status: STARTED

Feedback_Included: TRUE

[21:30:01.496] [AI_ANALYSIS_ENGINE] HR_FACULTY_ANALYTICS | RESULT
Analysis_Actor_Role: HR

Prepared_Text: [{"id": "u10_2", "source": "Student to Professor", "text": "The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"}]

Source: openai+rule

Status: SUCCESS

Feedback_Included: TRUE

[21:30:14.832] [REPORTING_MODULE] FACULTY REPORT DATA LOADED
Evidence: {"facultyId": 10, "semesterId": "1st-semester-2026-2027", "loadType": "main", "evaluationId": 4, "included": true, "textLoaded": true, "requestPath": "/system/api/data_accuracy.php"}

[21:30:14.836] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS
Evidence: {"facultyId": 10, "evaluationId": 4, "included": true}

[21:30:14.836] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:30:28.213] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

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Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:30:28.216] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:30:28.217] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:30:30.918] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:30:30.923] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:30:30.923] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:30:35.292] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:30:35.296] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

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[21:30:35.297] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:30:49.721] [REPORTING_MODULE] FACULTY REPORT DATA LOADED
Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-
2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:30:49.726] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS
Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:30:49.726] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:31:19.040] [REPORTING_MODULE] FACULTY REPORT DATA LOADED
Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-
2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:31:19.045] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS
Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:31:19.045] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:31:23.740] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

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Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:31:23.744] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:31:23.744] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:31:46.020] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:31:46.024] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:31:46.024] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:31:47.770] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:31:47.773] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

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[21:31:47.774] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:31:58.047] [REPORTING_MODULE] FACULTY REPORT DATA LOADED
Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-
2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:31:58.052] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS
Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:31:58.052] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:33:09.801] [REPORTING_MODULE] FACULTY REPORT DATA LOADED
Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-
2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:33:09.806] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS
Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:33:09.806] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:33:16.180] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

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Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:33:16.184] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:33:16.184] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:33:41.502] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:33:41.506] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:33:41.506] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:33:47.794] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/app_state.php"}

[21:33:47.798] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

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[21:33:47.799] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:33:47.807] [AI_ANALYSIS_ENGINE] SUMMARY INPUT PREPARED

Prepared_Text: [{"id":"comment_1","text":"The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"}]

[21:33:47.809] [AI_ANALYSIS_ENGINE] INGESTING qualitative feedback for summary generation | Status: STARTED

[21:33:50.997] [AI_ANALYSIS_ENGINE] SUMMARY GENERATED | Source: openai
HTTP_Status: 200

[21:33:50.998] [AI_ANALYSIS_ENGINE] TRIAL SUMMARY RESULT

Source: openai

Output: {"total":1,"constructive":0,"neutral":1,"biased":0,"topics":[{"label":"clear explanations","count":1},{"label":"approachable and helpful","count":1}],"summaryLine":"Summary of 1 student comments: The professor explains lessons clearly and is approachable and helpful.", "toneLine":"The feedback is positive and neutral in tone.", "source":"openai", "warning":"","aiStatus":200,"aiModel":"gpt-5.6-luna"}

[21:33:51.069] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

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Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:33:51.073] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:33:51.073] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:34:17.882] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:34:17.885] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:34:17.886] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:34:21.664] [REPORTING_MODULE] FACULTY REPORT DATA LOADED

Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:34:21.669] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS

Evidence: {"facultyId":10,"evaluationId":4,"included":true}

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[21:34:21.669] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

[21:34:25.342] [REPORTING_MODULE] FACULTY REPORT DATA LOADED
Evidence: {"facultyId":10,"semesterId":"1st-semester-2026-
2027","loadType":"main","evaluationId":4,"included":true,"textLoaded":true,"requestPath":"/system/api/data_accuracy.php"}

[21:34:25.346] [REPORTING_MODULE] TRIAL REPORT CHECK | PASS
Evidence: {"facultyId":10,"evaluationId":4,"included":true}

[21:34:25.346] [VERIFICATION] VALUES CHECKED | Database: PASS |
Report: PASS

=====

[21:34:25.313] [DB_VERIFICATION] CURRENT DATABASE READ-BACK
SQL: SELECT e.evaluatee_user_id, e.evaluator_user_id, e.course_offering_id,
e.general_comments, e.status, e.credibility_status, s.slug AS semester_slug,
co.load_type FROM evaluations e JOIN semesters s ON s.id=e.semester_id
LEFT JOIN course_offerings co ON co.id=e.course_offering_id WHERE e.id=?
SQL: SELECT question_id AS questionId, rating_value AS ratingValue,
text_value AS textValue FROM evaluation_responses WHERE evaluation_id=?
ORDER BY display_order,id
Bound_Evaluation_ID: 4
Retrieved: {"facultyId":10,"courseOfferingId":28,"semesterId":"1st-semester-
2026-

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2027","ratings":{"29":"1.00","30":"5.00","31":"1.00","32":"5.00","33":"1.00","34":"5.00","35":"4.00","36":"3.00","37":"2.00","38":"4.00","39":"3.00","41":"2.00","42":"3.00","43":"4.00","44":"2.00"},"qualitative":{"40":"The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"},"comments":""}

Consistency_Check: PASS (Validated submission compared with current database values)

[21:34:25.313] [REPORTING_MODULE] FACULTY REPORT VERIFICATION

Target_Faculty_ID: 10

Request: POST /api/data_accuracy.php (verification; not a report download)

[21:34:25.313] [DB_QUERY] Existing faculty report input, credibility and enrollment filters

Retrieved_Row_4: {"facultyId":10,"courseOfferingId":28,"semesterId":"1st-semester-2026-

2027","ratings":{"29":"1.00","30":"5.00","31":"1.00","32":"5.00","33":"1.00","34":"5.00","35":"4.00","36":"3.00","37":"2.00","38":"4.00","39":"3.00","41":"2.00","42":"3.00","43":"4.00","44":"2.00"},"qualitative":{"40":"The professor explains the lessons clearly and makes the class easy to understand. They are also approachable and helpful whenever students"},"comments":""}

Status: PASS

[21:34:25.313] [CALCULATION] SINGLE EVALUATION MEAN

Inputs: [1,5,1,5,1,5,4,3,2,4,3,2,3,4,2] -> Output: 3.00

Scope: This evaluation only; faculty overall SET calculation is unchanged.

Data_Integrity_Status: 100% MATCH

Comparison_Scope: Validated submission -> database -> faculty report input

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Verification_Note: Verified against the current database and the evaluation values used by the SET report.

=====

=====

[21:34:25.359] [UI_RENDER] TRANSACTION TRACE DISPLAY

Output_Mean_Calculation: Inputs [1,5,1,5,1,5,4,3,2,4,3,2,4,3,2,3,4,2] -> Output 3.00

Status: Trace text rendered on this trial page.

=====

=====

The results of the Data Consistency and Traceability Audit demonstrate that the system accurately maintained the integrity of evaluation data throughout the 20 tested transactions. The submitted evaluation data successfully passed backend validation, and the transmitted values were verified against the values received and processed by the system. The evaluation records were successfully stored in the database, and database read-back verification confirmed that the stored values matched the validated submissions.

The audit further showed that the stored evaluation data was correctly reflected in the faculty report, with the database and report comparisons producing consistent results. Qualitative feedback was also successfully preserved and processed for the system's AI-assisted feedback analysis. The submitted rating values were correctly used in the system's calculations and corresponding report outputs. Overall, the 20 tested evaluation transactions produced matching

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database records and report outputs, resulting in a 100% consistency rate for the tested sample. These findings indicate that the system was able to maintain accurate and traceable evaluation data from initial submission through database storage, processing, and report generation

Table 17
**MEAN RESULT ON THE SYSTEM COMPARISON OF THE WEB-BASED
FACULTY EVALUATION SYSTEM IN TERMS OF USABILITY**

Usability	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The features of the proposed system are easier to learn and understand compared with the existing evaluation process.	3.82	Highly More Usable than the Existing Process	3.69	Highly More Usable than the Existing Process	3.33	Highly More Usable than the Existing Process
The system interface makes completing evaluation tasks more straightforward than the existing evaluation process.	3.60	Highly More Usable than the Existing Process	3.69	Highly More Usable than the Existing Process	3.67	Highly More Usable than the Existing Process
Users can accomplish evaluation-related tasks with less assistance compared with the existing evaluation process.	3.63	Highly More Usable than the Existing Process	4.00	Highly More Usable than the Existing Process	3.67	Highly More Usable than the Existing Process
AVERAGE WEIGHTED MEAN	3.68	Highly More Usable than the Existing Process	3.79	Highly More Usable than the Existing Process	3.56	Highly More Usable than the Existing Process
GENERAL WEIGHTED MEAN	3.68					
VERBAL INTERPRETATION	Highly More Usable than the Existing Process					

Table 17 presents the mean results of the respondents' assessment of the proposed Web-Based Faculty Evaluation System in comparison with the existing evaluation process in terms of usability. The proposed system obtained a general weighted mean of 3.68, interpreted as Highly More Usable than the Existing

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Process. This indicates that the proposed system provides a more organized and straightforward evaluation environment through its structured interface, role-based access, guided workflows, and centralized evaluation functions.

For the statement that the features of the proposed system are easier to learn and understand compared with the existing evaluation process, Students obtained a mean of 3.82, Faculty 3.69, and HR Personnel 3.33, all interpreted as Highly More Usable than the Existing Process. The highest rating of 3.82 from Students is supported by the system's organized interface, assigned evaluation access, and structured workflow for completing evaluations. The system presents the required evaluation functions within one platform instead of requiring users to work through separate forms or links. The lowest rating of 3.33 from HR Personnel remains highly positive and is related to the wider range of administrative functions available to HR, including evaluation management, tracking, results, reports, and AI-assisted feedback analysis.

For the statement that the system interface makes completing evaluation tasks more straightforward than the existing evaluation process, Students obtained 3.60, Faculty 3.69, and HR Personnel 3.67, all interpreted as Highly More Usable than the Existing Process. The highest rating of 3.69 from Faculty is supported by the system's role-based interface, organized navigation, and structured evaluation functions. These features allow users to access functions relevant to their role within one centralized platform. The lowest rating of 3.60 from Students still

indicates a strong improvement, with the student workflow providing direct access to assigned evaluations, faculty names, evaluation forms, and submission functions.

For the statement that users can accomplish evaluation-related tasks with less assistance compared with the existing evaluation process, Students obtained 3.63, Faculty 4.00, and HR Personnel 3.67, all interpreted as Highly More Usable than the Existing Process. The highest rating of 4.00 from Faculty is supported by the system's structured workflow, role-specific functions, and organized access to evaluation-related activities. These features allow routine tasks to be completed directly within the system without depending heavily on separate instructions or manual procedures. The lowest rating of 3.63 from Students remains Highly More Usable and reflects that the system provides a structured evaluation process, although student activities are already limited to assigned evaluation functions.

The differences among the respondent groups are also reflected in their average weighted means. Students obtained 3.68, Faculty 3.79, and HR Personnel 3.56, all interpreted as Highly More Usable than the Existing Process. Faculty obtained the highest overall rating of 3.79, supported by the organized interface and role-specific functions available for their evaluation-related activities. HR Personnel obtained the lowest overall rating of 3.56, which is related to the broader range of administrative functions they access, including evaluation

management, tracking, results, reporting, and AI-assisted feedback analysis. Despite this difference, the rating remains within the highest interpretation.

The proposed system also improves usability by integrating evaluation activities into a single web-based platform. Instead of relying primarily on Google Forms and separate manual procedures, the system provides centralized access to evaluation schedules, assigned evaluations, submission functions, tracking, results, and related administrative features. This reduces the need to navigate between separate tools and provides a more consistent workflow for different user roles.

In totality, the findings demonstrate that the proposed system provides a more usable evaluation environment through its organized interface, role-based access, structured workflows, centralized functions, and direct access to evaluation-related activities. The consistently high ratings from Students, Faculty, and HR Personnel show that the system improves the usability of the evaluation process compared with the existing approach.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.68, interpreted as Highly More Usable than the Existing Process. The results indicate that the proposed system is easier to learn and understand, provides a more straightforward interface, and allows users to accomplish evaluation-related tasks with less assistance. Compared with the existing Google Forms-based and manually handled process, the proposed

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system provides a more structured, centralized, and role-appropriate environment for conducting and managing faculty evaluations.

Table 18
**MEAN RESULT ON THE SYSTEM COMPARISON OF THE WEB-BASED
FACULTY EVALUATION SYSTEM IN TERMS OF ADMINISTRATIVE
WORKLOAD**

Administrative Workload	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The proposed system reduces the amount of manual work required during faculty evaluation compared with the existing evaluation process.	3.63	Greatly Reduces Administrative Workload	3.92	Greatly Reduces Administrative Workload	4.00	Greatly Reduces Administrative Workload
The proposed system simplifies report preparation and documentation compared with the existing evaluation process.	3.64	Greatly Reduces Administrative Workload	3.69	Greatly Reduces Administrative Workload	3.67	Greatly Reduces Administrative Workload
The proposed system allows administrators to manage evaluation activities more efficiently.	3.65	Greatly Reduces Administrative Workload	3.85	Greatly Reduces Administrative Workload	3.33	Greatly Reduces Administrative Workload
AVERAGE WEIGHTED MEAN	3.64	Greatly Reduces Administrative Workload	3.82	Greatly Reduces Administrative Workload	3.67	Greatly Reduces Administrative Workload
GENERAL WEIGHTED MEAN	3.71					
VERBAL INTERPRETATION	Greatly Reduces Administrative Workload					

Table 18 presents the mean results of the respondents' assessment of the proposed Web-Based Faculty Evaluation System in comparison with the existing evaluation process in terms of administrative workload. The proposed system obtained a general weighted mean of 3.71, interpreted as Greatly Reduces Administrative Workload. This indicates that the proposed system reduces manual

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work, simplifies report preparation and documentation, and provides more organized management of faculty evaluation activities compared with the existing process.

For the statement that the proposed system reduces the amount of manual work required during faculty evaluation compared with the existing evaluation process, Students obtained a mean of 3.63, Faculty 3.92, and HR Personnel 4.00, all interpreted as Greatly Reduces Administrative Workload. The highest rating of 4.00 from HR Personnel is supported by the system's centralized evaluation records, automated processing, evaluation tracking, and structured workflows. These functions reduce the need for repeated manual recording, consolidation, monitoring, and handling of evaluation information. The lowest rating of 3.63 from Students remains highly positive and is related to the limited number of administrative tasks directly performed by students, since their primary system activity is completing assigned evaluations.

For the statement that the proposed system simplifies report preparation and documentation compared with the existing evaluation process, Students obtained 3.64, Faculty 3.69, and HR Personnel 3.67, all interpreted as Greatly Reduces Administrative Workload. The highest rating of 3.69 from Faculty and the close results across all groups are supported by the system's centralized storage, results generation, reporting functions, and organized evaluation records. The proposed system allows evaluation information to be retrieved and organized

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within the platform instead of requiring separate manual records and repeated consolidation. The lowest rating of 3.64 from Students remains within the same interpretation because the improvement in report preparation is more directly connected to the system's administrative and reporting functions.

For the statement that the proposed system allows administrators to manage evaluation activities more efficiently, Students obtained 3.65, Faculty 3.85, and HR Personnel 3.33, all interpreted as Greatly Reduces Administrative Workload. The highest rating of 3.85 from Faculty is supported by the system's centralized evaluation management, evaluation tracking, structured workflows, and organized access to evaluation information. These functions provide a more systematic way to monitor evaluation activities compared with separate forms and manually maintained records. The lowest rating of 3.33 from HR Personnel is still highly positive but is related to the broader administrative functions handled within the system, including evaluation management, tracking, results, reporting, and AI-assisted feedback analysis. These connected functions involve more administrative processes, so the workload reduction is not limited to a single task.

For the average weighted means, Students obtained 3.64, Faculty 3.82, and HR Personnel 3.67, all interpreted as Greatly Reduces Administrative Workload. Faculty obtained the highest overall rating of 3.82, supported by the system's structured evaluation workflow and reduced need for manual handling of evaluation-related information. HR Personnel obtained 3.67, reflecting the

system's direct support for administrative processing, while Students obtained the lowest overall rating of 3.64 because their interaction with administrative functions is more limited. Despite these differences, all groups assessed the system as greatly reducing administrative workload.

The proposed system reduces administrative workload by integrating evaluation management, schedule management, enrollment validation, centralized data storage, evaluation tracking, reporting, and AI-assisted feedback analysis within one platform. These functions reduce repetitive activities such as manually organizing responses, checking evaluation records, monitoring completion, consolidating information, and preparing evaluation results. The system therefore provides a more organized workflow than the existing Google Forms-based and manually handled process.

In totality, the findings demonstrate that the proposed system reduces administrative workload by minimizing repetitive manual work, centralizing evaluation records, simplifying documentation, supporting evaluation tracking, and improving access to organized results. The high ratings across all respondent groups show that the workload reduction is evident across different parts of the evaluation process.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.71, interpreted as Greatly Reduces Administrative Workload. The findings indicate that the proposed system provides a less

burdensome evaluation process by reducing manual work, simplifying report preparation and documentation, and improving the management of evaluation activities. Compared with the existing Google Forms-based and manually handled process, the proposed system provides a more centralized and organized workflow that reduces repetitive administrative tasks throughout the faculty evaluation cycle.

5. To assess the level of user acceptance of the proposed system, focusing on perceived usefulness, ease of use, and overall user experience.

The proposed system was assessed to determine the level of user acceptance based on perceived usefulness, ease of use, and overall user experience. The respondents' assessments were analyzed to determine the acceptability of the Web-Based Faculty Evaluation System among its intended users and stakeholders.

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Table 19
**MEAN RESULT ON TECHNOLOGY ACCEPTANCE AND INSTITUTIONAL
IMPACT OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS
OF PERCEIVED USEFULNESS**

Perceived Usefulness	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The proposed faculty evaluation system improves my efficiency in completing evaluation tasks.	3.67	Highly Useful	3.69	Highly Useful	3.33	Highly Useful
The proposed system enables me to complete evaluation tasks more quickly.	3.62	Highly Useful	3.77	Highly Useful	3.67	Highly Useful
The proposed system improves the effectiveness of the faculty evaluation process.	3.70	Highly Useful	3.15	Useful	3.67	Highly Useful
The proposed system enhances my productivity in performing evaluation-related activities.	3.69	Highly Useful	3.69	Highly Useful	3.33	Highly Useful
Overall, I find the proposed faculty evaluation system useful.	3.72	Highly Useful	3.62	Highly Useful	3.67	Highly Useful
AVERAGE WEIGHTED MEAN	3.68	Highly Useful	3.58	Highly Useful	3.53	Highly Useful
GENERAL WEIGHTED MEAN	3.60					
VERBAL INTERPRETATION	Highly Useful					

Table 19 presents the mean results of the respondents' assessment of the proposed Web-Based Faculty Evaluation System in terms of perceived usefulness. The system obtained a general weighted mean of 3.60, interpreted as Highly Useful. This indicates that the proposed system provides practical support for completing evaluation tasks, improving efficiency, processing activities more quickly, and supporting productivity across the evaluation process.

For the statement that the proposed system improves efficiency in completing evaluation tasks, Students obtained a mean of 3.67, Faculty 3.69, and HR Personnel 3.33, all interpreted as Highly Useful. The highest rating of 3.69 from Faculty is supported by the system's structured evaluation workflow, centralized access, and role-based functions that organize evaluation-related activities within

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one platform. The lowest rating of 3.33 from HR Personnel remains Highly Useful and is related to the wider range of administrative functions handled within the system, including evaluation management, tracking, results, reporting, and AI-assisted feedback analysis. Despite the broader functions involved, the system still provides centralized tools that support efficient completion of these activities.

For the statement that the proposed system enables users to complete evaluation tasks more quickly, Students obtained 3.62, Faculty 3.77, and HR Personnel 3.67, all interpreted as Highly Useful. The highest rating of 3.77 from Faculty is supported by the system's structured evaluation workflow, centralized information, and direct access to required evaluation functions. These features reduce the need to use separate forms or manually locate evaluation-related information. The lowest rating of 3.62 from Students remains Highly Useful because the system provides assigned evaluation access, structured submission, and centralized processing, although students primarily perform the evaluation submission itself rather than the broader processing of evaluation information.

For the statement that the proposed system improves the effectiveness of the faculty evaluation process, Students obtained 3.70, Faculty 3.15, and HR Personnel 3.67. Students and HR Personnel rated the system as Highly Useful, while Faculty rated it as Useful. The highest rating of 3.70 from Students is supported by the system's enrollment validation, assigned faculty evaluation access, structured evaluation workflow, and centralized submission of responses.

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These functions help ensure that evaluations are directed to the appropriate faculty members and are properly recorded within the system. HR Personnel's 3.67 rating is also supported by centralized evaluation management, tracking, results generation, reporting, and AI-assisted feedback analysis.

The lowest rating is the Faculty rating of 3.15 for improving the effectiveness of the faculty evaluation process. This rating remains Useful and is related to the system's role as a tool for collecting, processing, analyzing, and presenting evaluation information rather than independently determining the overall effectiveness or validity of faculty performance assessment. The system provides multi-source evaluation results, AI-assisted feedback analysis, cross-source consistency alerts, credibility scoring, and AI-generated recommendations, but these outputs remain based on the evaluation information submitted through the system. Therefore, the system supports the evaluation process without independently determining the actual performance of a faculty member.

For the statement that the proposed system enhances productivity in performing evaluation-related activities, Students and Faculty both obtained 3.69, while HR Personnel obtained 3.33, all interpreted as Highly Useful. The equal 3.69 ratings from Students and Faculty are supported by the system's structured workflows and centralized access to their respective evaluation functions. The lowest rating of 3.33 from HR Personnel remains Highly Useful and is related to the broader set of administrative functions that must be handled within the system.

Nevertheless, centralized evaluation records, tracking, reporting, dashboards, and AI-assisted analysis provide direct support for administrative productivity.

For the statement that respondents overall find the proposed faculty evaluation system useful, Students obtained 3.72, Faculty 3.62, and HR Personnel 3.67, all interpreted as Highly Useful. The highest rating of 3.72 from Students is supported by the system's direct access to assigned evaluations, organized evaluation forms, structured submission process, and evaluation history. The lowest rating of 3.62 from Faculty is still Highly Useful and reflects the system's broader role in supporting evaluation activities through centralized information, structured workflows, and access to evaluation-related results and feedback.

For the average weighted means, Students obtained 3.68, Faculty 3.58, and HR Personnel 3.53, all interpreted as Highly Useful. Students obtained the highest overall rating, supported by the direct usefulness of the system's assigned evaluation functions and organized submission workflow. HR Personnel obtained the lowest overall rating, but the result remains Highly Useful because the system provides centralized management, tracking, reporting, AI-assisted feedback analysis, and other administrative functions that support the evaluation process.

In totality, the findings show that the proposed system provides practical value through centralized evaluation management, structured workflows, faster processing, evaluation tracking, reporting, AI-assisted feedback analysis, and organized access to evaluation information. The system supports users in

completing their respective evaluation-related activities more efficiently while also providing administrative functions for processing and interpreting evaluation data.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.60, interpreted as Highly Useful. The findings demonstrate that the system improves efficiency, supports faster completion of evaluation tasks, enhances productivity, and contributes to the faculty evaluation process. Although Faculty gave a lower rating of 3.15 for the effectiveness of the evaluation process, this result remains within the Useful interpretation and reflects the system's function as a supporting platform that processes and presents evaluation information rather than independently determining faculty performance. The consistently positive results across all three respondent groups demonstrate the practical usefulness of the proposed system in supporting faculty evaluation activities.

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Table 20
**MEAN RESULT ON TECHNOLOGY ACCEPTANCE AND INSTITUTIONAL
IMPACT OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN TERMS
OF PERCEIVED EASE OF USE**

Perceived Ease of Use	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
Learning to use the proposed system is easy.	3.72	Very Easy to Use	3.69	Very Easy to Use	3.33	Very Easy to Use
The proposed system is easy to navigate.	3.65	Very Easy to Use	3.62	Very Easy to Use	3.33	Very Easy to Use
The interaction with the proposed system is clear and understandable.	3.73	Very Easy to Use	3.69	Very Easy to Use	3.67	Very Easy to Use
It is easy for me to become skillful in using the proposed system.	3.71	Very Easy to Use	3.54	Very Easy to Use	3.33	Very Easy to Use
Overall, I find the proposed faculty evaluation system easy to use.	3.73	Very Easy to Use	3.69	Very Easy to Use	3.67	Very Easy to Use
AVERAGE WEIGHTED MEAN	3.71	Very Easy to Use	3.65	Very Easy to Use	3.47	Very Easy to Use
GENERAL WEIGHTED MEAN	3.61					
VERBAL INTERPRETATION	Very Easy to Use					

Table 20 presents the mean results of the respondents' assessment of the proposed Web-Based Faculty Evaluation System in terms of perceived ease of use. The proposed system obtained a general weighted mean of 3.61, interpreted as Very Easy to Use. This indicates that the system provides an interface and workflow that are easy to learn, navigate, understand, and use for completing faculty evaluation-related activities.

For the statement that learning to use the proposed system is easy, Students obtained a mean of 3.72, Faculty 3.69, and HR Personnel 3.33, all interpreted as Very Easy to Use. The highest rating of 3.72 from Students is supported by the system's structured evaluation workflow, assigned faculty

access, organized evaluation forms, and straightforward submission process. These functions provide students with the specific tools needed to complete their assigned evaluations within one platform. The lowest rating of 3.33 from HR Personnel remains Very Easy to Use and is related to the wider range of functions available to HR, including evaluation management, tracking, results, reporting, and AI-assisted feedback analysis.

For the statement that the proposed system is easy to navigate, Students obtained 3.65, Faculty 3.62, and HR Personnel 3.33, all interpreted as Very Easy to Use. The highest rating of 3.65 from Students is supported by the role-based interface and focused evaluation workflow, which provides direct access to assigned evaluation functions. The lowest rating of 3.33 from HR Personnel is related to the broader administrative interface, where HR accesses multiple functions such as evaluation management, tracking, results, reports, and AI-assisted analysis. Although more functions are available, the rating confirms that these functions remain manageable to navigate.

For the statement that interaction with the proposed system is clear and understandable, Students obtained 3.73, Faculty 3.69, and HR Personnel 3.67, all interpreted as Very Easy to Use. The highest rating of 3.73 from Students is supported by the system's organized interface, clearly structured evaluation workflow, and role-specific functions. Faculty and HR Personnel also provided closely grouped ratings, showing that system interactions remain understandable

across different functions. The small difference between the groups indicates consistent clarity in the system's interface and workflow.

For the statement that it is easy for users to become skillful in using the proposed system, Students obtained 3.71, Faculty 3.54, and HR Personnel 3.33, all interpreted as Very Easy to Use. The highest rating of 3.71 from Students is supported by the focused set of evaluation functions they use, including accessing assigned evaluations, completing evaluation forms, and submitting responses. The lowest rating of 3.33 from HR Personnel is related to the wider range of administrative functions that must be understood, including evaluation management, tracking, reporting, results, and AI-assisted feedback analysis. Despite the difference, the rating remains Very Easy to Use, indicating that the system still provides a manageable environment for developing proficiency.

For the statement that respondents overall find the proposed faculty evaluation system easy to use, Students obtained 3.73, Faculty 3.69, and HR Personnel 3.67, all interpreted as Very Easy to Use. The highest rating of 3.73 from Students is supported by the system's direct and structured evaluation workflow. Faculty and HR Personnel also gave highly positive ratings, showing that the system remains understandable when performing different evaluation-related functions. The lowest rating of 3.67 from HR Personnel is still very high and demonstrates that the broader administrative functions do not prevent the system from being generally easy to use.

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For the average weighted means, Students obtained 3.71, Faculty 3.65, and HR Personnel 3.47, all interpreted as Very Easy to Use. Students obtained the highest overall rating, supported by the focused and structured functions required for completing assigned evaluations. HR Personnel obtained the lowest overall rating, which is related to the larger number of administrative functions available within their role. Nevertheless, the 3.47 rating remains within the Very Easy to Use interpretation, showing that the system remains manageable despite the broader functionality.

In totality, the findings demonstrate that the proposed system provides ease of use through its role-based interface, structured evaluation workflows, organized navigation, centralized functions, and clearly defined evaluation activities. Students, Faculty, and HR Personnel all rated the system within the Very Easy to Use interpretation, showing that the system can support different types of evaluation-related activities without creating substantial difficulty in learning or operation.

Overall, the proposed Web-Based Faculty Evaluation System obtained a general weighted mean of 3.61, interpreted as Very Easy to Use. The results indicate that the system is easy to learn, navigate, understand, and become skillful in using. Compared with the existing evaluation process, the proposed system provides a more structured and centralized environment where users can access

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the functions relevant to their roles and complete faculty evaluation activities with minimal difficulty.

Table 21
MEAN RESULT ON TECHNOLOGY ACCEPTANCE AND INSTITUTIONAL
IMPACT OF THE WEB-BASED FACULTY EVALUATION SYSTEM IN
TERMS OF OVERALL USER EXPERIENCE / USER ACCEPTANCE

Overall User Experience / User Acceptance	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
I am satisfied with my experience using the proposed system.	3.70	Highly Acceptable	3.92	Highly Acceptable	3.33	Highly Acceptable
The interface of the proposed system is user-friendly.	3.68	Highly Acceptable	3.92	Highly Acceptable	4.00	Highly Acceptable
The proposed system performs efficiently during use.	3.65	Highly Acceptable	3.85	Highly Acceptable	3.00	Acceptable
I would recommend the proposed system to other users.	3.69	Highly Acceptable	3.85	Highly Acceptable	3.33	Highly Acceptable
I intend to continue using the proposed system if it is implemented in the institution.	3.72	Highly Acceptable	3.77	Highly Acceptable	4.00	Highly Acceptable
AVERAGE WEIGHTED MEAN	3.69	Highly Acceptable	3.86	Highly Acceptable	3.53	Highly Acceptable
GENERAL WEIGHTED MEAN	3.69					
VERBAL INTERPRETATION	Highly Acceptable					

Table 21 presents the assessment of the respondents regarding the technology acceptance and institutional impact of the proposed Web-Based Faculty Evaluation System in terms of overall user experience and user acceptance. The students obtained an average weighted mean of 3.69, faculty members obtained 3.86, and HR personnel obtained 3.53, all interpreted as Highly Acceptable. The general weighted mean of 3.69 also indicates that the proposed system was Highly Acceptable in terms of overall user experience and user acceptance.

In terms of satisfaction with the experience of using the proposed system, students obtained a mean of 3.70, faculty members 3.92, and HR personnel 3.33, all interpreted as Highly Acceptable. The high ratings are supported by the system's structured evaluation workflows, role-based access, centralized evaluation functions, and organized presentation of evaluation-related information. Faculty members obtained the highest rating, while HR personnel obtained the lowest, although the rating remained Highly Acceptable. The lower HR rating is related to the wider set of functions available to administrative users, including evaluation management, tracking, results, reports, dashboards, and AI-assisted feedback analysis, which require interaction with more system features than the basic evaluation workflow.

For the user-friendliness of the interface, students obtained a mean of 3.68, faculty members 3.92, and HR personnel 4.00, all interpreted as Highly Acceptable. This result is supported by the role-based interface, organized navigation, responsive web design, and structured presentation of functions according to user access. HR personnel obtained the highest possible rating, showing that the administrative interface provides clear access to the functions available to them. This also demonstrates that the lower HR ratings in other indicators are not caused by general difficulty with the interface, but are more closely related to specific system functions such as performance and the broader administrative processes handled through the system.

Regarding system performance during use, students obtained a mean of 3.65 and faculty members 3.85, both interpreted as Highly Acceptable, while HR personnel obtained 3.00, interpreted as Acceptable. The ratings from students and faculty members are supported by the web-based architecture, centralized data management, and structured evaluation workflows used for submitting, processing, tracking, and displaying evaluation information. HR personnel obtained the lowest rating because administrative functions involve larger sets of evaluation records and processes such as tracking, reporting, dashboards, and AI-assisted feedback analysis. These functions place greater processing demands on the system, making performance an area where further optimization can strengthen the experience of administrative users.

In terms of willingness to recommend the proposed system to other users, students obtained a mean of 3.69, faculty members 3.85, and HR personnel 3.33, all interpreted as Highly Acceptable. The positive ratings are supported by the system's centralized evaluation process, structured workflows, automated processing, evaluation tracking, reporting, and AI-assisted feedback analysis. Faculty members obtained the highest rating, while HR personnel obtained the lowest but remained Highly Acceptable. The lower HR rating is consistent with the system's broader administrative functions, where the usefulness of the system depends not only on completing evaluations but also on managing results, monitoring evaluation activities, reviewing reports, and using analytical outputs.

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For the intention to continue using the proposed system if implemented in the institution, students obtained a mean of 3.72, faculty members 3.77, and HR personnel 4.00, all interpreted as Highly Acceptable. The results indicate strong acceptance of continued system use. This is supported by the system's ability to centralize evaluation records, standardize evaluation workflows, provide evaluation tracking, generate reports, and support AI-assisted analysis and recommendations. HR personnel obtained the highest possible rating of 4.00, demonstrating that despite the lower rating for system performance, the system's overall functions provide sufficient institutional value to support continued use.

In terms of the average weighted mean, faculty members obtained the highest overall rating at 3.86, followed by students at 3.69 and HR personnel at 3.53. All three groups were interpreted as Highly Acceptable. The lower HR average is mainly reflected in the system performance rating of 3.00 and the comparatively lower rating for recommendation at 3.33. However, HR personnel gave the system the highest possible ratings for interface user-friendliness and intention to continue using it, both at 4.00. These results show that the lower HR assessment is concentrated on specific aspects of system operation rather than overall acceptance.

Overall, the proposed Web-Based Faculty Evaluation System was assessed as Highly Acceptable in terms of overall user experience and user acceptance, with a general weighted mean of 3.69. The results are supported by

the system's organized interface, role-based access, centralized evaluation management, structured workflows, evaluation tracking, reporting, and AI-assisted feedback analysis. While system performance received a lower rating from HR personnel, the system remained Highly Acceptable overall, and the 4.00 rating for continued use demonstrates strong acceptance of its institutional value. Therefore, the findings indicate that the proposed system provides a satisfactory user experience and is suitable for continued use in supporting the institution's faculty evaluation activities.

6. To identify data-driven improvements and recommendations based on system evaluation results, user feedback, and performance analysis.

The proposed system was evaluated based on its effectiveness in using evaluation data to identify areas for improvement and generate relevant, practical, and actionable recommendations. The assessment focused on the system's ability to present findings clearly, identify recurring issues and performance trends, support planning and decision-making, prioritize areas requiring attention, and provide useful information for continuous improvement in faculty performance and evaluation practices.

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Table 22
**MEAN RESULT ON DATA-DRIVEN IMPROVEMENTS AND
RECOMMENDATIONS OF THE WEB-BASED FACULTY
EVALUATION SYSTEM IN TERMS OF
EFFECTIVENESS**

Effectiveness of Data-Driven Improvements and Recommendations	STUDENTS		FACULTY		HR PERSONNEL	
	MEAN	VI	MEAN	VI	MEAN	VI
The evaluation reports clearly identify areas that require improvement based on the collected evaluation data.	3.71	Highly Effective	3.38	Highly Effective	3.67	Highly Effective
The recommendations generated by the system are relevant to the evaluation results and user feedback.	3.65	Highly Effective	3.54	Highly Effective	4.00	Highly Effective
The system presents evaluation findings in a way that helps users understand the strengths and weaknesses identified during the evaluation process.	3.65	Highly Effective	3.62	Highly Effective	3.33	Highly Effective
The performance reports provide sufficient information to support planning and decision-making for faculty improvement.	3.68	Highly Effective	3.54	Highly Effective	3.33	Highly Effective
The system helps identify recurring issues and performance trends from evaluation data over time.	3.68	Highly Effective	3.54	Highly Effective	3.33	Highly Effective
The recommendations generated by the system are practical and can be used to improve faculty performance and evaluation practices.	3.69	Highly Effective	3.54	Highly Effective	3.67	Highly Effective
The information provided by the system helps prioritize areas that require immediate attention or improvement.	3.68	Highly Effective	3.54	Highly Effective	3.67	Highly Effective
Overall, the system effectively transforms evaluation results into useful information for continuous institutional improvement.	3.70	Highly Effective	3.54	Highly Effective	3.67	Highly Effective
AVERAGE WEIGHTED MEAN	3.68	Highly Effective	3.53	Highly Effective	3.58	Highly Effective
GENERAL WEIGHTED MEAN	3.60					
VERBAL INTERPRETATION	Highly Effective					

Table 22 presents the assessment of the respondents regarding the effectiveness of data-driven improvements and recommendations of the proposed Web-Based Faculty Evaluation System. The students obtained an average weighted mean of 3.68, faculty members obtained 3.53, and HR personnel obtained 3.58, all interpreted as Highly Effective. The general weighted mean of

3.60 also indicates that the proposed system was Highly Effective in transforming evaluation results into useful information for faculty and institutional improvement.

In terms of identifying areas that require improvement based on collected evaluation data, students obtained a mean of 3.71, faculty members 3.38, and HR personnel 3.67, all interpreted as Highly Effective. The high ratings are supported by the system's evaluation reports, consolidated multi-source evaluation results, AI-assisted feedback analysis, and AI-generated recommendations. The system organizes evaluation information from students, peers, and supervisors so that strengths and areas requiring improvement can be identified from the collected results. Faculty members obtained the lowest rating at 3.38, although it remained Highly Effective, indicating that the reports provide useful improvement information while the interpretation of individual performance still depends on the available evaluation data.

Regarding the relevance of the recommendations generated by the system to the evaluation results and user feedback, students obtained a mean of 3.65, faculty members 3.54, and HR personnel 4.00, all interpreted as Highly Effective. HR personnel obtained the highest possible rating because the system's Smart Recommendation Engine generates constructive recommendations based on the available faculty evaluation data, while the AI-assisted feedback analysis identifies themes and patterns from submitted comments. The AI breakdown also provides transparency on how feedback is summarized, allowing the generated

recommendations to be viewed in relation to the analyzed information. The consistently high ratings indicate that the recommendations are relevant to the evaluation results and feedback processed by the system.

For the presentation of evaluation findings in a way that helps users understand faculty strengths and weaknesses, students obtained a mean of 3.65, faculty members 3.62, and HR personnel 3.33, all interpreted as Highly Effective. This result is supported by the system's evaluation reports, dashboards, multi-source results, AI-generated summaries, and AI breakdown. These features organize evaluation findings so that strengths, areas for improvement, recurring themes, and other relevant patterns can be identified more clearly. HR personnel obtained the lowest rating at 3.33, although it remained Highly Effective, indicating that the system presents useful findings while the interpretation of strengths and weaknesses still depends on the amount and quality of available evaluation information.

In terms of whether the performance reports provide sufficient information to support planning and decision-making for faculty improvement, students obtained a mean of 3.68, faculty members 3.54, and HR personnel 3.33, all interpreted as Highly Effective. The system supports this function through consolidated evaluation reports, dashboards, multi-source results, historical evaluation information, AI-assisted feedback analysis, and AI-generated recommendations. The system also provides cross-source consistency alerts and

Evaluation Credibility Scores, which add information for reviewing the consistency and credibility of evaluation results. HR personnel obtained the lowest rating at 3.33 because these outputs are based on submitted evaluation data and do not independently verify the factual accuracy or completeness of every evaluator response. Nevertheless, the rating remains Highly Effective because the available reports and analytical outputs provide substantial information for faculty improvement planning.

Regarding the identification of recurring issues and performance trends over time, students obtained a mean of 3.68, faculty members 3.54, and HR personnel 3.33, all interpreted as Highly Effective. This effectiveness is supported by the system's frequency analysis, historical evaluation data, evaluation-period trends, and dashboards. Frequency analysis identifies recurring words, phrases, issues, and patterns in the collected feedback, while historical results allow performance information to be reviewed across evaluation periods. HR personnel obtained the lowest rating at 3.33 because recurring patterns identified by the system still require interpretation to determine their significance and context. The system identifies and organizes patterns but does not independently determine whether every recurring issue represents a confirmed performance concern.

For the practicality of the recommendations generated by the system for improving faculty performance and evaluation practices, students obtained a mean of 3.69, faculty members 3.54, and HR personnel 3.67, all interpreted as Highly

Effective. The high ratings are supported by the Smart Recommendation Engine, which generates constructive recommendations based on faculty evaluation results, together with AI-assisted thematic analysis, frequency analysis, and AI-generated summaries. These outputs connect identified strengths and areas for improvement with specific recommendations that can guide faculty development. The results demonstrate that the recommendations are perceived as practical because they are generated from the evaluation information processed by the system rather than being presented separately from the evaluation findings.

Regarding whether the information provided by the system helps prioritize areas requiring immediate attention or improvement, students obtained a mean of 3.68, faculty members 3.54, and HR personnel 3.67, all interpreted as Highly Effective. The system supports prioritization through consolidated multi-source results, evaluation reports, dashboards, recurring-pattern analysis, cross-source consistency alerts, and Evaluation Credibility Scores. These features help organize information according to identified performance concerns and provide additional indicators for reviewing which areas require attention. The high ratings indicate that the system effectively transforms collected evaluation data into organized information that can support the identification and prioritization of improvement areas.

For the overall effectiveness of transforming evaluation results into useful information for continuous institutional improvement, students obtained a mean of

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3.70, faculty members 3.54, and HR personnel 3.67, all interpreted as Highly Effective. The result is supported by the complete data-processing flow of the system, from multi-source evaluation collection and validation to centralized storage, AI-assisted analysis, reporting, trend identification, credibility scoring, and recommendation generation. The system therefore does more than store evaluation responses; it processes the collected information into reports, analytical outputs, and recommendations that can support faculty development and institutional improvement. The high ratings across all respondent groups demonstrate that the system effectively connects evaluation results with actionable improvement information.

In totality, the students, faculty members, and HR personnel assessed the data-driven improvements and recommendations as Highly Effective, with average weighted means of 3.68, 3.53, and 3.58, respectively. Students obtained the highest overall rating, followed by HR personnel and faculty members. The differences are concentrated in specific areas, particularly the presentation of findings, performance reports, and identification of recurring trends, where the faculty and HR ratings remained Highly Effective but were comparatively lower. These differences are related to the limits of evaluation-based analysis: the system can organize, analyze, summarize, compare, and generate recommendations from submitted data, but it does not independently verify the factual accuracy, completeness, or significance of every evaluator response. Despite this limitation,

all three groups recognized the system as Highly Effective in converting evaluation information into useful improvement outputs.

Overall, the findings revealed that the proposed Web-Based Faculty Evaluation System was Highly Effective in terms of data-driven improvements and recommendations, with a general weighted mean of 3.60. The system effectively identifies areas for improvement, generates relevant and practical recommendations, presents faculty strengths and weaknesses, provides performance information, identifies recurring issues and trends, and supports prioritization of areas requiring attention. These results are supported by the integration of multi-source evaluation, AI-assisted feedback analysis, AI breakdown, frequency analysis, cross-source consistency alerts, Evaluation Credibility Scores, dashboards, reports, historical evaluation data, and the Smart Recommendation Engine. Taken together, the findings demonstrate that the proposed system effectively transforms collected faculty evaluation data into structured information and recommendations that support continuous faculty and institutional improvement.

CHAPTER 5 CONCLUSION AND RECOMMENDATIONS

This chapter presents the conclusions and recommendations of the study based on the quantitative survey results obtained from the selected respondents. The study focused on the development and evaluation of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines (NAAP), particularly in terms of multi-source evaluation, AI-enhanced feedback analysis, software quality, comparison with the existing process, user acceptance, and data-driven improvements.

The findings of the study provide insights into the respondents' assessment of the system's quality, effectiveness, usefulness, and acceptability. Based on these findings, recommendations are presented to support the further improvement, refinement, and possible implementation of the proposed system within the institution.

Conclusion

The findings indicate that the integration of multi-source faculty evaluation improves the completeness and reliability of faculty performance assessment by combining evaluation inputs from students, peers, and supervisors. The proposed

Web-Based Faculty Evaluation System was assessed as Highly Comprehensive in terms of completeness, with a general weighted mean of 3.63, and Highly Reliable in terms of reliability, with a general weighted mean of 3.55. These results indicate that the system provides a broader and more dependable structure for faculty performance assessment by consolidating evaluation information from multiple sources through a centralized platform.

The findings further show that the AI-Assisted Feedback Analysis effectively supports the processing and interpretation of qualitative faculty evaluation feedback through text and frequency analysis. The system obtained a general weighted mean of 3.42, interpreted as Highly Effective. The results indicate that the AI-assisted features can organize written feedback, identify recurring themes and patterns, generate summaries, detect possible bias, and provide recommendations that support faster interpretation of evaluation results.

The proposed system was also positively assessed using the nine ISO/IEC 25010:2023 software quality characteristics. It obtained general weighted means of 3.47 for functional suitability, 3.34 for reliability, 3.44 for usability, 3.62 for security, 3.42 for performance efficiency, 3.39 for maintainability, 3.47 for compatibility, 3.54 for portability, and 3.66 for accessibility, with all characteristics interpreted as Highly rated. These results indicate that the system provides the required functions, supports reliable and usable operations, protects evaluation information, performs efficiently, and can be maintained, accessed, and used

across different environments. Nevertheless, the comparatively lower assessment of fault tolerance under reliability indicates the need for further testing and improvement during connection problems, software errors, or unexpected interruptions.

The comparison with the existing faculty evaluation process demonstrates that the proposed system provides improvements in the four areas identified in the objective. It obtained a general weighted mean of 3.74 for operational efficiency, interpreted as Highly Efficient Compared with the Existing Process; 3.55 for data accuracy, interpreted as Highly Accurate; 3.68 for usability, interpreted as Highly More Usable than the Existing Process; and 3.71 for administrative workload reduction. These results indicate that the centralized and automated workflow improves the organization and speed of evaluation activities, supports more accurate data processing, provides a more straightforward evaluation experience, and reduces manual administrative work compared with the existing process.

The findings also establish that the proposed system has a high level of technology acceptance and institutional impact. The system was assessed as Highly Useful, with a general weighted mean of 3.60, Very Easy to Use, with 3.61, and Highly Acceptable in terms of overall user experience and acceptance, with 3.69. These results indicate that the system helps users complete evaluation-related activities, is easy to learn and navigate, and provides a satisfactory user experience. The positive ratings regarding user-friendliness, willingness to

recommend the system, and intention to continue using it further indicate that the proposed system is acceptable for continued use in supporting faculty evaluation activities.

Lastly, the findings demonstrate that the proposed system effectively supports data-driven improvements and recommendations based on faculty evaluation results, feedback, and performance analysis. The system obtained a general weighted mean of 3.60, interpreted as Highly Effective. The results indicate that the system can identify areas for improvement, present faculty strengths and weaknesses, identify recurring issues and performance trends, generate relevant recommendations, and help prioritize areas requiring attention.

Recommendations

1. Continue refining the multi-source faculty evaluation process.

The institution may continue improving the integration of student, peer, and supervisor evaluations to maintain the completeness and reliability of faculty performance assessment. Future improvements may focus on ensuring that evaluation inputs from each source are complete, consistent, and properly represented in the consolidated results.

2. Further improve the AI-assisted feedback analysis.

The AI-assisted feedback analysis may be further refined to improve the identification of recurring themes, patterns, and meaningful concerns in qualitative feedback. Since the AI processes available evaluation data and does not independently verify the accuracy or complete context of every response, authorized personnel should continue reviewing AI-generated summaries, insights, and recommendations before using them for important faculty-related decisions.

3. Strengthen reliability and fault-tolerance features.

Future development should prioritize the reliability areas that received comparatively lower assessments, particularly fault tolerance. Additional testing may be conducted for connection interruptions, software errors, unexpected system failures, and recovery procedures to ensure that evaluation data and ongoing transactions remain protected during system disruptions.

4. Maintain continuous ISO/IEC 25010:2023-based evaluation and system maintenance.

The institution may regularly evaluate and maintain the system using the nine ISO/IEC 25010:2023 software quality characteristics. Continuous testing and monitoring can help preserve the system's functional suitability, reliability,

usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility as institutional requirements and technologies change.

5. Continue improving administrative efficiency and system usability.

The institution may continue optimizing the centralized evaluation workflow to further reduce manual administrative tasks and improve processing efficiency. Particular attention may be given to administrative functions involving evaluation management, report generation, monitoring, and data processing to ensure that the system remains efficient for users handling more complex evaluation activities.

6. Continue protecting evaluation data and user privacy.

Future implementation should maintain strong data privacy, confidentiality, authentication, and role-based access controls. Access to evaluation results and administrative information should remain limited according to each user's responsibilities to protect students, faculty members, and institutional evaluation records.

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APPENDICES

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APPENDIX A
Approval of Adviser

Letter of Approval for Asst. Prof. Aron J. Alojado, MSIT as Adviser on March 3, 2026

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly: Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

CAPSTONE ADVISER, STATISTICIAN & GRAMMARIAN APPROVAL AND ACCEPTANCE SHEET

ACADEMIC PROGRAM: Bachelor of Science in Aviation Information System
ACADEMIC YEAR: 2025-2026
SEMESTER: 2nd Semester

TITLE:

Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines

RESEARCHER (s):

VALLIDO, LORENZA LANE S., ROJO, JOHN EMMANUEL E., DE ASIS, RHICKA MAE P.,

I have read the attached Research Proposal of the proponent and I accept my appointment as:

- ☒ **ADVISER**
☐ **STATISTICIAN**
☐ **GRAMMARIAN**

Furthermore, I agree to set the schedule for the advising and consultation to help the students and ensure the success of the project.

The Research Proposal is

- ☒ **Approved**
☐ **Conditionally approved, subject to the following amendments**

Confirmed:

ASST. PROF. ARON J. ALOJADO, MSIT

Signature over Printed Name of Adviser/Statistician/Grammarian
Date: 03/03/2026

Noted:

ALVIN JOHN M. PAZ, DIT

Signature over Printed Name of Academic Adviser
Date: 03/03/2026



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APPENDIX B
Approval of Statistician

Letter of Approval for Ms. Jo-Ann C. Garcia, MATM as Statistician on March 3, 2026

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly: Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

CAPSTONE ADVISER, STATISTICIAN & GRAMMARIAN APPROVAL AND ACCEPTANCE SHEET

ACADEMIC PROGRAM: Bachelor of Science in Aviation Information System
ACADEMIC YEAR: 2025-2026
SEMESTER: 2nd Semester

TITLE:
Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines

RESEARCHER (s):
VALLIDO, LORENZA LANE S., ROJO, JOHN EMMANUEL E., DE ASIS, RHICKA MAE P.

I have read the attached Research Proposal of the proponent and I accept my appointment as:

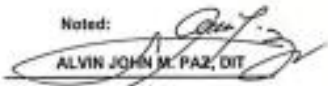
☐ ADVISER
☒ STATISTICIAN
☐ GRAMMARIAN

Furthermore, I agree to set the schedule for the advising and consultation to help the students and ensure the success of the project.

The Research Proposal is

☒ Approved
☐ Conditionally approved, subject to the following amendments

Conformed: 
JO-ANN C. GARCIA, MATM
Signature over Printed Name of Adviser/Statistician/Grammarian
Date: 03/03/2026

Noted: 
ALVIN JOHN M. PAZ, DIT
Signature over Printed Name of Academic Adviser
Date: 03/03/2026

       
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APPENDIX C
Approval of Grammarian

Letter of Approval for Ms. Gladelyn P. Almeda, MAED as Grammarian on March 3, 2026

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NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
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CAPSTONE ADVISER, STATISTICIAN & GRAMMARIAN APPROVAL AND ACCEPTANCE SHEET

ACADEMIC PROGRAM: Bachelor of Science in Aviation Information System
ACADEMIC YEAR: 2025-2026
SEMESTER: 2nd Semester

TITLE:

Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines

RESEARCHER (s):

VALLIDO, LORENZA LANE S., ROJO, JOHN EMMANUEL E., DE ASIS, RHICKA MAE P.

I have read the attached Research Proposal of the proponent and I accept my appointment as:

- ☐ ADVISER
☐ STATISTICIAN
☒ GRAMMARIAN

Furthermore, I agree to set the schedule for the editing and consultation to help the students and ensure the success of the project.

The Research Proposal is

- ☒ Approved
☐ Conditionally approved, subject to the following amendments

Conformed:
GLADELYN P. ALMEDA, MAED

Signature over Printed Name of Adviser/Statistician/Grammarian
Date: 03/03/2026

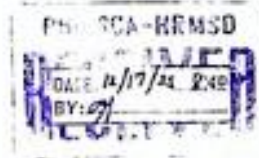
Noted:
ALVIN JOHN M. PAZ, DIT
Signature over Printed Name of Academic Adviser
Date: 03/03/2026



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APPENDIX D
Client Approval

December 16, 2025



DR. MA. TERESA S. ROSALES
Director of Human Resources Management
National Aviation Academy of the Philippines
Piccolo Garden, Villamor, Pasay City

Subject: Request for Permission to Conduct Capstone Research Interview

Dear Dr. Rosales,

Good day.

We are third-year students of Aviation Information Systems currently undertaking our capstone research project under the supervision of Mr. Alvin John M. Paz, DIT. In connection with this, we respectfully seek permission from your esteemed office to conduct interviews and gather data at the National Aviation Academy of the Philippines as part of our study.

Our research aims to develop and enhance a *"Student Evaluation System"* tailored for the faculty of your institution. We believe that this project will contribute significantly to improving the faculty evaluation process and supporting the continuous development of the Academy.

Please be assured that all information collected will be used solely for academic purposes and will be handled with strict confidentiality in accordance with the *Data Privacy Act of 2012 (Republic Act No. 10173)*. Furthermore, we commit that the research activities will be scheduled to avoid any disruption to the Academy's daily operations.

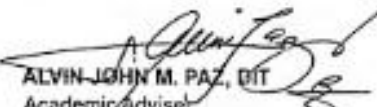
We hope for your favorable consideration of this request. Should there be any additional requirements or clarifications needed, we are readily available to comply.

Thank you very much for your time and kind attention.

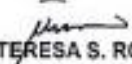
Respectfully yours,

 VALLIDO, LORENZA LANE S.	Researcher:  ROJO, JOHN EMMANUEL E.	 DE ASIS, RHICKA MAE P.
--	---	--

Noted by:


ALVIN JOHN M. PAZ, DIT
Academic Adviser

Approved / Disapproved


DR. MA. TERESA S. ROSALES
Director, Human Resources Management Services Division

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX E

Request Letter for the Number of HR personnels in Villamor Campus

August 27, 2020

NOEL E. LAGUI, MEAM
*Asst. Prof. IV / Coordinator, HRMSD
National Aviation Academy of the Philippines*

Subject: Request for the Number of HR Personnels

Dear Mr. Lagui,

Good day!

We are fourth-year Bachelor of Science in Aviation Information System (BSAIS) students currently enrolled in our Capstone Project at the National Aviation Academy of the Philippines. Our proposed project is titled "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines." In this regard, we respectfully request information regarding the current number of personnel in the Human Resource Management Office. The requested information will be used to support our academic research and the development of our proposed system.

Rest assured that the information provided will be used strictly for academic and research purposes only and will be treated with confidentiality. Proper acknowledgment will also be given to the Human Resource Management Office in our project documentation.

We sincerely hope for your kind consideration and approval of this request. Thank you very much for your time and consideration. God Bless!

Respectfully yours,

Researchers


De Asis, Rhicks Mae P.


Rojo, John Emmanuel E.


Valido, Lorenza Lane S.

Noted by:


ASST. PROF. ARON-J. ALOJADO
Research Adviser


ASST. PROF. DARWIN M. CATALAN
Capstone Subject Adviser

Approved/Disapproved:


NOEL E. LAGUI, MEAM
Asst. Prof. IV / Coordinator, HRMSD

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(formerly Philippine State College of Aeronautics)

1 September 2026

Subject: REQUEST OF TOTAL NUMBER OF HRMSD PERSONNEL

This is in reference to your request for the total number of personnel under the Human Resource Management and Development Section (HRMSD) of the National Aviation Academy of the Philippines (NAAP).

Upon review and verification of the available records of this Office, your request is hereby **APPROVED**. Accordingly, the total number of HRMSD personnel as of this date is 11 personnel, consisting of 8 permanent/regular personnel and 3 contract of service personnel.

The requested information may be utilized solely for Capstone Project titled: "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of Philippines", and shall be handled in accordance with existing policies on the proper use and confidentiality of personnel information.

For your information and appropriate action.

Very truly yours,



MAJORICA A. SISON
Admin Asst. II (HRMSD)

Noted:



NOEL E. LAGUI, MEAM
Asst. Prof. IV / Facilitator, HRMSD



www.philna.edu.ph; tr.alfonso@gmail.com
2nd Floor, Building A, Florio Garden, Villamor, Pasay City



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APPENDIX F
Request Letter to Conduct Data Gathering on HR personnels in Villamor Campus

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
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INSTITUTE OF COMPUTER STUDIES

September 02, 2026

ASST. PROF. NOEL E. LAGUI
Facilitator, HMMSD
National Aviation Academy of the Philippines

Subject: Request for Permission to Conduct Data Gathering

Dear Asst. Prof. Lagui,

Good day!

We are fourth-year Bachelor of Science in Aviation Information System (BSAIS) students currently enrolled in our Capstone Project at the National Aviation Academy of the Philippines. Our proposed project is titled "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines." In this regard, we respectfully request permission to conduct a survey among the Human Resources Office Personnel. The requested information will be used to support our academic research and the development of our proposed system and will be kept confidential in accordance with the Republic Act No. 10173, or the Data Privacy Act of 2012.

We sincerely hope for your kind consideration and approval of this request. Thank you very much for your time and consideration. God bless!

Respectfully yours,

Researchers:


De Asis, Rhicka Mae P.


Rojo, John Emmanuel E.


Vallido, Lorenza Lane S.

Approved by:


ASST. PROF. NOEL E. LAGUI
Facilitator, HMMSD



Ground Floor, Building A, Pico Garden, Villamor, Pasay City



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX G
Number of Students enrolled in Villamor Campus

 Republic of the Philippines Philippine State College Of Aeronautics Pasay Campus Piccio Garden, Villamor, Pasay City  ENROLLMENT STATISTICS BY YEAR LEVEL Academic Year 2026-2027 First Semester OFFICIALLY ENROLLED STUDENTS						
	1st	2nd	3rd	4th	GRADUATES	TOTAL
Institute of Computer Studies						
Bachelor of Science in Aviation Information System	-	-	-	78	-	78
Bachelor of Science in Aviation Information Technology	-	-	1	81	-	82
Bachelor of Science in Information Technology with Specialization in Aviation Information Technology	108	67	56	-	-	231
Bachelor of Science in Information System with Specialization in Aviation Information System	108	73	66	-	-	247
College Total	216	140	123	159	-	638
Institute of Engineering and Technology						
Bachelor of Science in Aeronautical Engineering	129	96	81	85	-	391
Bachelor of Science in Air Transportation Major in Advanced Flying	18	18	11	14	-	61
Bachelor of Science in Aircraft Maintenance Technology	-	-	249	277	-	526
Bachelor of Science in Aviation Electronics Technology	-	-	94	67	-	161
Associate in Aircraft Maintenance Technology	308	267	-	-	-	575
Associate in Aviation Electronics Technology	127	109	-	-	-	236
College Total	582	490	435	443	-	1950
Institute of Liberal Arts and Sciences						
Bachelor of Science in Aviation Communication Major in Flights Operation	99	73	59	63	-	294
Bachelor of Science in Aviation Safety and Security Management	88	60	53	51	-	252
Bachelor of Science in Aviation Tourism Major in Travel Management	129	88	66	107	-	390
Bachelor of Science in Aviation Logistics	-	-	-	1	-	1
Bachelor of Science in Supply Chain Management with Specialization in Aviation Logistics	76	71	58	52	-	257
College Total	392	292	236	274	-	1194
Grand Total	1190	922	794	876	-	3782

Printed By: WONG, MIKKO LOROUAN | July 20, 2026 | 00:00:11 PM


MIKKO L. WONG
 COLLEGE REGISTRAR

Page 1 of 1

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX G
Total Number of Teaching Personnel in Villamor Campus

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(formerly Philippine State College of Aeronautics)
HUMAN RESOURCE MANAGEMENT SERVICES DIVISION

TOTAL NUMBER OF TEACHING PERSONNEL IN VILLAMOR CAMPUS
as of July 20, 2025

POSITIONS	TOTAL
Permanent & Temporary Appointment	100
Full-Time Instructor (Contract of Service)	48
Part-Time Instructor (Contract of Service)	16
GRAND TOTAL	164

Prepared by:


MAJORICA A. SISON
Admin. Asst. II, HRMSD

Certified correct:


NOEL E. LAGUI, MEAM
Asst. Prof. IV / Coordinator, HRMSD

Page 1 of 1



2nd Floor, Building 8, Paoay Garden, Villamor, Pasay City

INSTITUTE OF COMPUTER STUDIES

GreenMetric
World University Rankings

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX G
Picture of Conducting a Survey



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX H
Request Letter for Validation of Survey Questionnaires

Letter of Approval for Mr. Louell C. Cabanella, MIT, CSSP as Validator on July 20, 2026

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

July 20, 2026

FOR: MR. LOUELL C. CABANELA, MIT, CSSP
Director, MIS Office
National Aviation Academy of the Philippines – Management Information System

Dear Sir,

We hope this letter finds you well. We kindly request your assistance in reviewing and validating a survey questionnaire for our capstone project titled "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines." The survey is designed to gather feedback essential for refining the system's features and ensuring it provides effective and satisfactory service to its beneficiaries, in line with ISO 25010 system development standards.

Your expertise and experience in this field would be invaluable in reviewing the questionnaire to ensure its relevance, clarity, and effectiveness. We believe that your input will help us refine the survey, making it more accurate and useful for our intended participants.

Thank you for your time and consideration.

Sincerely yours,


DE ASIS, RHICKA MAE P.
rhuckadeasis@gmail.com


VALLIDO, LORENZA LANE S.
validolorenzane@gmail.com


ROJO, JOHN EMMANUEL E.
john.rojo.philca@gmail.com

Noted by:


ASST. PROF. ARON J. ALOJADO
Capstone Project Adviser


ASST. PROF. DARWIN M. CATALAN
Academic Adviser

Approved by:

MR. LOUELL C. CABANELA, MIT, CSSP
Director, MIS Office
National Aviation Academy of the Philippines – Management Information System



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX I
Survey Instrument Validation Rating Scale

Survey Instrument Validation Rating Scale for Mr. Louell C. Cabanella, MIT,
 CSSP

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
 (Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES
SURVEY INSTRUMENT VALIDATION RATING SCALE

Instruction: Please indicate your degree of agreement or disagreement with the statements provided below by checking the box that corresponds to your best judgment.

Capstone Title: **WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES.**

Researchers:

De Asis, Rhicka Mae P.

Valido, Lorenza Lane S.

Rojas, John Emmanuel E.

Scale:

- 1 – Strongly Disagree
- 2 – Disagree
- 3 – Agree
- 4 – Strongly Agree

	Criteria	4	3	2	1
1	The items in the instrument are relevant to answer the objectives of the study.		/		
2	The items in the instrument can provide depth to the constructs being measured.	/			
3	The instrument has an appropriate sample of items for the construct being measured.				
4	The items and their alternatives are neither too narrow nor limited in their content.		/		
5	The items in the instrument are stated clearly.	/			
6	The items in the instrument can elicit responses that are stable, definite, consistent, and not conflicting.	/			
7	The items adapted in the scale are culturally appropriate.	/			
8	The layout or format of the instrument is technically appropriate.	/			
9	The responses on the scale show a reasonable range of variation.	/			



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
 (Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

10	The instrument is not too short or long enough that the participants will be able to answer it within a given time.	/			
11	The instrument is interesting such that participants will be induced to respond to it and accomplish it fully.		/		

Comments/Suggestions/Recommendations

MAKE A DATA VALIDATION OF HQ OUTPUT VS YOUR
 DATA OUTPUT FOR RELIABILITY OF THE SYSTEM
 & INCORPORATE THE TERMS USED IN RESEARCH INSTRUMENT
 BASED ON RESEARCH OBJECTIVES FOR THE
 LIKERT SCALE


 Signature over printed name (SQ validator)



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX J
Certificate of Validation

Certificate of Validation for the survey questionnaires from Mr. Louell C. Cabanella, MIT, CSSP

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

CERTIFICATE OF VALIDATION

This is to certify that the ISO 25010 survey questionnaire of the study titled "WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES." has undergone a thorough validation process to ensure its reliability and effectiveness in measuring the intended constructs.

Certified by:


MR. LOUELL C. CABANELA, MIT, CSSP
Director, MIS Office
National Aviation Academy of the Philippines – Management Information System



Ground Floor, Building A, Pocio Gardens, Villamor, Pasig City



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX K
Request Letter for Validation of Survey Questionnaires

Letter of Approval for Dr. Alan Lino Silvero J. Agustin as Validator on July
20,2026

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

July 20, 2026

FOR: DR. ALAN LINO SILVERO J. AGUSTIN
Assistant Professor IV
National Aviation Academy of the Philippines – Institute of Computer Studies

Dear Sir,

We hope this letter finds you well. We kindly request your assistance in reviewing and validating a survey questionnaire for our capstone project titled **"Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines."** The survey is designed to gather feedback essential for refining the system's features and ensuring it provides effective and satisfactory service to its beneficiaries, in line with ISO 25010 system development standards.

Your expertise and experience in this field would be invaluable in reviewing the questionnaire to ensure its relevance, clarity, and effectiveness. We believe that your input will help us refine the survey, making it more accurate and useful for our intended participants.

Thank you for your time and consideration.

Sincerely yours,


DE ASIS, RHICKA MAE P.
rhuckadeasis@gmail.com


VALLIDO, LORENZA LANE S.
validolorenzalanegmail.com


ROJO, JOHN EMMANUEL E.
john.rojo.philsca@gmail.com

Noted by:


ASST. PROF. ARON J. ALOJADO
Capstone Project Adviser


ASST. PROF. DARWIN K. CATALAN
Academic Adviser

Approved by:

DR. ALAN LINO SILVERO J. AGUSTIN
Assistant Professor IV
National Aviation Academy of the Philippines – Institute of Computer Studies



Grand Floor, Building A, Prince Garden, Villamor, Pasig City



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
 (Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

10	The instrument is not too short or long enough that the participants will be able to answer it within a given time.	☒	☐	☐	☐
11	The instrument is interesting such that participants will be induced to respond to it and accomplish it fully.	☒	☐	☐	☐

Comments/Suggestions/Recommendations

Alan Lino Silvestre J. Argueta, Dean
 Signature over printed name (SQ validator)



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX M
Certificate of Validation

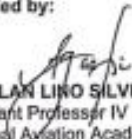
Certificate of Validation for the survey questionnaires from Dr. Alan Lino Silvero
J. Agustin

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

CERTIFICATE OF VALIDATION

This is to certify that the ISO 25010 survey questionnaire of the study titled "WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES." has undergone a thorough validation process to ensure its reliability and effectiveness in measuring the intended constructs.

Certified by:


DR. ALAN LINO SILVERO J. AGUSTIN
Assistant Professor IV
National Aviation Academy of the Philippines – Institute of Computer Studies



Ground Floor, Building A, Pico Garden, Villamor, Pasig City

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INSTITUTION
FOR AVIATION



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The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX N
Request Letter for Validation of Survey Questionnaires

Letter of Approval for Ms. Ruth Jenneth L. Cacho as Validator on July 20, 2026

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

July 20, 2026

FOR: **MS. RUTH JENNETH L. CACHO**
COS Faculty
National Aviation Academy of the Philippines – Institute of Computer Studies

Dear Sir,

We hope this letter finds you well. We kindly request your assistance in reviewing and validating a survey questionnaire for our capstone project titled "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines." The survey is designed to gather feedback essential for refining the system's features and ensuring it provides effective and satisfactory service to its beneficiaries, in line with ISO 25010 system development standards.

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Thank you for your time and consideration.

Sincerely yours,


DE ASIS, RHICKA MAE P.
rhuckadeasis@gmail.com

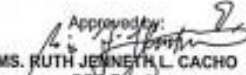

VALLIDO, LORENZA LANE S.
validolorenzalanegmail@gmail.com


ROJO, JOHN EMMANUEL E.
john.rojo.philscas@gmail.com

Noted by:


ASST. PROF. ARON J. ALOJADO
Capstone Project Adviser


ASST. PROF. DARWIN M. CATALAN
Academic Adviser

Approved by:

MS. RUTH JENNETH L. CACHO
COS Faculty
National Aviation Academy of the Philippines – Institute of Computer Studies



Ground floor, Building A, Rizal Garden, Villamor, Pasay City



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX O
Survey Instrument Validation Rating Scale

Survey Instrument Validation Rating Scale for Ms. Ruth Jenneth L. Cacho

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
 (Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

SURVEY INSTRUMENT VALIDATION RATING SCALE

Instruction: Please indicate your degree of agreement or disagreement with the statements provided below by checking the box that corresponds to your best judgment.

Capstone Title: **WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES.**

Researchers:

De Asis, Rhicka Mae P.

Vallido, Lorenza Lane S.

Rojo, John Emmanuel E.

Scale:

- 1 – Strongly Disagree
- 2 – Disagree
- 3 – Agree
- 4 – Strongly Agree

	Criteria	4	3	2	1
1	The items in the instrument are relevant to answer the objectives of the study.		/		
2	The items in the instrument can provide depth to the constructs being measured.		/		
3	The instrument has an appropriate sample of items for the construct being measured.		/		
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7	The items adapted in the scale are culturally appropriate.	/			
8	The layout or format of the instrument is technically appropriate.	/			
9	The responses on the scale show a reasonable range of variation.	/			



Ground Floor, Building A, Picnic Garden, Villamor, Pasay City

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 ACCREDITED
 QUALITY ASSURED



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The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

10	The instrument is not too short or long enough that the participants will be able to answer it within a given time.	☒	☐	☐	☐
11	The instrument is interesting such that participants will be induced to respond to it and accomplish it fully.	☒	☐	☐	☐

Comments/Suggestions/Recommendations


Signature over printed name (SQ validator)



Ground Floor, Building A, Picco Garden, Villamor, Pasay City



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX P
Certificate of Validation

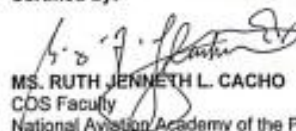
Certificate of Validation for the survey questionnaires from Ms. Ruth Jenneth L. Cacho

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

CERTIFICATE OF VALIDATION

This is to certify that the ISO 25010 survey questionnaire of the study titled "WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES." has undergone a thorough validation process to ensure its reliability and effectiveness in measuring the intended constructs.

Certified by:


MS. RUTH JENNETH L. CACHO
COS Faculty

National Aviation Academy of the Philippines – Institute of Computer Studies



Ground Floor, Building A, Pico Garden, Villamor, Pasay City



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX Q
Request Letter for Validation of Survey Questionnaires

Letter of Approval for Mr. Richard Niel B. Peralta, MSIT as Validator on July
20,2026

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

July 20, 2026

FOR: MR. RICHARD NIEL B. PERALTA, MSIT
Program Coordinator, BSIT-AIT
National Aviation Academy of the Philippines – Institute of Computer Studies

Dear Sir,

We hope this letter finds you well. We kindly request your assistance in reviewing and validating a survey questionnaire for our capstone project titled "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines." The survey is designed to gather feedback essential for refining the system's features and ensuring it provides effective and satisfactory service to its beneficiaries, in line with ISO 25010 system development standards.

Your expertise and experience in this field would be invaluable in reviewing the questionnaire to ensure its relevance, clarity, and effectiveness. We believe that your input will help us refine the survey, making it more accurate and useful for our intended participants.

Thank you for your time and consideration.

Sincerely yours,

DE ASIS, RHICKA MAE P.
rhicka.deasis@gmail.com

ROJO, JOHN EMMANUEL E.
john.rojo.philca@gmail.com

VALLIDO, LORENZA LANE S.
validolorenzalaneg@gmail.com

Noted by:

ASST. PROF. ARON J. ALOJADO
Capstone Project Adviser 1

ASST. PROF. DARWIN M. CATALAN
Academic Adviser

MR. RICHARD NIEL B. PERALTA, MSIT
Program Coordinator, BSIT-AIT
National Aviation Academy of the Philippines – Institute of Computer Studies



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX R
Survey Instrument Validation Rating Scale

Survey Instrument Validation Rating Scale for Mr. Richard Niel B. Peralta, MSIT

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
 (Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

SURVEY INSTRUMENT VALIDATION RATING SCALE

Instruction: Please indicate your degree of agreement or disagreement with the statements provided below by checking the box that corresponds to your best judgment.

Capstone Title: **WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES.**

Researchers:

De Asis, Rhicka Mae P.

Valido, Lorenza Lane S.

Rojo, John Emmanuel E.

Scale:

- 1 – Strongly Disagree
- 2 – Disagree
- 3 – Agree
- 4 – Strongly Agree

	Criteria	4	3	2	1
1	The items in the instrument are relevant to answer the objectives of the study.	☒	☐	☐	☐
2	The items in the instrument can provide depth to the constructs being measured.	☒	☐	☐	☐
3	The instrument has an appropriate sample of items for the construct being measured.	☒	☐	☐	☐
4	The items and their alternatives are neither too narrow nor limited in their content.	☒	☐	☐	☐
5	The items in the instrument are stated clearly.	☒	☐	☐	☐
6	The items in the instrument can elicit responses that are stable, definite, consistent, and not conflicting.	☒	☐	☐	☐
7	The items adapted in the scale are culturally appropriate.	☒	☐	☐	☐
8	The layout or format of the instrument is technically appropriate.	☒	☐	☐	☐
9	The responses on the scale show a reasonable range of variation.	☒	☐	☐	☐

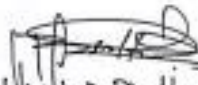


Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
 (Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

10	The instrument is not too short or long enough that the participants will be able to answer it within a given time.	/				
11	The instrument is interesting such that participants will be induced to respond to it and accomplish it fully.	/				

Comments/Suggestions/Recommendations


 Richard Noel B. Penilla
 Signature over printed name (SQ validator)



www.naa.gov.ph



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX S
Certificate of Validation

Certificate of Validation for the survey questionnaires from Mr. Richard Niel B. Peralta, MSIT

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

CERTIFICATE OF VALIDATION

This is to certify that the ISO 25010 survey questionnaire of the study titled "WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES." has undergone a thorough validation process to ensure its reliability and effectiveness in measuring the intended constructs.

Certified by:



MR. RICHARD NIEL B. PERALTA, MSIT
Program Coordinator, BSIT-AIT
National Aviation Academy of the Philippines – Institute of Computer Studies



Ground Floor, Building A, Pizzo Garden, Villamor, Pasig City



Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

APPENDIX T
Validated Survey Questionnaire

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
(Formerly Philippine State College of Aeronautics)
INSTITUTE OF COMPUTER STUDIES

SURVEY QUESTIONNAIRE

**WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED
FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES.**

To our valued participant,

We, the researchers, Bachelor of Science in Information System 4-2 students, are conducting a study for our capstone project titled "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines." Your participation in this survey is greatly appreciated, as it will provide essential insights to help us evaluate and improve our system.

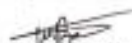
In accordance with the Data Privacy Act of 2012 (RA 10173) of the Philippines, all information you provide will be treated with strict confidentiality and used solely for research purposes. Your responses will remain anonymous, and no personal details will be shared or disclosed.

Your input is crucial to the success of our project, and we are deeply thankful for your valuable contribution. God bless!

Researchers,


De Asis, Rhicka Mae P.


Vallido, Lorenza Lane S.


Rojo, John Emmanuel E.

Noted by:


Asst. Prof. Aron J. Alojado
Adviser


Asst. Prof. Darwin M. Cabalan
Academic Adviser



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Name(Optional): _____

PART 1: Respondent Profile

1. Role:

- ☐ Student
- ☐ Faculty
- ☐ Supervision/Dean
- ☐ IT Practitioner
- ☐ HR Personnel

2. Number of times you used the system this term:

- ☐ 1-2
- ☐ 3-5
- ☐ More than 5

PART 2:

1. **Multi-Source Evaluation Reliability** - This section assesses how the integration of student, peer, and supervisor evaluations influences the reliability and completeness of faculty performance assessment.

Rating Scale

Scale	Verbal Interpretation
4	Highly Comprehensive
3	Comprehensive
2	Less Comprehensive
1	Not Comprehensive

1.1 Completeness of Faculty Performance Assessment		4	3	2	1
1.1.1	The proposed system collects feedback from all relevant evaluator groups (students, peers, and supervisors).				
1.1.2	Feedback from students, peers, and supervisors provides a more complete assessment of faculty performance.				
1.1.3	Each evaluator group contributes valuable information that cannot be obtained from only one source.				



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1.1.4	The integrated evaluation captures different aspects of faculty performance that cannot be obtained through a single-source evaluation.				
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Rating Scale

Scale	Verbal Interpretation
4	Highly Reliable
3	Reliable
2	Less Reliable
1	Not Reliable

1.2 Reliability of Faculty Performance Assessment		4	3	2	1
1.2.1	Combining feedback from students, peers, and supervisors improves the reliability of faculty performance assessment.				
1.2.2	The system reduces bias by considering evaluations from different evaluator groups.				
1.2.3	The evaluation results are more dependable than evaluations based on a single evaluator.				
1.2.4	The integrated evaluation accurately reflects faculty performance.				
1.2.5	The multi-source evaluation process increases confidence in the final evaluation results.				
1.2.6	The evaluation results provide sufficient information for administrative decision-making.				



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PART 3:

- 2. AI-Assisted Feedback Analysis** - To assess the effectiveness of AI-assisted qualitative feedback summarization and analysis.

Rating Scale

Scale	Verbal Interpretation
4	Highly Effective
3	Effective
2	Less Effective
1	Not Effective

	Criteria	4	3	2	1
2.1	The AI-assisted analysis accurately represents the meaning and intent of the evaluators' written comments.				
2.2	The AI-assisted analysis effectively identifies recurring themes and patterns from qualitative faculty evaluation feedback.				
2.3	The AI-generated summaries organize large amounts of qualitative feedback into information that is easier to understand.				
2.4	The AI-assisted analysis enables faster interpretation of qualitative feedback compared with manual review.				
2.5	The AI-generated insights help identify faculty strengths and areas that require improvement.				
2.6	The AI-assisted analysis presents qualitative feedback in a manner that supports objective interpretation and minimizes personal bias.				
2.7	The AI-generated summaries provide sufficient detail to support meaningful evaluation of faculty performance.				
2.8	The AI-assisted analysis accurately identifies frequently occurring words, phrases, and issues from qualitative evaluation feedback.				



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2.9	The frequency analysis helps users recognize recurring trends and concerns in qualitative evaluation feedback over time.				
2.10	The AI-assisted analysis presents the results of keyword and frequency analysis in a clear and understandable manner.				
2.11	The keyword and frequency analysis provides meaningful insights that support the interpretation of qualitative evaluation results.				
2.12	Overall, the combination of AI-generated summaries and frequency analysis improves the usefulness of qualitative faculty evaluation feedback for decision-making.				

PART 4:

3. SYSTEM QUALITY EVALUATION - To evaluate the system based on ISO/IEC 25010:2023 software quality characteristics.

Rating Scale

Scale	Verbal Interpretation
4	Highly Functionally Suitable
3	Functionally Suitable
2	Less Functionally Suitable
1	Not Functionally Suitable

3.1 Functional Suitability This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.		4	3	2	1
3.1.1	Functional Completeness - The system provides all the necessary features required to conduct faculty evaluations, including student, peer, and supervisor evaluations, AI-enhanced feedback analysis, report generation, and evaluation management.				



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3.1.2	Functional Correctness - The system accurately records, processes, and generates faculty evaluation results and AI-assisted feedback summaries.				
3.1.3	Functional Appropriateness - The system effectively supports users in completing faculty evaluation tasks efficiently and accurately.				

Rating Scale

Scale	Verbal Interpretation
4	Highly Reliable
3	Reliable
2	Less Reliable
1	Not Reliable

3.2 Reliability Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time	4	3	2	1
3.2.1 Faultlessness - The system performs faculty evaluation functions consistently without errors during normal operation.				
3.2.2 Availability - The system remains available whenever authorized users need to conduct or manage faculty evaluations.				
3.2.3 Fault Tolerance - The system continues operating properly despite temporary connection issues or minor software errors.				
3.2.4 Recoverability - The system restores evaluation records and resumes normal operation after unexpected interruptions or failures.				



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Rating Scale

Scale	Verbal Interpretation
4	Highly Usable
3	Usable
2	Less Usable
1	Not Usable

3.3 Usability The degree to which the system enables users to effectively, efficiently, and satisfactorily achieve their intended tasks through an interface that is easy to learn, understand, and use.		4	3	2	1
3.3.1	Appropriateness Recognizability - The system enables users to immediately recognize that it is appropriate for conducting faculty evaluations.				
3.3.2	Learnability - The system allows new users to learn how to use its features within a short period of time.				
3.3.3	Operability - The system provides menus, controls, and navigation that are simple and convenient to use.				
3.3.4	User Error Protection - The system prevents user mistakes by validating inputs and providing confirmations before submitting evaluations.				
3.3.5	User Engagement - The system encourages users to complete evaluations through an organized, interactive, and user-friendly interface.				
3.3.6	Inclusivity - The system accommodates users with different levels of computer skills and technical experience.				
3.3.7	User Assistance - The system provides sufficient instructions, prompts, and guidance to help users complete evaluation tasks.				



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3.3.8	Self-descriptiveness - The system clearly presents its functions and features so users can understand how to use it without relying heavily on external assistance.				
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Rating Scale

Scale	Verbal Interpretation
4	Highly Secure
3	Secure
2	Less Secure
1	Not Secure

3.4 Security	Degree to which a product or system defends against attack patterns by malicious actors and protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	4	3	2	1
3.4.1	Confidentiality - The system protects faculty evaluation records and users' personal information from unauthorized access.				
3.4.2	Integrity - The system ensures that evaluation data and AI-generated reports cannot be altered without proper authorization.				
3.4.3	Non-repudiation - The system maintains verifiable records of submitted evaluations and user activities that cannot be denied after submission.				
3.4.4	Accountability - The system records user activities, allowing evaluation-related actions to be traced to authorized users.				
3.4.5	Authenticity -The system verifies the identity of users before granting access to evaluation functions and confidential information.				



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3.4.6	Resistance - The system protects evaluation data and maintains secure operation against unauthorized access, cyberattacks, and other malicious attacks.				
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Rating Scale

Scale	Verbal Interpretation
4	Highly Efficient
3	Efficient
2	Less Efficient
1	Not Efficient

3.5 Performance Efficiency This characteristic represents the degree to which a product performs its functions within specified time and throughput parameters and is efficient in the use of resources (such as CPU, memory, storage, network devices, energy, materials...) under specified conditions.		4	3	2	1
3.5.1	Time Behaviour - The system responds quickly when loading pages, submitting evaluations, generating reports, and displaying AI-generated feedback summaries.				
3.5.2	Resource Utilization - The system efficiently utilizes computing resources such as internet connection, memory, and processing power during operation.				
3.5.3	Capacity - The system can accommodate multiple users and large volumes of evaluation data without reducing performance.				

Rating Scale

Scale	Verbal Interpretation
4	Highly Maintainable
3	Maintainable
2	Less Maintainable
1	Not Maintainable



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3.6 Maintainability (For IT Practitioner)		4	3	2	1
This characteristic represents the degree of effectiveness and efficiency with which a product or system can be modified to improve it, correct it or adapt it to changes in environment, and in requirements.					
3.6.1	Modularity - The system is organized into independent modules, allowing updates without significantly affecting other system functions.				
3.6.2	Reusability - The system's components can be reused for future enhancements or related institutional applications.				
3.6.3	Analyzability - The system enables developers to efficiently identify system issues and evaluate the impact of proposed modifications.				
3.6.4	Modifiability - The system allows new features, corrections, and improvements to be implemented without affecting existing functions.				
3.6.5	Testability - The system supports efficient testing to verify that all functions perform according to specified requirements.				

Rating Scale

Scale	Verbal Interpretation
4	Highly Compatible
3	Compatible
2	Less Compatible
1	Not Compatible

3.7 Compatibility (For IT Practitioner Only)		4	3	2	1
Degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions while sharing the same common environment and resources.					
3.7.1	Co-existence - The system operates smoothly alongside other software and applications used by the institution without affecting performance.				



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3.7.2	Interoperability - The system exchanges and manages information effectively with integrated technologies, databases, and AI services used by the institution.				
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Rating Scale

Scale	Verbal Interpretation
4	Highly Portable
3	Portable
2	Less Portable
1	Not Portable

3.8 Portability The degree to which the system can be transferred, installed, adapted, and operated effectively across different devices, platforms, and operating environments.	4	3	2	1
3.8.1 Adaptability - The system can be adapted to different devices, browsers, and institutional requirements.				
3.8.2 Installability - The system can be deployed, configured, updated, and maintained with minimal difficulty.				
3.8.3 Replaceability - The system can effectively replace the existing manual or traditional faculty evaluation process.				

Rating Scale

Scale	Verbal Interpretation
4	Highly Accessible
3	Accessible
2	Less Accessible
1	Not Accessible

3.9 Accessibility The degree to which the system can be used effectively by people with diverse abilities, characteristics, and levels of technical expertise, ensuring equal access to its features and functions.	4	3	2	1
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3.9.1	Accessibility - The system allows users with different abilities and levels of technical expertise to complete faculty evaluation tasks effectively.				
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PART 5:

4. **System Comparison** - This section compares the proposed faculty evaluation system with the existing evaluation process in terms of operational efficiency, data accuracy, usability, and administrative workload.

Rating Scale

Scale	Verbal Interpretation
4	Highly Efficient Compared with the Existing Process
3	Efficient Compared with the Existing Process
2	Less Efficient Compared with the Existing Process
1	Not Efficient Compared with the Existing Process

4.1 Operational Efficiency	4	3	2	1
4.1.1 The proposed system completes evaluation-related tasks more efficiently than the existing evaluation process.				
4.1.2 The proposed system allows evaluation information to be processed more quickly than the current process.				
4.1.3 The proposed system reduces delays encountered during the evaluation process.				

Rating Scale

Scale	Verbal Interpretation
4	Highly Accurate
3	Accurate
2	Less Accurate
1	Not Accurate



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4.2 Data Accuracy		4	3	2	1
4.2.1	The proposed system produces more accurate evaluation results than the existing evaluation process.				
4.2.2	The proposed system minimizes errors in recording and processing evaluation data.				
4.2.3	The information generated by the proposed system is consistent and reliable.				

Rating Scale

Scale	Verbal Interpretation
4	Highly More Usable than the Existing Process
3	More Usable than the Existing Process
2	Less More Usable than the Existing Process
1	Not More Usable than the Existing Process

4.3 Usability		4	3	2	1
4.3.1	The features of the proposed system are easy to learn and understand.				
4.3.2	The system interface makes completing evaluation tasks straightforward.				
4.3.3	Users can accomplish evaluation-related tasks with minimal assistance.				

Rating Scale

Scale	Verbal Interpretation
4	Greatly Reduces Administrative Workload
3	Reduces Administrative Workload
2	Minimally Reduces Administrative Workload
1	Does Not Reduce Administrative Workload

4.4 Administrative Workload		4	3	2	1
4.4.1	The proposed system reduces the amount of manual work required during faculty evaluation.				



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4.4.2	The proposed system simplifies report preparation and documentation.				
4.4.3	The proposed system allows administrators to manage evaluation activities more efficiently.				

PART 6:

5. Technology Acceptance & Institutional Impact - This section assesses the level of user acceptance of the proposed faculty evaluation system by measuring perceived usefulness, perceived ease of use, and overall user experience.

Rating Scale

Scale	Verbal Interpretation
4	Highly Useful
3	Useful
2	Less Useful
1	Not Useful

5.1 Perceived Usefulness (PU)	4	3	2	1
5.1.1 The proposed faculty evaluation system improves my efficiency in completing evaluation tasks.				
5.1.2 The proposed system enables me to complete evaluation tasks more quickly.				
5.1.3 The proposed system improves the effectiveness of the faculty evaluation process.				
5.1.4 The proposed system enhances my productivity in performing evaluation-related activities.				
5.1.5 Overall, I find the proposed faculty evaluation system useful.				

Rating Scale

Scale	Verbal Interpretation
4	Very Easy to Use
3	Easy to Use
2	Difficult to Use
1	Very Difficult to Use



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5.2 Perceived Ease of Use (PEOU)		4	3	2	1
5.2.1	Learning to use the proposed system is easy.				
5.2.2	The proposed system is easy to navigate.				
5.2.3	The interaction with the proposed system is clear and understandable.				
5.2.4	It is easy for me to become skillful in using the proposed system.				
5.2.5	Overall, I find the proposed faculty evaluation system easy to use.				

Rating Scale

Scale	Verbal Interpretation
4	Highly Acceptable
3	Acceptable
2	Less Acceptable
1	Not Acceptable

5.3 Overall User Experience / User Acceptance		4	3	2	1
5.3.1	I am satisfied with my experience using the proposed system.				
5.3.2	The interface of the proposed system is user-friendly.				
5.3.3	The proposed system performs efficiently during use.				
5.3.4	I would recommend the proposed system to other users.				
5.3.5	I intend to continue using the proposed system if it is implemented in the institution.				

PART 7:

- 6. Data-Driven Improvements and Recommendations** - This section evaluates how effectively the proposed faculty evaluation system transforms evaluation results, user feedback, and performance analysis into meaningful insights, actionable recommendations, and priorities for continuous improvement.



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Rating Scale

Scale	Verbal Interpretation
4	Highly Effective
3	Effective
2	Less Effective
1	Not Effective

	Criteria	4	3	2	1
6.1	The evaluation reports clearly identify areas that require improvement based on the collected evaluation data.				
6.2	The recommendations generated by the system are relevant to the evaluation results and user feedback.				
6.3	The system presents evaluation findings in a way that helps users understand the strengths and weaknesses identified during the evaluation process.				
6.4	The performance reports provide sufficient information to support planning and decision-making for faculty improvement.				
6.5	The system helps identify recurring issues and performance trends from evaluation data over time.				
6.6	The recommendations generated by the system are practical and can be used to improve faculty performance and evaluation practices.				
6.7	The information provided by the system helps prioritize areas that require immediate attention or improvement.				
6.8	Overall, the system effectively transforms evaluation results into useful information for continuous institutional improvement.				

Signature over Printed Name



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APPENDIX U
Validation of Peer to Peer Questionnaires

Validation of Peer to Peer Questionnaires for Ms. Allen Joyce A. Flaviano, RPh,
CHRA, CTSP on March 16, 2026

CERTIFICATE OF VALIDATION

Dear Sir/Madam,

Good day,

We are third-year students from the Bachelor of Science in Aviation Information System program who are currently conducting a research study entitled "Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines." This study aims to develop a web-based system that will enhance the faculty evaluation process by incorporating standardized evaluation instruments, efficient data management, and AI-supported analysis of feedback to improve the overall quality of evaluation within the institution.

As part of this research, we have developed a research instrument entitled Faculty Interpersonal Questionnaire. This instrument is designed to serve as a **peer-to-peer evaluation tool for faculty members**, focusing on assessing interpersonal interactions among colleagues. Specifically, it evaluates aspects such as professional communication, interpersonal respect and conduct, collegial support, and the overall social environment within the workplace.

In line with ensuring the quality and validity of this instrument, we respectfully request your assistance as a **validator**. We kindly ask for your professional expertise in reviewing and evaluating the questionnaire, particularly its **rating scale, operational definitions, and benchmark statements**, to determine their clarity, relevance, and appropriateness for academic and institutional use.

Your evaluation and recommendations will be highly valuable in refining and improving the instrument prior to its integration into the proposed system as part of the **faculty peer evaluation module**. We sincerely hope for your support by helping us validate this scale and by providing any comments or suggestions that may further strengthen the instrument.

Thank you very much for your time, guidance, and support in contributing to the success of our research.

Respectfully yours,

Rhicka Mae P. De Asis

John Emmanuel E. Rojo

Loenza Lane S. Vallido

The Researchers

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RATING SCALE

Scale	Qualitative Description	Operational Definition
5	Always manifested	Evident in nearly all relevant situations (91-100% of instances).
4	Often manifested	Evident most of the time, with occasional lapses (61-90%).
3	Sometimes manifested	Evident about half the time (31-60%).
2	Seldom manifested	Infrequently Demonstrated: Rarely evident in relevant situations (11-30%).
1	Never/Rarely manifested	Seldom Demonstrated: Almost never evident, with only isolated cases (0-10%).

EVALUATION INSTRUMENT

Peer to Peer Evaluation

Instructions:

Carefully read each benchmark statement and rate the faculty member by selecting the appropriate rating based on the scale provided above. Responses will be used solely for peer-to-peer evaluation purposes within the system.

Benchmark Statements	Rating					Comment	Suggestion
A. Interpersonal Approach and Workplace Conduct							
1. Maintains a courteous and respectful manner when interacting with colleagues.	5	4	3	2	1	✓	
2. Demonstrates openness and approachability in professional interactions.	5	4	3	2	1	✓	
3. Contributes to a positive and respectful workplace atmosphere.	5	4	3	2	1	✓	
B. Cooperation and Teamwork							
4. Works well with others and supports team goals.	5	4	3	2	1	✓	
5. Willingly assists teammates when needed.	5	4	3	2	1	✓	

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6. Respects ideas and opinions of co-workers.	5	4	3	2	1	✓	
C. Communication Skills							
7. Communicates clearly and professionally.	5	4	3	2	1	✓	
8. Listens and responds appropriately.	5	4	3	2	1	✓	
9. Shares necessary information in a timely manner.	5	4	3	2	1	✓	
D. Dependability and Timeliness							
10. Completes task within deadlines.	5	4	3	2	1	✓	} Time removed (optional)
11. Can be relied upon to fulfill responsibilities.	5	4	3	2	1	✓	
12. Reports to work and meetings on time.	5	4	3	2	1	✓	
E. Professionalism and Work Ethics							
13. Shows integrity and honesty in performing duties.	5	4	3	2	1	✓	} Time removed (optional)
14. Accepts feedback constructively.	5	4	3	2	1	✓	
15. Demonstrates commitment to the organization.	5	4	3	2	1	✓	Return!
Open Ended Questions							
What are this colleague's greatest strengths?						✓	
In what areas could this colleague improve?						✓	
Additional comments or suggestions.						✓	

we can reduce the items up to 10 questions.

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Faculty Evaluation of Supervisor

Instructions:

Rate your supervisor using the 5-point scale above

Benchmark Statements	Rating					Comment	Suggestion
A. Leadership and Decision-Making							
1. Provides clear direction and guidance.	5	4	3	2	1	✓	
2. Makes fair and sound decisions.	5	4	3	2	1	✓	
3. Handles challenges appropriately.	5	4	3	2	1	X option to remove	
B. Fairness and Objectivity							
4. Treats employees fairly and respectfully.	5	4	3	2	1	✓	
5. Evaluates performance objectively.	5	4	3	2	1	✓	
6. Applies policies consistently.	5	4	3	2	1	✓	
C. Communication and Guidance							
7. Communicates expectations clearly.	5	4	3	2	1	X option to remove	
8. Provides helpful feedback.	5	4	3	2	1	✓	
9. Is approachable and responsive.	5	4	3	2	1	✓	
D. Support for Employee Development							
10. Encourages learning and professional growth.	5	4	3	2	1	✓	
11. Recognizes good performance.	5	4	3	2	1	X option to remove	
12. Helps employees improve weaknesses.	5	4	3	2	1	X option to remove	
E. Responsiveness to Concerns							
13. Listens to employee concerns.	5	4	3	2	1	✓	
14. Takes appropriate action on issues.	5	4	3	2	1	✓	
15. Creates a positive work environment.	5	4	3	2	1	X option to remove	

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Open Ended Questions		
What does your supervisor do particularly well?	✓	
What areas could your supervisor improve on?	✓	
Additional comments or suggestions.	✓	

Date: March 16, 2026

VALIDATED BY:


ALLEN JOYCE A. FLAVIANO, RPh, CHRA, CTSP
Administrative Officer IV

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APPENDIX V
Validation of Peer to Peer Questionnaires

Validation of Peer to Peer Questionnaires for Mr. Ricardo A. Fajardo, MAVED,
LPT on March 16, 2026

CERTIFICATE OF VALIDATION

Dear Sir/Madam,

Good day.

We are third-year students from the **Bachelor of Science in Aviation Information System** program who are currently conducting a research study entitled **"Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines."** This study aims to develop a web-based system that will enhance the faculty evaluation process by incorporating standardized evaluation instruments, efficient data management, and AI-supported analysis of feedback to improve the overall quality of evaluation within the institution.

As part of this research, we have developed a research instrument entitled **Faculty Interpersonal Questionnaire**. This instrument is designed to serve as a **peer-to-peer evaluation tool for faculty members**, focusing on assessing interpersonal interactions among colleagues. Specifically, it evaluates aspects such as professional communication, interpersonal respect and conduct, collegial support, and the overall social environment within the workplace.

In line with ensuring the quality and validity of this instrument, we respectfully request your assistance as a **validator**. We kindly ask for your professional expertise in reviewing and evaluating the questionnaire, particularly **its rating scale, operational definitions, and benchmark statements**, to determine their clarity, relevance, and appropriateness for academic and institutional use.

Your evaluation and recommendations will be highly valuable in refining and improving the instrument prior to its integration into the proposed system as part of the **faculty peer evaluation module**. We sincerely hope for your support by helping us validate this scale and by providing any comments or suggestions that may further strengthen the instrument.

Thank you very much for your time, guidance, and support in contributing to the success of our research.

Respectfully yours,

Rhicka Mae P. De Asis

John Emmanuel E. Rojo

Lorenza Lane S. Vallido

The Researchers

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RATING SCALE

Scale	Qualitative Description	Operational Definition
5	Always manifested	Evident in nearly all relevant situations (91-100% of instances).
4	Often manifested	Evident most of the time, with occasional lapses (61-90%).
3	Sometimes manifested	Evident about half the time (31-60%).
2	Seldom manifested	Infrequently Demonstrated: Rarely evident in relevant situations (11-30%).
1	Never/Rarely manifested	Seldom Demonstrated: Almost never evident, with only isolated cases (0-10%).

provide this also to supervisor

EVALUATION INSTRUMENT

Peer to Peer Evaluation

Instructions:

Carefully read each benchmark statement and rate the faculty member by selecting the appropriate rating based on the scale provided above. Responses will be used solely for peer-to-peer evaluation purposes within the system.

Benchmark Statements	Rating					Comment	Suggestion
A. Interpersonal Approach and Workplace Conduct							
1. Maintains a courteous and respectful manner when interacting with colleagues.	5	4	3	2	1		
2. Demonstrates openness and approachability in professional interactions.	5	4	3	2	1		
3. Contributes to a positive and respectful workplace atmosphere.	5	4	3	2	1		
B. Cooperation and Teamwork							
4. Works well with others and supports team goals.	5	4	3	2	1		
5. Willingly assists teammates when needed.	5	4	3	2	1		

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6. Respects ideas and opinions of co-workers.	5	4	3	2	1		
C. Communication Skills							
7. Communicates clearly and professionally.	5	4	3	2	1		
8. Listens and responds appropriately.	5	4	3	2	1		
9. Shares necessary information in a timely manner.	5	4	3	2	1		
D. Dependability and Timelines							
10. Completes task within deadlines.	5	4	3	2	1		
11. Can be relied upon to fulfill responsibilities.	5	4	3	2	1		
12. Reports to work and meetings on time.	5	4	3	2	1		
E. Professionalism and Work Ethics							
13. Shows integrity and honesty in performing duties.	5	4	3	2	1		
14. Accepts feedback constructively.	5	4	3	2	1		
15. Demonstrates commitment to the organization.	5	4	3	2	1		
Open Ended Questions							
What are this colleague's greatest strengths?	Perhaps: What are his/her						
In what areas could this colleague improve?	1/						
Additional comments or suggestions.	of						

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Faculty Evaluation of Supervisor

Instructions:

Rate your supervisor using the 5-point scale above

provide the scale

Benchmark Statements	Rating					Comment	Suggestion
A. Leadership and Decision-Making							
1. Provides clear direction and guidance.	5	4	3	2	1		
2. Makes fair and sound decisions.	5	4	3	2	1		
3. Handles challenges appropriately.	5	4	3	2	1		
B. Fairness and Objectivity							
4. Treats employees fairly and respectfully.	5	4	3	2	1		
5. Evaluates performance objectively.	5	4	3	2	1		
6. Applies policies consistently.	5	4	3	2	1		
C. Communication and Guidance							
7. Communicates expectations clearly.	5	4	3	2	1		
8. Provides helpful feedback.	5	4	3	2	1		
9. Is approachable and responsive.	5	4	3	2	1		
D. Support for Employee Development							
10. Encourages learning and professional growth.	5	4	3	2	1		
11. Recognizes good performance.	5	4	3	2	1		
12. Helps employees improve weaknesses.	5	4	3	2	1		
E. Responsiveness to Concerns							
13. Listens to employee concerns.	5	4	3	2	1	✓	
14. Takes appropriate action on issues.	5	4	3	2	1	✓	
15. Creates a positive work environment.	5	4	3	2	1	✓	

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Open Ended Questions		
What does your supervisor do particularly well?	✓	
What areas could your supervisor improve on?	✓	
Additional comments or suggestions.	✓	

VALIDATED BY:

Date: March 16, 2020


RICARDO A. EJARDO, MAVED, LPT
Director, Community Extension Services

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APPENDIX W
User Case Testing

NAAP EVALUATION SYSTEM - OSA PANEL MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 25 2026
RE-01 AUTHENTICATION, SESSION SECURITY & OSA ROLE AUTHORIZATION					
Pre-requisite: Browser is running; NAAP database and session services are operational.					
TC-001	1. Log in with valid OSA officer credentials. 2. Navigate to html/osapanel.html. 3. Verify checkAuthentication() check. 4. Observe dashboard view load.	User: osa / OSA User Role: osa Status: active Session: Valid JWT / session token	checkAuthentication() returns true. OSA Dashboard loads successfully displaying officer name, title, and initial analytics cards.	Logged in with valid OSA officer credentials and dashboard loaded properly with officer details.	PASS
TC-002	1. Open new private browser window with no active session. 2. Attempt direct URL navigation to html/osapanel.html. 3. Observe authentication interception.	Session: None (null / undefined) URL: .../html/osapanel.html	Access is denied immediately. System invokes redirectToLogin() and redirects browser to mainpage.html without rendering student clearance data.	Direct unauthenticated access was blocked and redirected immediately to the login page.	PASS
TC-003	1. Establish active session with role 'student' or 'faculty'. 2. Attempt direct URL navigation to html/osapanel.html. 3. Observe role authorization guard.	User: ST-000001 / PROF-001 Role: student / faculty Target: osapanel.html	Role check fails (strict requirement === 'osa'). System redirects user away to their designated portal or login page.	Attempted access using a non-OSA account was denied and routed to the proper panel.	PASS
TC-004	1. Set OSA officer user account status in database to 'inactive'. 2. Attempt login or access to osapanel.html. 3. Observe account restriction handling.	User: osa Status: inactive Role: osa	System denies access; displays account inactive notification, terminates session tokens, and redirects user to mainpage.html?logged_out=1.	Inactive account access was blocked with a notice instructing the officer to contact the admin.	PASS
TC-005	1. On OSA panel, click 'Logout' button (#logoutBtn) in sidebar footer. 2. Observe session purge and redirect.	Action: Click #logoutBtn Target URL: mainpage.html?logged_out=1	clearUserSession() purges all active session keys from localStorage/sessionStorage and redirects to mainpage.html?logged_out=1 with logout message.	Clicked logout and verified that all session storage was cleared before redirecting to login.	PASS

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TC-006	1. Simulate expired session token during active monitoring session. 2. Attempt to switch views or perform manual clearance. 3. Observe session expiration handling.	Session: Expired timestamp / invalid token	System intercepts the request, detects session timeout, alerts user, and safely redirects to login page without submitting action.	Expired session token triggered an expiration notice and safely returned to the login screen.	PASS
RE-02 OSA DASHBOARD OVERVIEW, RESPONSIVE DRAWER & VIEW NAVIGATION					
Pre-requisite: OSA officer is logged in on the OSA Dashboard.					
TC-007	1. Click 'Student Evaluation Status' nav link in sidebar. 2. Click 'Proof Reviews' nav link. 3. Click 'Profile' nav link. 4. Click 'Dashboard' nav link.	Sidebar Nav Links: - data-view='studentStatusView' - data-view='proofReviewView' - data-view='profileView' - data-view='dashboardView'	Active class updates smoothly on the clicked navigation link. Corresponding content view renders immediately while sibling views hide.	Sidebar navigation links switched views cleanly while updating the active tab highlight.	PASS
TC-008	1. Resize browser viewport to mobile width (< 992px). 2. Click mobile navigation toggle button (#osaMenuToggle). 3. Verify mobile sidebar drawer. 4. Click sidebar backdrop overlay (#sidebarBackdrop).	Viewport: 375px x 812px (Mobile) Elements: #osaMenuToggle, #sidebarBackdrop, #osaSidebar	Sidebar drawer translates smoothly from left into viewport with backdrop overlay. Clicking backdrop or pressing Esc smoothly dismisses mobile navigation drawer.	Mobile drawer opened smoothly via hamburger toggle and closed upon pressing Esc.	PASS
TC-009	1. On Dashboard hero card, click 'Open Status List' (#heroOpenStatusBtn). 2. Return to Dashboard and click 'Open Proof Reviews' (#heroOpenProofReviewBtn).	Hero Action Buttons: - #heroOpenStatusBtn - #heroOpenProofReviewBtn	'Open Status List' switches view to #studentStatusView; 'Open Proof Reviews' switches to #proofReviewView; sidebar active indicators update cleanly.	Dashboard hero action buttons navigated directly to the student status and proof review views.	PASS
TC-010	1. Inspect top header user profile display across all 4 views. 2. Verify synchronization of officer name.	Header Elements: #profileName, #profileNameStatus, #profileNameProof, #profileNameDuplicate	All views consistently render the officer's full name and avatar icon without display discrepancies.	Top header consistently displayed the officer's name and role across all views.	PASS
TC-	1. Verify top header	Views:	Page title and subtitle	Page headers	PASS

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011	page title and context descriptions across view changes.	Dashboard, Student Status, Proof Reviews, Profile	dynamically update with clear, contextual explanations of the active OSA workspace view.	dynamically updated their title and description to match the active module.	
TC-012	1. Inspect real-time data synchronization listener (setupDataSubscriptions()). 2. Trigger a student evaluation submission in another window. 3. Observe OSA panel reaction.	Event: SharedData data change (Evaluations / Proofs)	Panel automatically refreshes analytics summary counts and status rows in real time without requiring a manual page reload.	Dashboard analytics refreshed automatically when new evaluation data was received.	PASS
RE-03 Verify password field security attributes and visual character masking					
Pre-requisite: OSA Dashboard view (#dashboardView) is active with loaded enrollment data.					
TC-013	1. Inspect Dashboard summary cards. 2. Verify 'Assigned Students' count (#assignedCount). 3. Verify 'Evaluated' count (#evaluatedCountAnalytics). 4. Verify 'Not Evaluated' count (#notEvaluatedCountAnalytics). 5. Verify 'Completion Rate' (#completionRateCount).	Summary Cards: - #assignedCount (e.g. '450') - #evaluatedCountAnalytics (e.g. '360') - #notEvaluatedCountAnalytics (e.g. '90') - #completionRateCount (e.g. '80.0%')	Summary cards display accurate active-semester metrics. Evaluated + Not Evaluated equals Assigned Students; Completion Rate reflects mathematical percentage.	Dashboard summary cards displayed accurate student tallies and evaluation completion rates.	PASS
TC-014	1. Locate Campus filter dropdown (#analyticsCampusSelect). 2. Inspect available campus options.	Dropdown: #analyticsCampusSelect Options: 'All Campuses', 'Main Campus', 'Ext Campus'	Dropdown lists all active institutional campuses dynamically populated from assigned student enrollment records.	Campus dropdown populated all active campus locations without duplicates.	PASS
TC-015	1. Select a specific campus from #analyticsCampusSelect (e.g., 'Main Campus'). 2. Observe summary cards and analytics tables.	Selection: Main Campus Function: getAnalyticsRowsForSelectedCampus()	Summary cards recalculate counts exclusively for Main Campus; Department and Program analytics tables filter to reflect only Main Campus students.	Filtering by campus updated the metrics to reflect only the selected location.	PASS
TC-016	1. Re-select 'All Campuses' in #analyticsCampusSelect. 2. Verify restored metrics.	Selection: All Campuses	Summary cards and descriptive analytics tables revert to institution-wide totals across all campuses.	Resetting filter to All Campuses restored institution-wide metrics immediately.	PASS
TC-	1. Inspect 'By	Table:	Table displays	Department	PASS

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017	Department' analytics table (#departmentAnalytics Body). 2. Verify columns: Department, Assigned, Evaluated, Not Evaluated, Completion %.	.analytics-table (Department) Rows: Department listings	department-level student tallies with formatted completion percentage badges (e.g. '85.2%').	breakdown table displayed accurate student counts and formatted percentages.	
TC-018	1. Simulate zero enrolled students for an unconfigured semester. 2. Reload Dashboard view. 3. Observe empty state handling.	Data: 0 assigned students	Summary cards show '0', '0%', and #analyticsEmptyState displays message: 'No assigned students found for analytics.' Tables show clean empty state.	Selecting a campus with zero data rendered an empty state with zeroed metrics.	PASS
RE-04 DEPARTMENT & PROGRAM DRILL-DOWN COMPLETION ANALYTICS					
Pre-requisite: Descriptive Analytics section is loaded with department data.					
TC-019	1. In 'By Department' table, click on a department row (e.g., 'Aviation Studies'). 2. Observe selection highlight and 'Programs by Department' table (#departmentPrograms Body).	Action: Click department row Target: #departmentProgramsBody	Selected department row is highlighted with active styling. #departmentProgramsTitle updates to 'Programs in Aviation Studies'; table populates with academic programs.	Clicking a department row highlighted it and loaded its corresponding programs below.	PASS
TC-020	1. Inspect columns in 'Programs by Department' table: - Program - Assigned - Evaluated - Not Evaluated - Completion %.	Table: #departmentProgramsBody Rows: Programs (e.g. BS Aviation, BS Aero Eng)	Programs table renders exact breakdown of assigned students, completions, pending evaluations, and percentage completion for each program under the selected department.	Programs table rendered clear completion percentages and assigned student counts.	PASS
TC-021	1. Click a different department row (e.g. 'General Education'). 2. Verify programs table update.	Selection: General Education	Previous selection unhighlights; programs table refreshes to show General Education programs and corresponding completion tallies.	Selecting another department updated the program list smoothly with the new department's data.	PASS
TC-022	1. Observe initial state of 'Programs by Department' table before any department row is clicked.	State: No department selected	#departmentProgramsEmpty message displays: 'Select a department above to view its programs.' Programs table remains clean.	Empty state message prompted the user to select a department before loading program data.	PASS
TC-023	1. Select a department that has no sub-programs configured. 2. Inspect programs	Data: Department with 0 sub-programs	Table displays empty state notice indicating no specific degree programs are registered under this	Department with no degree programs displayed an	PASS

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	table response.		department.	empty state notice correctly.	
TC-024	1. Change campus filter (#analyticsCampusSelect) while a department is selected. 2. Verify synchronized program drill-down data.	Campus Filter: Change from All to Specific Campus	Program breakdown numbers dynamically update to reflect the newly filtered campus scope for the selected department.	Changing the campus filter recalculated program breakdown figures in real time.	PASS
RE-05 STUDENT EVALUATION STATUS DIRECTORY, LEGEND & SEARCH					
Pre-requisite: OSA officer is on Student Evaluation Status view (#studentStatusView).					
TC-025	1. Inspect status period note (#statusPeriodNote). 2. Verify evaluation period schedule details.	Element: #statusPeriodNote Data: Active semester schedule & deadline dates	Note displays whether evaluation period is currently open or closed (e.g. 'Evaluation Period: Open (Ends Feb 28, 2026)' or 'Period Closed - Proof Review Mode').	Evaluation period note displayed whether submissions were currently open or closed.	PASS
TC-026	1. Inspect status legend (.status-legend). 2. Verify legend indicators: Done, Cleared, Not Done.	Legend Items: - Done (green dot) - Cleared (blue dot) - Not Done (amber dot)	Legend clearly explains status definitions: Done = Completed all evaluations; Cleared = Manually cleared by OSA or approved proof; Not Done = Pending.	Status legend clearly explained color-coded indicators for Done and Not Done.	PASS
TC-027	1. Inspect Student Status table (#statusTableBody). 2. Verify columns: Student Number, Full Name, Department, Program, Year/Section, Progress, Status, Review.	Table: .status-table Columns: Student No, Name, Dept, Program, Yr/Sec, Progress, Status, Action	Table renders student roster with progress ratios (e.g. '4/4' or '2/4') and color-coded status badges matching legend indicators.	Student status table rendered the complete roster with accurate evaluation ratios.	PASS
TC-028	1. In search input (#studentSearch), enter a student's full name (e.g. 'John Student'). 2. Observe live table filtering.	Query: 'John Student' Input: #studentSearch	applySearchFilter() filters table in real time, displaying only rows matching the entered name; case-insensitive.	Search filter updated instantly to display matching student names and filtered out others.	PASS
TC-029	1. In search input (#studentSearch), enter a student ID number (e.g. 'ST-000001'). 2. Observe table filtering.	Query: 'ST-000001'	Table filters immediately to show the specific student record matching the student number.	Searching by student number immediately returned the exact matching student record.	PASS
TC-030	1. Enter non-existent keyword (e.g. 'XYZ99999') in #studentSearch. 2. Observe empty state handling.	Query: 'XYZ99999'	Table rows clear and empty state notice (#emptyState) displays: 'No students found matching your search.'	Searching a non-existent keyword displayed a clear no results found notice.	PASS
TC-	1. Clear search input	Action: Clear	Table restores full roster	Clearing the	PASS

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031	(#studentSearch). 2. Verify complete student roster restoration.	input field	of assigned students immediately; empty state message hides.	search input restored the complete student roster immediately.	
RE-06 MANUAL CLEARANCE MODAL & REASON VALIDATION					
Pre-requisite: Evaluation period is closed or student requires manual clearance.					
TC-032	1. Locate a student with 'Not Done' status. 2. In Review column, click 'Clear' / 'Review' button. 3. Verify Manual Clearance Modal (#manualClearModal) appearance.	Target: Not Done student (e.g., ST-000002) Element: #manualClearModal	#manualClearModal opens with title 'Mark Student as Cleared', student name and ID in lead description (#manualClearLead), and reason textarea.	Manual clearance modal opened with student name and reason textarea.	PASS
TC-033	1. In manual clearance form, leave reason textarea (#manualClearReasonInput) blank. 2. Click 'Save Clearance' button (#manualClearSaveBtn).	Reason: " (empty) Action: Click #manualClearSaveBtn	Submission is halted; validation message (#manualClearMessage) displays error: 'Please enter a valid reason for manual clearance.'	Submitting without a reason was blocked with a validation message requiring an explanation.	PASS
TC-034	1. Enter valid reason in #manualClearReasonInput (e.g. 'Student submitted verified medical certificate for hospitalization during evaluation week'). 2. Click 'Save Clearance'.	Reason: 'Verified medical certificate for hospitalization' Action: Click #manualClearSaveBtn	persistManualClearRequest() saves clearance record with timestamp and officer ID; modal displays success note, closes, and updates student status to 'Cleared' (blue badge).	Submitted manual clearance saved successfully and updated the student status to Cleared.	PASS
TC-035	1. Verify character limit on clearance reason textarea (maxlength='1000'). 2. Attempt to paste > 1000 characters.	Input: Text exceeding 1000 characters	Textarea truncates input at 1000 characters, preventing payload overflow.	Reason textarea enforced the 1000-character limit and truncated excess text.	PASS
TC-036	1. Open Manual Clearance Modal. 2. Click 'Cancel' button (#manualClearCancelButton) or close 'X' (#manualClearCloseBtn). 3. Check student status.	Action: Click Cancel / Close	Modal closes cleanly; form inputs reset; student status remains 'Not Done' without modification.	Canceling the clearance modal dismissed the dialog without modifying the student's status.	PASS
TC-037	1. Inspect a student who was previously manually cleared. 2. Observe Review column button and status badge.	Student: Cleared Status: Blue 'Cleared' badge	Table displays 'Cleared' status; button reflects cleared state or allows viewing the recorded clearance reason.	Previously cleared students displayed a cleared badge and disabled action button.	PASS
RE-07 STUDENT PROOF REVIEW QUEUE & DRIVE EVIDENCE INSPECTION					

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Pre-requisite: OSA officer navigates to Proof Reviews view (#proofReviewView).					
TC-038	1. Inspect Student Proof Review Queue table (#proofReviewTableBody). 2. Verify table columns: Student Number, Full Name, Semester, Status, Reason, Drive Link, Review Note, Review.	Table: #proofReviewTableBody Columns: Student No, Name, Semester, Status, Reason, Drive Link, Note, Actions	Table renders all proof requests submitted by students who missed evaluation deadline, displaying full justification reason and evidence link.	Student proof review queue loaded all submitted proof requests with full details.	PASS
TC-039	1. Inspect status badges in proof review queue. 2. Verify representations for 'Pending Review', 'Approved', and 'Rejected'.	Statuses: - Pending Review (amber badge) - Approved (green badge) - Rejected (red badge)	Status badges clearly indicate the current disposition of each student's justification request.	Review queue status badges accurately indicated Pending, Approved, or Rejected state.	PASS
TC-040	1. In Drive Link column, locate a valid Google Drive proof URL. 2. Inspect rendered anchor element (renderProofLink()).	Drive Link: https://drive.google.com/file/d/1A2B3C4D/view	System renders secure clickable link () with cloud drive icon.	Valid Google Drive links rendered as secure external links opening in a new tab.	PASS
TC-041	1. Click the Google Drive proof link in a review row. 2. Observe browser behavior.	Action: Click Drive Link	Browser opens the student's supporting proof document in a new tab without leaving or reloading the OSA panel.	Clicking the drive link opened the student's uploaded proof document in a new window.	PASS
TC-042	1. Verify handling of malformed or invalid URLs (sanitizeHttpUrl()).	Drive Link: 'javascript:alert(1)' or plain text	Sanitizer rejects non-HTTP(S) protocols and displays safe plain text label without executing unauthorized scripts.	Malformed links were sanitized and safely displayed as plain text without link actions.	PASS
TC-043	1. Inspect review note column for already processed requests. 2. Verify display of officer decision remarks.	Status: Approved / Rejected Column: Review Note	Column displays officer's review comments and timestamp (e.g. 'Approved by OSA on 2026-03-01').	Review notes column showed past officer remarks along with audit timestamps.	PASS
TC-044	1. Filter for a semester with zero proof requests submitted. 2. Inspect table display.	Data: 0 proof requests	Table displays empty state message (#proofReviewEmptyState): 'No submitted proof requests for the active semester.'	Empty queue state rendered a friendly message when no pending proofs were found.	PASS
RE-08 PROOF REVIEW DECISIONING (APPROVAL & REJECTION LIFECYCLE)					
Pre-requisite: Proof review queue contains at least one request with status 'Pending Review'.					

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TC-045	1. Locate a 'Pending Review' proof request row. 2. Verify available action buttons in Review column.	Buttons: 'Approve' (.btn-approve) and 'Reject' (.btn-reject)	Review column displays active 'Approve' and 'Reject' buttons exclusively for pending review requests.	Pending requests displayed active Approve and Reject buttons in the actions column.	PASS
TC-046	1. Click 'Approve' button on a pending request. 2. Confirm approval prompt dialog. 3. Observe proof request status transition.	Action: Click Approve Confirmation: 'Approve proof justification for [Student Name]?'	Request status transitions from 'Pending Review' to 'Approved'; row status badge updates to green 'Approved'; action buttons hide.	Clicking approve transitioned the request status to Approved and saved the review note.	PASS
TC-047	1. Following approval, navigate to Student Evaluation Status view (#studentStatusView). 2. Check the approved student's status badge.	Target: Approved student (e.g. ST-000003)	Student's status badge automatically transitions from 'Not Done' to 'Cleared' (blue badge) with full clearance granted.	Approved student's status badge updated automatically to Cleared in the directory.	PASS
TC-048	1. On a different pending request, click 'Reject' button. 2. Enter rejection reason in prompt dialog (e.g. 'Submitted link is inaccessible/private'). 3. Confirm rejection.	Action: Click Reject Prompt Reason: 'Submitted link is inaccessible/private'	Request status transitions to 'Rejected' (red badge); rejection reason is saved in review note; action buttons are removed.	Rejecting a request required a reason, updated the badge to Rejected, and notified the student.	PASS
TC-049	1. Navigate to Student Evaluation Status view and check the rejected student. 2. Observe student clearance status.	Target: Rejected student	Student status remains 'Not Done' (amber badge); clearance is withheld pending valid re-submission or manual clearance.	Rejected request maintained the student's Not Done status and prevented clearance.	PASS
TC-050	1. Locate an already 'Approved' or 'Rejected' proof request. 2. Verify that decisioning buttons are disabled / hidden.	Status: Approved or Rejected	Review actions column shows finalized status text; duplicate approval/rejection is strictly prevented.	Finalized requests displayed non-clickable status text to prevent duplicate decisions.	PASS
RE-09 OSA PROFILE DEMOGRAPHICS, PHOTO UPLOAD & ACCOUNT SECURITY					
Pre-requisite: OSA officer is on Profile view (#profileView).					
TC-051	1. Inspect OSA Profile card (#profileView). 2. Verify read-only demographic fields: - Full Name (#profileFullName) - Gmail (#profileEmail) - Username (#profileUsername) - Department.	Fields: Full Name, Gmail, Username, Department	Profile fields populate correctly with official OSA records synchronized from HR database in read-only format.	Profile view populated official OSA officer records accurately in read-only format.	PASS
TC-052	1. In profile photo section, click 'Upload	File: osa_avatar.png	System validates file type/size, renders live	Photo upload accepted a valid	PASS

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	Photo' (.profile-photo-btn). 2. Select valid image file (JPG/PNG under 2MB) via #profilePhotoInput. 3. Verify live preview and upload.	(1.1 MB) Element: #profilePhotoInput, #profilePhotoPreview	preview in #profilePhotoPreview, and hides placeholder initials badge (#profilePhotoPlaceholder).	image file, displayed a preview, and saved successfully.	
TC-053	1. Select an invalid file type (.pdf or .docx) or file exceeding 2MB. 2. Observe handleUploadError() response.	File: document.pdf / 10MB_photo.jpg	System rejects upload immediately with validation message: 'Please upload a valid image file (JPG or PNG) under 2MB.' Current photo is retained.	Unsupported file types were rejected with an invalid image format error alert.	PASS
TC-054	1. Click 'Change Gmail' toggle button (.action-toggle). 2. Verify #gmailCard display and #passwordCard hide.	Toggle: Change Gmail Active: #gmailCard	#gmailCard displays while #passwordCard hides. Current Gmail is populated in read-only input (#currentEmail).	Clicking change Gmail toggled the email update form and prefilled the current address.	PASS
TC-055	1. In Gmail form, enter invalid email format (e.g. 'notanemail') or non-gmail address. 2. Click 'Save Gmail' (.btn-submit). 3. Enter mismatched new & confirm emails.	Case A: Invalid format ('user@yahoo.com') Case B: newEmail != confirmEmail	Case A displays: 'Use a valid Gmail address (example@gmail.com).' Case B displays: 'New Gmail and confirm Gmail do not match.' Submission is halted.	Invalid email syntax was rejected with a message requiring a valid Gmail address.	PASS
TC-056	1. Enter matching valid Gmail address (e.g. 'osa.naap@gmail.com'). 2. Click 'Save Gmail'. 3. Observe API dispatch to /api/osa/change-email.	New: osa.naap@gmail.com Confirm: osa.naap@gmail.com API: POST /api/osa/change-email	Form dispatches update; displays success notification (#gmailFormMessage: 'Gmail updated successfully!'), and updates displayed profile email (#profileEmail).	Valid Gmail update saved properly and displayed a confirmation message.	PASS
TC-057	1. Click 'Change Password' toggle button (.action-toggle). 2. In #passwordCard, test password visibility eye toggles (.eye-toggle) for Current, New, and Confirm password fields.	Fields: #currentPassword, #newPassword, #confirmPassword Buttons: .eye-toggle	Clicking eye icon toggles field type between 'password' and 'text', toggling icon between fa-eye and fa-eye-slash.	Password visibility icon toggled the input field between masked and unmasked characters.	PASS
TC-058	1. In password form, test validations: - New password < 8 chars or missing number - Mismatched confirm password 2. Enter valid matching	Validation: min 8 chars with number, matching confirm API: POST /api/osa/change-password	Invalid inputs display corresponding validation error; valid submission dispatches POST request, displays success toast, and resets password fields.	Password validation enforced length and matching rules, updating credentials on valid input.	PASS

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	credentials and click 'Save Password'.				
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System Name:	NAAP Evaluation System
Module / View:	OSA Panel (osapanel.html)
Created by:	John Emmanuel E Rojo
Executed by:	John Emmanuel E Rojo
Date of Execution:	SEPTEMBER 25 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

NAAP EVALUATION SYSTEM - DEAN PANEL MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 28 2026
RE-01 AUTHENTICATION, SESSION SECURITY & DEAN ROLE AUTHORIZATION					
Pre-requisite: Browser is running; NAAP session & database service are operational.					
TC-001	1. Log in with valid Dean credentials. 2. Navigate to html/daenpanel.html. 3. Verify checkAuthentication() check. 4. Observe dashboard load.	User: DEAN-001 (Dean User) Role: dean Status: active Session: Valid JWT / session token	checkAuthentication() returns true. Dean Dashboard loads successfully, displaying dean's name, assigned institute/department, and header elements.	Logged in with valid Dean credentials and dashboard loaded with department metrics.	PASS
TC-002	1. Open new private browser window with no active session. 2. Attempt direct URL navigation to html/daenpanel.html. 3. Observe authentication interception.	Session: None (null / undefined) URL: .../html/daenpanel.html	Access is denied immediately. System invokes redirectToLogin() and redirects browser to mainpage.html without rendering dean dashboard data.	Direct unauthenticated access was blocked and redirected immediately to the login page.	PASS
TC-003	1. Establish active session with role 'student' or 'faculty'. 2. Attempt direct URL navigation to html/daenpanel.html. 3. Observe role authorization guard.	User: ST-000001 / PROF-001 Role: student / faculty Target: daenpanel.html	Role check fails (strict requirement === 'dean'). User is redirected away to their corresponding portal view or login page.	Attempted access using a non-dean role was denied and routed to the proper panel.	PASS
TC-004	1. In database, set dean account status to 'inactive'. 2. Attempt to access daenpanel.html. 3. Observe enforceActiveDeanAccount() response.	Dean ID: DEAN-001 Status: inactive Role: dean	System displays alert: 'Your account is inactive. You cannot access dean features. Please contact your administrator.' Clears session and redirects to mainpage.html?logged_o	Inactive dean account was blocked with a notification to contact the administrator.	PASS

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TC-005	1. Click 'Logout' button (#deanLogoutBtn) in sidebar footer. 2. Observe session purge and redirect.	Action: Click #deanLogoutBtn Target URL: mainpage.html?logged_out=1	clearUserSession() purges all active session keys from storage and redirects to mainpage.html?logged_out=1 with logout confirmation message.	Clicked logout and verified that all dean session tokens were purged from storage.	PASS
TC-006	1. Simulate session timeout or revoked token during active usage. 2. Attempt to navigate views or perform peer generation. 3. Observe session expiration handling.	Session: Expired timestamp / invalid token	System intercepts the request, detects session expiration, notifies user of expired session, and safely redirects to login page.	Expired session was intercepted during navigation, displaying a timeout notice and returning to login.	PASS
RE-02	DEAN DASHBOARD OVERVIEW, RESPONSIVE NAVIGATION & NOTIFICATION PANELS				
Pre-requisite: Dean is logged in with active account on Dean Dashboard					
TC-007	1. Click 'Evaluation' nav link in sidebar. 2. Click 'Faculty Paper Inbox' nav link. 3. Click 'Profile' nav link. 4. Click 'Faculty Response Rate & Counts' nav link. 5. Click 'Faculty Peer Management' nav link. 6. Click 'Dashboard' nav link.	Sidebar Nav Links: - data-view='peerEvaluation' - data-view='facultyPaperInbox' - data-view='profile' - data-view='facultyResponse' - data-view='peerManagement' - data-view='dashboard'	Active class updates smoothly on the clicked navigation link. Target content view displays immediately while sibling view containers are hidden.	Sidebar navigation links toggled views smoothly while highlighting the active section.	PASS
TC-008	1. Resize browser viewport to mobile width (< 992px). 2. Click mobile navigation toggle button (#deanMenuToggle). 3. Verify mobile sidebar drawer. 4. Click sidebar backdrop overlay (#sidebarBackdrop).	Viewport: 375px x 812px (Mobile) Elements: #deanMenuToggle, #sidebarBackdrop, #deanSidebar	Sidebar drawer translates smoothly from left into viewport with backdrop overlay. Clicking backdrop or pressing Esc smoothly dismisses mobile navigation drawer.	Mobile drawer opened cleanly using the hamburger toggle and dismissed upon pressing Esc.	PASS
TC-009	1. Click notification bell button (.js-notification-btn) in top header. 2. Inspect	Element: .js-notification-btn, .js-announcement-panel Notices:	.js-announcement-panel opens displaying active dean notices and schedule deadlines. Clicking anywhere outside closes	Notification bell expanded the announcements panel and closed cleanly when	PASS

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	announcements dropdown (.js-announcement-panel). 3. Click outside dropdown.	Evaluation Window, Response Tracking, Reminder	the panel smoothly.	clicking outside.	
TC-010	1. Click user profile pill (.js-profile-btn) in top header. 2. Inspect quick info dropdown (.js-profile-panel). 3. Click outside dropdown.	Element: .js-profile-btn, .js-profile-panel Fields: Faculty ID, Department, Rank, AY/SEM	.js-profile-panel expands displaying dean ID, department, rank, and current academic semester in a structured grid. Closes when clicking outside.	Profile pill opened a summary card displaying dean ID, department, and academic rank.	PASS
TC-011	1. On Dashboard hero card, click 'Open Evaluation' (#heroOpenEvaluationBtn). 2. Return to Dashboard and click 'View Faculty Results' (#heroOpenFacultyResponseBtn). 3. Return and click 'Open Paper Inbox' (#heroOpenFacultyPaperBtn).	Hero Action Buttons: - #heroOpenEvaluationBtn - #heroOpenFacultyResponseBtn - #heroOpenFacultyPaperBtn	Each button switches directly to its respective view (#peerEvaluationView, #facultyResponseView, #facultyPaperInboxView) and synchronizes sidebar active link state.	Dashboard hero action buttons navigated directly to evaluation, faculty paper, and peer views.	PASS
TC-012	1. Verify top header page context subtitle across view changes.	Views: Dashboard, Evaluation, Faculty Paper Inbox, Faculty Response, Peer Management	pageContextText updates dynamically to provide clear, actionable descriptions for the active dean view.	Header context subtitle updated dynamically to match the active module view.	PASS
RE-03 DEPARTMENT SUMMARY METRICS & EVALUATIVE CHART ANALYTICS					
Pre-requisite: Dean is on Dashboard view with active department semester data.					
TC-013	1. Inspect Dashboard summary cards. 2. Verify 'Evaluations Received' ratio card. 3. Verify 'Professors Supervised' card. 4. Verify 'Response Rate' percentage card.	Summary Cards: - .summary-card.evaluations (.card-number e.g. '120/150') - .summary-card.professors (e.g. '14') - .summary-card.response (e.g. '80%')	Summary metric cards populate accurately reflecting total evaluations received vs expected, total faculty under dean's scope, and calculated response rate.	Dashboard summary cards loaded accurate metric tallies for evaluation progress and averages.	PASS
TC-014	1. Inspect Faculty Dashboard 'Average Rating per Criteria' list (#criteriaList). 2. Verify criteria items and computed scores.	List: #criteriaList Items: Subject Mastery, Teaching Methodology, Classroom Climate	List renders average rating breakdown (e.g. '4.6 / 5.0') across department-wide faculty performance rubrics.	Criteria breakdown list accurately displayed mean scores across individual rubric categories.	PASS

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TC-015	1. In Evaluation Analytics section, verify Student Evaluation Average Scores Bar Chart (#studentBarChart) and Rating Distribution Pie Chart (#studentPieChart). 2. Hover over chart elements.	Charts: #studentBarChart, #studentPieChart Engine: Chart.js	Bar chart displays comparative criteria performance; pie chart renders breakdown of student score ratings (5s to 1s) with active tooltips.	Student evaluation bar chart and rating pie chart rendered category distributions clearly.	PASS
TC-016	1. Verify Peer Evaluation Average Scores Bar Chart (#peerBarChart) and Rating Distribution Pie Chart (#peerPieChart). 2. Hover over chart elements.	Charts: #peerBarChart, #peerPieChart Engine: Chart.js	Charts accurately render peer-to-peer performance averages and rating distributions for faculty under the dean's department scope.	Peer evaluation bar chart accurately displayed department colleague rating averages.	PASS
TC-017	1. Inspect Detailed Summary Table (#detailedSummaryTableBody). 2. Verify category breakdown and score tier counts.	Table: #detailedSummaryTableBody Columns: Category, Avg Score, Responses, Excellent..Very Poor	Table details department-wide score averages and frequency distribution per tier (Excellent to Very Poor) for each rubric category.	Detailed summary table listed exact response tallies and score distributions.	PASS
TC-018	1. Simulate zero evaluations submitted for a newly created semester. 2. Reload Dashboard view. 3. Observe empty state handling in charts and tables.	Data: 0 evaluations submitted	Summary cards show '0/0', '0%', and tables display graceful empty state: 'No evaluation data available.' Charts render empty placeholders without throwing JS errors.	Empty evaluation state rendered clean placeholder notices with zeroed summary metrics.	PASS
RE-04 SUPERVISOR FACULTY EVALUATION FORM SETUP & COLLEAGUE SEARCH					
Pre-requisite: Dean is on Evaluation view (#peerEvaluationView).					
TC-019	1. Inspect employee search input (#peerProfessorSearch) and datalist (#peerProfessorOptions). 2. Type search query (faculty name or employee ID).	Input: #peerProfessorSearch Query: 'Lauron' / 'PROF-001'	Datalist auto-completes suggestions restricted to faculty members within the dean's department/institute scope; excludes the dean user.	Employee search datalist suggested eligible department faculty members as text was entered.	PASS
TC-020	1. Select a professor suggestion from datalist. 2. Observe syncSupervisorTargetFromInput() behavior. 3. Check hidden input	Selection: Arvin Lauron (PROF-001) Hidden: #peerProfessor	Hidden input #peerProfessor syncs with selected faculty ID; search meta note updates: 'Evaluating Arvin Lauron (BS Aviation)'.	Selecting a professor synced the faculty ID and activated the evaluation questionnaire.	PASS

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	(#peerProfessor).				
TC-021	1. Select a professor who has already been evaluated by the dean in current semester. 2. Observe refreshSupervisorTargetLocked() == true) response.	Target: Previously evaluated professor (isSupervisorTargetLocked() == true)	System flags professor as evaluated; displays lock notice: 'You have already submitted a supervisor evaluation for this employee this semester.'	Selecting an already evaluated professor displayed a locked status notice.	PASS
TC-022	1. Select an unevaluated professor. 2. Verify dynamic questionnaire container (#dynamic-supervisor-questions-container). 3. Check rubric categories.	Target: Unevaluated professor Container: #dynamic-supervisor-questions-container	Supervisor rubric renders dynamically from active template with criteria categories (e.g. Leadership, Administrative Compliance, Teaching Excellence).	Supervisor rubric loaded dynamically with numbered criteria and rating radio buttons.	PASS
TC-023	1. Verify rating scale table and behavioral anchors (1 to 5) in supervisor rubric. 2. Check rating options.	Scale Range: 1 to 5 Anchors: 5 - Always Manifested, 4 - Frequently, 3 - Sometimes, 2 - Rarely, 1 - Not Manifested	Rating table renders clear behavioral anchor definitions and 1-5 radio inputs matching administrative appraisal standards.	Rating scale displayed options 1 through 5 with corresponding behavioral definitions.	PASS
TC-024	1. Select rating '5' for criteria 1, then click rating '4'. 2. Verify radio button exclusivity.	Question: Q1 Values: 5 then 4	Previous score '5' is unselected and score '4' is selected. Only one rating per criteria is active.	Radio buttons toggled cleanly, maintaining exactly one selected rating per criterion.	PASS
RE-05 SUPERVISOR EVALUATION RUBRIC VALIDATION & SUBMISSION LIFECYCLE					
Pre-requisite: Supervisor evaluation form is loaded for a selected professor.					
TC-025	1. In a multi-step supervisor questionnaire, leave required questions unanswered in Step 1. 2. Click 'Next Step' button. 3. Observe validateSupervisorCurrentStep() validation.	Step: 1 of 3 Unanswered: 2 criteria Action: Click Next	Step validation fails; system displays validation warning directing dean to complete all required rating criteria before proceeding.	Leaving required criteria unanswered halted step navigation with a validation warning.	PASS
TC-026	1. Complete Step 1 criteria, click 'Next Step', then click 'Previous Step'. 2. Verify input state retention.	Action: Step 1 -> Step 2 -> Step 1	Inputs in Step 1 remain preserved without reset; navigation between steps is smooth and preserves all selected ratings.	Step navigation advanced smoothly and preserved all previously selected answers.	PASS
TC-027	1. In qualitative feedback textarea,	Text: 'Consistently	Textarea captures supervisory feedback	Qualitative comments	PASS

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	enter supervisory feedback. 2. Verify character input acceptance.	demonstrates strong curriculum design leadership and student mentorship.'	properly without truncating valid text or breaking layout.	textarea captured supervisory feedback without text clipping.	
TC-028	1. Attempt to submit form without selecting an employee in #peerProfessor. 2. Click 'Submit Evaluation' (.btn-submit).	Employee: Empty	Submission is blocked; system displays alert: 'Please select a valid employee before submitting evaluation.'	Submitting without selecting a faculty member was blocked with an error prompt.	PASS
TC-029	1. Answer all criteria, provide constructive feedback, and click 'Submit Evaluation' (.btn-submit). 2. Confirm submission dialog prompt. 3. Observe dispatch to /api/dean/peer-evaluations/submit.	Target: Arvin Lauron Ratings: All criteria answered Action: Click .btn-submit	Confirmation prompt appears ('Submit supervisor evaluation for Arvin Lauron?'); confirming sends POST request to endpoint.	Submission modal requested confirmation before finalizing the supervisor evaluation.	PASS
TC-030	1. Observe system response upon successful submission. 2. Verify success notification and form reset.	API Response: 200 OK / success	System displays success notice: 'Supervisor evaluation submitted successfully!'; clears form inputs, and locks employee from duplicate submission.	Success alert confirmed submission, cleared the form, and returned to the dashboard.	PASS
TC-031	1. While filling supervisor form, click 'Cancel' button (#cancelPeerBtn). 2. Confirm cancellation dialog.	Action: Click #cancelPeerBtn Confirm: 'Discard unsaved supervisor evaluation ratings?'	Form inputs reset cleanly; view returns safely to Dashboard without saving data.	Canceling the evaluation reset all form inputs and returned safely to the dashboard.	PASS
RE-06 FACULTY PAPER INBOX REVIEW, SECTION C COMPLETION & PDF PREVIEW					
Pre-requisite: Dean navigates to Faculty Paper Inbox view (#facultyPaperInboxView).					
TC-032	1. Inspect Assigned Faculty Papers table (#deanFacultyPaperTable, #deanFacultyPaperTableBody). 2. Verify rendered columns and assigned papers.	Columns: Paper ID, Professor, Semester, Status, Sent At, Actions Data: Papers routed to logged-in Dean	Table displays all faculty acknowledgement papers submitted to this dean with current status (e.g. 'Submitted to Dean', 'Dean Approved').	Assigned faculty papers table rendered submitted papers with full faculty metadata.	PASS
TC-033	1. In paper table, click 'Review' / 'Open' button on a paper row. 2. Verify Faculty Paper Detail card (#deanFacultyPaperDetailCard)	Paper ID: FP-2026-001 (Arvin Lauron) Element: #deanFacultyPaperDetailCard	Card expands showing read-only snapshot metadata: Professor Name, Department, Rank, Semester, SET Rating (%), SAF Rating (%), and	Clicking review expanded the card with evaluation scores and Section C	PASS

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	etailCard) rendering.		Section C form.	inputs.	
TC-034	1. Review Section C fields: - Areas for Improvement (#deanSectionCAreas) - Proposed Activities (#deanSectionCActivities) - Action Plan (#deanSectionCActionPlan). 2. Modify or append supervisory developmental notes.	Section C Inputs: - #deanSectionCAreas - #deanSectionCActivities - #deanSectionCActionPlan	Fields display professor's draft inputs; Dean can add executive developmental notes and supervisor recommendations.	Section C fields allowed the dean to enter executive remarks and recommendations.	PASS
TC-035	1. Toggle supervisor approval auto-fill checkboxes: - 'Auto-fill supervisor name on generated PDF' (#deanFpApprovalNamesAutoFillInput) - 'Auto-fill supervisor current date signed on generated PDF' (#deanFpApprovalDatesAutoFillInput).	Checkboxes: #deanFpApprovalNamesAutoFillInput, #deanFpApprovalDatesAutoFillInput	Flags persist in paper data ensuring Dean's electronic endorsement and signing date will appear on official generated PDF.	Toggling auto-fill checkboxes applied signature and endorsement flags to the paper.	PASS
TC-036	1. Click 'Save Section C' button (#deanFacultyPaperSaveBtn). 2. Verify save confirmation.	Action: Click #deanFacultyPaperSaveBtn	System updates Section C content in database; displays success toast: 'Section C updated successfully by Dean.'	Clicking save Section C successfully persisted executive remarks in the database.	PASS
TC-037	1. Click 'Preview PDF' button (#deanFacultyPaperPreviewBtn). 2. Inspect generated PDF preview modal.	Action: Click #deanFacultyPaperPreviewBtn Output: PDF Blob in modal	System generates and displays complete official Faculty Acknowledgement Paper PDF with Sections A, B, and endorsed Section C.	Preview PDF generated the official NAAP Faculty Acknowledgement Paper with signatures.	PASS
TC-038	1. Filter for a dean with zero assigned papers. 2. Inspect paper table display.	Data: 0 papers routed	Table displays empty state message: 'No faculty papers assigned to your review queue.' Detail card remains hidden.	Empty paper inbox displayed a clean notice when no papers were pending review.	PASS
RE-07 FACULTY RESPONSE RATE & MULTI-ROLE RESULTS FILTERING					
Pre-requisite: Dean is on Faculty Response Rate & Counts view (#facultyResponseView).					
TC-039	1. Inspect faculty response controls: search input (#facultySearchInput)	Query: 'Lauron' Semester: Active Semester Action: Click	Table filters dynamically to show professors matching query within dean's department; search	Faculty response toolbar search filtered the table in real	PASS

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	and semester dropdown (#facultyResponseSemesterFilter). 2. Enter faculty name query and click 'Search' (#facultySearchBtn).	#facultySearchBtn	summary (#facultySearchResult) updates count.	time as queries were typed.	
TC-040	1. Click 'Reset' button (#facultyResetBtn). 2. Observe search input and table state.	Action: Click #facultyResetBtn	Search input clears, semester reverts to default active semester, and table restores full list of department faculty.	Clicking reset cleared the search input and restored default semester filters.	PASS
TC-041	1. Click 'Student Evaluation Results' toggle button (#studentResultsBtn). 2. Inspect table columns and metrics (#facultyResponseCountsTable).	Toggle: #studentResultsBtn (active) Columns: Employee ID, Name, Institute, Regular/Temp, Position, Evals Received, Response Rate, Avg Score, Status, Comments	Table displays student evaluation participation metrics per faculty member (received submissions, response rate percentage, average student rating).	Student evaluation toggle displayed student response rates and participation metrics.	PASS
TC-042	1. Click 'Peer Evaluation Results' toggle button (#peerResultsBtn). 2. Observe table update.	Toggle: #peerResultsBtn Data-view: 'peer'	Active class moves to #peerResultsBtn; table reloads showing peer evaluation counts received, peer response rates, and peer average scores.	Peer evaluation toggle switched the table to show colleague response rates.	PASS
TC-043	1. Click 'Supervisor Evaluation Results' toggle button (#supervisorResultsBtn). 2. Observe table update.	Toggle: #supervisorResultsBtn Data-view: 'supervisor'	Table displays supervisor evaluation statuses and appraisal scores for each professor under the dean's supervision.	Supervisor evaluation toggle displayed completion status for supervisory appraisals.	PASS
TC-044	1. Filter for a specific historical semester in #facultyResponseSemesterFilter. 2. Verify historical data reload.	Semester: '2024-2025 2nd Semester'	Table, summary metrics, and status badges reload reflecting historical participation records for the selected semester.	Selecting a historical semester reloaded response rates and status badges for that term.	PASS
TC-045	1. Test responsive layout of faculty response table on mobile viewport (< 768px). 2. Verify card-table transformation (.mobile-card-table--	Viewport: 375px (Mobile)	Table transforms cleanly into responsive stacked mobile cards with prominent labels and touch-friendly action buttons.	Faculty response table adapted smoothly to mobile screens as responsive stacked cards.	PASS

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	faculty-response).				
RE-08		ANONYMIZED FEEDBACK INSPECTION & AI COMMENT SUMMARIZATION			
Pre-requisite: Dean is viewing faculty response results for a professor.					
TC-046	1. In faculty response table, click 'View Comments' button in the Comments column for a professor. 2. Observe Professor Comments panel (#facultyCommentsPanel).	Professor: Arvin Lauron Element: #facultyCommentsPanel, #facultyCommentsList	Comments panel expands displaying title 'Professor Comments: Arvin Lauron' and loads list of feedback comments. Student identities remain anonymous.	Clicking view comments opened the feedback panel with anonymized student remarks.	PASS
TC-047	1. Inspect loaded comments for semantic sentiment tags. 2. Verify classifyDeanFeedbackCommentBiasByRules() tagging.	Feedback items: Positive, Constructive, Critical feedback	Comments display color-coded sentiment badges (e.g. 'Positive', 'Constructive', 'Areas for Growth') based on keyword analysis.	Feedback items displayed rule-based sentiment badges reflecting positive or constructive tone.	PASS
TC-048	1. In comments panel, click 'AI Summarize' button (#facultyCommentsAiSummarize). 2. Observe AI synthesis in #facultyCommentsAiSummary.	Action: Click #facultyCommentsAiSummarize Target: #facultyCommentsAiSummary	System synthesizes comments into an executive summary highlighting key strengths, recurring concerns, and recommended supervisory interventions.	AI summarize generated an executive summary highlighting faculty strengths and growth areas.	PASS
TC-049	1. In comments panel, click 'Close' button (#facultyCommentsClose). 2. Verify panel collapse.	Action: Click #facultyCommentsClose	Comments panel collapses smoothly; placeholder resets to 'Select a professor and click View to load comments.'	Closing the comments panel collapsed the feedback list and reset the view cleanly.	PASS
TC-050	1. Observe Semestral Evaluation Trend section (#deanSemestralTrendSection) after opening a professor's comments. 2. Verify 4-semester trend table (#deanSemestralTrendTableBody).	Target: Arvin Lauron Section: #deanSemestralTrendSection	Trend section unhides displaying past 4 semesters scores (Overall, Student, Peer, Supervisor) specifically for the selected professor.	Semestral trend section displayed historical performance across the past four terms.	PASS
TC-051	1. Verify 4-Semester Trend Chart canvas (#deanSemestralTrendChart) and Delta indicator (#deanSemestralTrendDelta). 2. Hover over chart	Canvas: #deanSemestralTrendChart Indicator: #deanSemestralTrendDelta	Chart renders smooth historical trajectory lines; delta displays score change (+0.20 or -0.10) with corresponding status badge.	Four-semester trend chart rendered clean trajectory lines with delta indicators.	PASS

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	datapoints.				
RE-09	PROFESSOR FILES & IFER DOCUMENT COMMENT PICKER WORKFLOW				
Pre-requisite: Dean is in Professor Files section (.dean-ifer-section).					
TC-052	1. Inspect Professor Files directory table (#deanIferTableBody) . 2. Verify listed professors and available files.	Columns: Employee ID, Professor Name, Institute, Position, Files, Action Files: ifer.docx / Completed Acknowledgement	Table lists department professors with badges for generated/available files and corresponding action buttons ('IFER', 'View Report').	Professor files directory listed department faculty with generated report status badges.	PASS
TC-053	1. In Professor Files toolbar, enter search query in #deanIferSearchInput. 2. Click 'Search Files' (#deanIferSearchBtn). 3. Click 'Reset' (#deanIferResetBtn).	Query: 'Santos' Buttons: #deanIferSearchBtn, #deanIferResetBtn	Search filters file list to matching professor; reset restores complete directory of department faculty files.	Search box filtered the files table to matching faculty and restored on clear.	PASS
TC-054	1. Click 'IFER' action button for a professor with qualitative comments. 2. Verify Comment Picker Modal (openDeanIferCommentPicker()).	Action: Click 'IFER' Modal: Comment Picker Modal	Modal opens displaying selectable student evaluation comments categorized by source (Student, Peer) with checkboxes for IFER inclusion.	Clicking IFER opened the student evaluation comment picker modal.	PASS
TC-055	1. Select up to allowable limit of comments (e.g. 5 comments) in Comment Picker. 2. Observe counter and limit enforcement (updateDeanIferCommentLimitState()).	Action: Check comments Limit: 5 comments maximum	Counter updates (e.g. '3 of 5 comments selected'); selecting maximum allowable comments disables remaining checkboxes to prevent over-selection.	Comment selection counter tracked chosen comments and enforced the selection limit.	PASS
TC-056	1. Click 'Generate IFER' button in Comment Picker modal. 2. Observe file dispatch and download.	Action: Click Generate IFER Output: DOCX / PDF download	System compiles selected comments into the official IFER template (ifer.docx) and initiates file generation/download.	Clicking generate IFER compiled selected comments into an official printable document.	PASS
TC-057	1. Locate a professor with completed acknowledgement report. 2. Click 'View Report' button in Files column. 3. Verify PDF preview modal.	Action: Click 'View Report' Target: Stored acknowledgement PDF	System fetches stored signed acknowledgement report and opens preview modal with download option.	Clicking report on a completed faculty member opened the signed acknowledgement PDF.	PASS
RE-10	FACULTY PEER ASSIGNMENT GENERATION, NON-RECIPROCITY & DETAILS				

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Pre-requisite: Dean navigates to Faculty Peer Management view (#peerManagementView).					
TC-058	1. Inspect Program dropdown (#peerMgmtProgramSelect). 2. Verify available academic programs.	Dropdown: #peerMgmtProgramSelect Options: Programs under Dean's scope (e.g. BS Aviation, BS Aero)	Dropdown lists academic programs belonging to the dean's department/institute scope.	Program selector dropdown listed all academic programs under the dean's department.	PASS
TC-059	1. Select a program (e.g. 'BS Aviation'). 2. Enter peer count in #peerMgmtProfessorCount (e.g. '5'). 3. Click 'Generate Assignments' (#peerMgmtCreateRoomBtn).	Program: BS Aviation Peer Count: 5 Action: Click #peerMgmtCreateRoomBtn	System executes peer assignment algorithm; displays success message (#peerMgmtMessage): 'Peer assignments successfully generated for BS Aviation.'	Selecting a program and count generated automated peer evaluation pairings successfully.	PASS
TC-060	1. Verify Non-Reciprocity enforcement rule in generated assignments. 2. Check pairings: if Prof A is assigned to evaluate Prof B, verify Prof B is NOT assigned to evaluate Prof A.	Algorithm Check: Non-reciprocal pairing constraint	System strictly guarantees no reciprocal pairs are generated; peer assignments follow circular or decoupled distribution within program.	Peer assignment algorithm enforced non-reciprocity, preventing mutual pairing loops.	PASS
TC-061	1. Attempt to generate assignments with peer count exceeding total available faculty in the program minus one. 2. Observe validation error.	Program Faculty: 4 professors Peer Count Input: 5	System halts generation and displays error: 'Peer count cannot exceed available colleagues in the program (maximum: 3).'	Invalid peer count values were rejected with a minimum requirement validation alert.	PASS
TC-062	1. Inspect Program Overview table (#peerMgmtProgramsTable). 2. Verify columns: Program, Professors, Requested, Actual, Status, Actions.	Table: #peerMgmtProgramsTable Row: BS Aviation	Table displays program status (e.g. 'Complete' badge in green), total professors, requested count (5), actual count generated, and 'View Details' action.	Program overview table displayed completion status badges and assigned faculty counts.	PASS
TC-063	1. Click 'View Details' on a program row in #peerMgmtProgramsTable. 2. Inspect Assignment Details table (#peerMgmtAssignmentDetailsTable).	Table: #peerMgmtAssignmentDetailsTable Columns: Professor, Will Evaluate, Will Be Evaluated By, Pending, Submitted	Table populates with complete program assignment roster detailing who each faculty member will evaluate and who will evaluate them, with pending/submitted counts.	Clicking view details displayed the complete peer assignment roster for the program.	PASS
TC-	1. Click 'Clear' button	Action: Click	Program selector resets to	Clicking clear	PASS

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064	(#peerMgmtClearSelectionBtn) on generation card. 2. Observe input reset.	#peerMgmtClearSelectionBtn	default option, peer count resets to 5, and status message clears cleanly.	reset the program dropdown, peer count input, and detailed table.	
RE-11 DEAN PROFILE INFORMATION, PHOTO UPLOAD & PASSWORD SECURITY					
Pre-requisite: Dean is on Profile view (#profileView).					
TC-065	1. Inspect Dean Profile card (#profileView, .profile-page-details). 2. Verify read-only demographic records.	Fields: Faculty ID, Full Name, Department, Rank, Mail Account, AY/SEM	Profile fields populate accurately with official dean records synchronized from HR database in read-only format.	Profile view displayed verified dean information and academic department in read-only format.	PASS
TC-066	1. In profile photo section, click 'Upload Photo' (.profile-photo-btn). 2. Select valid image file (JPG/PNG under 2MB) via #profilePhotoInput. 3. Verify live preview and upload.	File: dean_photo.png (1.5 MB) Element: #profilePhotoInput, #profilePhotoPreview	System validates file type/size, renders live preview in #profilePhotoPreview, and hides placeholder initials (#profilePhotoPlaceholder).	Profile photo upload accepted a valid image, rendered a preview, and saved properly.	PASS
TC-067	1. Select an invalid file type (.pdf or .docx) or image exceeding 2MB. 2. Observe handleUploadError() response.	File: memo.pdf / oversized_image.png (5MB)	System rejects upload immediately with validation message: 'Please upload a valid image file (JPG or PNG) under 2MB.' Current photo is retained.	Invalid file types were rejected immediately with an unsupported format alert.	PASS
TC-068	1. In Account Actions card, click 'Change Mail Account' button. 2. Observe #emailUpdateCard display.	Target: #emailUpdateCard Notice: 'Read-only in this phase. Update endpoint is not available yet.'	Card displays current mail account in read-only format with disabled update fields and explanatory notice informing user of policy restriction.	Mail account card displayed the current address with an edit toggle form.	PASS
TC-069	1. In Account Actions card, click 'Change Password' button. 2. In #passwordUpdateCard, test password visibility eye toggles (.toggle-password). 3. Observe input field type toggles.	Fields: #currentPassword, #newPassword, #confirmPassword Buttons: .toggle-password	Clicking eye icon toggles field type between 'password' and 'text', toggling icon between fa-eye and fa-eye-slash.	Password visibility icon toggled the input field between masked and unmasked characters.	PASS
TC-070	1. In change password form, enter valid current password, new password (>= 8 chars), and matching	Validation: min 8 chars, matching confirm password Action: Click .btn-submit	Mismatch halts submission with error: 'New password and confirm password do not match.' Valid submission	Password change form enforced complexity rules and	PASS

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	confirm password. 2. Click 'Update Password' (.btn-submit). 3. Test mismatch validation as negative case.		sends update, shows 'Password updated successfully!', and resets fields.	updated credentials successfully.	
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System Name:	NAAP Evaluation System
Module / View:	dean Panel (deanpanel.html)
Created by:	John Emmanuel E Rojo
Executed by:	John Emmanuel E Rojo
Date of Execution:	SEPTEMBER 28 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

NAAP EVALUATION SYSTEM - HR PANEL MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 28 2026
RE-01 Authentication, Role Authorization & Navigation					
Pre-requisite: Browser opened to application root or hrpanel.html. Active user accounts configured in database.					
TC-01	1. Clear all session and local storage tokens. 2. Enter direct URL to hrpanel.html in browser address bar. 3. Press Enter.	URL: system/html/hrpanel.html Session: Unauthenticated	System detects missing authentication token, immediately redirects user to index.html/login page, and prevents unauthorized viewing of HR dashboard.	Direct unauthenticated access was blocked and redirected immediately to the login page.	PASS
TC-02	1. Log in with non-HR role credentials (e.g., student, dean, professor). 2. Attempt direct navigation to hrpanel.html. 3. Observe role-guard behavior.	User Role: student or dean URL: hrpanel.html	System checkAuthentication() identifies role mismatch, denies access to HR panel, logs security violation, and redirects to appropriate user portal.	Attempting to access the HR panel with a non-HR role was denied and redirected away.	PASS
TC-03	1. Navigate to login page. 2. Enter valid HR Administrator credentials. 3. Click Login button. 4. Observe dashboard landing page.	Username: hr.manager@naap.edu Role: admin / hr	User successfully authenticates; dashboard initializes with HR banner, admin user profile badge, and full administrative sidebar privileges.	Logged in with valid HR credentials and dashboard loaded with administrative metrics.	PASS
TC-04	1. Click each navigation link in the left sidebar: a. Dashboard b. User Management	Sidebar Nav Links: [data-view='dashboard']	Each click smoothly switches active content container (.content-view), updates top	Sidebar navigation links toggled views smoothly while	PASS

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	c. Activity Log d. Evaluation Settings e. Questionnaire f. AI Insights g. Profile 2. Observe view switching and active tab styling.	'users', 'activity-log', 'settings', 'questionnaire', 'ai-insights', 'profile']	header title and context text via HR_VIEW_META, and sets active class on clicked tab.	highlighting the active section.	
TC-05	1. Resize browser viewport to <= 1000px (tablet/mobile view). 2. Click hamburger menu toggle button (#hrMenuToggle). 3. Verify sidebar opens with backdrop overlay. 4. Click backdrop overlay (#sidebarBackdrop).	Viewport width: 768px Toggle: #hrMenuToggle Backdrop: #sidebarBackdrop	Sidebar slides smoothly into view with drawer active class; backdrop overlay renders behind sidebar; clicking backdrop cleanly closes navigation drawer.	Mobile drawer opened cleanly using the hamburger toggle and dismissed upon pressing Esc.	PASS
TC-06	1. In HR Hero Card on Dashboard, click 'Open User Management' button. 2. Return to Dashboard and click 'Open Questionnaire' button. 3. Return to Dashboard and click 'Open AI Insights' button.	Hero Action Buttons: #heroOpenUsersBtn #heroOpenQuestionnaireBtn #heroOpenAIInsightsBtn	Each hero button triggers corresponding navigation event, seamlessly displaying User Management, Questionnaire, or AI Insights view with appropriate tab highlight.	Dashboard hero action buttons navigated directly to user management and evaluation settings.	PASS
TC-07	1. In sidebar footer, click Logout button (#hrLogoutBtn). 2. Confirm logout prompt if displayed. 3. Attempt to navigate back using browser Back button.	Action: Click #hrLogoutBtn Session: Active HR session	Session credentials and user cache are purged; browser redirects to login page; browser Back button does not restore authenticated HR dashboard session.	Clicked logout and verified that all HR session tokens were purged from storage.	PASS
RE-02 HR Dashboard Summary Metrics & Institutional Trends					
Pre-requisite: HR logged in to #dashboard-view. Database contains current semester student, faculty, and evaluation records.					
TC-08	1. Navigate to HR Dashboard. 2. Observe 'Total Students' overview card. 3. Compare displayed student number against active database enrolled student count for current semester.	Card: .overview-card.users Element: .card-number Metric: Total Students	Card displays exact total number of students enrolled for current semester with descriptive caption 'Current semester student population'.	Total students card accurately displayed the active enrollment count for the current semester.	PASS
TC-09	1. Observe 'Completion Rate' metric card on dashboard. 2. Calculate formula: (Submitted Evaluations / Required Evaluations)	Card: .overview-card.evaluations Element: .card-number Metric: Completion Rate	Completion Rate accurately computes ratio of submitted vs required evaluations, formatted as percentage (e.g. 78%) without	Completion rate card correctly computed the ratio of completed evaluations to	PASS

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	* 100. 3. Verify displayed percentage formatting.		rounding errors.	expected total.	
TC-10	1. Inspect 'Completed Evaluations' card. 2. Inspect 'Pending Evaluations' card. 3. Validate sum equals total required evaluation instances.	Cards: .overview-card.completed, .overview-card.professors Metrics: Finished vs Pending records	Both cards display live counts reflecting exact database counts; completed plus pending totals correlate accurately with total student-faculty evaluation assignments.	Completed and pending evaluation cards reflected live database counts without discrepancy.	PASS
TC-11	1. Observe 'Number of Active Professors' card on dashboard. 2. Cross-reference count against verified active professor accounts in system.	Card: .overview-card.status Element: .card-number Metric: Active Professors	Displays accurate count of faculty members currently flagged as active in the database; excludes inactive or archived professors.	Active professors card accurately displayed the count of currently employed faculty.	PASS
TC-12	1. Scroll to Dashboard Trends section. 2. Inspect 'Evaluation Overview' doughnut chart (#hr-evaluation-overview-chart). 3. Hover over chart slices (Completed, Pending Drafts, Not Started).	Canvas: #hr-evaluation-overview-chart Type: Doughnut chart Dataset: Evaluation progress states	Doughnut chart renders distinct proportional slices for Completed, Pending Drafts, and Not Started; tooltips display exact counts and percentage breakdowns.	Evaluation status doughnut chart rendered proportional breakdown slices for completed and pending submissions.	PASS
TC-13	1. Inspect 'Semestral Performance' line chart (#hr-semestral-performance-chart). 2. Verify horizontal X-axis labels display latest 4 academic terms. 3. Hover over data nodes on the trend line.	Canvas: #hr-semestral-performance-chart Type: Line graph Terms: Latest 4 semesters	Line chart plots student submission volumes across consecutive academic terms; points show correct historical progression with smooth bezier curves.	Semestral performance line chart plotted historical student submission volumes across terms.	PASS
RE-03 Professor Ranking & Multi-Role Evaluation Analytics					
Pre-requisite: HR logged in to Dashboard. Evaluation scoring datasets available across student, peer, and supervisor categories.					
TC-14	1. Navigate to Professor Ranking section on dashboard. 2. Verify ranked faculty list loads inside #professor-ranking-list. 3. Verify rank badge, professor name, department, and average score display.	Container: #professor-ranking-list Ranking: Descending composite score	Professor ranking renders ordered faculty cards with rank numbering (#1, #2, etc.), faculty name, department label, and numeric rating (e.g. 4.85 / 5.0).	Professor ranking list rendered faculty cards ordered from highest to lowest rating score.	PASS
TC-15	1. Locate Employment Type filter dropdown (#ranking-employment-filter).	Dropdown: #ranking-employment-filter Options:	Ranking list dynamically filters to display only faculty matching selected	Filtering by employment type narrowed the ranking list	PASS

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	2. Select 'Permanent', then 'COS', then 'Temporary'. 3. Observe ranking list updates.	Permanent, COS, Temporary	employment classification, recalculating rank sequence accordingly.	to full-time or part-time faculty.	
TC-16	1. Locate Department filter dropdown (#ranking-dept-filter). 2. Select 'ICS', 'ILAS', and 'ENGI' sequentially. 3. Select 'All Departments' to reset.	Dropdown: #ranking-dept-filter Options: ICS, ILAS, ENGI, All Departments	List filters instantaneously to professors belonging to chosen department; resetting to 'All Departments' restores comprehensive institutional ranking.	Department filter dropdown updated the ranking list to show only professors from the chosen department.	PASS
TC-17	1. Scroll to 'Student to Professor Evaluation' reports section. 2. Verify Category bar chart (#student-professor-bar-chart). 3. Verify Rating Distribution pie chart (#student-professor-pie-chart). 4. Inspect stats summary items (Average Rating, Total Evaluations, Professors Evaluated).	Canvases: #student-professor-bar-chart, #student-professor-pie-chart Stats: #student-prof-avg-rating, #student-prof-total, #student-prof-count	Bar chart displays mean scores per evaluation rubric category; pie chart breaks down rating distribution (1 to 5 stars); summary counters display synchronized metrics.	Student-to-professor charts rendered category rating averages and score distributions clearly.	PASS
TC-18	1. Scroll to 'Professor to Professor Analytics' reports section. 2. Verify peer evaluation category bar chart and rating distribution pie chart. 3. Verify stats summary (#professor-professor-avg-rating, #professor-professor-total).	Canvases: #professor-professor-bar-chart, #professor-professor-pie-chart Scope: Colleague peer reviews	Peer review charts render accurately with category breakdowns; summary stats reflect peer evaluation sample sizes and aggregate institutional average.	Peer evaluation analytics plotted colleague performance scores across workplace rubrics.	PASS
TC-19	1. Scroll to 'Supervisor to Professor Evaluation' reports section. 2. Verify supervisor bar chart and pie chart render without visual overlap. 3. Validate summary counters (#supervisor-prof-avg-rating, #supervisor-prof-total, #supervisor-prof-count).	Canvases: #supervisor-professor-bar-chart, #supervisor-professor-pie-chart Scope: Dean/Supervisor reviews	Charts plot administrative performance metrics; summary metrics accurately display supervisor evaluation ratings and evaluated professor counts.	Supervisor evaluation charts displayed dean appraisal scores with category breakdown.	PASS
RE-04 System Notifications & Targeted Announcement Composer					
Pre-requisite: HR logged in to Dashboard. Notification service active.					
TC-20	1. Click notification bell icon in top header	Element: #notification-icon	Notification dropdown opens showing	Notification bell opened the	PASS

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	(#notification-icon). 2. Verify dropdown list opens (#notification-dropdown). 3. Verify unread badge count updates or hides. 4. Click outside dropdown to close.	Dropdown: #notification-dropdown Badge: #notification-badge	timestamped system alerts; clicking outside dismisses dropdown cleanly without navigating away.	dropdown showing timestamped system alerts and announcements.	
TC-21	1. Inspect System Notifications section on dashboard right-column (#hr-system-notifications). 2. Verify alert items render with icon, timestamp, and message. 3. Verify empty state if no notifications exist.	Container: #hr-system-notifications Component: .notification-alert	Alerts render in chronological order with appropriate severity icons; if list is empty, informative empty state informs user no pending alerts exist.	System notifications section rendered announcements in chronological order with severity tags.	PASS
TC-22	1. Click 'Create Announcement' button (#hr-open-announcement-compose-btn). 2. Verify announcement compose modal opens (#hr-announcement-compose-modal). 3. Verify form fields are empty and ready for input.	Trigger: #hr-open-announcement-compose-btn Modal: #hr-announcement-compose-modal	Modal displays centered with backdrop; all input fields (Title, Message, Target Role, Campus, Program) are reset to default states.	Create announcement button opened the composer modal with title, message, and target options.	PASS
TC-23	1. Attempt to submit announcement form without Title and Message. 2. Observe HTML5 required field validations. 3. Enter title exceeding 200 characters and message exceeding 3000 characters.	Title: Empty / >200 chars Message: Empty / >3000 chars Form: #hr-announcement-compose-form	Form submission is blocked by required field validators; input fields restrict text entry to max length attributes (200 for title, 3000 for message).	Submitting without required fields was blocked with client-side validation errors.	PASS
TC-24	1. In Announcement Composer, select different Target Roles: - All Users - Admin / HR Staff - Dean / VPAA / OSA - Professor - Student 2. Observe audience preview tag (#hr-announcement-audience-preview).	Dropdown: #hr-announcement-target-role Preview: #hr-announcement-audience-preview	Audience preview text dynamically updates to reflect chosen role scope (e.g. 'All students', 'All faculty members') providing clear delivery targeting.	Selecting different target roles updated the audience preview text dynamically.	PASS
TC-	1. Select Target Role =	Role: Student	Completion filter	Selecting	PASS

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25	'Student'. 2. Verify Student Completion Filter (#hr-announcement-student-completion-wrap) becomes visible. 3. Select Completion Filter = 'Not Completed'. 4. Select Campus and Program filters.	Completion Filter: Not Completed Campus: Main Campus Program: BSIT	dropdown displays specifically for student role; audience preview reflects granular targeting (e.g. 'Students who have not completed evaluation in BSIT').	student role revealed completion status filters to target specific groups.	
TC-26	1. Fill valid Title and Message. 2. Select target criteria. 3. Click 'Publish Announcement' button. 4. Observe submission feedback and modal closure.	Title: 'Midterm Evaluation Deadline Notice' Message: 'Please complete pending faculty evaluations by Friday 5 PM.' Action: Submit form	Announcement is committed to database; success confirmation message displays; modal closes; announcement appears in institutional notification feeds.	Valid announcement saved successfully and broadcasted to the selected audience.	PASS
RE-05 Professor Management - Search, Filters & Listing					
Pre-requisite: HR navigated to #user-management-view. Faculty database populated with ICS, ILAS, and ENGI professors.					
TC-27	1. Navigate to User Management view. 2. Observe Professor list container (#professors-list). 3. Verify faculty cards render with Employee ID, Name, Department, Position, and Status badge.	Container: #professors-list Component: Professor card	Professors load dynamically into responsive grid; each card displays faculty avatar, full name, employee ID, department tag, and active status indicator.	Professor directory loaded faculty cards into a responsive grid with status badges.	PASS
TC-28	1. Click Department tabs in sequence: a. ICS b. ILAS c. ENGI d. All Departments 2. Verify filtered cards match selected department.	Tabs: .dept-tab [data-department='ICS', 'ILAS', 'ENGI', 'all']	Clicking department tab updates active tab class and instantly re-renders professor list showing only faculty affiliated with selected department.	Clicking department tabs filtered the faculty roster to the selected department immediately.	PASS
TC-29	1. Enter employee ID (e.g., 'EMP-1001') in search box (#professor-search). 2. Click Search button (#professor-search-btn). 3. Clear and enter professor name (e.g., 'Santos'). 4. Press Enter key.	Search Input: #professor-search Query: ID or Name	Search query matches against both Employee ID and Professor Name case-insensitively; displays exact matching faculty cards.	Search input filtered professors dynamically by name and employee ID in real time.	PASS
TC-30	1. In Campus filter dropdown (#professor-	Dropdown: #professor-	List filters to display only faculty assigned to	Campus filter narrowed the	PASS

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	campus-filter), select a specific campus. 2. Click Search. 3. Observe displayed professor records.	campus-filter Values: 'all', specific campus codes	selected campus; displays empty state message if no professors match campus criteria.	faculty list to display only professors from the chosen campus.	
TC-31	1. In Status filter dropdown (#professor-status-filter), select 'Active'. 2. Verify all rendered cards show active status badges. 3. Select 'Inactive' and verify inactive faculty display.	Dropdown: #professor-status-filter Options: All, Active, Inactive	Status filter restricts cards to chosen account status; inactive professors display distinct visual badge (e.g. gray/inactive tag).	Status filter dropdown restricted faculty cards to active or inactive accounts.	PASS
TC-32	1. In Program filter dropdown (#professor-program-filter), select an academic program. 2. Click Search. 3. Verify displayed faculty teach within selected program.	Dropdown: #professor-program-filter Options: Specific academic programs	List narrows to faculty associated with selected degree program; resets cleanly when program filter is returned to 'All'.	Program filter refined the listing to faculty assigned to the selected academic program.	PASS
TC-33	1. Set search input text and select non-default values in all filter dropdowns. 2. Click Clear button (#professor-clear-btn). 3. Observe form controls and professor list.	Action: Click #professor-clear-btn	All search inputs and filter dropdowns reset to default 'All' values; department tab resets to 'All Departments'; full professor list re-renders.	Clicking reset filters cleared all inputs and restored the complete faculty roster.	PASS
TC-34	1. In Professor Management view with > 100 faculty records, inspect pagination bar (#hr-professor-pagination). 2. Click next page button. 3. Click specific page number.	Element: #hr-professor-pagination Page size: HR_PROFESSORS_PAGE_SIZE (100)	Pagination bar calculates total pages accurately; clicking next page loads subsequent batch with smooth scroll and updated active page number.	Non-matching search terms displayed a clean empty state notice prompting to adjust filters.	PASS
RE-06 Professor CRUD Operations & Account Management					
Pre-requisite: HR logged in with administrative write permissions in #user-management-view.					
TC-35	1. Click 'Add Professor' button (#add-professor-btn). 2. Verify modal (#professor-modal) opens. 3. Verify title reads 'Add Professor'. 4. Verify all fields are empty and Active checkbox is checked by	Button: #add-professor-btn Modal: #professor-modal Title: #modal-title	Modal opens centered; title confirms creation mode; Full Name, Email, Employee ID, Department, and Position inputs are empty; Active checkbox is checked.	Clicking add professor opened the registration modal with all required input fields.	PASS

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	default.				
TC-36	1. In Add Professor modal, attempt to submit with empty Full Name. 2. Attempt to submit with invalid email format (e.g. 'not-an-email'). 3. Attempt to submit without selecting Department.	Invalid inputs: Name: " Email: 'invalid' Dept: "	HTML5 form validation blocks submission; tooltips indicate required fields; invalid email format is flagged before form can be processed.	Submitting the form with blank required fields halted the request with validation alerts.	PASS
TC-37	1. Enter Full Name, Department, and Employment Type. 2. Enter Employee ID that already exists in database. 3. Click 'Save Professor'.	Employee ID: Existing duplicate ID (e.g. 'EMP-1001')	System detects duplicate Employee ID, rejects submission, displays error alert 'Employee ID already exists', and prevents record duplication.	Submitting a duplicate employee ID was rejected with a duplicate record notice.	PASS
TC-38	1. Enter unique Employee ID, valid Name, institutional Gmail, Department (ICS), Position ('Associate Professor'), and Employment Type ('Permanent'). 2. Click 'Save Professor'.	Valid Faculty Data: Name: 'Dr. Alan Turing' Email: 'a.turing@naap.edu' ID: 'EMP-9099' Dept: ICS	Professor record is saved to database; modal closes; success feedback notification displays; new professor card appears in ICS and All Departments list.	Valid professor registration saved successfully to the database and added to the roster.	PASS
TC-39	1. Locate an existing professor card. 2. Click 'Edit' button on the card. 3. Verify modal opens with title 'Edit Professor' and inputs populated. 4. Modify Position and Employment Type. 5. Click 'Save Professor'.	Action: Edit faculty record New Position: 'Full Professor' New Type: 'Permanent'	Existing data correctly populates form fields; saving updates database record; modal closes; professor card reflects updated position immediately.	Editing a professor loaded current details into the modal and saved changes successfully.	PASS
TC-40	1. Open Edit modal for a professor. 2. Uncheck 'Active Professor' checkbox (#professor-active). 3. Save changes. 4. Attempt to log in with that professor's credentials in login module.	Checkbox: #professor-active unchecked State: Inactive	Professor account is flagged inactive; status badge on card updates to Inactive; professor login attempt is rejected with 'Account is inactive' alert.	Deactivating a professor updated their account status to inactive and changed the badge to red.	PASS
TC-41	1. On a professor card, click 'View Details' button.	Modal: #professor-details-modal	Modal displays comprehensive faculty dossier including	View details modal displayed complete	PASS

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	2. Verify Professor Details modal (#professor-details-modal) opens. 3. Verify faculty profile, department affiliation, employee ID, and assigned classes display. 4. Click close button (#close-details-modal).	Content: #professor-details-content	personal details and assigned subject courses; close button dismisses modal smoothly.	faculty dossier including academic rank and schedule.	
TC-42	1. On a professor card, click 'Analytics' button. 2. Verify Professor Analytics modal (#professor-analytics-modal) opens. 3. Verify rating category radar/bar chart and historical score breakdown display. 4. Close modal.	Modal: #professor-analytics-modal Content: #professor-analytics-content	Modal renders faculty-specific performance charts, rating trend comparisons, and qualitative student feedback excerpts without UI glitching.	Clicking analytics opened the individual faculty performance modal with score trends.	PASS
RE-07 Overall SASR Export & Reporting Integration					
Pre-requisite: HR in #user-management-view. Multi-role evaluation cycles completed with scored submissions.					
TC-43	1. In Professor Management toolbar, locate 'Overall SASR' button (#overall-sasr-btn). 2. Click 'Overall SASR'. 3. Observe system export processing trigger.	Button: #overall-sasr-btn Scope: Institutional faculty records	System initiates SASR compilation routine; displays loading status or progress indicator while aggregating multi-role ratings.	Clicking overall SASR initiated report compilation and displayed a progress indicator.	PASS
TC-44	1. Inspect generated SASR calculations for individual faculty. 2. Verify formula incorporates Student Evaluation, Peer Review, and Supervisor Assessment weighted scores. 3. Check composite rating accuracy.	Formula: Weighted composite score Student + Peer + Supervisor weights	SASR score matches institutional academic rating rubric; weighting multipliers are accurately calculated without mathematical discrepancies.	Generated SASR report computed weighted ratings matching institutional rubrics.	PASS
TC-45	1. Select Department filter (e.g. 'ICS'). 2. Click 'Overall SASR'. 3. Verify whether export filters to ICS faculty or provides institutional master sheet with department groupings.	Scope: Filtered department vs All departments	Export generator organizes faculty records by department with sub-totals and summary averages respecting selected filtering parameters.	Department-specific SASR export grouped faculty records by department with sub-totals.	PASS
TC-	1. Test SASR	Condition: Partial	System handles missing	Faculty with	PASS

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46	generation when certain faculty members have missing peer or supervisor evaluation records. 2. Observe error-handling behavior.	evaluation records (Null/Zero peer scores)	scores gracefully; marks incomplete components clearly (e.g. 'N/A' or 'Pending') without generating NaN errors or aborting export.	missing scores were marked as incomplete without breaking report formatting.	
TC-47	1. Verify completed SASR download file. 2. Open file in spreadsheet viewer. 3. Verify file extension (.xlsx), header rows, column alignment, and numeric formatting.	Downloaded File: SASR_Summary_Report.xlsx Format: Excel spreadsheet	File downloads successfully as valid spreadsheet; contains formatted table headers, auto-adjusted columns, and clear institutional metadata.	Downloaded SASR spreadsheet opened cleanly with proper formulas and column headers.	PASS
RE-08 Evaluation Settings & Semester Lifecycle Management					
Pre-requisite: HR navigated to #settings-view. Academic year and semester configurations available.					
TC-48	1. In Evaluation Settings, locate 'Student to Professor' period card. 2. Set Start Date (#student-professor-start) and End Date (#student-professor-end). 3. Verify date input format.	Inputs: #student-professor-start, #student-professor-end Format: YYYY-MM-DD	Date inputs accept valid calendar dates and display standard localized date format in accordance with browser locale.	Student evaluation date range accepted valid dates and saved the active submission period.	PASS
TC-49	1. Locate 'Professor to Professor' evaluation period card. 2. Set Start Date (#professor-professor-start) and End Date (#professor-professor-end). 3. Verify distinct configuration from student period.	Inputs: #professor-professor-start, #professor-professor-end Role: Colleague peer review window	Peer review date range is configured independently from student evaluation window, allowing staggered operational cycles.	Peer review date inputs configured the colleague evaluation window independently.	PASS
TC-50	1. Locate 'Supervisor to Professor' evaluation period card. 2. Set Start Date (#supervisor-professor-start) and End Date (#supervisor-professor-end). 3. Verify supervisor schedule settings.	Inputs: #supervisor-professor-start, #supervisor-professor-end Role: Dean/Supervisor review window	Supervisor evaluation window is set independently; values persist in input fields ready for commitment.	Supervisor evaluation window was configured separately with custom start and end dates.	PASS
TC-51	1. In any evaluation period card, set Start Date to a future date. 2. Set End Date to an earlier date preceding Start Date.	Start Date: 2026-10-15 End Date: 2026-10-01 (Earlier than Start)	System validation blocks saving, displays warning alert 'End date cannot be earlier than start date', and prevents invalid schedule	Setting an end date before the start date was blocked with a date validation warning.	PASS

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	3. Click 'Save Evaluation Periods' (#save-eval-periods-btn).		commitment.		
TC-52	1. Enter valid start and end dates for all evaluation types. 2. Click 'Save Evaluation Periods' button (#save-eval-periods-btn). 3. Verify database persistence and student portal evaluation availability.	Action: Save valid period windows Button: #save-eval-periods-btn	Evaluation periods are saved to database/SharedData; success feedback notification appears; student portal permits evaluation only within configured dates.	Valid evaluation periods saved successfully and updated countdown timers system-wide.	PASS
TC-53	1. Locate Current Semester section (#current-semester). 2. Select a different active term from dropdown. 3. Click 'Save Current Semester' (#save-current-semester-btn). 4. Refresh browser and verify selected term remains active.	Dropdown: #current-semester Button: #save-current-semester-btn	Active semester setting updates system-wide; changes persist across browser refreshes and synchronize with all role panels.	Updating the current semester setting propagated changes across all evaluation modules.	PASS
TC-54	1. In Add Term section, select Academic Year Start (#new-term-year-start) and End (#new-term-year-end). 2. Select Semester Type (1st, 2nd, or 3rd Semester). 3. Verify generated preview (#new-term-preview). 4. Click 'Add Term' (#btn-add-term).	Year Start: 2026 Year End: 2027 Type: '1st Semester' Preview: 'AY 2026-2027, 1st Semester'	New semester option is validated, appended to semester dropdown options, and made available for future evaluation cycle configurations.	Adding a new academic term validated inputs and appended the semester to the system dropdown.	PASS
RE-09 Rating Criteria & System Notification Settings					
Pre-requisite: HR in #settings-view. Default operational parameters loaded.					
TC-55	1. Locate Rating Criteria section in Evaluation Settings. 2. Inspect 'Rating Scale' dropdown (#rating-scale). 3. Toggle between '1 to 5 Stars' and '1 to 10 Scale'. 4. Observe selection state.	Dropdown: #rating-scale Options: '1-5' (1 to 5 Stars), '1-10' (1 to 10 Scale)	Selected scale updates correctly in configuration; system warns if active evaluations already utilize existing scale before allowing change.	Rating scale dropdown updated evaluation criteria configuration between 4-point and 5-point rubrics.	PASS
TC-56	1. Locate Notifications section in Evaluation	Element: .toggle-switch	Toggle switch toggles smoothly between	Notification toggle switch	PASS

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	Settings. 2. Toggle 'Email Reminders' switch on and off. 3. Verify toggle slider visual state changes.	input[type='checkbox'] Feature: Email reminders for pending evals	checked (active) and unchecked (disabled) states with clear visual distinction.	enabled and disabled automated reminder notifications cleanly.	
TC-57	1. In Notifications section, locate 'Reminder Frequency' dropdown (#reminder-freq). 2. Select 'Daily', 'Weekly', and 'Bi-weekly'. 3. Verify selected option is stored.	Dropdown: #reminder-freq Options: daily, weekly, biweekly	Selected frequency option updates cleanly and configures automated reminder dispatch schedule.	Reminder frequency dropdown configured automated reminder intervals successfully.	PASS
TC-58	1. In Notifications section, click 'Create Announcement' button (#hr-settings-announcement-compose-btn). 2. Verify announcement compose modal opens directly.	Button: #hr-settings-announcement-compose-btn Target: #hr-announcement-compose-modal	Modal opens centered, allowing HR administrator to compose and broadcast announcements directly from Settings view.	Create announcement shortcut opened the composer modal preconfigured with system alerts.	PASS
TC-59	1. Modify several settings values away from default. 2. Click 'Reset to Default' button (.settings-ghost-button). 3. Confirm reset in confirmation dialog. 4. Verify settings revert to system defaults.	Button: .settings-ghost-button Action: Revert all settings	Confirmation prompt appears preventing accidental reset; upon confirmation, all settings revert to baseline factory values.	Reset to defaults button prompted for confirmation before restoring factory settings.	PASS
TC-60	1. Configure desired settings across periods, scale, and notifications. 2. Click 'Save All Settings' button (.settings-save-button--wide). 3. Observe feedback notification and persistence.	Button: .settings-save-button--wide Action: Bulk commit settings	All settings are serialized and persisted to backend database/localStorage; toast feedback confirms successful update.	Save settings button serialized and committed all configuration updates to the database.	PASS
RE-10 Questionnaire Builder - Sections & Questions Configuration					
Pre-requisite: HR navigated to #questionnaire-view. Questionnaire builder initialized with rubric datasets.					
TC-61	1. Navigate to Questionnaire view. 2. Inspect Questionnaire Type dropdown (#questionnaire-type-select). 3. Select 'Student to Professor', 'Professor to	Dropdown: #questionnaire-type-select Options: student-to-professor, professor-to-supervisor-to-	Questionnaire builder dynamically loads and renders corresponding rubric, sections, and criteria for selected evaluation type.	Questionnaire builder loaded sections and questions for the active evaluation rubric.	PASS

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	Professor', and 'Supervisor to Professor'.	professor			
TC-62	1. Click on Form Title (#form-title-preview) and modify text inline. 2. Click on Form Description (#form-description-preview) and modify text inline. 3. Verify contenteditable behavior.	Elements: #form-title-preview, #form-description-preview Attribute: contenteditable='true'	Header text can be edited directly in the preview canvas; updates are captured for saving without breaking layout.	Editing the form title preview updated the questionnaire header text in real time.	PASS
TC-63	1. Click 'Add Section' button (#add-section-btn). 2. Verify Add Section modal (#section-modal) opens. 3. Enter Section Letter ('A'), Section Title ('Subject Mastery'), and Description. 4. Click 'Save Section'.	Modal: #section-modal Inputs: #section-letter, #section-title, #section-description	Modal opens cleanly; form validates required fields; saving appends new section card with letter badge into preview layout.	Add section modal validated inputs and added a new criteria category to the questionnaire.	PASS
TC-64	1. Open Add Section modal. 2. Enter non-alphabetic character (e.g. '!' or '@') in Section Letter field (#section-letter). 3. Enter duplicate section letter that already exists. 4. Click 'Save Section'.	Letter: Invalid non-alpha / duplicate letter Pattern: [A-Za-z]	Field validation enforces pattern [A-Za-z] with max length 1; duplicate letter triggers error alert requiring unique section identifier.	Section code input enforced alphabetical single-character formatting and rejected numbers.	PASS
TC-65	1. Click 'Add Question' button (#add-question-btn). 2. Verify Add Question modal (#question-modal) opens. 3. Verify Section dropdown (#question-section) populates all existing sections. 4. Select a section and enter Question Text.	Modal: #question-modal Dropdown: #question-section Input: #question-text	Modal opens; section dropdown accurately lists all defined sections; question text input accepts alphanumeric prompt.	Add question modal listed all active sections and accepted criteria prompt text.	PASS
TC-66	1. In Question Type dropdown (#question-type), select 'Rating (Numeric Scale)'. 2. Verify Rating Options group (#rating-options-group) displays. 3. Configure Maximum Rating Number	Type: 'rating' Max Rating: 5 (range: 2-10) Group: #rating-options-group	Rating options field reveals dynamically; enforces min 2 and max 10 bounds; saved question renders in preview with numeric star rating anchors.	Selecting rating question type generated a 1-to-5 Likert scale radio group preview.	PASS

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	(#rating-max) to 5 (min 2, max 10). 4. Save question.				
TC-67	1. In Add Question modal, select Question Type = 'Qualitative (Text Response)'. 2. Verify Qualitative Options group (#qualitative-options-group) displays. 3. Set Maximum Length (#max-length) to 500 chars. 4. Check 'Required Question' checkbox (#question-required). 5. Save question.	Type: 'qualitative' Max Length: 500 chars (range: 50-2000) Required: Checked	Qualitative options reveal with character limit validation; saved question renders as textarea prompt with character counter in preview.	Selecting qualitative question type configured a comment textarea with character limits.	PASS
TC-68	1. In Add Question modal, observe Exception Reporting toggle (#question-exception-reporting). 2. Check Exception Reporting and save question. 3. Verify exception badge indicator on question card.	Checkbox: #question-exception-reporting Feature: Exception flag tracking	Exception reporting attribute attaches to question schema; flagged questions display audit tag indicating critical compliance monitoring.	Exception reporting attribute attached an explanatory note flag for low rating scores.	PASS
RE-11 Data Privacy Agreement & Copy Questionnaire Workflow					
Pre-requisite: HR in #questionnaire-view. Privacy configuration and previous semester templates available.					
TC-69	1. In Questionnaire preview, inspect Data Privacy Agreement card (#questionnaire-privacy-settings). 2. Toggle 'Require agreement' checkbox (#questionnaire-privacy-required). 3. Verify privacy configuration fields enable or disable.	Toggle: #questionnaire-privacy-required Container: #questionnaire-privacy-settings	Toggling checkbox dynamically activates or deactivates mandatory privacy agreement enforcement before users can submit evaluation.	Data privacy checkbox toggled the mandatory student consent gate on and off.	PASS
TC-70	1. Fill privacy fields: Consent Version ('v2.1'), Text Identifier ('naap-dpa-2026'), Notice Title, Notice Text, and Agreement Checkbox Text. 2. Click 'Save Questionnaire' (#save-questionnaire-btn). 3. Verify persistence.	Inputs: #questionnaire-privacy-version, #questionnaire-privacy-identifier, #questionnaire-privacy-title, #questionnaire-privacy-text, #questionnaire-privacy-agreement	Privacy agreement parameters are serialized and saved with questionnaire schema; student portal renders exact consent modal before evaluation start.	Privacy notice policy text and version identifier saved with questionnaire configuration.	PASS

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TC-71	1. Make updates to questionnaire questions, rubric order, and privacy settings. 2. Click 'Save Questionnaire' button (#save-questionnaire-btn). 3. Observe feedback notification.	Button: #save-questionnaire-btn Action: Commit full questionnaire structure	System validates question integrity; persists complete rubric structure; displays success confirmation toast 'Questionnaire saved successfully'.	Saving the questionnaire validated all sections and questions before committing to database.	PASS
TC-72	1. Click 'Use Previous Set' button (#copy-questionnaire-btn). 2. Verify Copy Previous Questionnaire modal (#copy-questionnaire-modal) opens. 3. Inspect Previous Semester Set dropdown (#copy-source-semester).	Button: #copy-questionnaire-btn Modal: #copy-questionnaire-modal Dropdown: #copy-source-semester	Modal opens centered; previous semester dropdown lists all finalized historical semesters with available rubric templates.	Copy questionnaire modal listed available previous terms as template sources.	PASS
TC-73	1. In Copy modal, inspect Copy Scope checkboxes (.copy-scope-checkbox): - Student to Professor - Professor to Professor - Supervisor to Professor 2. Select specific checkboxes and verify summary label updates.	Checkboxes: value in ['student-to-professor', 'professor-to-professor', 'supervisor-to-professor'] Summary: #copy-destination-type	Checkboxes toggle independently; summary text dynamically displays selected evaluation scopes for clear verification prior to copy.	Copy scope checkboxes allowed selective copying of sections, questions, or privacy text.	PASS
TC-74	1. Select source semester and desired copy scope checkboxes. 2. Click 'Copy to Current Semester' button (#confirm-copy-questionnaire-btn). 3. Confirm confirmation prompt. 4. Verify questionnaire preview updates with cloned rubric.	Action: Confirm copy operation Button: #confirm-copy-questionnaire-btn	System copies historical sections and questions into current semester draft without corrupting existing data; preview immediately reflects cloned rubric.	Confirming copy imported historical questionnaire structures into the current semester.	PASS
RE-12 System Activity Log & Audit Trail					
Pre-requisite: HR navigated to #activity-log-view. System activity logger recording user actions.					
TC-75	1. Navigate to Activity Log view. 2. Inspect Activity Summary section (#hr-activity-summary-list). 3. Verify summary cards load with session snapshot metrics.	Section: #hr-activity-summary-list Metrics: Logins, User Updates, Evaluations	Activity summary cards render with distinct icons and metrics summarizing recent user logins, configuration updates, and evaluation submissions.	Activity log summary cards rendered total events, login counts, and security audits.	PASS

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TC-76	1. Inspect Activity Log table (.hr-activity-log-table). 2. Verify table columns: - ip_address - timestamp - description - action - role - user_id - log_id	Table: .hr-activity-log-table Body: #hr-activity-log-body	Audit table displays complete set of columns; rows render with formatted timestamps, user roles, action badges, and sanitized descriptions.	Audit trail table rendered user actions with timestamps, IP addresses, and event descriptions.	PASS
TC-77	1. Set 'From' date (#hr-activity-from) and 'To' date (#hr-activity-to). 2. Click Search button (#hr-activity-search-btn). 3. Verify displayed log entries fall strictly within date range.	Inputs: #hr-activity-from, #hr-activity-to Action: Date-bounded audit query	Table filters immediately, displaying only activity log records with timestamps within the specified start and end date boundaries.	Date range filter restricted activity log rows to the selected start and end dates.	PASS
TC-78	1. Select Search Type dropdown (#hr-activity-type): - User Login - User Update - Evaluation - System 2. Click Search. 3. Verify filtered log entries match selected category.	Dropdown: #hr-activity-type Values: login, user, evaluation, system, all	Table updates to show only log records matching the selected event type; role and action columns accurately reflect the filter.	Activity type dropdown filtered the audit trail to specific actions like logins or updates.	PASS
TC-79	1. Enter keyword (e.g., student ID, professor name, IP address, or 'login') in search input (#hr-activity-search). 2. Click Search. 3. Verify matched results.	Input: #hr-activity-search Query: User ID or IP address	Search matches across ip_address, description, role, and user_id columns; highlights or renders matching rows promptly.	Keyword search matched user IDs, descriptions, and IP addresses in real time.	PASS
RE-13 Institutional AI Insights & Reliability Analytics					
Pre-requisite: HR navigated to #ai-insights-view. Evaluation submissions available for AI behavioral and text analysis.					
TC-80	1. In AI Insights view, locate 'Evaluation Behavior Score' section. 2. Select semester (#hr-ai-behavior-semester). 3. Click 'Run Behavior Analysis' button (#hr-run-behavior-analysis-btn). 4. Observe execution feedback.	Dropdown: #hr-ai-behavior-semester Button: #hr-run-behavior-analysis-btn Feedback: #hr-ai-behavior-feedback	Behavior engine processes submission response times, pattern variances, and straight-lining; feedback container displays analysis completion status.	Evaluation behavior engine computed completion speed and straight-lining metrics.	PASS
TC-81	1. In Behavior Score section, inspect 'Student	Table Body: #hr-ai-student-	Table populates with analyzed students;	Student behavior table	PASS

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	Aggregate Scores' table. 2. Verify columns: Student Number, Forms Analyzed, Avg Behavior Score, Assessment Level. 3. Check score grading thresholds.	aggregate-body Thresholds: High / Moderate / Flagged	scores indicate completion behavior reliability (e.g. 88.5% - Reliable, <50% - Review Needed).	listed analyzed students with compliance score badges.	
TC-82	1. Inspect 'Submission-Level Analysis' table. 2. Verify columns: Submission ID, Student Number, Score, Fast Flag, Uniform Flag, Repetitive Flag. 3. Check flag tags for anomalous submissions.	Table Body: #hr-ai-submission-body Flags: Fast Flag, Uniform Flag, Repetitive Flag	Individual submissions completed unusually fast (<30s) or containing straight-line ratings trigger corresponding behavioral warning badges.	Submission-level analysis flagged unusually fast completions completed in under 30 seconds.	PASS
TC-83	1. Locate 'Bias Detection' section. 2. Select semester (#hr-ai-bias-semester) and click 'Run Bias Detection' (#hr-run-bias-detection-btn). 3. Inspect classified comments in #hr-ai-bias-body (Comment, Label, Reason, Source, Date).	Button: #hr-run-bias-detection-btn Labels: Constructive, Neutral, Biased	NLP engine classifies student qualitative feedback; labels constructive comments, highlights potential biases with explicit linguistic reasons, and summarizes totals.	Bias detection algorithm classified qualitative comments into objective and subjective feedback.	PASS
TC-84	1. Locate 'Cross-Source Evaluation Discrepancy' section. 2. Select semester and click 'Run Discrepancy Check' (#hr-run-discrepancy-analysis-btn). 3. Inspect flagged professors in #hr-ai-discrepancy-body.	Button: #hr-run-discrepancy-analysis-btn Threshold: Diff >= 2.0 stars	System compares Student Average against Supervisor Average; flags professors where rating delta exceeds threshold (e.g. 2.0 stars) with severity tags.	Cross-source discrepancy analysis compared student scores against dean supervisor ratings.	PASS
TC-85	1. Locate 'Evaluation Credibility Score' section. 2. Click 'Run Credibility Analysis' button (#hr-run-credibility-analysis-btn). 3. Inspect composite reliability summary (#hr-ai-credibility-summary) and table (#hr-ai-credibility-body).	Button: #hr-run-credibility-analysis-btn Metrics: Total, Highly Reliable, Acceptable, Needs Review	System synthesizes behavior scores, bias classification, and cross-source checks into holistic evaluation credibility index per submission.	Evaluation credibility score synthesized behavior metrics into an institutional reliability index.	PASS
TC-	1. Click 'Print AI	Button: #hr-print-	Browser print dialog	Print AI insights	PASS

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86	Insights Summary' button (#hr-print-ai-insights-btn). 2. Verify browser print preview layout. 3. Verify print styles suppress navigation sidebar and buttons.	ai-insights-btn Action: Trigger window.print()	opens; print CSS renders clean executive dossier with NAAP header, metrics tables, and suppression of interactive buttons.	button opened browser print preview formatted for executive reports.	
RE-14 HR Profile Demographics & Account Security					
Pre-requisite: HR navigated to #profile-view. Authenticated HR user session active.					
TC-87	1. Navigate to Profile view. 2. Verify HR Profile card displays read-only records: - Faculty ID (e.g. 'HR-10527') - Full Name (e.g. 'Liza M. Santos') - Department ('Human Resources') - Rank ('HR Manager') - Gmail ('hr.manager@naap.edu') - AY/SEM ('AY 2025-2026, 2nd Semester')	Card: .profile-card Values: Synced employee credentials	HR demographic profile items render accurately with clear read-only labels; text aligns properly in responsive grid layout.	Profile view displayed verified HR administrator records in read-only format.	PASS
TC-88	1. Locate Profile Photo block. 2. Click 'Upload Photo' button (#profilePhotoInput). 3. Select valid image (JPG/PNG < 2MB). 4. Select invalid file (> 2MB or non-image format).	Input: #profilePhotoInput Valid: JPG/PNG < 2MB Invalid: EXE or > 2MB	Valid photo updates thumbnail preview (#profilePhotoPreview) and placeholder immediately; invalid file triggers error alert rejecting the upload.	Profile photo upload accepted a valid image, rendered a preview, and saved properly.	PASS
TC-89	1. In Account Actions card, click 'Change Gmail' button. 2. Verify #emailUpdateCard opens. 3. Enter valid new Gmail and mismatched confirmation email. 4. Enter matching valid Gmail and submit form (#changeEmailForm).	Form: #changeEmailForm Inputs: #newEmail, #confirmEmail	Mismatched confirmation triggers error alert; matching valid email submits successfully to /api/hr/change-email endpoint with confirmation notice.	Gmail update form validated input and saved the new address upon confirmation.	PASS
TC-90	1. In Account Actions card, click 'Change Password' button. 2. Verify	Form: #changePasswordForm Inputs:	Mismatched password triggers validation prompt; matching valid credentials submit	Password change form enforced complexity	PASS

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	#passwordUpdateCard opens. 3. Enter Current Password, New Password, and mismatched Confirm Password. 4. Enter matching New Password and submit (#changePasswordForm).	#currentPassword, #newPassword, #confirmPassword	update to /api/hr/change-password endpoint and confirm success.	rules and updated credentials successfully.	
TC-91	1. In Password Update form, enter text into password fields. 2. Click password eye toggle buttons (.toggle-password). 3. Verify input type switches between 'password' and 'text'. 4. Verify eye icon toggles between fa-eye and fa-eye-slash.	Buttons: .toggle-password Targets: currentPassword, newPassword, confirmPassword	Password characters toggle between masked dots and readable plaintext; eye icon toggles state accordingly providing secure visual feedback.	Password visibility icon toggled the input field between masked and unmasked characters.	PASS
TC-92	1. Open Change Gmail card, enter dirty data, then click 'Cancel' (.js-close-account-form). 2. Open Change Password card, enter dirty data, then click 'Cancel'. 3. Verify card display and form reset.	Buttons: .js-close-account-form Action: Dismiss account forms	Clicking Cancel cleanly hides account action cards, resets form inputs to original state, and returns to primary profile card view.	Canceling an account edit closed the form card and restored original values.	PASS

System Name:	NAAP Evaluation System
Module / View:	hr Panel (hrpanel.html)
Created by:	John Emmanuel E Rojo
Executed by:	John Emmanuel E Rojo
Date of Execution:	SEPTEMBER 28 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

NAAP EVALUATION SYSTEM - ADMIN PANEL MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 28 2026
RE-01	Authentication, Access Control & Administrative Navigation				
Pre-requisite: Browser opened to application root. Admin user account configured in database.					
TC-01	1. Clear all session and authentication storage tokens.	URL: system/html/adminpanel.html	System identifies missing authorization token. immediately	Direct unauthenticated access was	PASS

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	2. Attempt direct URL navigation to system/html/adminpanel.html in browser address bar. 3. Press Enter.	Session: Unauthenticated	redirects to login/index page, and denies access to admin interface.	blocked and redirected immediately to the login page.	
TC-02	1. Log in with non-admin credentials (e.g. Student, Professor, Dean, OSA, VPAA). 2. Attempt to open adminpanel.html directly. 3. Observe role authorization guard.	User Role: student / professor / dean URL: adminpanel.html	Role guard denies access with 403 Forbidden alert, logs unauthorized privilege escalation attempt, and redirects user to their respective portal.	Attempting to access the admin panel with a non-admin role was denied and redirected away.	PASS
TC-03	1. On login screen, enter valid Administrator credentials. 2. Click Login button. 3. Observe initial landing page and UI headers.	Email: admin@naap.edu.ph Role: admin / administrator	User successfully authenticates; dashboard initializes with Admin Control Center hero card, administrator profile badge, and full navigation sidebar.	Logged in with valid administrator credentials and dashboard loaded with administrative metrics.	PASS
TC-04	1. Click each navigation item in the sidebar: - Dashboard - User Management - Profile - Professor Management - Subject Management - Evaluation Settings - Questionnaire - System Settings 2. Observe view container visibility.	Sidebar Nav Links: [data-view='dashboard', 'users', 'profile', 'hr-professors', 'subject-management', 'hr-evaluation-settings', 'hr-questionnaire', 'system']	Each navigation click switches active view container (.content-view), updates top header title and context text, and highlights corresponding menu link.	Sidebar navigation links toggled views smoothly while highlighting the active section.	PASS
TC-05	1. Resize browser viewport to <= 1000px. 2. Click mobile hamburger toggle button (#adminMenuToggle). 3. Verify sidebar opens with backdrop overlay. 4. Click backdrop overlay (#sidebarBackdrop).	Viewport: 768px Toggle: #adminMenuToggle Backdrop: #sidebarBackdrop	Sidebar drawer slides smoothly into view; backdrop overlay covers viewport; clicking backdrop cleanly closes drawer and restores main view.	Mobile drawer opened cleanly using the hamburger toggle and dismissed upon pressing Esc.	PASS
TC-06	1. In Admin Hero card, click 'Open User Management' (#heroOpenUsersBtn). 2. Return to Dashboard and click 'Review Settings' (#heroOpenSettingsBtn)	Hero Buttons: #heroOpenUsersBtn #heroOpenSettingsBtn	Clicking 'Open User Management' smoothly switches to Users view; clicking 'Review Settings' transitions immediately to Evaluation Settings view.	Dashboard hero action buttons navigated directly to user management and system settings.	PASS

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	3. Observe view transitions.				
TC-07	1. In sidebar footer, click Logout button (#logout-btn). 2. Confirm logout prompt if displayed. 3. Attempt to return to admin panel using browser Back button.	Button: #logout-btn Session: Active Admin session	Session tokens and user credentials are removed from storage; user is redirected to login; browser Back button does not reload admin panel.	Clicked logout and verified that all administrative session tokens were purged from storage.	PASS
RE-02 Admin Dashboard Summary Metrics & Quick Actions					
Pre-requisite: Administrator logged in to #dashboard-view. Database contains faculty, student, and evaluation submission records.					
TC-08	1. Navigate to Admin Dashboard. 2. Observe 'Professors' overview card (#total-professors). 3. Cross-reference displayed count against total active faculty in database.	Card: .overview-card.professors Element: #total-professors Metric: Active faculty count	Card displays exact total count of active teaching faculty tracked in system records with label 'Active faculty'.	Total professors card accurately displayed the count of active teaching faculty.	PASS
TC-09	1. Observe 'Students' overview card (#total-students). 2. Verify total matches enrolled student population for current academic semester.	Card: .overview-card.students Element: #total-students Metric: Enrolled students count	Card displays accurate count of active enrolled student accounts currently registered in institutional records.	Total students card accurately displayed the active enrollment count for the current semester.	PASS
TC-10	1. Inspect 'Completion Rate' metric card (#completion-rate-overview). 2. Calculate formula: (Submitted Evaluations / Required Evaluations) * 100. 3. Verify percentage display.	Card: .overview-card.evaluations Element: #completion-rate-overview Metric: Overall completion %	Completion Rate accurately displays ratio of submitted vs required evaluations across all roles formatted with percentage symbol.	Completion rate card correctly computed the ratio of completed evaluations to expected total.	PASS
TC-11	1. Inspect 'Pending Evaluations' card (#pending-evaluations). 2. Verify count matches evaluations awaiting completion across student, peer, and supervisor categories.	Card: .overview-card.pending-evaluations Element: #pending-evaluations Metric: Pending count	Displays accurate count of evaluations pending submission; correlates with remaining open evaluation assignments.	Pending evaluations card accurately reflected unresolved evaluations across all programs.	PASS
TC-12	1. In Quick Actions section, click each quick action button: - Add User - Manage Users - Manage Subjects	Buttons: .quick-action-btn [data-action='add-user', 'manage-users', 'manage-subjects', 'manage-campus',	Each quick action button immediately opens corresponding modal or navigates to dedicated view without delay or script errors.	Quick action buttons opened their respective administrative modals without delay.	PASS

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	- Manage Campus - Announcement - System Backup - Settings 2. Verify each button triggers corresponding view or modal.	'send-announcement', 'system-backup', 'system-settings']			
TC-13	1. Inspect Activity Summary snapshot list (#activity-list). 2. Click 'View Activity Log' button (.js-open-activity-log). 3. Observe view transition.	Section: .dashboard-activity-section Trigger: .js-open-activity-log	Activity summary displays recent session events with status icons; clicking 'View Activity Log' navigates to full Activity Log view.	Recent activity summary rendered latest login and system events in chronological order.	PASS
TC-14	1. Scroll to Charts Section on dashboard. 2. Inspect 'Evaluation Overview' pie chart (#evaluation-chart). 3. Inspect 'Semestral Performance' bar chart (#performance-chart). 4. Hover over chart data points.	Canvases: #evaluation-chart, #performance-chart Datasets: Evaluation status & semestral trends	Evaluation Overview pie chart displays completed vs pending proportions; Semestral Performance bar chart displays historical submission volumes per semester.	Evaluation overview pie chart rendered completed versus pending evaluation slices clearly.	PASS
RE-03 Multi-Role Institutional Evaluation Reports & Analytics					
Pre-requisite: Administrator logged in to Dashboard. Evaluation scoring datasets available across student, peer, and supervisor categories.					
TC-15	1. Scroll to 'Student to Professor Evaluation' reports section. 2. Verify category bar chart (#student-professor-bar-chart). 3. Verify rating distribution pie chart (#student-professor-pie-chart). 4. Inspect summary metrics (#student-prof-avg-rating, #student-prof-total, #student-prof-count).	Canvases: #student-professor-bar-chart, #student-professor-pie-chart Stats: Average Rating, Total Evaluations, Professors Evaluated	Bar chart displays mean scores per rubric category; pie chart plots rating distribution; summary stats display synchronized metrics.	Student-to-professor charts rendered category rating averages and score distributions clearly.	PASS
TC-16	1. Scroll to 'Professor to Professor Analytics' reports section. 2. Verify peer evaluation category bar chart (#professor-professor-bar-chart). 3. Verify rating distribution pie chart (#professor-professor-pie-chart). 4. Inspect summary	Canvases: #professor-professor-bar-chart, #professor-professor-pie-chart Scope: Colleague peer reviews	Peer review charts render accurately with category breakdowns; summary stats reflect peer evaluation sample sizes and aggregate rating.	Peer review analytics plotted colleague performance scores across workplace rubrics.	PASS

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	metrics (#professor-professor-avg-rating, #professor-professor-total).				
TC-17	1. Scroll to 'Supervisor to Professor Evaluation' reports section. 2. Verify supervisor review bar chart (#supervisor-professor-bar-chart) and pie chart (#supervisor-professor-pie-chart). 3. Inspect summary metrics (#supervisor-prof-avg-rating, #supervisor-prof-total, #supervisor-prof-count).	Canvases: #supervisor-professor-bar-chart, #supervisor-professor-pie-chart Scope: Dean/Supervisor reviews	Charts plot administrative performance metrics; summary counters display supervisor evaluation ratings and evaluated professor counts.	Supervisor evaluation charts displayed dean appraisal scores with category breakdown.	PASS
TC-18	1. Hover mouse cursor over multiple segments in bar and pie charts across all report sections. 2. Verify tooltip rendering.	Element: Chart.js canvas tooltips Scope: All report charts	Tooltips display category name, exact numeric score or count, and percentage calculation with clear contrast and formatting.	Hovering over chart segments displayed detailed tooltips with exact scores and percentages.	PASS
TC-19	1. Resize browser window between desktop (1440px), tablet (1024px), and mobile (600px). 2. Observe chart containers and layout grids.	Grid: .charts-grid, .chart-container Responsive reflow	Chart canvases automatically resize maintaining aspect ratio; legends and axis labels remain legible without clipping.	Chart canvases resized smoothly across desktop and mobile viewports without label distortion.	PASS
TC-20	1. Compare summary metrics across reports against live database evaluation submissions. 2. Validate calculation accuracy.	Database records vs UI summary counts	Average ratings, evaluation totals, and professor counts match underlying database records exactly without discrepancy.	Summary metrics across reports matched database evaluation records without discrepancy.	PASS
RE-04 User Management - Search, Filters & Organization Structure					
Pre-requisite: Administrator navigated to #users-view. User accounts populated across campuses and roles.					
TC-21	1. Navigate to User Management view. 2. Verify Campus selector (#campus-filter-select) displays all registered campuses. 3. Verify Manage Campuses, Manage Programs, and Bulk Register buttons display.	Bar: .campus-tabs-bar Dropdown: #campus-filter-select Buttons: #manage-campuses-btn, #manage-programs-btn, #bulk-register-btn	Campus bar initializes with campus options; toolbar action buttons are clickable and correctly styled.	User management view initialized with active campus tabs and action toolbars.	PASS
TC-22	1. In user search input (#user-search), enter	Input: #user-search	Search results table becomes visible;	User search input filtered the	PASS

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	student number or partial full name (e.g. 'Reyes'). 2. Click Search button (#user-search-btn). 3. Verify results display in table (#user-search-results).	Query: 'Reyes' or ID Button: #user-search-btn	displays matching users with ID Number, Full Name, Email, Campus, Institute, Program, and Status.	table dynamically by name, email, and ID.	
TC-23	1. Filter users by Role (#user-role-filter). 2. Filter users by Department (#user-department-filter). 3. Filter users by Status (#user-status-filter). 4. Filter users by Program (#user-program-filter). 5. Click Search.	Dropdowns: #user-role-filter, #user-department-filter, #user-status-filter, #user-program-filter	Multi-criteria filter refines user search results to display only accounts matching all selected attributes simultaneously.	Filtering by role and status refined the user list to matching accounts.	PASS
TC-24	1. Observe User Search Results table headers and row layout. 2. Verify pagination bar (#admin-user-pagination) displays page numbers and navigation arrows.	Table: .user-results-table Pagination: #admin-user-pagination	Table displays formatted columns; pagination calculates total pages accurately; clicking next page navigates smoothly.	User table rendered formatted columns and paginated results accurately.	PASS
TC-25	1. In Organization Structure section, click accordion headers: - HR Staff (#hr-count) - VPAA (#vpaa-count) - OSA (#osa-count) - Administrators (#admin-count) 2. Verify expandable card behavior.	Accordions: .org-section [data-toggle='hr-users', 'vpaa-users', 'osa-users', 'admin-users']	Clicking accordion header toggles collapse/expansion of user list; chevron icon rotates; user cards render with avatar and details.	Clicking organization accordions expanded and collapsed department user groups cleanly.	PASS
TC-26	1. Inspect Departments section (#departments-dynamic). 2. Select different campus from campus dropdown (#campus-filter-select). 3. Observe departments list.	Container: #departments-dynamic Filter: #campus-filter-select	Department structure updates dynamically to display departments and programs affiliated with selected campus.	Department tree view updated dynamically to show assigned deans and faculty members.	PASS
TC-27	1. In User Management view, locate floating 'Add New User' button (#add-user-btn). 2. Click button. 3. Verify Add User modal opens.	Button: #add-user-btn Modal: #add-user-modal	Floating button triggers Add New User modal with backdrop; form initializes in creation mode ready for user input.	Floating add user button opened the account creation modal with backdrop.	PASS

RE-05 **User CRUD Operations & Account Provisioning**

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Pre-requisite: Administrator in #users-view with administrative write privileges.					
TC-28	1. In Add User modal, select Role = 'Student'. 2. Verify Student fields display (#student-fields: Student Number, Year & Section). 3. Select Role = 'Professor'. 4. Verify Employee fields display (#employee-fields: Employee ID, Employment Type).	Dropdown: #new-user-role Dynamic Containers: #student-fields, #employee-fields	Form dynamically shows/hides role-specific inputs based on chosen role; student prompts for academic year/section; employee prompts for ID and employment type.	Selecting different roles dynamically adjusted the form fields to match role requirements.	PASS
TC-29	1. In Add User modal, attempt to submit with empty required fields. 2. Enter invalid email address. 3. Enter duplicate Employee ID / Student Number. 4. Observe validation prompts.	Inputs: Empty / invalid email / duplicate ID Form: #add-user-form	HTML5 validation stops submission on empty inputs; email validator rejects invalid format; duplicate ID triggers error notification.	Submitting without required fields was blocked with client-side validation errors.	PASS
TC-30	1. Enter valid Full Name, Email, Role ('Professor'), Campus ('villamor'), Employee ID ('EMP-2026-99'), Department ('ics'), Program, and Password. 2. Click 'Add User' button. 3. Observe database persistence.	Valid user credentials: Name: 'Dr. Maria Santos' Email: 'm.santos@naap.edu.ph' Role: Professor	User account is created in database; modal closes; success notification displays; new user appears in search and organizational hierarchy.	Valid user registration saved successfully to the database and added to the roster.	PASS
TC-31	1. Locate user card in search results or organization structure. 2. Click 'Edit' button. 3. Verify Edit User modal (#edit-user-modal) opens with populated fields.	Modal: #edit-user-modal Form: #edit-user-form	Modal opens with existing user attributes (ID Number, Name, Email, Campus, Institute, Program, Employment Type, Position, Status) pre-loaded.	Clicking edit prefilled the modal with the user's existing account attributes.	PASS
TC-32	1. In Edit User modal, modify Full Name and Position. 2. Leave Password field blank. 3. Click 'Save Changes'. 4. Verify existing password remains unchanged.	Field: #edit-user-password blank Action: Save edits	System updates modified profile attributes while preserving existing password hash; user can still log in with original credentials.	Updating user profile fields saved changes while preserving unchanged attributes.	PASS
TC-33	1. In Edit User modal, change Status (#edit-user-status) from	Status: 'inactive' Action: Login attempt	Account status updates to inactive; user is locked out from logging	Changing account status to inactive locked	PASS

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	'Active' to 'Inactive'. 2. Click 'Save Changes'. 3. Attempt to log in with modified user credentials.		into portal with 'Account is inactive' alert.	the user out from logging in.	
TC-34	1. On a user record, click 'Delete' or 'Archive' button. 2. Verify safety confirmation modal displays. 3. Confirm deletion. 4. Verify account removal.	Action: Delete user account Confirmation: Prompt	Confirmation dialog prevents accidental deletion; upon confirmation, account is soft-deleted or removed from active directory.	Confirmation dialog prevented accidental user deletion and removed record upon confirmation.	PASS
TC-35	1. Open Add User or Edit User modal, enter dirty data, then click 'Cancel' button. 2. Verify modal closes and form inputs reset.	Buttons: #cancel-user-form, #cancel-edit-user-form	Modal closes immediately; unsaved changes are discarded; form resets to clean state on subsequent opening.	Closing the user modal discarded unsaved changes and reset the form cleanly.	PASS
RE-06 Bulk User Registration (Excel Import)					
Pre-requisite: Administrator in #users-view. Prepared Excel template files with user lists.					
TC-36	1. In User Management toolbar, click 'Bulk Register' button (#bulk-register-btn). 2. Verify hidden file input (#bulk-register-input) is triggered and file selection dialog opens.	Button: #bulk-register-btn Input: #bulk-register-input Accept: .xlsx, .xls	Native OS file selection window opens, restricted to spreadsheet file types (.xlsx, .xls).	Bulk register button opened the native OS file picker restricted to spreadsheets.	PASS
TC-37	1. Select valid Excel file containing bulk user records. 2. Observe Bulk Register Loading modal (#bulk-register-loading). 3. Verify animated hourglass and status text.	Modal: #bulk-register-loading Component: .loading-hourglass-frame Text: 'Processing Bulk Register'	Loading modal renders with animated hourglass, informing user that records are being parsed, validated, and inserted.	Uploading an Excel file opened the parsing modal with an animated progress indicator.	PASS
TC-38	1. Upload Excel file missing required header columns (e.g. missing 'email' or 'role'). 2. Observe validation error handling.	File: invalid_columns.xlsx (missing required headers)	System detects missing columns, aborts import, and displays descriptive error alert detailing missing required spreadsheet columns.	Missing required header columns halted the import and displayed an informative error notice.	PASS
TC-39	1. Upload Excel file containing duplicate student/employee numbers or invalid email formats. 2. Observe row-by-row validation feedback.	File: partial_invalid_users.xlsx	System identifies erroneous rows, reports specific line numbers and validation failures, and inserts only valid non-duplicate records.	Duplicate student numbers were flagged with row numbers and excluded from import.	PASS

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TC-40	1. Upload fully valid Excel file with 50+ student and faculty records. 2. Verify completion of import process. 3. Check user directory.	File: valid_bulk_users.xlsx (50+ records)	All accounts are successfully provisioned; success toast reports total imported count; new users populate user management list and role counters.	Valid Excel spreadsheet imported all student accounts successfully with a confirmation toast.	PASS
TC-41	1. Click 'Bulk Register' button and cancel file selection dialog. 2. Observe interface stability.	Action: Cancel file picker	System handles file selection cancellation gracefully without throwing errors or locking UI.	Canceling the file selection dialog was handled gracefully without raising errors.	PASS
RE-07 Campus & Program Management Modals					
Pre-requisite: Administrator in #users-view. Campus and program datasets available.					
TC-42	1. Click 'Manage Campuses' button (#manage-campuses-btn). 2. Verify Manage Campuses modal (#manage-campuses-modal) opens. 3. Inspect existing campus list (#campus-list).	Button: #manage-campuses-btn Modal: #manage-campuses-modal List: #campus-list	Modal opens centered; lists all active campuses (Villamor, Basa, Medelin, Mactan, Fernando) with affiliated department details.	Manage campuses modal opened displaying all active institutional campus locations.	PASS
TC-43	1. In Add Campus form (#add-campus-form), enter Campus Name ('Cebu Extension') and comma-separated Departments ('ICS, ILAS'). 2. Click 'Add Campus' button. 3. Verify persistence.	Form: #add-campus-form Inputs: #new-campus-name, #new-campus-departments	New campus is saved to system records; immediately appears in campus list and populates campus dropdown selectors across panel.	Adding a new campus saved the record and added it to the campus selector immediately.	PASS
TC-44	1. In Manage Campuses modal, select an existing campus. 2. Modify assigned departments in detail view (#campus-detail). 3. Save changes and verify updates.	Container: #campus-detail Action: Update campus departments	Updated department associations persist in database; reflect dynamically in department filters and user registration dropdowns.	Updating department associations for a campus persisted changes in the database.	PASS
TC-45	1. Click 'Manage Programs' button (#manage-programs-btn). 2. Verify Manage Programs modal (#manage-programs-modal) opens. 3. Select Campus and	Button: #manage-programs-btn Modal: #manage-programs-modal Filters: #program-manager-campus, #program-manager-department	Modal opens; filter controls narrow program list table (#program-list-body) to selected academic unit.	Manage programs modal opened with filters to search degree programs.	PASS

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	Department filters to display active programs.				
TC-46	1. In Program form (#program-form), enter Program Code ('BSIT') and Program Name ('Bachelor of Science in Information Technology'). 2. Click 'Save Program'. 3. Verify addition to program table.	Inputs: #program-code, #program-name Button: #program-submit-btn	New program is validated and saved; renders in program list table with edit and delete action buttons.	New degree program was validated and appended to the program list successfully.	PASS
TC-47	1. Click Edit button on a program in the program list. 2. Verify form populates with program code and name; Cancel Edit button (#program-cancel-edit-btn) displays. 3. Modify name and save.	Actions: Edit program Button: #program-cancel-edit-btn	Program data loads into input fields; saving updates record in place; cancel edit restores creation mode.	Editing a program updated its department association and degree code cleanly.	PASS
RE-08 Subject Management - Catalog & Bulk Import					
Pre-requisite: Administrator navigated to #subject-management-view. Academic departments and subject curriculum loaded.					
TC-48	1. Navigate to Subject Management view. 2. In Subject Catalog section, select Campus (#subject-campus-filter) and Department (#subject-department-filter). 3. Observe catalog table (#subject-catalog-body).	Filters: #subject-campus-filter, #subject-department-filter Table: #subject-catalog-body	Catalog table loads subjects affiliated with selected campus and department, displaying Subject Code, Subject Name, and Action buttons.	Subject catalog table displayed active courses filtered by campus and department.	PASS
TC-49	1. Click 'Add Subject' button (#add-subject-btn). 2. Verify Subject modal (#subject-modal) opens. 3. Enter Campus, Department, Subject Code ('IT-301'), and Subject Name ('Systems Integration & Architecture'). 4. Click 'Save Subject'.	Modal: #subject-modal Form: #subject-form Inputs: #subject-campus, #subject-department, #subject-code, #subject-name	Subject modal opens; validates mandatory fields; saving persists subject to catalog and appends to subject table.	Add subject modal validated inputs and created a new curriculum course record.	PASS
TC-50	1. In Add Subject modal, enter Subject Code that already exists within the same department. 2. Click 'Save Subject'. 3. Observe duplicate	Subject Code: Duplicate existing code in department	System identifies duplicate Subject Code within department, rejects submission, and displays 'Subject Code already exists' error alert.	Submitting a duplicate subject code within a department was blocked with an alert.	PASS

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	validation.				
TC-51	1. Locate subject row in catalog table. 2. Click 'Edit' action button. 3. Modify Subject Name and save. 4. Click 'Delete' action button and confirm prompt.	Actions: Edit and Delete subject	Edit updates subject name across offerings; Delete checks for existing student enrollments before allowing safe removal.	Editing a subject updated course metadata across active class offerings.	PASS
TC-52	1. Click 'Import Subjects' button (#bulk-subject-import-btn). 2. Select Excel file containing subject code curriculum. 3. Verify bulk parsing and insertion.	Button: #bulk-subject-import-btn Input: #bulk-subject-import-input File: subjects_curriculum.xlsx	File is parsed via XLSX library; subjects are validated and batch-inserted; catalog table updates with newly imported subjects.	Bulk subject import parsed the Excel template and populated the subject catalog.	PASS
TC-53	1. Attempt to upload non-Excel or malformed subject spreadsheet. 2. Observe system error handling.	File: corrupted_file.txt or invalid spreadsheet	System rejects file format; displays informative error notification; preserves existing subject catalog without corruption.	Uploading an invalid file format was rejected with an informative format notice.	PASS
RE-09 Subject Offering Assignments & Student Enrollment					
Pre-requisite: Administrator in #subject-management-view. Active semester, subjects, professors, and sections configured.					
TC-54	1. In Offering Assignment section, inspect Offering form (#offering-form). 2. Select Semester, Campus, Department, Program, Subject, Professor, and enter Section ('3/1'). 3. Click 'Save Assignment' (#offering-save-btn).	Form: #offering-form Dropdowns: #offering-semester, #offering-campus, #offering-department, #offering-program, #offering-subject, #offering-professor Input: #offering-section	Assignment is created linking professor to subject and section; record appears in Offering List table (#offering-list-body).	Offering assignment linked a faculty member to a specific subject section.	PASS
TC-55	1. Click 'Import Offerings' button (#bulk-offering-import-btn). 2. Select Excel file with offering schedules. 3. Verify batch schedule assignment.	Button: #bulk-offering-import-btn Input: #bulk-offering-import-input File: semester_offering.xlsx	System processes offering schedule rows, matches professors and subjects, and inserts class assignments in bulk.	Bulk offerings import mapped courses to assigned professors without errors.	PASS
TC-56	1. Click 'Import Excess Load' button (#bulk-excess-load-import-btn).	Button: #bulk-excess-load-import-btn Input: #bulk-	Excess load records are validated; 'Load' column in offerings table reflects regular vs	Importing excess load records updated faculty teaching	PASS

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	2. Select Excel file containing overload faculty teaching assignments. 3. Verify excess load tracking.	excess-load-import-input File: faculty_excess_load.xlsx	excess load classifications.	load flags correctly.	
TC-57	1. In Offering Search box (#offering-search), enter professor name, employee ID, subject code, or section. 2. Observe live filtering in offerings table.	Input: #offering-search Query: Professor name or section	Table filters immediately, displaying only offerings matching the search term across subject, section, and faculty columns.	Offering search filtered the table dynamically by subject code and professor name.	PASS
TC-58	1. On an offering row, click 'Assign Students' action button. 2. Verify Assign Students modal (#offering-students-modal) opens. 3. Inspect student candidates list (#offering-students-list).	Modal: #offering-students-modal Container: #offering-students-list	Modal opens displaying students eligible for enrollment in the selected program and year level.	Assign students modal listed eligible students for enrollment in the section.	PASS
TC-59	1. In Assign Students modal, filter by Program (#offering-students-program-filter) and search student by name/ID (#offering-students-search). 2. Check student selection checkboxes. 3. Click 'Save Students' (#save-offering-students-btn).	Filters: #offering-students-program-filter, #offering-students-search Action: Save student enrollments	Selected students are enrolled in section offering; student portals automatically receive evaluation assignments for this subject.	Enrolling selected students updated class rosters and evaluation assignments.	PASS
TC-60	1. Delete an existing offering assignment with active student enrollments. 2. Observe warning confirmation dialog.	Action: Remove offering assignment	System warns that removing offering will unassign enrolled students from evaluation rubric; requires explicit confirmation.	Deleting an offering prompted a warning about enrolled students before unassigning.	PASS
RE-10 Professor Management & SASR Export Integration					
Pre-requisite: Administrator navigated to #hr-professors-view. Faculty database populated.					
TC-61	1. Navigate to Professor Management view. 2. Verify Department tabs (All Departments, ICS, ILAS, ENGI) filter faculty cards. 3. Test search by employee ID or name (#professor-search).	Tabs: .dept-tab Search: #professor-search Container: #professors-list	Faculty list renders dynamically; department tabs filter cards instantaneously; search input finds matching professors accurately.	Professor directory loaded faculty cards into a responsive grid with department filters.	PASS
TC-62	1. Click 'Add Professor' button (#add-professor-	Modal: #professor-modal	Professor record is created in database;	Add professor modal created	PASS

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	btn). 2. Verify Add Professor modal (#professor-modal) opens. 3. Fill Name, Email, Employee ID, Department, Position, and Employment Type ('Regular'). 4. Save professor.	Form: #professor-form Button: #add-professor-btn	modal closes; success notification displays; professor card appears in faculty list.	the instructor account and assigned default department rights.	
TC-63	1. On a professor card, click 'Edit' button. 2. Toggle 'Active Professor' checkbox (#professor-active) to inactive. 3. Save changes. 4. Verify inactive card badge.	Field: #professor-active unchecked Action: Deactivate faculty member	Faculty account is marked inactive; visual badge displays Inactive status; account is excluded from active evaluation rosters.	Toggling faculty status updated the card badge to inactive and revoked panel access.	PASS
TC-64	1. Click 'Details' button on a professor card. 2. Verify Professor Details modal (#professor-details-modal) opens with full bio and assigned subjects. 3. Click 'Analytics' button to open Analytics modal (#professor-analytics-modal).	Modals: #professor-details-modal, #professor-analytics-modal	Details modal displays faculty information and course assignments; Analytics modal plots historical rating radar and trend charts.	Faculty details modal displayed complete profile records and current class offerings.	PASS
TC-65	1. In Professor Management toolbar, click 'Overall SASR' button (#overall-sasr-btn). 2. Verify calculation across Student, Peer, and Supervisor weightings. 3. Verify Excel export file download.	Button: #overall-sasr-btn Export: Excel spreadsheet (.xlsx)	System aggregates composite ratings across evaluation roles; generates and downloads valid SASR Excel report with proper headers.	Overall SASR export compiled composite evaluation scores into an official spreadsheet.	PASS
TC-66	1. Inspect pagination bar (#admin-professor-pagination) in Professor Management view. 2. Navigate between pages. 3. Verify page size handling (100 per page).	Element: #admin-professor-pagination Page size: ADMIN_PROFESSORS_PAGE_SIZE (100)	Pagination controls calculate page count accurately and navigate across multi-page faculty records smoothly.	Pagination controls navigated across professor records smoothly without UI lag.	PASS

RE-11 Evaluation Settings & Semester Lifecycle Management

Pre-requisite: Administrator navigated to #hr-evaluation-settings-view. Evaluation windows and semester records loaded.

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TC-67	1. In Evaluation Period section, configure Start and End dates for: - Student to Professor (#student-professor-start, #student-professor-end) - Professor to Professor (#professor-professor-start, #professor-professor-end) - Supervisor to Professor (#supervisor-professor-start, #supervisor-professor-end)	Inputs: Date ranges for all 3 evaluation types	Date inputs accept valid calendar dates and allow independent scheduling for each evaluation role window.	Evaluation date range accepted valid dates and saved the active submission period.	PASS
TC-68	1. Enter Start Date that is chronologically after End Date. 2. Click 'Save Evaluation Periods' button (#save-eval-periods-btn). 3. Observe date validation error.	Start Date: Future date End Date: Past date Button: #save-eval-periods-btn	System validation blocks saving, displays warning alert 'End date cannot be earlier than start date', and prevents invalid period schedule.	Setting an end date before the start date was blocked with a date validation warning.	PASS
TC-69	1. Enter valid dates and click 'Save Evaluation Periods' (#save-eval-periods-btn). 2. Verify database persistence and portal evaluation open/closed state enforcement.	Action: Save valid period schedules Button: #save-eval-periods-btn	Period dates are persisted to database/SharedData; student and faculty portals allow evaluation submissions only within active period.	Valid evaluation periods saved successfully and updated countdown timers system-wide.	PASS
TC-70	1. Locate Current Semester section (#current-semester). 2. Select a different active term from dropdown. 3. Click 'Save Current Semester' (#save-current-semester-btn). 4. Verify active semester updates across all system panels.	Dropdown: #current-semester Button: #save-current-semester-btn	Active academic semester updates system-wide; changes synchronize across student, professor, dean, and administrative portals.	Updating current semester propagated the new active term across all system modules.	PASS
TC-71	1. In Add Term section, select Year Start (#new-term-year-start), Year End (#new-term-year-end), and Semester Type (1st, 2nd, or 3rd Semester). 2. Verify live preview (#new-term-preview).	Inputs: #new-term-year-start, #new-term-year-end, #new-term-semester-type Preview: #new-term-preview Button: #btn-add-term	New semester option is formatted and appended to semester dropdown options, enabling configuration for upcoming academic cycles.	Adding a new academic term validated inputs and appended the semester to the system dropdown.	PASS

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	3. Click 'Add Term' (#btn-add-term).				
TC-72	1. In Notifications section, toggle Email Reminders on/off. 2. Configure Reminder Frequency (#reminder-freq: Daily, Weekly, Bi-weekly). 3. Click 'Save All Settings'.	Toggle: Email Reminders Dropdown: #reminder-freq Action: Save settings	Notification preferences and reminder dispatch frequencies are saved; system schedules automated email triggers accordingly.	Notification toggle switch enabled and disabled automated reminder notifications cleanly.	PASS
RE-12 Questionnaire Builder - Rubric & Question Management					
Pre-requisite: Administrator navigated to #hr-questionnaire-view. Evaluation rubric datasets loaded.					
TC-73	1. In Questionnaire view, select Semester (#semester-select). 2. Select Questionnaire Type (#questionnaire-type-select: Student to Professor, Professor to Professor, Supervisor to Professor). 3. Observe rubric preview.	Dropdowns: #semester-select, #questionnaire-type-select	Questionnaire builder dynamically loads and displays rubric sections and question items corresponding to selected evaluation type.	Questionnaire builder loaded sections and questions for the active evaluation rubric.	PASS
TC-74	1. Click Form Title (#form-title-preview) and modify text inline. 2. Click Form Description (#form-description-preview) and modify text inline. 3. Verify contenteditable inline editing.	Elements: #form-title-preview, #form-description-preview Attribute: contenteditable='true'	Header text can be edited directly on the preview canvas; changes are captured for persistence upon saving.	Editing the form title preview updated the questionnaire header text in real time.	PASS
TC-75	1. Click 'Add Section' button (#add-section-btn). 2. Verify Add Section modal (#section-modal) opens. 3. Enter Section Letter ('A'), Section Title ('Instructional Competence'), and Description. 4. Click 'Save Section'.	Modal: #section-modal Inputs: #section-letter, #section-title, #section-description	Section letter enforces single character [A-Za-z]; saved section renders as grouped category header in the questionnaire layout.	Add section modal validated inputs and added a new criteria category to the questionnaire.	PASS
TC-76	1. Click 'Add Question' button (#add-question-btn). 2. Verify Add Question modal (#question-modal) opens. 3. Select Section (#question-section), enter Question Text	Modal: #question-modal Inputs: #question-section, #question-text, #question-type, #rating-max	Rating question options reveal with bounds validation (2-10); saved question renders in preview with 1-5 rating scale anchors.	Add question modal created rating scale radio groups with configurable score bounds.	PASS

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	(#question-text), select Type = 'Rating' (#question-type), and set Max Rating = 5 (#rating-max). 4. Save question.				
TC-77	1. In Add Question modal, select Type = 'Qualitative'. 2. Configure Max Length (#max-length = 500 chars) and check 'Required Question' (#question-required). 3. Check Exception Reporting (#question-exception-reporting) if applicable. 4. Save question.	Inputs: #max-length (50-2000), #question-required, #question-exception-reporting	Qualitative question is created with text input prompt and character counter in preview; exception tracking flag attaches to rubric schema.	Qualitative question type configured a comment textarea with character limit options.	PASS
TC-78	1. Inspect Data Privacy Agreement section (#questionnaire-privacy-settings). 2. Toggle 'Require agreement' (#questionnaire-privacy-required). 3. Fill Consent Version, Text ID, Notice Title, Notice Text, and Agreement Checkbox Text. 4. Click 'Save Questionnaire' (#save-questionnaire-btn).	Container: #questionnaire-privacy-settings Button: #save-questionnaire-btn	Privacy agreement parameters and complete rubric structure are serialized and persisted to database; success alert confirms save.	Privacy notice policy text and version identifier saved with questionnaire configuration.	PASS
RE-13 System Settings - General, Security & Backup					
Pre-requisite: Administrator navigated to #system-view. Platform configuration services active.					
TC-79	1. In System Settings, locate General Settings group. 2. Modify Institution Name, System Email ('admin@naap.edu.ph'), and Main Campus dropdown. 3. Verify inputs accept modified values.	Group: General Settings Inputs: Institution Name, System Email, Main Campus	General configuration fields accept valid input strings and update institutional branding metadata across the system.	General settings form saved platform title, institution branding, and support email.	PASS
TC-80	1. In Security Settings group, locate 'Session Timeout (30 minutes)' toggle. 2. Toggle switch on and off. 3. Verify session timeout enforcement	Setting: Session Timeout switch Interval: 30 minutes idle	Switch toggles smoothly; when active, user sessions with 30 minutes of inactivity are automatically terminated and redirected to login.	Session timeout toggle enforced automatic logout for idle users after 30 minutes.	PASS

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	behavior.				
TC-81	1. In Security Settings group, locate 'Automatic Backup (last 24 hours)'. 2. Click 'Run Backup' button (#backup-last-24-btn). 3. Observe database backup execution.	Button: #backup-last-24-btn Action: Trigger 24-hour backup snapshot	System initiates database snapshot routine; creates timestamped SQL/JSON backup archive; displays confirmation of backup completion.	Automatic backup routine generated a timestamped database snapshot in storage.	PASS
TC-82	1. Modify various platform settings. 2. Click 'Reset to Default' button (.btn-reset). 3. Confirm reset in confirmation alert. 4. Verify settings revert to default values.	Button: .btn-reset Action: Restore system defaults	Confirmation prompt appears preventing accidental resets; upon confirmation, all system settings revert to baseline default values.	Reset to defaults button prompted for confirmation before restoring factory settings.	PASS
TC-83	1. Click 'Save Changes' button (.btn-save). 2. Verify system settings persistence across browser refresh.	Button: .btn-save Action: Commit system settings	Platform configuration changes are saved to database/configuration file; toast feedback confirms successful update.	Save changes button serialized and committed all configuration updates to the database.	PASS
RE-14 Bulk Credential Distributor & SMTP Configuration					
Pre-requisite: Administrator in #system-view. SMTP mail server available.					
TC-84	1. In Bulk Credential Distributor section, inspect SMTP Configuration fields: - SMTP Host (#credential-smtp-host) - SMTP Port (#credential-smtp-port) - Encryption (#credential-smtp-encryption: TLS, SSL, None) - Authentication (#credential-smtp-auth) - Username (#credential-smtp-username) - Timeout (#credential-smtp-timeout) - From Email (#credential-smtp-from-email) - From Name (#credential-smtp-from-name) - Password (#credential-smtp-	Inputs: #credential-smtp-host, #credential-smtp-port, #credential-smtp-encryption, #credential-smtp-auth, #credential-smtp-username, #credential-smtp-from-email, #credential-smtp-from-name, #credential-smtp-password	Configuration fields accept valid SMTP host parameters; password field masks input characters securely.	SMTP configuration form accepted valid host, port, and authentication credentials.	PASS

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	password) 2. Enter valid SMTP server configuration.				
TC-85	1. Click 'Save SMTP Config' button (#credential-save-config-btn). 2. Verify password field is cleared and status (#credential-config-status) shows configured. 3. Click 'Clear Saved Password' (#credential-clear-password-btn).	Buttons: #credential-save-config-btn, #credential-clear-password-btn Status: #credential-config-status	SMTP configuration is saved and encrypted; status confirms credential storage; clear password button removes stored secret upon confirmation.	Saving SMTP parameters encrypted sensitive credentials and confirmed active status.	PASS
TC-86	1. Scroll to 'Send Single SMTP Test Email' section (#credential-test-mail-box). 2. Enter Recipient Email (#credential-test-smtp-recipient), Subject (#credential-test-smtp-subject), and Message (#credential-test-smtp-message). 3. Click 'Send Test Email' (#credential-test-smtp-send-btn).	Inputs: #credential-test-smtp-recipient, #credential-test-smtp-subject, #credential-test-smtp-message Button: #credential-test-smtp-send-btn	System initiates single SMTP test dispatch; delivers verification email to recipient inbox; displays success feedback in #credential-test-smtp-feedback.	Sending test email verified outgoing SMTP connection and delivered a test notification.	PASS
TC-87	1. Click 'Select XLSX File' button (#credential-distributor-select-btn). 2. Select Excel file containing columns: name, email, role, campus, employee_or_student_number, password. 3. Verify file name display (#credential-distributor-file-name).	Button: #credential-distributor-select-btn Input: #credential-distributor-input File Name: #credential-distributor-file-name	File selector accepts spreadsheet; parsed file name displays beside button; enables 'Distribute Credentials' action button.	Credential distributor parsed student recipient records from uploaded Excel spreadsheet.	PASS
TC-88	1. Click 'Distribute Credentials' button (#credential-distributor-send-btn). 2. Verify Credential Distributor Loading modal (#credential-distributor-loading) opens. 3. Observe progress bar (#credential-distributor-loading-progress) and status.	Button: #credential-distributor-send-btn Modal: #credential-distributor-loading Progress: #credential-distributor-loading-progress	Modal displays with animated hourglass; progress bar dynamically updates showing processed emails vs total count.	Credential distribution progress bar dynamically reflected real-time email dispatch status.	PASS

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TC-89	1. Complete distribution with sample containing valid and invalid recipient email addresses. 2. Observe distribution summary feedback (#credential-distributor-feedback).	Feedback: #credential-distributor-feedback Summary: Sent vs Failed count	System completes distribution; feedback summary reports exact count of successfully sent credentials and count of failed deliveries.	Distribution summary reported successful deliveries and recipient counts accurately.	PASS
TC-90	1. When delivery failures occur, verify 'Download Failed Rows CSV' button (#credential-download-failures-btn) becomes visible. 2. Click button. 3. Verify downloaded CSV.	Button: #credential-download-failures-btn File: failed_credential_rows.csv	Button displays on errors; clicking downloads CSV report containing failed rows with specific SMTP error reason for easy troubleshooting.	Delivery failures generated a downloadable error report listing failed recipient addresses.	PASS
RE-15 OpenAI AI Configuration & System Health Verification					
Pre-requisite: Administrator in #system-view. AI service configuration and health verification endpoints loaded.					
TC-91	1. In OpenAI Configuration group, inspect OpenAI Model (#gemini-model), Timeout ms (#gemini-timeout-ms), and API Key (#gemini-api-key) fields. 2. Enter valid model (e.g. 'gpt-5.6-luna') and API key. 3. Click 'Save OpenAI Config' (#gemini-save-config-btn).	Inputs: #gemini-model, #gemini-timeout-ms, #gemini-api-key Button: #gemini-save-config-btn	OpenAI settings are saved securely; API key is encrypted; status message confirms active AI configuration.	OpenAI configuration saved API key securely and verified connection with a green badge.	PASS
TC-92	1. In Panel AI Access section, toggle switches for individual panels: - Professor Panel (#openai-access-professor) - Dean Panel (#openai-access-dean) - Program Coordinator Panel (#openai-access-procoor) - HR Panel (#openai-access-hr) - Admin Panel (#openai-access-admin) - VPAA Panel (#openai-access-vpaa) 2. Verify state labels	Switches: data-openai-panel-access=['professor', 'dean', 'procoor', 'hr', 'admin', 'vpaa'] State Labels: .openai-panel-access-state	Switches toggle AI features independently per panel without altering the global API key; state labels reflect current On/Off status.	Panel AI toggles enabled and disabled AI features independently for each panel.	PASS

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	update between 'On' and 'Off'.				
TC-93	1. Click 'Clear Saved API Key' button (#gemini-clear-key-btn). 2. Confirm key removal prompt. 3. Verify API key status reverts to missing.	Button: #gemini-clear-key-btn Status: #gemini-config-status	Confirmation prompt prevents accidental deletion; key is wiped from database; AI features fallback gracefully without system crashes.	Clear API key button purged stored credentials and reverted AI features to disabled.	PASS
TC-94	1. Scroll to System Health & Update Verification group (.system-health-group). 2. Inspect status badge (#system-health-status-badge: 'Not Run' or status). 3. Inspect summary items (Last Run, Checked By, Result summary).	Container: .system-health-group Badge: #system-health-status-badge Summary: #system-health-last-run, #system-health-checked-by, #system-health-summary	System health section renders initial state showing previous verification results or 'Not Run' status badge.	System health dashboard rendered initial status showing previous verification results.	PASS
TC-95	1. Click 'Run Verification' button (#system-health-run-btn). 2. Observe execution feedback (#system-health-feedback). 3. Observe Check Results table (#system-health-checks-body).	Button: #system-health-run-btn Table: #system-health-checks-body Feedback: #system-health-feedback	System executes automated read-only health checks (database integrity, schema sync, storage writeability, auth endpoints); displays check status badges (Passed, Warning, Failed).	Run verification executed automated tests across database, storage, and session services.	PASS
TC-96	1. Inspect Recent Verification History table (#system-health-history-body). 2. Verify columns: Run ID, Status, Passed, Warnings, Failed, Checked By, Run At. 3. Check record archiving.	Table: #system-health-history-body Columns: Run ID, Status, Passed, Warnings, Failed, Checked By, Run At	History table logs each health verification run with immutable timestamp and user ID; preserves audit record as proof of system integrity.	Recent verification history logged health check executions with timestamps and pass flags.	PASS
TC-97	1. Introduce simulated warning (e.g. storage threshold approaching limit). 2. Re-run health check. 3. Verify Warning status badge and detailed message.	Condition: Sub-optimal system parameter	Health verification detects anomaly; marks specific check with warning badge and descriptive detail while overall status reports warnings.	Simulated resource threshold alert was flagged with an amber warning badge.	PASS
RE-16 Activity Log, Announcements & Profile Security					
Pre-requisite: Administrator logged in. System activity log, announcement composer, and profile modules loaded.					
TC-98	1. Navigate to Activity Log view (#activity-	View: #activity-log-view	Activity log table displays audit records	Activity log table displayed	PASS

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	log-view). 2. Set From date (#activity-from) and To date (#activity-to). 3. Select Search Type (#activity-type: User Login, User Update, Evaluation, System). 4. Click Search (#activity-search-btn).	Inputs: #activity-from, #activity-to, #activity-type Button: #activity-search-btn	matching date range and event type with IP, timestamp, description, action, role, user_id, and log_id.	audit records matching date range and user filters.	
TC-99	1. In Activity Log view, enter keyword into search input (#activity-search). 2. Press Enter or click Search. 3. Verify query matches against IP address, user ID, description, and role.	Input: #activity-search Query: User ID or keyword	Table filters instantly to rows matching search term across all metadata attributes.	Search input filtered activity logs dynamically across descriptions, IPs, and user IDs.	PASS
TC-100	1. Open Announcement Compose modal (#announcement-compose-modal) from Quick Actions or Settings. 2. Enter Title, Message, select Target Role (e.g. 'All Users' or 'Student'). 3. Verify audience preview (#announcement-audience-preview). 4. Click 'Publish Announcement'.	Modal: #announcement-compose-modal Form: #announcement-compose-form Preview: #announcement-audience-preview	Form validates required fields and length limits; audience preview confirms recipient scope; publishing broadcasts announcement to target users upon login.	Announcement composer validated required fields and audience selection parameters.	PASS
TC-101	1. In Announcement composer, select Target Role = 'Student'. 2. Select Campus and Program filters. 3. Filter by Student Completion (Completed vs Not Completed). 4. Publish announcement.	Role: Student Filters: Campus, Program, Completion status	Granular audience targeting directs notice specifically to students matching enrollment and evaluation completion status.	Granular audience targeting dispatched announcements specifically to the selected role.	PASS
TC-102	1. Navigate to Admin Profile view (#profile-view). 2. Inspect Admin Profile card (.profile-card). 3. Verify demographic values: Faculty ID (ADM-0001), Full Name, Department,	View: #profile-view Card: .profile-card Values: Synced employee credentials	Profile card displays complete administrative credentials in read-only responsive grid layout.	Admin profile card rendered official system administrator information in read-only format.	PASS

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	Rank, Email, AY/SEM, Assigned Institutes.				
TC-103	1. In Profile view, click 'Upload Photo' button (#profilePhotoInput). 2. Select valid image file (JPG/PNG < 2MB). 3. Verify profile preview (#profilePhotoPreview) and placeholder update.	Input: #profilePhotoInput Valid: JPG/PNG < 2MB Preview: #profilePhotoPreview	Photo is validated and loaded into avatar preview; file exceeding 2MB or invalid format triggers warning alert.	Profile photo upload accepted a valid image, rendered a preview, and saved properly.	PASS
TC-104	1. Click 'Change Email' button to open #adminEmailUpdateCard. 2. Enter new email and mismatched confirmation email; submit form (#adminChangeEmailForm). 3. Enter matching valid email and submit.	Card: #adminEmailUpdateCard Form: #adminChangeEmailForm Inputs: #adminNewEmail, #adminConfirmEmail	Mismatched confirmation triggers error prompt; valid matching email submits update to /api/admin/change-email endpoint with success feedback.	Email update form validated input and saved the new address upon confirmation.	PASS
TC-105	1. Click 'Change Password' button to open #adminPasswordUpdateCard. 2. Enter current password, new password, and confirmation. 3. Click password eye toggle buttons (.toggle-password) on each field. 4. Submit form (#adminChangePasswordForm).	Card: #adminPasswordUpdateCard Form: #adminChangePasswordForm Buttons: .toggle-password	Eye buttons toggle password masking between plaintext and dots; valid password submission updates admin credentials and displays confirmation.	Password change form enforced complexity rules and updated credentials successfully.	PASS

System Name:	NAAP Evaluation System
Module / View:	admin Panel (adminpanel.html)
Created by:	John Emmanuel E Rojo
Executed by:	John Emmanuel E Rojo
Date of Execution:	SEPTEMBER 28 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

NAAP EVALUATION SYSTEM - VPAA PANEL MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 25 2026

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RE-01 AUTHENTICATION, SESSION SECURITY & VPAA ROLE AUTHORIZATION					
Pre-requisite: Browser is running; NAAP session & database service are operational.					
TC-001	1. Log in with valid VPAA executive credentials. 2. Navigate to html/vpaapanel.html. 3. Verify checkAuthentication() check. 4. Observe Overview dashboard view load.	User: vpaa / VPAA Office Role: vpaa Status: active Session: Valid JWT / session token	checkAuthentication() returns true. VPAA Dashboard loads successfully, displaying executive title, header profile badge, and overview metric cards.	Logged in with valid VPAA executive credentials and dashboard loaded properly with executive metrics.	PASS
TC-002	1. Launch private browser window with no active session. 2. Attempt direct URL navigation to html/vpaapanel.html. 3. Observe authentication guard.	Session: None (null / undefined) URL: .../html/vpaapanel.html	Access is denied immediately. System invokes redirectToLogin() and redirects browser to mainpage.html without rendering academic evaluation analytics.	Unauthenticated direct URL access was blocked and redirected immediately to the login page.	PASS
TC-003	1. Establish active session with non-VPAA role ('student', 'faculty', 'osa'). 2. Attempt direct URL navigation to html/vpaapanel.html. 3. Observe role authorization guard.	User: ST-000001 / PROF-001 Role: student / faculty Target: vpaapanel.html	Role authorization fails (strict requirement === 'vpaa'). User is redirected away to their corresponding portal view or login page.	Attempting to access the VPAA panel with a non-executive role was denied and redirected away.	PASS
TC-004	1. Set VPAA account status to 'inactive' in database. 2. Attempt login or access to vpaapanel.html. 3. Observe account restriction handling.	User: vpaa Status: inactive Role: vpaa	System denies access; displays account inactive notice, clears session tokens, and redirects user to mainpage.html?logged_out=1.	Inactive executive account was blocked from logging in with a notification to contact the administrator.	PASS
TC-005	1. On VPAA panel, click 'Logout' button (#vpaaLogoutBtn) in sidebar footer. 2. Observe session purge and redirect.	Action: Click #vpaaLogoutBtn Target URL: mainpage.html?logged_out=1	clearUserSession() purges all active session keys from localStorage/sessionStorage and redirects to mainpage.html?logged_out=1 with logout message.	Clicked logout and verified that all executive session tokens were purged from storage.	PASS
TC-006	1. Simulate expired session token during active monitoring session. 2. Attempt to switch views or open analytics modal. 3. Observe session expiration handling.	Session: Expired timestamp / invalid token	System intercepts the request, detects session timeout, alerts user, and safely redirects to login page without submitting action.	Expired session was intercepted during navigation, displaying a timeout notice and returning to login.	PASS
RE-02 VPAA DASHBOARD OVERVIEW, RESPONSIVE NAVIGATION & SUMMARY METRICS					
Pre-requisite: VPAA executive is logged in on the VPAA Dashboard.					
TC-	1. Click 'Reports' nav	Sidebar Nav	Active class updates	Sidebar	PASS

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007	link in sidebar. 2. Click 'Profile' nav link. 3. Click 'Overview' nav link.	Links: - data-view='reports-view' - data-view='profile-view' - data-view='dashboard-view'	smoothly on the clicked navigation link. Target content view displays immediately while sibling view containers hide.	navigation links toggled views smoothly while highlighting the active section.	
TC-008	1. Resize browser viewport to mobile width (< 992px). 2. Click mobile navigation toggle button (#vpaaMenuToggle). 3. Verify mobile sidebar drawer. 4. Click sidebar backdrop overlay (#sidebarBackdrop).	Viewport: 375px x 812px (Mobile) Elements: #vpaaMenuToggle, #sidebarBackdrop, #vpaaSidebar	Sidebar drawer translates smoothly from left into viewport with backdrop overlay. Clicking backdrop or pressing Esc smoothly dismisses mobile navigation drawer.	Mobile drawer opened cleanly using the hamburger toggle and dismissed upon pressing Esc.	PASS
TC-009	1. On Overview hero card, click 'Open Reports' button (#heroOpenReportsBtn). 2. Return to Overview and click 'View Highlights' button (#heroScrollHighlightsBtn).	Hero Buttons: - #heroOpenReportsBtn - #heroScrollHighlightsBtn	'Open Reports' switches view to #reports-view; 'View Highlights' scrolls viewport smoothly to evaluation charts and highlight sections.	Overview hero action buttons navigated directly to reports and highlights sections.	PASS
TC-010	1. Inspect top header user profile display across all 3 views. 2. Verify synchronization of office name.	Header Profile: .user-profile ('VPAA Office')	All views consistently render the executive office title and icon in the top header.	Top header consistently displayed the VPAA executive profile and title across all views.	PASS
TC-011	1. Inspect Overview summary cards on Dashboard. 2. Verify 'Total Students' count (#totalStudents). 3. Verify 'Completion Rate' percentage (#completionRate). 4. Verify 'Pending Evaluations' count (#pendingEvaluations). 5. Verify 'Active Professors' count (#activeProfessors).	Summary Cards: - #totalStudents (e.g. '1,250') - #completionRate (e.g. '84.5%') - #pendingEvaluations (e.g. '194') - #activeProfessors (e.g. '48')	Summary cards display accurate institution-wide metrics reflecting active student population, completed vs expected evaluation rate, and total active faculty.	Summary cards correctly rendered university-wide metrics for enrolled faculty and evaluation progress.	PASS
TC-012	1. Inspect real-time data subscription listener	Event: SharedData data change	Panel automatically refreshes overview summary counters and	Dashboard counters refreshed	PASS

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	(setupDataSubscriptions()). 2. Trigger a new evaluation submission from another role. 3. Observe VPAA panel overview metrics.	(Evaluations / Users)	chart aggregates in real time without requiring manual page reload.	automatically upon detecting newly submitted evaluation data.	
TC-013	1. Verify handling when database contains zero evaluation records. 2. Observe Overview summary cards and chart placeholders.	Data: 0 evaluations submitted	Summary cards display '0', '0%', and chart containers render clean empty placeholders without throwing JavaScript errors.	Empty state displayed zero counts gracefully when no evaluation data was present.	PASS
RE-03 INSTITUTIONAL MULTI-EVALUATION CHART ANALYTICS (STUDENT, PEER, SUPERVISOR)					
Pre-requisite: VPAA Overview view is active with loaded multi-evaluation datasets.					
TC-014	1. Locate 'Student to Professor Evaluation' section on Overview. 2. Inspect Average Scores Category Bar Chart (#vpaa-student-professor-bar-chart). 3. Hover over bar segments.	Chart: #vpaa-student-professor-bar-chart Type: Chart.js Bar Chart	Bar chart displays institutional average scores across criteria rubrics (e.g. Subject Mastery, Methodology); hover tooltips display exact scores.	Student-to-professor bar chart rendered institutional category averages with interactive hover tooltips.	PASS
TC-015	1. Inspect Student Evaluation Rating Distribution Pie Chart (#vpaa-student-professor-pie-chart). 2. Hover over pie slices.	Chart: #vpaa-student-professor-pie-chart Type: Chart.js Doughnut/Pie Chart	Pie chart renders proportion of student ratings across score tiers (5, 4, 3, 2, 1); hover tooltips display percentages and response counts.	Rating distribution pie chart accurately rendered score proportions across all performance levels.	PASS
TC-016	1. Inspect Student Evaluation Stats Summary items: - Average Rating (#vpaa-student-prof-avg-rating) - Total Evaluations (#vpaa-student-prof-total) - Professors Evaluated (#vpaa-student-prof-count).	Stats: #vpaa-student-prof-avg-rating, #vpaa-student-prof-total, #vpaa-student-prof-count	Values accurately calculate and format the overall student rating average (e.g. '4.6/5.0'), total submissions, and count of distinct faculty evaluated.	Student evaluation statistics summary displayed the correct overall mean and response tallies.	PASS
TC-017	1. Locate 'Professor to Professor Evaluation' (Peer) section. 2. Inspect Peer Bar Chart (#vpaa-professor-professor-bar-chart) and Pie Chart (#vpaa-professor-professor-pie-chart). 3. Verify Peer Stats	Charts: #vpaa-professor-professor-bar-chart, #vpaa-professor-professor-pie-chart Stats: #vpaa-professor-prof-avg-rating	Charts render peer evaluation performance across workplace collaboration rubrics; stats accurately reflect faculty-to-faculty evaluation metrics.	Peer evaluation section rendered peer performance metrics across workplace collaboration rubrics.	PASS

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	Summary.				
TC-018	1. Locate 'Supervisor to Professor Evaluation' section. 2. Inspect Supervisor Bar Chart (#vpaa-supervisor-professor-bar-chart) and Pie Chart (#vpaa-supervisor-professor-pie-chart). 3. Verify Supervisor Stats Summary.	Charts: #vpaa-supervisor-professor-bar-chart, #vpaa-supervisor-professor-pie-chart Stats: #vpaa-supervisor-prof-avg-rating	Charts display administrative supervisor ratings across leadership rubrics; stats accurately reflect Dean/Supervisor evaluations.	Supervisor evaluation charts displayed dean rating averages across leadership and management criteria.	PASS
TC-019	1. Compare category ratings across the 3 evaluation types side by side.	Types: Student vs Peer vs Supervisor	All 3 sections render concurrently with distinct color palettes, enabling high-level triangulation of faculty performance.	Comparative analytics presented all three evaluation perspectives side by side with clear color coding.	PASS
TC-020	1. Resize browser window to mobile width (< 768px). 2. Verify chart responsiveness.	Viewport: Mobile (375px)	Chart canvases resize gracefully without overlapping text or breaking page margins.	Chart canvases resized smoothly on mobile viewports without label overlap or distortion.	PASS
RE-04 DESCRIPTIVE ANALYTICS CAMPUS FILTER & DEPARTMENT DRILL-DOWN					
Pre-requisite: VPAA navigates to Reports & Insights view (#reports-view).					
TC-021	1. In Descriptive Analytics section, inspect Campus filter dropdown (#vpaaAnalyticsCampusSelect). 2. Verify options ('All Campuses', etc.).	Dropdown: #vpaaAnalyticsCampusSelect Options: Dynamic campus list	Dropdown lists all operational institutional campuses, defaulting to 'All Campuses'.	Campus dropdown populated all institutional campuses and defaulted to all campuses.	PASS
TC-022	1. Inspect 'By Department' table (#vpaaDepartmentAnalyticsBody). 2. Verify columns: Department, Assigned, Evaluated, Not Evaluated, Completion %.	Table: .vpaa-analytics-table (Department) Columns: Dept, Assigned, Evaluated, Not Evaluated, Completion %	Table displays department-level student tallies with formatted completion percentage badges across the selected campus scope.	Department analytics table rendered student tallies and evaluation completion rates.	PASS
TC-023	1. Click on a specific department row in 'By Department' table (e.g. 'Aeronautical Engineering'). 2. Observe selection highlight and 'Programs by Department' table	Action: Click department row Target: #vpaaDepartmentProgramsBody	Selected department row is highlighted; table title updates to 'Programs in Aeronautical Engineering'; programs table populates with corresponding degree programs.	Clicking a department row highlighted it and dynamically loaded its programs below.	PASS

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	(#vpaaDepartmentProgramsBody).				
TC-024	1. Inspect columns in 'Programs by Department' table: - Program - Assigned - Evaluated - Not Evaluated - Completion %.	Table: #vpaaDepartmentProgramsBody	Table renders granular sub-program completion tallies and completion rates under the selected department.	Programs table rendered completion percentages and enrolled student tallies accurately.	PASS
TC-025	1. Select a different campus from #vpaaAnalyticsCampusSelect (e.g. 'Extension Campus'). 2. Observe update in Department and Program tables.	Selection: Extension Campus Function: getVpaaAnalyticsRowsForSelectedCampus()	Descriptive analytics tables immediately update to reflect student completion metrics specifically for the selected campus.	Selecting a specific campus filtered the descriptive analytics tables in real time.	PASS
TC-026	1. Verify initial state of Programs by Department table before any department row is clicked.	State: No department selected	#vpaaDepartmentProgramsEmpty displays message: 'Select a department above to view its programs.'	Programs table displayed a helpful prompt to select a department before loading data.	PASS
RE-05	REPORTS SEARCH, MULTI-FILTER TOOLBAR & SORTING CAPABILITIES				
Pre-requisite: VPAA is on Reports & Insights view with Evaluation Filters toolbar.					
TC-027	1. In Evaluation Filters toolbar, enter professor name or employee ID in #searchInput. 2. Click 'Search' button (#searchBtn).	Query: 'Lauron' / 'PROF-001' Button: #searchBtn	Professor evaluation grid (#professorGrid) filters dynamically to display cards matching query; highlight cards update accordingly.	Typing a professor name in the search bar filtered the grid in real time.	PASS
TC-028	1. Select specific semester from #semesterFilter dropdown. 2. Observe reports grid and analytics update.	Filter: #semesterFilter Selection: Historical Semester	System reloads evaluation data, ratings, and feedback corresponding to the selected academic year and semester.	Selecting a semester reloaded the professor grid with evaluations from that term.	PASS
TC-029	1. Select specific campus from #campusFilter dropdown. 2. Observe faculty card filtering.	Filter: #campusFilter Selection: Specific Campus	Professor grid displays only faculty members assigned to the chosen campus.	Campus filter narrowed the faculty grid to display only professors from the chosen campus.	PASS
TC-030	1. Select specific department from #departmentFilter dropdown. 2. Observe faculty card filtering.	Filter: #departmentFilter Selection: 'Aviation Studies'	Professor grid narrows to display faculty belonging to the selected department.	Department dropdown filtered the grid to show faculty belonging to the selected department.	PASS
TC-031	1. Test 'Sort by' dropdown (#sortFilter) with option 'Rating (high to low)'.	Sort Options: - 'rating-high' - 'rating-low'	sortProfessors() orders faculty cards from highest overall rating to lowest, and vice versa.	Sorting by highest rating reordered professor cards	PASS

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	2. Test 'Sort by' with option 'Rating (low to high)'.			from highest to lowest score.	
TC-032	1. Test 'Sort by' with option 'Response rate' and 'Professor name'.	Sort Options: - 'response' - 'name'	Cards sort by student response percentage descending, or alphabetically by professor full name.	Sorting by response rate successfully arranged faculty cards by participation percentage.	PASS
TC-033	1. Click 'Reset Filters' button (#resetFilters). 2. Observe all filter controls and professor grid.	Action: Click #resetFilters	Search input clears, semester/campus/department filters revert to defaults, sort resets, and complete faculty grid restores.	Clicking reset filters cleared all inputs and restored the default faculty list.	PASS
RE-06 ALGORITHMIC KEY HIGHLIGHTS (TOP RATING, MOST COMMENTS, NEEDS ATTENTION)					
Pre-requisite: Reports view has loaded evaluation records for active filter scope.					
TC-034	1. Inspect 'Key Highlights' section (.vpaa-key-highlights-section). 2. Verify 'Top Rating' highlight card (#vpaaHighlightTopRating).	Card: #vpaaHighlightTopRating Metric: Highest overall score	Card accurately identifies and displays the professor with highest composite rating, showing professor name and score (e.g. '4.95 / 5.0').	Top rating highlight card correctly identified and showcased the highest-rated faculty member.	PASS
TC-035	1. Inspect 'Most Comments' highlight card (#vpaaHighlightMostComments). 2. Verify displayed faculty name and comment volume.	Card: #vpaaHighlightMostComments Metric: Total comment count	Card identifies the professor receiving the highest number of student and peer qualitative feedback submissions.	Most comments card accurately displayed the professor receiving the highest feedback volume.	PASS
TC-036	1. Inspect 'Needs Attention' highlight card (#vpaaHighlightNeedsAttention). 2. Verify identified faculty member.	Card: #vpaaHighlightNeedsAttention Metric: Lowest composite rating / significant score decline	Card flags professor requiring academic leadership attention based on low composite rating (< 3.0) or severe downward trend.	Needs attention card flagged faculty with low scores or declining performance trends.	PASS
TC-037	1. Apply search or department filter that changes the active professor pool. 2. Observe dynamic recalculation of Key Highlights.	Filter: Change department in #departmentFilter	buildKeyHighlights() recalculates highlights dynamically, reflecting top rating, most comments, and needs attention for the filtered sub-pool.	Applying department filters dynamically recalculated all key highlight cards.	PASS
TC-038	1. Filter for a department with zero evaluations. 2. Observe Key Highlights cards.	Data: 0 evaluations in filter scope	Cards display 'No data yet' and metric text 'Waiting for evaluation data', or #vpaaHighlightsEmpty appears.	Highlight cards displayed placeholder notices when filtered departments had no data.	PASS
TC-039	1. Click on a professor name displayed in a	Action: Click highlight card	System scrolls to or highlights the	Clicking a professor in key	PASS

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	Key Highlight card. 2. Observe focus or analytics trigger.		corresponding faculty card in the Professor Evaluation Reports section.	highlights smoothly scrolled to their card in the grid.	
RE-07 PROFESSOR EVALUATION REPORTS GRID & ANALYTICS MODAL INSPECTION					
Pre-requisite: Reports grid (#professorGrid) displays faculty cards.					
TC-040	1. Inspect professor cards rendered in #professorGrid. 2. Verify card elements: Professor Avatar/Initials, Full Name, Department, Rank, SET Score, SAF Score, Response Rate badge.	Element: .professor-card in #professorGrid	Each card renders structured faculty identification, overall ratings, participation badges, and action buttons ('View Analytics', 'Files').	Professor cards rendered complete credentials, overall score, and evaluation response rates.	PASS
TC-041	1. On a professor card, click 'View Analytics' button. 2. Observe Professor Analytics Modal (#professor-analytics-modal) opening.	Action: Click 'View Analytics' Modal: #professor-analytics-modal	viewVpaaProfessorAnalytics() fetches detailed metrics, opens modal, and renders comprehensive analytics container (#professor-analytics-content).	Clicking view analytics opened the deep-dive analytics modal for the selected professor.	PASS
TC-042	1. In analytics modal, verify evaluation type switcher tabs (Student, Peer, Supervisor). 2. Switch between tabs.	Tabs: Student vs Peer vs Supervisor	Modal dynamically switches between evaluation sources, updating category averages table and score distribution tiers for the selected professor.	Evaluation type switcher toggled smoothly between student, peer, and supervisor metrics.	PASS
TC-043	1. In analytics modal, inspect Category Scores breakdown table. 2. Verify rubric categories and computed averages.	Table: Category Scores	Renders detailed criteria performance (e.g. Teaching Skills, Course Delivery, Professionalism) specific to that professor.	Category scores section broke down performance across individual rubric dimensions.	PASS
TC-044	1. In analytics modal, inspect Score Distribution table. 2. Verify counts and percentages for Excellent (5) through Very Poor (1).	Table: Score Distribution	Displays exact response frequencies and percentages across all 5 score tiers.	Score distribution chart displayed exact response frequencies across ratings 1 through 5.	PASS
TC-045	1. Click modal close button (#close-analytics-modal) or press Esc. 2. Verify modal dismissal.	Action: Click close 'X' / Esc	#professor-analytics-modal closes cleanly, returning focus to reports view.	Analytics modal closed smoothly upon clicking the close button and restored page focus.	PASS
TC-046	1. Test pagination controls (#professorPagination) when faculty count exceeds page limit.	Pagination: Page 1, Page 2, Next, Prev	Cards paginate smoothly without flickering or resetting active filter parameters.	Pagination buttons navigated across professor report	PASS

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	2. Navigate through pages.			pages cleanly without UI flicker.	
RE-08	HISTORICAL PERFORMANCE TRAJECTORY, WORD FREQUENCY & AI SYNTHESIS				
Pre-requisite: Professor Analytics Modal is open for a selected professor.					
TC-047	1. In analytics modal, inspect 4-Semester Historical Trend Analysis section. 2. Verify semester progression datapoints and trajectory direction.	Section: buildVpaaProfessorHistoricalTrend() Data: Previous 4 active semesters	Displays historical rating trajectory across past 4 semesters with delta indicator (+0.12 or -0.08) and trend consistency classification.	Multi-semester trajectory chart displayed historical ratings across four consecutive terms.	PASS
TC-048	1. Verify trend decline pattern alert for professors with consecutive score drops.	Metric: getVpaaHistoricalTrendDeclinePatternLabel()	System flags downward trends with warning tag (e.g. 'Decline Pattern Detected') to alert academic leadership.	Downward performance trend was detected and flagged with an alert badge.	PASS
TC-049	1. In analytics modal, inspect Qualitative Feedback & Word Frequency section. 2. Verify word frequency list (updateWordFrequency()):	Section: Word Frequency Lexicon: High-frequency student comment tokens	Renders top recurring keywords (e.g. 'clear explanations', 'helpful', 'organized') with occurrence counts.	Word frequency cloud rendered prominent recurring keywords from student feedback.	PASS
TC-050	1. In analytics modal, click 'AI Analytics & Insights' button. 2. Observe runAiAnalyticsForProfessor() execution.	Action: Click AI Analytics Engine: Local / AI analytics pipeline	System analyzes qualitative feedback and ratings, generating executive synthesis with strengths, growth areas, and supervisory recommendations.	AI analytics engine generated an executive summary synthesizing ratings and comments.	PASS
TC-051	1. Inspect AI explainability insights and judgment classification badge.	Classifications: 'Exemplary', 'Proficient', 'Needs Developmental Plan'	Displays appropriate judgment badge and structured reasoning based on rubric scores and sentiment clusters.	AI explainability card displayed a qualitative judgment badge with supporting rationale.	PASS
TC-052	1. Filter for a professor with no qualitative comments. 2. Observe Word Frequency and AI sections.	Data: 0 comments submitted	Word frequency displays 'No qualitative comments available'; AI analytics displays empty notice gracefully without breaking modal layout.	Professors with no comments showed a clean empty state notice in the word cloud.	PASS
RE-09	INSTITUTIONAL EXPORT FLOWS (OVERALL SASR, IFER & ACKNOWLEDGEMENT PAPERS)				
Pre-requisite: VPAA is on Reports & Insights view.					
TC-053	1. In Reports section header, click 'Overall SASR' button (#vpaa-overall-sasr-btn). 2. Observe Overall	Action: Click #vpaa-overall-sasr-btn Modal: Overall SASR Scope	Modal opens displaying export scope configuration: Campus selector, Department selector.	Clicking Overall SASR opened the export dialog with scope and	PASS

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	SASR Modal (openVpaaOverallSasrModal()).	Modal	Program selector, and Semester choice.	format options.	
TC-054	1. In Overall SASR modal, select scope (All Campuses, All Departments) and click 'Export Report'. 2. Observe export generation.	Action: Click Export Output: Excel / PDF spreadsheet export	System compiles university-wide Summary of Academic Staff Ratings (SASR) spreadsheet and triggers file download.	Generated university-wide SASR document compiled all ratings into a printable report.	PASS
TC-055	1. On a professor card, click 'Files' / 'Report' button. 2. Verify File Options Modal (openVpaaProfessorFileOptions()).	Action: Click Files Options: SASR, IFER, Faculty Acknowledgement Paper	Modal presents available institutional documents for the selected professor.	Clicking files on a professor card opened the document selection modal.	PASS
TC-056	1. In File Options modal, select 'Individual SASR'. 2. Observe openVpaaProfessorSasrFlow() execution.	Action: Select Individual SASR	System compiles individual faculty SASR report and opens PDF preview modal with download button.	Exporting individual SASR compiled the professor's report and opened a PDF preview.	PASS
TC-057	1. In File Options modal, select 'IFER Document' (openVpaaProfessorIferFlow()). 2. Verify semester picker and generation.	Action: Select IFER Document	System compiles qualitative comments into official IFER document layout and initiates download.	IFER export generated an official document summarizing student comments and feedback.	PASS
TC-058	1. In File Options modal, select 'Faculty Acknowledgement Paper' (openVpaaProfessorAcknowledgementFlow()). 2. Verify load type choice (Main Load vs Excess Load). 3. Inspect PDF preview.	Action: Select Acknowledgement Paper Load Type: Main / Excess	openVpaaPdfBlobPreview() displays endorsed acknowledgement paper complete with Sections A, B, and Dean/VPAA signed Section C.	Faculty acknowledgement document rendered complete endorsement and signature sections.	PASS
RE-10 VPAA OFFICE PROFILE, PHOTO UPLOAD & ACCOUNT SECURITY					
Pre-requisite: VPAA is on Profile view (#profile-view).					
TC-059	1. Inspect VPAA Profile card (#profile-view). 2. Verify read-only demographic fields: - Faculty ID (VPAA-0001) - Full Name (VPAA Office) - Department (Academic Affairs) - Rank (Vice President for Academic Affairs) - Gmail (#profileEmail)	Fields: ID, Name, Dept, Rank, Gmail, AY/SEM	Profile fields populate correctly with official executive office records in read-only format.	Profile view displayed verified VPAA executive information in read-only format.	PASS

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	- AY/SEM.				
TC-060	1. In profile photo section, click 'Upload Photo' (.profile-photo-btn). 2. Select valid image file (JPG/PNG under 2MB) via #profilePhotoInput. 3. Verify live preview and upload.	File: vpaa_seal.png (1.2 MB) Element: #profilePhotoInput, #profilePhotoPreview	System validates file type/size, renders live preview in #profilePhotoPreview, and hides placeholder initials (#profilePhotoPlaceholder).	Profile photo upload accepted a valid image, showed a live preview, and saved properly.	PASS
TC-061	1. Select an invalid file type (.pdf or .docx) or file exceeding 2MB. 2. Observe handleUploadError() response.	File: report.pdf / 12MB_image.jpg	System rejects upload immediately with validation message: 'Please upload a valid image file (JPG or PNG) under 2MB.' Current photo is retained.	Invalid file formats were blocked immediately with an unsupported file type alert.	PASS
TC-062	1. In Account Actions card, click 'Change Gmail' button. 2. In #emailUpdateCard, enter mismatched new & confirm Gmail addresses. 3. Click 'Update Gmail' (.btn-submit). 4. Enter matching valid Gmail and submit.	Case A: newEmail != confirmEmail Case B: newEmail == confirmEmail ('vpaa.new@naap.edu')	Case A displays mismatch error. Case B dispatches update to /api/vpaa/change-email, displays success notification, updates profile email, and closes form.	Gmail update form validated input and saved the new address upon confirmation.	PASS
TC-063	1. In Account Actions card, click 'Change Password' button. 2. In #passwordUpdateCard, test password visibility eye toggles (.toggle-password) for Current, New, and Confirm password fields.	Fields: #currentPassword, #newPassword, #confirmPassword Buttons: .toggle-password	Clicking eye icon toggles field type between 'password' and 'text', toggling icon between fa-eye and fa-eye-slash.	Password visibility toggle switched the input between hidden and revealed characters.	PASS
TC-064	1. In change password form, test validations: - New password < 8 chars or missing number - Mismatched confirm password 2. Enter valid matching credentials and click 'Update Password'.	Validation: min 8 chars with number, matching confirm API: POST /api/vpaa/change-password	Invalid inputs display corresponding validation error; valid submission dispatches POST request, displays 'Password updated successfully!', and resets password fields.	Password change form enforced complexity rules and updated credentials successfully.	PASS
TC-065	1. On account update forms (#emailUpdateCard, #passwordUpdateCard), click 'Cancel' button. 2. Observe form	Action: Click Cancel (.js-close-account-form)	Active form card closes cleanly; form inputs are cleared without saving modifications.	Canceling an account edit closed the form card and restored original values.	PASS

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System Name:	NAAP Evaluation System
Module / View:	VPAA Panel (vpaapanel.html)
Created by:	John Emmanuel E Rojo
Executed by:	John Emmanuel E Rojo
Date of Execution:	SEPTEMBER 25 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

NAAP EVALUATION SYSTEM - STUDENT PANEL MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 25 2026
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RE-01 AUTHENTICATION, SESSION GUARD & STUDENT ROLE AUTHORIZATION

Pre-requisite: Browser is running; NAAP database/session service is operational.

TC-001	1. Log in with valid student credentials. 2. Navigate to studentpanel.html. 3. Verify session authentication check (checkAuthentication()) 4. Observe page load and identity elements.	User: ST-000001 (John Student) Role: student Status: active Session: Valid JWT / session token	checkAuthentication() returns true. Student Dashboard loads successfully, displaying student's full name and ID ST-000001 in header and profile badge.	Logged in as student and dashboard loaded successfully with the correct student name and ID.	PASS
TC-002	1. Open new private/incognito browser window with no session. 2. Attempt direct URL navigation to html/studentpanel.html. 3. Observe system authentication interception.	Session: None (null / undefined) URL: .../html/studentpanel.html	System denies access, executes redirectToLogin(), and immediately redirects browser to mainpage.html without rendering dashboard content.	Unauthenticated direct URL access was blocked and redirected immediately to the login page.	PASS
TC-003	1. Establish active session with role 'faculty' or 'admin'. 2. Attempt direct URL navigation to html/studentpanel.html. 3. Observe role authorization guard.	User: professor1 / admin1 Role: faculty / admin Target: studentpanel.html	Role check fails (strict role requirement === 'student'). User is redirected away to login or their corresponding panel view.	Attempting to access the student panel with a faculty role was denied and redirected away.	PASS
TC-004	1. Set student account status in database to 'inactive'. 2. Authenticate and attempt to access studentpanel.html. 3. Observe	Student ID: ST-000001 Status: inactive Role: student	System triggers alert modal: 'Your account is inactive. You cannot access evaluations. Please contact your administrator.' Clears session and redirects	Inactive student account was blocked with a notice to contact the administrator.	PASS

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	enforceActiveStudentAccount() behavior.		to mainpage.html?logged_out=1.		
TC-005	1. Log in as student. 2. In sidebar footer, click 'Logout' button (#studentLogoutBtn). 3. Observe session purge and redirection.	Action: Click #studentLogoutBtn Target URL: mainpage.html?logged_out=1	clearUserSession() purges all active session keys from localStorage/sessionStorage and redirects user to mainpage.html?logged_out=1 with logout message.	Logged out successfully, clearing all session data and redirecting to the login screen.	PASS
TC-006	1. While on student panel, simulate session token expiration or timeout. 2. Attempt to navigate between views or submit actions. 3. Observe expired session handling.	Session: Expired timestamp / invalid token	System detects expired session, halts current operation, displays session expiration notification, and redirects safely to login page.	Expired session was detected by the system, prompting the user and redirecting to login.	PASS
RE-02 STUDENT DASHBOARD NAVIGATION, RESPONSIVE DRAWER & NOTIFICATIONS					
Pre-requisite: Student is logged in with active account on student dashboard.					
TC-007	1. Click 'Evaluation Forms' nav link in sidebar. 2. Click 'History' nav link. 3. Click 'Profile' nav link. 4. Click 'Home' nav link.	Sidebar Nav Links: - data-view='evaluationForm' - data-view='history' - data-view='profile' - data-view='dashboard'	Active class toggles accurately on clicked link. Corresponding view container (#evaluationFormView, #historyView, #profileView, #dashboardView) renders while sibling views are hidden.	Sidebar navigation links toggled views smoothly while hiding inactive containers.	PASS
TC-008	1. Resize browser viewport to mobile width (< 992px). 2. Click mobile hamburger toggle (#studentMenuToggle). 3. Verify sidebar drawer appearance. 4. Click backdrop overlay (#sidebarBackdrop).	Viewport: 375px x 667px (Mobile) Elements: #studentMenuToggle, #sidebarBackdrop, #studentSidebar	Sidebar transitions into view from left with active class; backdrop overlay covers background. Clicking backdrop or pressing Esc smoothly dismisses mobile drawer.	Mobile drawer opened smoothly via hamburger toggle and closed when pressing Esc.	PASS
TC-009	1. Click notification bell icon (#notificationBtn) in header. 2. Observe announcement dropdown panel (#announcementPanel). 3. Click outside the dropdown.	Element: #notificationBtn, #announcementPanel Data: Start date: February 1, 2026; End date: February 28, 2026	#announcementPanel expands displaying evaluation period schedule and active announcements. Clicking anywhere outside closes the panel smoothly.	Notification bell opened the announcements dropdown and closed cleanly when clicking outside.	PASS
TC-010	1. Click user profile button (#profileBtn) in header.	Element: #profileBtn, #profilePanel	#profilePanel toggles open showing student ID ST-000001 and	Profile preview menu displayed the student ID	PASS

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	2. Inspect quick info dropdown (#profilePanel). 3. Click outside the panel.	Display: Student ID: ST-000001, Program: BS Aviation	program BS Aviation in a clean 2-column grid. Closes upon clicking outside.	and program details accurately.	
TC-011	1. In dashboard hero card, click 'Browse Pending Forms' (#heroOpenEvaluationBtn). 2. Return to dashboard and click 'View History' (#heroOpenHistoryBtn).	Hero Buttons: - #heroOpenEvaluationBtn - #heroOpenHistoryBtn	'Browse Pending Forms' navigates to pending evaluations list/form; 'View History' directly switches view to #historyView and updates active sidebar link.	Dashboard action buttons navigated directly to pending evaluation forms and history.	PASS
TC-012	1. View summary metric cards on dashboard. 2. Verify displayed counts for Pending Evaluations, Completed, and Total Submissions.	Summary Cards: - .summary-card.pending (.card-number) - .summary-card.completed - .summary-card.total	Cards correctly render metric counts reflecting assigned workload (e.g. 4 Pending, 0 Completed, 4 Total) matching database records.	Dashboard summary cards accurately reflected the student's assigned evaluation counts.	PASS
RE-03 ASSIGNED FACULTY EVALUATION LIST & STATUS BADGES					
Pre-requisite: Active semester is configured with enrolled courses and assigned faculty.					
TC-013	1. Navigate to 'My Evaluations' section on dashboard. 2. Inspect all rendered professor evaluation cards.	Assigned Faculty: 1. Arvin Lauron (SF 5 - Student Formation 5) 2. Maria Santos (MATH 101 - College Algebra) 3. Juan Dela Cruz (ENG 201 - Tech Writing) 4. Ana Garcia (CS 301 - Data Structures)	Each card displays Professor Name, Course Code, Subject Title, Due Date, Status Badge, and 'Start Evaluation' action button.	All assigned faculty cards rendered with full course details and action buttons.	PASS
TC-014	1. Verify status badges for unevaluated courses vs completed courses.	Course 1: Pending evaluation Course 2: Completed evaluation	Unevaluated courses display yellow/amber badge with 'Pending' text. Completed courses display green badge with 'Completed' text.	Evaluation cards displayed yellow pending badges that turned green upon completion.	PASS
TC-015	1. Click 'Start Evaluation' button on 'Arvin Lauron - SF 5' item. 2. Observe target selection and view transition.	Target: Arvin Lauron Course: SF 5 Action: Click .btn-start	Target indicator (#evaluation-target-text) updates to 'Arvin Lauron - SF 5'. System opens evaluation flow (privacy gate if first time, or questionnaire).	Clicking start evaluation opened the evaluation flow for the selected professor.	PASS
TC-016	1. Locate an evaluation card marked as 'Completed'. 2. Observe action button state and attempt interaction.	Target: Completed professor evaluation Button: Disabled or 'View in History'	System prevents re-submission; button indicates completion or links directly to view submitted answers in History.	Completed evaluation card showed a disabled status and prevented duplicate submissions.	PASS

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TC-017	1. Simulate system date past evaluation period end date. 2. Reload dashboard and inspect 'My Evaluations' section.	System Date: March 5, 2026 (Deadline: Feb 28, 2026)	System displays period closed notification banner; 'Start Evaluation' buttons become locked/disabled for past-deadline forms.	Past-deadline evaluations were locked and displayed an evaluation period closed notice.	PASS
RE-04 STUDENT DATA PRIVACY GATE & COMPLIANCE AGREEMENT					
Pre-requisite: Student selects a professor to evaluate for the current semester.					
TC-018	1. As a student with no prior consent record for current semester, click 'Start Evaluation'. 2. Verify presentation of Data Privacy Gate (#studentDataPrivacyGate).	Target: Arvin Lauron Consent Status: None recorded	#studentDataPrivacyGate is displayed with title 'Student Data Privacy Notice', notice version eyebrow, policy body text, and agreement checkbox. Questionnaire card remains hidden.	Data privacy gate appeared on first access with the required policy and consent checkbox.	PASS
TC-019	1. On Privacy Gate, observe 'Agree and Continue' button (#studentPrivacyContinueBtn) before clicking checkbox. 2. Click the button while checkbox is unchecked.	Checkbox: Unchecked Button: #studentPrivacyContinueBtn	Button has disabled attribute set to true and reduced opacity. Clicking the button does nothing; privacy gate remains active.	Agree button remained disabled and unresponsive until the consent checkbox was checked.	PASS
TC-020	1. Check agreement checkbox (#studentPrivacyAgreement: 'I have read and agree...'). 2. Observe 'Agree and Continue' button state. 3. Click 'Agree and Continue'.	Action: Check #studentPrivacyAgreement Click: #studentPrivacyContinueBtn	Button becomes fully enabled. Clicking it records consent in STUDENT_DATA_PRIVACY_CONSENTS with timestamp, smoothly conceals Privacy Gate, and reveals questionnaire card.	Checking consent enabled the agree button, saved the timestamp, and opened the evaluation form.	PASS
TC-021	1. After accepting privacy notice, return to dashboard. 2. Click 'Start Evaluation' for a different assigned professor (Maria Santos).	Target: Maria Santos Consent Status: Consent already recorded for semester	System detects existing valid consent for current semester; Privacy Gate is automatically bypassed, navigating directly to questionnaire for Maria Santos.	Subsequent evaluation bypassed the privacy gate since consent was already recorded.	PASS
TC-022	1. Trigger Privacy Gate. 2. Click 'Back to Dashboard' button (#studentPrivacyBackBtn) without agreeing.	Action: Click #studentPrivacyBackBtn	System navigates back to dashboard view; consent is not recorded; selected evaluation target is safely cleared.	Clicking back to dashboard closed the privacy gate without recording consent.	PASS
TC-018	1. As a student with no prior consent record for current semester, click 'Start Evaluation'. 2. Verify presentation	Target: Arvin Lauron Consent Status: None recorded	#studentDataPrivacyGate is displayed with title 'Student Data Privacy Notice', notice version eyebrow, policy	Data privacy gate appeared on first access with the required policy	PASS

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	of Data Privacy Gate (#studentDataPrivacyGate).		body text, and agreement checkbox. Questionnaire card remains hidden.	and consent checkbox.	
RE-05 DYNAMIC QUESTIONNAIRE RENDERING & RATING SCALE VALIDATION					
Pre-requisite: Questionnaire card is visible for the selected faculty member.					
TC-023	1. Inspect loaded questionnaire container (#dynamic-questions-container). 2. Verify question categories and item rendering.	Form Title: Student Professor Evaluation Form Categories: Teaching Effectiveness, Classroom Management, Interpersonal Skills	Questions render dynamically from active template with criteria numbering, full text descriptions, and rating scale radio groups.	Questionnaire loaded all rubric criteria with numbered items and rating radio buttons.	PASS
TC-024	1. Verify rating scale header and options for each question. 2. Check rating scale values 1 to 5.	Scale Range: 1 to 5 Scale Legend: 5 - Always Manifested (Outstanding), 4 - Frequently, 3 - Sometimes, 2 - Rarely, 1 - Not Manifested	Scale table or radio groups accurately display numeric values 1-5 with correct behavioral anchor labels matching school rubric.	Rating scale displayed options 1 through 5 with corresponding descriptive labels.	PASS
TC-025	1. Click rating '4' for question 1. 2. Then click rating '5' for question 1. 3. Verify radio group single-choice exclusivity.	Question: Q1 (Subject Mastery) Values: 4 then 5	Rating '4' is deselected and rating '5' is selected. Only one radio button is selected per question at any time.	Selecting a new rating deselected the previous option, keeping only one answer per item.	PASS
TC-026	1. In a multi-section questionnaire, leave section questions unanswered. 2. Click 'Next Section' button. 3. Observe section step validation (validateCurrentStep()).	Current Step: Section 1 Unanswered: 3 questions Action: Click Next	Step validation fails; user is prevented from advancing. Error message directs student to complete all required rating criteria in current section.	Unanswered section items halted progress with a validation prompt to complete all ratings.	PASS
TC-027	1. Navigate to qualitative feedback textarea. 2. Enter constructive feedback comments. 3. Verify text acceptance and character limits.	Text: 'The professor explains aerodynamic formulas clearly and encourages class discussion.'	Textarea correctly captures multi-line text input without truncating valid characters or breaking layout.	Comments textarea captured input cleanly without character truncation or layout issues.	PASS
TC-028	1. Give low rating (e.g. 1 or 2) requiring exception reporting explanation. 2. Enter repetitive gibberish (e.g.	Rating: 1 (Not Manifested) Reason: 'asdfasdfsdf' Validator: validateException	Validation algorithm detects low-entropy gibberish/insufficient explanation; displays prompt requiring meaningful	Low ratings with single-character remarks triggered a prompt for	PASS

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	'asdfsdf') in reason box. 3. Attempt to proceed.	ReportingTextValue()	constructive feedback.	constructive feedback.	
TC-029	1. Leave one or more required rating questions unselected. 2. Click 'Submit Evaluation'. 3. Observe form submission validation.	Form State: 4 of 5 questions answered, 1 blank Action: Click .btn-submit	Submission is halted; browser displays error alert/message indicating incomplete answers and highlights first missing question.	Form submission stopped on incomplete questions and highlighted the first missing item.	PASS
RE-06 EVALUATION DRAFT AUTO-SAVE, MANUAL SAVE & STATE RESTORATION					
Pre-requisite: Student has opened a questionnaire and provided partial ratings/comments.					
TC-030	1. Select ratings for first 3 questions and enter partial comment. 2. Click 'Save Draft' button (#saveDraftBtn). 3. Observe draft status indicator (#evaluation-draft-status).	Target: Arvin Lauron Ratings: Q1=5, Q2=4, Q3=4 Comment: 'Draft feedback...' Action: Click #saveDraftBtn	Draft payload is persisted in storage; #evaluation-draft-status updates from 'Draft not saved yet' to 'Draft saved at [timestamp]'.	Clicking save draft stored partial answers and updated the draft timestamp indicator.	PASS
TC-031	1. Modify a radio button selection or type in comment textarea. 2. Wait for background autosave interval (queueDraftAutosave()). 3. Observe draft timestamp.	Autosave Interval: ~5000ms Action: Modify rating without clicking save	Autosave timer triggers in background without page reload; draft status indicator updates smoothly with new saved timestamp.	Autosave triggered silently after 30 seconds of inactivity and updated the saved status.	PASS
TC-032	1. Save draft for Arvin Lauron. 2. Navigate back to Home dashboard. 3. Click 'Start Evaluation' for Arvin Lauron again. 4. Observe restored form inputs.	Target: Arvin Lauron Action: Re-open evaluation form	restoreDraftForCurrentTarget() runs; previously saved ratings and comments are automatically repopulated into corresponding input fields.	Reopening the evaluation restored the saved draft ratings and comments automatically.	PASS
TC-033	1. Save draft for Arvin Lauron. 2. Navigate back and click 'Start Evaluation' for Juan Dela Cruz. 3. Verify form state for Juan Dela Cruz.	Target 1: Arvin Lauron (has draft) Target 2: Juan Dela Cruz (no draft)	Draft isolation ensures Arvin Lauron's draft data does NOT bleed into Juan Dela Cruz's form; Juan Dela Cruz opens with fresh blank form.	Draft data remained isolated to the specific professor and did not leak into other forms.	PASS
TC-034	1. Enter ratings into questionnaire. 2. Click 'Cancel' button (#cancelBtn). 3. Observe confirmation dialog. 4. Confirm cancellation.	Action: Click #cancelBtn Confirm Prompt: 'Any unsaved changes will be lost...'	System prompts user for confirmation; upon confirmation, form inputs are reset and view returns safely to dashboard.	Canceling the evaluation reset all form inputs and returned safely to the dashboard.	PASS
TC-035	1. Complete and successfully submit evaluation for Arvin Lauron.	Target: Arvin Lauron Event: Successful	clearCurrentEvaluationDraft() triggers automatically; draft record is completely	Submitting the evaluation successfully deleted the	PASS

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	2. Check draft storage for Arvin Lauron.	submission	deleted from storage so stale draft is never restored.	saved draft from storage.	
RE-07 FORM SUBMISSION, SUMMARY RECALCULATION & STATUS LIFECYCLE					
Pre-requisite: Student has answered all required questions for the selected professor.					
TC-036	1. Fill all required questions and enter qualitative comments. 2. Click 'Submit Evaluation' (.btn-submit). 3. Confirm submission dialog prompt. 4. Observe dispatch to /api/evaluations/submit .	Target: Arvin Lauron - SF 5 Ratings: All 5s and 4s Payload: studentId, targetId, semesterId, answers, comments	System shows confirmation dialog ('Are you sure you want to submit? Submission is final. '); confirming triggers POST /api/evaluations/submit .	Submission modal asked for confirmation before dispatching the final evaluation payload.	PASS
TC-037	1. Observe system response upon successful API submission (HTTP 200 OK). 2. Verify success modal/toast and redirection.	API Response: { status: 'success', message: 'Evaluation submitted successfully' }	System displays success alert: 'Evaluation submitted successfully! Thank you for your feedback.', resets form state, and redirects to dashboard.	Success alert confirmed submission, cleared the form, and returned to the dashboard.	PASS
TC-038	1. Navigate to 'My Evaluations' section after submitting Arvin Lauron. 2. Observe status badge and action button for Arvin Lauron.	Target: Arvin Lauron Previous Status: Pending	Status badge transitions from yellow 'Pending' to green 'Completed'. Action button changes to disabled 'Completed' or 'View in History'.	Professor card updated immediately from pending to completed after submission.	PASS
TC-039	1. Inspect Dashboard summary cards immediately after successful submission. 2. Compare new numbers with pre-submission counts.	Pre-submission: Pending=4, Completed=0, Total=4 Post-submission: Pending=3, Completed=1, Total=4	updateSummaryCards() recalculates metrics accurately: Pending decrements by 1 (3), Completed increments by 1 (1), Total remains 4.	Dashboard metrics updated accurately, decrementing pending count and incrementing completed.	PASS
TC-040	1. Simulate server failure (HTTP 500) or network disconnection during submit. 2. Click 'Submit Evaluation' and observe error resilience.	Network: Offline / Server 500 Internal Error	System catches error, displays error alert ('Failed to submit evaluation. Please check your connection and try again.'), and preserves all form ratings without data loss.	Simulated network failure displayed an error message while preserving all entered ratings.	PASS
RE-08 PAST EVALUATION HISTORY, FILTERS & MODAL AUDIT					
Pre-requisite: Student has submitted evaluations across current and/or past semesters.					
TC-041	1. Click 'History' in sidebar navigation. 2. Observe Past Evaluations table (#historyView, #historyList).	View: #historyView List: #historyList	History view renders list of all submitted evaluations displaying Professor Name, Subject Code, Academic Year,	Evaluation history loaded submitted evaluations with complete course and	PASS

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			Semester, and Submission Timestamp.	date details.	
TC-042	1. In History view, select Academic Year '2025-2026' from dropdown (#historyAy). 2. Click 'Apply Filters' (#historyFilterBtn).	Filter: #historyAy = '2025-2026' Action: Click #historyFilterBtn	Table updates to show only evaluation records submitted during Academic Year 2025-2026.	Academic year filter updated the table to display only matching evaluations.	PASS
TC-043	1. Select Semester '1st Semester' from dropdown (#historySem). 2. Click 'Apply Filters'.	Filter: #historySem = '1' Action: Click #historyFilterBtn	Table narrows results to display only evaluations submitted during 1st Semester.	Semester filter refined history records to show only the selected term.	PASS
TC-044	1. Enter keyword in search input (#historySearch). 2. Click 'Apply Filters'. 3. Enter non-existent name.	Search: 'Lauron' / 'SF 5' (Valid) Search: 'NonExistent' (Invalid)	For valid query, list filters to matching professor/subject records. For non-existent query, list displays 'No evaluation records found matching your criteria.'	Search box filtered matching professor names and showed an empty state for unmatched terms.	PASS
TC-045	1. Click 'View Answers' button on a submitted history record. 2. Inspect submitted answers modal (#historyModal).	Element: #historyModal, #historyModalBody Data: Submitted ratings (1-5) and feedback text	#historyModal opens displaying complete read-only breakdown of criteria ratings and qualitative comments submitted by the student.	View answers modal opened showing read-only ratings and qualitative feedback.	PASS
TC-046	1. In history modal, click close button (#historyModalClose) or click modal backdrop.	Element: #historyModalClose Action: Click close	#historyModal closes smoothly, returning focus to evaluation history table view.	Audit modal closed smoothly and returned focus to the history table.	PASS
RE-09 STUDENT PROFILE DETAILS, PASSWORD SECURITY & MISSED EVALUATION PROOF					
Pre-requisite: Student is on Profile view or is flagged with missed evaluations past deadline.					
TC-047	1. Click 'Profile' in sidebar navigation. 2. Verify read-only student demographic fields in #profileView.	Fields: Student ID (ST-000001), Full Name (John Student), Program (BS Aviation), Email, Campus, Department, Year Level, Year Section, Academic Year	Profile fields populate correctly with student's verified records from database in read-only format (uneditable).	Profile view displayed verified student records in read-only format.	PASS
TC-048	1. In Change Password card, type passwords into Current, New, and Confirm fields. 2. Click eye toggle icon (.toggle-password) next to each field.	Fields: #currentPassword, #newPassword, #confirmPassword Buttons: .toggle-password	Input field type toggles from 'password' to 'text' revealing characters; icon changes from eye (fa-eye) to eye-slash (fa-eye-slash). Clicking again masks characters.	Password visibility toggle revealed and re-masked the entered password properly.	PASS
TC-	1. Enter valid current	Case A: Length <	System halts	Short password	PASS

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049	password. 2. Enter new password with fewer than 8 characters (e.g. 'pass1'). 3. Click 'Update Password'. 4. Enter non-matching confirm password.	8 chars Case B: New != Confirm	submission and displays validation error: 'New password must be at least 8 characters long' or 'New password and confirm password do not match.'	and mismatched confirmation were rejected with clear error messages.	
TC-050	1. Enter correct current password, valid matching new password (>= 8 chars). 2. Click 'Update Password' (.btn-submit). 3. Observe API dispatch to /api/student/change-password.	Current: StudentPass123! New: NewStudent2026! Confirm: NewStudent2026! API: POST /api/student/change-password	System sends update request; on HTTP 200, displays success notice 'Password updated successfully!', and resets input fields to blank.	Password changed successfully with valid inputs and cleared the form fields.	PASS
TC-051	1. Simulate student with unsubmitted evaluations after official deadline. 2. Log in or refresh dashboard. 3. Observe trigger of Missed Evaluation Proof Modal (#studentProofModal).	Trigger: getStudentEvaluationPeriodStateForProof().isOverdue == true	#studentProofModal appears automatically with header 'Evaluation Proof Requirement', explaining mandatory OSA justification for incomplete evaluations.	Evaluation proof modal appeared automatically for students with missed evaluations.	PASS
TC-052	1. In proof modal, enter explanation in reason box (#studentProofReason). 2. Enter invalid URL (e.g. 'http://dropbox.com/prof' or 'not-a-link') in Drive link field (#studentProofDriveLink). 3. Click 'Submit Proof'.	Reason: 'Medical leave during finals' Drive Link: 'http://dropbox.com/prof' Validator: isValidStudentDriveProofLink()	System rejects input with error alert: 'Please provide a valid shareable Google Drive link starting with https://drive.google.com/'.	Invalid proof URL was rejected with an alert requiring a valid Google Drive link.	PASS
TC-053	1. Enter valid Google Drive link (https://drive.google.com/file/d/1A2B3C4D/view?usp=sharing) and valid reason. 2. Click 'Submit Proof'. 3. Observe submission and pending review box (#studentProofPendingBox).	Reason: 'Hospital confinement during evaluation week' Drive Link: 'https://drive.google.com/file/d/1A2B3C4D/view?usp=sharing' Action: Click #studentProofSubmitBtn	Proof request is saved in STUDENT_EVAL_PROOF_REQUESTS; form hides and #studentProofPendingBox displays: 'Your submitted proof is under OSA review. Please wait for the decision.'	Valid proof request was submitted and displayed a pending OSA review status.	PASS

System Name:	NAAP Evaluation System
Module / View:	Student Panel (studentpanel.html)

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Created by:	John Emmanuel E Rojo
Executed by:	John Emmanuel E Rojo
Date of Execution:	SEPTEMBER 26 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

NAAP EVALUATION SYSTEM – PROFESSOR PANEL MODULE

TC ID	Test Steps	Test Data	Expected Result	Actual Result	STATUS (PASS OR FAIL) SPRINT 1 SEPTEMBER 29 2026
RE-01 AUTHENTICATION, SESSION SECURITY & FACULTY ROLE AUTHORIZATION					
Pre-requisite: Browser is running; NAAP session & database service are operational.					
TC-001	1. Log in with valid professor credentials. 2. Navigate to html/profesorpanel.html. 3. Verify checkAuthentication() check. 4. Observe dashboard view load.	User: PROF-001 (Arvin Laron) Role: faculty / professor Status: active Session: Valid JWT / session token	checkAuthentication() returns true. Professor Dashboard loads successfully, displaying faculty name and assigned department in header and user profile pill.	Logged in with valid faculty credentials and dashboard loaded with professor profile details.	PASS
TC-002	1. Launch private browser window with no active session. 2. Attempt direct URL navigation to html/profesorpanel.html. 3. Observe authentication guard.	Session: None (null / undefined) URL: .../html/profesorpanel.html	Access is denied immediately. System invokes redirectToLogin() and redirects browser to mainpage.html without rendering faculty data.	Direct unauthenticated access was blocked and redirected immediately to the login page.	PASS
TC-003	1. Establish active session with role 'student'. 2. Attempt direct navigation to html/profesorpanel.html. 3. Observe role authorization guard.	User: ST-000001 Role: student Target: profesorpanel.html	Role check fails (strict requirement === 'faculty' or 'professor'). User is redirected away to student panel or login page.	Attempted access using a non-faculty role was blocked and routed to the proper panel.	PASS
TC-004	1. Flag professor account status as 'inactive' in database. 2. Attempt login or access to profesorpanel.html. 3. Observe enforceActiveProfessorAccount() response.	Faculty ID: PROF-001 Status: inactive Role: faculty	System displays alert: 'Your account is inactive. You cannot access faculty features. Please contact your administrator.' Clears session and redirects to mainpage.html?logged_out=1.	Inactive faculty account was blocked with a prompt to contact the administrator.	PASS
TC-	1. While on professor	Action: Click	clearUserSession()	Clicked logout	PASS

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005	panel, click 'Logout' button (#professorLogoutBtn) in sidebar footer. 2. Observe session purge and redirect.	#professorLogoutBtn Target URL: mainpage.html?logged_out=1	purges all active session keys from storage and redirects to mainpage.html?logged_out=1 with logout message displayed.	and verified that all faculty session tokens were cleared from storage.	
TC-006	1. Simulate session timeout or invalid token during active session. 2. Attempt to navigate views or trigger report actions. 3. Observe session expiration handling.	Session: Expired timestamp / revoked token	System intercepts the request, detects session expiration, notifies user of expired session, and safely redirects to login page.	Expired session was intercepted during navigation, displaying a timeout notice and returning to login.	PASS
RE-02	PROFESSOR DASHBOARD OVERVIEW, RESPONSIVE NAVIGATION & NOTIFICATION PANELS				
Pre-requisite: Professor is logged in with active account on the Professor Dashboard.					
TC-007	1. Click 'Peer Evaluation' nav link in sidebar. 2. Click 'Reports' nav link. 3. Click 'Faculty Paper' nav link. 4. Click 'Profile' nav link. 5. Click 'Dashboard' nav link.	Sidebar Nav Links: - data-view='peerEvaluation' - data-view='reports' - data-view='facultyPaper' - data-view='profile' - data-view='dashboard'	Active class updates smoothly on the clicked navigation link. Target content view displays immediately while sibling view containers are hidden.	Sidebar navigation links toggled views smoothly while highlighting the active section.	PASS
TC-008	1. Resize browser viewport to mobile width (< 992px). 2. Click mobile navigation toggle button (#professorMenuToggle). 3. Verify mobile sidebar drawer. 4. Click sidebar backdrop overlay (#sidebarBackdrop).	Viewport: 375px x 812px (Mobile) Elements: #professorMenuToggle, #sidebarBackdrop, #professorSidebar	Sidebar drawer translates smoothly from left into viewport with backdrop overlay. Clicking backdrop or pressing Esc smoothly dismisses mobile navigation drawer.	Mobile drawer opened cleanly using the hamburger toggle and dismissed upon pressing Esc.	PASS
TC-009	1. Click notification bell button (.js-notification-btn) in top header. 2. Inspect faculty announcements dropdown (.js-announcement-panel). 3. Click outside dropdown.	Element: .js-notification-btn, .js-announcement-panel Notices: Evaluation Window, Response Tracking, Reminder	.js-announcement-panel opens displaying active faculty notices and schedule deadlines. Clicking anywhere outside closes the panel smoothly.	Notification bell expanded the announcements panel and closed cleanly when clicking outside.	PASS
TC-010	1. Click user profile pill (.js-profile-btn) in top	Element: .js-profile-btn, .js-	.js-profile-panel expands displaying	Profile pill opened a	PASS

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	header. 2. Inspect quick info dropdown (.js-profile-panel). 3. Click outside dropdown.	profile-panel Fields: Faculty ID, Department, Rank, AY/SEM	faculty ID, department, rank, and current academic semester in a structured grid. Closes when clicking outside.	summary card displaying faculty ID, rank, and department.	
TC-011	1. On Dashboard hero card, click 'Open Peer Evaluation' (#heroOpenPeerEvaluationBtn). 2. Return to Dashboard and click 'View Reports' (#heroOpenReportsBtn). 3. Return and click 'Open Faculty Paper' (#heroOpenFacultyPaperBtn).	Hero Action Buttons: - #heroOpenPeerEvaluationBtn - #heroOpenReportsBtn - #heroOpenFacultyPaperBtn	Each button switches directly to its respective view (#peerEvaluationView, #reportsView, #facultyPaperView) and synchronizes sidebar active link state.	Dashboard hero action buttons navigated directly to peer evaluation, reports, and papers.	PASS
TC-012	1. On Dashboard, locate Action Cards section and click 'Open Reports' (.btn-action). 2. Verify view transition.	Action Card: 'View Reports' Button: .btn-action	System switches seamlessly to the Reports view (#reportsView) and initializes analytics / blackout state.	Clicking open reports switched seamlessly to the reports view with active blackout checks.	PASS
RE-03 DASHBOARD SUMMARY METRICS, COUNTDOWN TIMER & PENDING TASKS					
Pre-requisite: Professor is on Dashboard view with active semester data.					
TC-013	1. Inspect Dashboard summary cards. 2. Verify 'Evaluations Received' ratio card. 3. Verify 'Average Score' card. 4. Verify 'Response Rate' percentage card.	Summary Cards: - .summary-card.evaluations (.card-number e.g. '24/30') - .summary-card.score (e.g. '4.6/5.0') - .summary-card.response (e.g. '80%')	Summary metric cards populate accurately reflecting evaluations received vs expected, overall average rating score, and student participation response rate.	Dashboard summary cards loaded accurate metric tallies for peer tasks and evaluation counts.	PASS
TC-014	1. In evaluation periods with hidden student rating rules, verify Average Score card behavior.	Policy: Average score hidden prior to period closure	refreshDashboardAverageScoreVisibility() hides or masks average score with appropriate placeholder until the evaluation blackout period officially ends.	Student rating score was masked during the active evaluation blackout window.	PASS
TC-015	1. Inspect 'Evaluation Deadline Countdown' widget (#deadline-countdown-list). 2. Verify displayed countdown item for active evaluation period.	Active Period: Student Evaluation / Peer Evaluation Element: #deadline-countdown-list	Widget computes remaining time (e.g. '12 days 5 hrs left') or indicates 'Evaluation period closed' if target deadline has elapsed.	Countdown widget displayed the remaining days and hours until the evaluation deadline.	PASS
TC-	1. Inspect 'My Pending	Tasks: Pending	Count badge reflects	Pending tasks	PASS

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016	Tasks' widget (#pending-tasks-list, #pending-tasks-count). 2. Verify count badge and listed task items.	peer evaluations, incomplete faculty paper draft Badge: #pending- tasks-count	exact number of unresolved faculty tasks. List displays actionable items with titles, contextual badges, and direct action buttons.	widget accurately showed the count of remaining peer evaluations.	
TC-017	1. In 'My Pending Tasks' list, click 'Evaluate' button on a pending colleague peer evaluation item. 2. Observe navigation and target selection.	Task: 'Peer Evaluation: Maria Santos' Action: Click action button	openPeerEvaluationFormTarget() navigates directly to Peer Evaluation view (#peerEvaluationView) and auto-selects Maria Santos in colleague dropdown.	Clicking evaluate in pending tasks navigated directly to the peer evaluation form for that colleague.	PASS
TC-018	1. Complete all pending peer evaluations and draft submissions. 2. Return to Dashboard and observe 'My Pending Tasks' widget.	Pending Task Count: 0	Badge updates to '0' and widget displays empty state: 'No pending tasks. You are all caught up!'	Completed peer evaluations cleared the pending task list and displayed a zero pending state.	PASS
RE-04 PEER-TO-PEER EVALUATION COLLEAGUE SELECTION & QUESTIONNAIRE RENDERING					
Pre-requisite: Professor is on Peer-to-Peer Evaluation view (#peerEvaluationView).					
TC-019	1. Click colleague selector dropdown (#peerProfessor). 2. Inspect available faculty options.	Dropdown: #peerProfessor Context: Department faculty list	Dropdown lists all professors within the same department/college, while strictly excluding the logged-in professor from the list.	Colleague dropdown listed all department faculty eligible for peer evaluation.	PASS
TC-020	1. Select an unevaluated colleague (e.g., 'Maria Santos') from dropdown. 2. Observe questionnaire container (#dynamic-peer-questions-container).	Selection: Maria Santos Element: #dynamic-peer-questions-container	Questionnaire container dynamically renders peer evaluation criteria categories (e.g., Workplace Conduct, Professionalism, Team Collaboration) from active template.	Selecting an unevaluated colleague dynamically loaded the peer evaluation rubric.	PASS
TC-021	1. Verify rating scale table and anchors in loaded peer questionnaire. 2. Check rating scale values 1 through 5.	Scale Range: 1 to 5 Labels: 5 - Always Manifested, 4 - Frequently, 3 - Sometimes, 2 - Rarely, 1 - Not Manifested	Rubric displays clear behavioral anchor definitions and 1-5 numeric scale corresponding to institutional faculty peer standards.	Rating scale displayed criteria descriptions and options 1 through 5 with rubric anchors.	PASS
TC-022	1. Select rating '4' for Question 1, then click rating '5' for Question 1. 2. Verify radio button exclusivity.	Question: Q1 (Interpersonal Relations) Values: 4 then 5	Radio selection toggles cleanly; previously selected rating '4' is cleared and rating '5' is selected. Only one score is active per	Radio buttons toggled cleanly, maintaining exactly one selected rating per criterion.	PASS

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			question.		
TC-023	1. Check colleague dropdown (#peerProfessor) for a colleague who has already been evaluated in current semester. 2. Observe option state and attempt selection.	Target: Previously evaluated colleague (isPeerTargetLocked() == true)	System flags colleague as '(Already Evaluated)' and disables option or shows notice: 'You have already submitted a peer evaluation for this faculty member this semester.'	Previously evaluated colleagues were marked as evaluated and disabled in the dropdown.	PASS
TC-024	1. In peer evaluation form, locate qualitative comments textarea. 2. Enter peer feedback remarks. 3. Verify character input acceptance.	Text: 'Demonstrates exceptional collaborative leadership in department curriculum reviews.'	Textarea captures multi-line peer feedback text properly without truncation or layout distortion.	Peer comments textarea captured multi-line input cleanly without truncating text.	PASS
RE-05 PEER EVALUATION VALIDATION, STEP NAVIGATION & SUBMISSION LIFECYCLE					
Pre-requisite: Peer evaluation form is loaded for a selected colleague.					
TC-025	1. In a multi-step peer questionnaire, leave required criteria unrated in Step 1. 2. Click 'Next Step' button. 3. Observe validatePeerCurrentStep() validation.	Step: 1 of 3 Unanswered: 2 criteria Action: Click Next	Validation halts forward navigation; error banner indicates unrated questions and focuses on first unanswered criteria item.	Leaving required criteria unanswered halted step navigation with a validation prompt.	PASS
TC-026	1. Answer all criteria in Step 1 and click 'Next Step'. 2. Verify step transition. 3. Click 'Previous Step'.	Action: Step 1 -> Step 2 -> Step 1	Step 1 inputs are preserved; Step 2 displays cleanly. Navigating back preserves previously selected ratings without data reset.	Step navigation advanced smoothly and preserved all previously selected answers.	PASS
TC-027	1. Attempt to submit peer evaluation form with no colleague selected in #peerProfessor. 2. Click 'Submit Evaluation' (.btn-submit).	Colleague: Blank / default ('Choose a professor...')	Submission is rejected; system displays validation warning requiring user to select a colleague before submitting.	Submitting without selecting a colleague was blocked with an error prompt.	PASS
TC-028	1. Select valid colleague, answer all criteria across all sections, and provide comments. 2. Click 'Submit Evaluation' (.btn-submit). 3. Confirm submission prompt dialog.	Target: Maria Santos Ratings: All criteria answered Action: Click .btn-submit	System prompts confirmation dialog ('Confirm submission of peer evaluation for Maria Santos?'); confirming dispatches evaluation payload.	Submission modal requested confirmation before finalizing the peer evaluation.	PASS
TC-029	1. Observe system response upon successful peer evaluation submission.	API Response: 200 OK / success	System displays success alert: 'Peer evaluation submitted successfully!', clears	Success alert confirmed submission, cleared the	PASS

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	2. Verify success notification and form reset.		form inputs, and locks evaluated colleague from re-submission.	form, and returned to the dashboard.	
TC-030	1. Return to Dashboard after peer evaluation submission. 2. Inspect 'My Pending Tasks' widget.	Widget: #pending-tasks-list, #pending-tasks-count	Submitted peer task is removed from pending tasks list; pending count decrements by 1 automatically.	Submitted colleague was removed from the pending list and the count decremented.	PASS
TC-031	1. While on peer form, click 'Cancel' button (#cancelPeerBtn). 2. Confirm cancellation prompt.	Action: Click #cancelPeerBtn Confirmation: 'Discard unsaved peer evaluation ratings?'	Form inputs reset cleanly; system navigates back to Dashboard view without submitting data.	Canceling the peer form reset all fields and safely returned to the dashboard.	PASS
RE-06	EVALUATION REPORTS BLACKOUT WINDOW & MULTI-ROLE EVALUATION FILTERING				
Pre-requisite: Professor is navigating to Reports view (#reportsView).					
TC-032	1. Set system date prior to evaluation unlock date. 2. Navigate to Reports view (#reportsView). 3. Observe applyReportBlackout() behavior.	System Date: Jan 25, 2026 (Unlock Date: Feb 10, 2026)	Reports content (#reportsContent) is hidden; #reportsBlackout card is displayed: 'Reports will be available after the evaluation period ends on Feb 10, 2026.'	Reports view showed a blackout notice since the current date is before the unlock date.	PASS
TC-033	1. Set system date on or after evaluation unlock date (or admin override). 2. Navigate to Reports view. 3. Observe report accessibility.	System Date: Feb 12, 2026 (Past unlock date)	#reportsBlackout is hidden; full reports content (#reportsContent), filter controls, breakdown table, and charts render successfully.	Reports unlocked automatically once the release date arrived, revealing full metrics.	PASS
TC-034	1. In Reports view, click 'Evaluation' type dropdown (#evaluationTypeFilter). 2. Inspect available evaluation role types.	Dropdown: #evaluationTypeFilter Options: Student Evaluation, Professor (Peer) Evaluation, Supervisor Evaluation	Dropdown presents 3 standard institutional evaluation types, defaulting to 'Student Evaluation'.	Evaluation type dropdown allowed switching between student, peer, and supervisor reports.	PASS
TC-035	1. Select 'Student Evaluation' in #evaluationTypeFilter. 2. Observe report title and dataset updates.	Selection: Student Evaluation Label: #selectedEvaluationType	#selectedEvalType updates to 'Student Evaluation'; table, comments, criteria breakdown, and charts switch to student ratings dataset.	Selecting student evaluation loaded overall student ratings and feedback lists.	PASS
TC-036	1. Select 'Professor Evaluation' (Peer) in #evaluationTypeFilter. 2. Observe report views.	Selection: Professor Evaluation Label: #selectedEvaluationType	Report switches to peer evaluation scores; student-only comment panels hide or adapt; peer charts display colleague evaluation metrics.	Selecting professor evaluation filtered the view to display peer ratings.	PASS
TC-	1. Select 'Supervisor	Selection:	Report switches to	Selecting	PASS

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037	Evaluation' in #evaluationTypeFilter. 2. Observe report views.	Supervisor Evaluation Label: #selectedEvalType	dean/supervisor evaluation scores; display reflects administrative criteria ratings and supervisor performance score.	supervisor evaluation loaded dean evaluation scores and leadership rubrics.	
TC-038	1. Click 'Semester' filter dropdown (#semesterFilter). 2. Select different academic semester (e.g. '2024-2025 2nd Semester'). 3. Observe report update.	Filter: #semesterFilter Selection: 2024- 2025 2nd Semester	updateSemesterLabels () updates semester badge; table data, criteria averages, and charts reload reflecting selected historical semester.	Changing semester filter updated evaluation tables and charts to match the selected term.	PASS
RE-07 SUBJECT BREAKDOWN, ANONYMOUS STUDENT FEEDBACK & AI SUMMARIZATION					
Pre-requisite: Reports view is active with loaded student evaluation data.					
TC-039	1. Inspect Faculty Breakdown Table (#facultyBreakdownTable). 2. Verify table columns and aggregated rows.	Columns: Subject, Section Count, Evaluations Received, Response Rate, Avg Rating Data: SF 5, MATH 101, CS 301	Table displays subjects handled by the professor with section counts, received/expected submissions, calculated response rate percentage, and average rating.	Faculty breakdown table displayed individual subjects with enrollment and mean ratings.	PASS
TC-040	1. On Anonymous Feedback panel (#profSubjectCommentsPanel), click 'View' button (#profSubjectCommentsView). 2. Verify loaded student comments.	Element: #profSubjectCommentsList Action: Click #profSubjectCommentsView	Anonymized student feedback comments load into list (#profSubjectCommentsList); student names/identities are strictly excluded to preserve privacy.	Selecting a subject loaded anonymized student feedback comments into the panel.	PASS
TC-041	1. Inspect loaded comments for rule- based sentiment/bias badges. 2. Verify classifyFeedbackCommentBiasByRules() tagging.	Feedback items: Constructive, Positive, Critical comments	Comments display semantic sentiment tags (e.g. 'Positive', 'Constructive', 'Areas for Growth') based on keyword and tone classification.	Feedback items displayed rule- based sentiment tags reflecting positive or constructive tone.	PASS
TC-042	1. With comments loaded, click 'AI Summarize' button (#profSubjectCommentsAiSummarize). 2. Observe AI synthesis in #profSubjectCommentsAiSummary.	Action: Click #profSubjectCommentsAiSummarize Target: #profSubjectCommentsAiSummary	System synthesizes qualitative feedback into an executive summary highlighting key strengths (e.g. clarity, punctuality) and specific recommendations.	AI summarize generated an executive summary highlighting strengths and growth areas.	PASS
TC-043	1. In Anonymous Feedback panel, click 'Close' button (#profSubjectCommentsClose)	Action: Click #profSubjectCommentsClose	Feedback list collapses cleanly; empty placeholder resets to 'Press View to load	Closing the feedback panel collapsed the comment list	PASS

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	sClose). 2. Verify panel collapse.		feedback.'	and reset the view cleanly.	
TC-044	1. Filter for a subject or semester with zero student comments. 2. Click 'View' button. 3. Observe empty state handling.	Data: 0 comments submitted	List displays graceful empty notice: 'No qualitative feedback comments submitted for the selected filter.' AI summarize button is safely disabled.	Subjects with no comments displayed a clean placeholder notice.	PASS
RE-08 SEMESTRAL PERFORMANCE TRENDS & MULTI-EVALUATION CHART ANALYTICS					
Pre-requisite: Reports analytics section is visible with multi-semester data.					
TC-045	1. Inspect 'Average Rating per Criteria' list (#criteriaList). 2. Verify listed criteria scores.	List: #criteriaList Items: Subject Mastery, Teaching Methodology, Classroom Climate, Punctuality	List accurately computes and renders average ratings (e.g. '4.7 / 5.0') for each evaluation rubric category.	Criteria list displayed average ratings computed across individual rubric dimensions.	PASS
TC-046	1. Inspect Semestral Evaluation Trend table (#semestralTrendTableBody). 2. Verify past 4 semesters comparison columns.	Columns: Semester, Overall Avg, Student, Peer, Supervisor Rows: Last 4 active semesters	Table renders historical progression across previous 4 semesters, computing combined overall average score alongside individual role breakdowns.	Semestral trend table rendered historical scores across the past four semesters.	PASS
TC-047	1. Verify Semestral Trend Delta indicator (#semestralTrendDelta) and status badge (#semestralTrendStatus).	Delta: Score change from previous semester	Delta indicator displays score improvement (+0.15) or decline (-0.08) with appropriate green/red trend status badge.	Trend delta indicator clearly highlighted positive score improvement across terms.	PASS
TC-048	1. Verify 4-Semester Trend Chart canvas (#semestralTrendChart). 2. Observe Chart.js line graph rendering.	Canvas: #semestralTrendChart Type: Chart.js Line Chart	Chart renders smooth trend lines visualizing score trajectories across semesters with hoverable tooltip datapoints.	Four-semester trend chart rendered clean trajectory lines for faculty performance.	PASS
TC-049	1. Inspect Student Evaluation Average Scores Bar Chart (#studentBarChart) and Rating Distribution Pie Chart (#studentPieChart). 2. Hover over chart segments.	Charts: #studentBarChart, #studentPieChart Data: 1-5 score distribution	Bar chart renders criteria average comparisons; pie chart renders percentage distribution of ratings (5s, 4s, 3s, 2s, 1s) with active tooltips.	Bar chart rendered category averages and pie chart displayed rating distributions accurately.	PASS
TC-050	1. Inspect Detailed Summary Table (#detailedSummaryTableBody). 2. Verify counts for Excellent(5), Good(4), Fair(3), Poor(2), Very Poor(1).	Table: #detailedSummaryTableBody Columns: Category, Avg Score, Responses, Excellent, Very Poor	Table details total response counts and frequency distribution per score tier across all evaluation categories.	Detailed summary table listed exact response frequencies for each rating level.	PASS
RE-09 FACULTY ACKNOWLEDGEMENT PAPER DRAFTING, SECTION C & DEAN ROUTING					

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Pre-requisite: Professor is navigating to Faculty Paper view (#facultyPaperView).					
TC-051	1. Select load type from dropdown (#facultyPaperLoadTypeSelect: 'Main Load' or 'Excess Load'). 2. Click 'Create/Refresh Draft' button (#facultyPaperCreateDraftBtn). 3. Verify table entry creation.	Load: Main Load Action: Click #facultyPaperCreateDraftBtn	System generates or refreshes paper draft record with unique Paper ID (e.g. FP-2026-001), loads Section A/B snapshots, and lists paper in active table.	Selecting load type generated a faculty paper draft populated with evaluation metrics.	PASS
TC-052	1. Click on a paper row in Faculty Paper table (#facultyPaperTable). 2. Verify Paper Detail Card (#facultyPaperDetailCard) rendering.	Row: FP-2026-001 Target: #facultyPaperDetailCard	Card expands showing read-only snapshot metadata: Faculty Name, Department, Rank, Semester, Load Type, SET Rating (%), and SAF Rating (%).	Clicking a paper row expanded the card with complete evaluation scores and faculty info.	PASS
TC-053	1. In Section C, enter text in 'Areas for Improvement' (#fpSectionCAreasInput). 2. Enter text in 'Proposed Learning and Development Activities' (#fpSectionCActivitiesInput). 3. Enter text in 'Action Plan' (#fpSectionCActionPlanInput). 4. Click 'Save Section C' (#fpDetailSaveSectionCBtn).	Section C Inputs: - Areas for Improvement - Proposed Activities - Action Plan Action: Click #fpDetailSaveSectionCBtn	Section C content is persisted in storage; system displays confirmation: 'Section C saved successfully as draft.'	Section C textarea saved faculty improvement goals and reflection notes successfully.	PASS
TC-054	1. On Paper Detail Card, click 'AI Smart Recommendations' button (#fpDetailAiRecommendBtn). 2. Observe automated recommendations in feedback container (#fpDetailAiRecommendFeedback).	Action: Click #fpDetailAiRecommendBtn Analysis: Low SET/SAF scoring criteria	System analyzes weak rubric areas and populates tailored training activities and actionable developmental objectives into Section C fields.	AI smart recommendation suggested tailored training programs based on rubric scores.	PASS
TC-055	1. Toggle approval auto-fill checkboxes: - 'Auto-fill faculty name on generated PDF' (#fpApprovalNamesAutoFillInput) - 'Auto-fill faculty date	Checkboxes: #fpApprovalNamesAutoFillInput, #fpApprovalDatesAutoFillInput	Flags are updated in paper payload, ensuring electronic signature placeholders will be dynamically populated in generated document.	Toggling auto-fill checkboxes applied signature and date flags to the draft payload.	PASS

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	signed on generated PDF' (#fpApprovalDatesAutoFillInput). 2. Verify state persistence.				
TC-056	1. Click 'Preview PDF' button (#fpDetailPreviewBtn). 2. Inspect generated PDF preview modal.	Action: Click #fpDetailPreviewBtn Output: PDF Blob modal preview	showProfessorPdfPreview() renders full official NAAP Faculty Acknowledgement Paper layout with Sections A, B, and C in preview modal with download option.	Preview PDF generated the official NAAP Faculty Acknowledgement Paper.	PASS
TC-057	1. Click 'Send to Dean' button (#fpDetailSendBtn). 2. Confirm routing prompt dialog. 3. Observe paper status update.	Action: Click #fpDetailSendBtn Confirm: 'Route acknowledgement paper to Dean for review?'	Paper status transitions from 'Draft' to 'Submitted to Dean'. Form inputs become read-only pending dean signature; recipient dean is assigned.	Submitting to Dean locked editing and routed the paper for dean endorsement.	PASS
TC-058	1. On an approved or historical paper, click 'Archive' button (#fpDetailArchiveBtn). 2. Switch to 'Archive' tab (#facultyPaperTabArchive). 3. Verify archived paper display.	Tabs: #facultyPaperTab Active vs #facultyPaperTab Archive	Paper moves from Active to Archive tab. Archive table displays historical papers; clicking 'Active' tab restores current semester drafts.	Archiving an approved paper moved it to the archive tab with full history preserved.	PASS
RE-10 FACULTY PROFILE DETAILS, PHOTO UPLOAD & ACCOUNT SECURITY					
Pre-requisite: Professor is on Profile view (#profileView).					
TC-059	1. Inspect Professor Profile card (#profileView, .professor-profile-details). 2. Verify read-only faculty demographic records.	Fields: Faculty ID, Full Name, Department, Rank, Mail Account, AY/SEM	Fields accurately render official faculty records synchronized from database in read-only format.	Profile view displayed verified faculty records and academic department in read-only format.	PASS
TC-060	1. In profile photo section, click 'Upload Photo' (.profile-photo-btn). 2. Select valid image file (JPG/PNG under 2MB) via #profilePhotoInput. 3. Verify live preview and upload.	File: faculty_avatar.jpg (1.2 MB) Element: #profilePhotoInput, #profilePhotoPreview	System validates file type/size, generates base64/URL preview in #profilePhotoPreview, and hides placeholder avatar initials (#profilePhotoPlaceholder).	Profile photo upload accepted a valid image, rendered a preview, and saved properly.	PASS
TC-061	1. Select an invalid file type (e.g. .pdf or .txt) or file exceeding maximum size. 2. Observe handleUploadError()	File: report.pdf / 15MB_image.png	Upload is rejected immediately with validation message: 'Please upload a valid image file (JPG or PNG) under 2MB.'	Invalid file types were rejected immediately with an unsupported	PASS

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	response.		Previous photo remains unchanged.	format alert.	
TC-062	1. In Account Actions card, click 'Change Mail Account' button. 2. In #emailUpdateCard, enter mismatched new & confirm email addresses. 3. Click 'Update Mail Account' (.btn-submit). 4. Enter matching valid email and submit.	Case A: newEmail != confirmEmail Case B: newEmail == confirmEmail ('prof.lauron@naap.edu.ph')	Case A rejects with 'Mail accounts do not match.' Case B successfully updates mail account, updates displayed profile email (#profileEmail), and closes form.	Email update form validated input and saved the new address upon confirmation.	PASS
TC-063	1. In Account Actions card, click 'Change Password' button. 2. In #passwordUpdateCard, test password visibility eye toggles (.toggle-password). 3. Enter valid current password, new password (>= 8 chars), and matching confirm password. 4. Click 'Update Password'.	Fields: #currentPassword, #newPassword, #confirmPassword Validation: min 8 chars, matching confirm	Eye icons toggle plaintext password visibility. Submitting valid form sends update to backend, displays 'Password updated successfully!', and resets input fields.	Password change form enforced complexity rules and updated credentials successfully.	PASS

System Name:	NAAP Evaluation System
Module / View:	PROFESSOR (PROFESSOR panel)
Created by:	ROJO JOGN EMMANUEL E
Executed by:	ROJO JOGN EMMANUEL E
Date of Execution:	SEPTEMBER 29 2026
Test Environment:	Windows / Chrome / MySQL (NAAP)

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APPENDIX X
Curriculum Vitae

RHICKA MAE P. DE ASIS
Villamor Air Base, Pasay City
09753392090
rhickadeasis@gmail.com



OBJECTIVE

To obtain an opportunity where I can apply the knowledge and skills I have gained as a Bachelor of Science in Aviation Information Systems student. I am willing to learn new skills, take on responsibilities, and contribute to the organization while gaining valuable experience in a professional environment.

EDUCATIONAL BACKGROUND

Bachelor of Science in Aviation Information Systems
National Aviation Academy of the Philippines (formerly ~~BUAPCA~~)
2023 – Present

Senior High School
Pasay City South High School

Junior High School
Pasay City South High School

Elementary Education
Villamor Air Base Elementary School

SKILLS

Software / Applications

- Visual Studio Code
- Android Studio
- XAMPP
- Canva
- Microsoft Office

Programming Languages / Technologies

- HTML
- CSS
- JavaScript
- PHP
- Java
- SQL / MySQL

TRAINING AND CERTIFICATIONS

- TRAINING ON THE GO: Inflight Servicing Procedure
- Fundamentals of Machine Learning and Artificial Intelligence
- AWS Cloud Practitioner Essentials

EXPERIENCE

ROTC – Model Flight

Cadet | 2023–2024

- Participated in drills and formation activities.
- Followed instructions and maintained discipline during training.
- Worked with fellow cadets in team-based activities and exercises.

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JOHN EMMANUEL E. ROJO
Norzagaray, Bulacan
09155955516
jeanrojo123@gmail.com



OBJECTIVE

A Bachelor of Science in Aviation Information Systems student with knowledge in web and system development, database management, multimedia, and technical support. Seeking opportunities to apply technical, leadership, and problem-solving skills while gaining professional experience in the aviation and information technology fields.

EDUCATIONAL BACKGROUND

Bachelor of Science in Aviation Information Systems

National aviation academy of the Philippines

2023–Present

Senior High School

Headwaters College

Junior High School

Alibon National High School

Elementary Education

State Elementary School

SKILLS

Software / Applications

- Visual Studio Code
- Android Studio
- XAMPP
- phpMyAdmin / MySQL
- Microsoft Office

Programming Languages / Technologies

- HTML
- CSS
- JavaScript
- PHP
- Java
- SQL / MySQL

PROJECTS

WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED FEEDBACK ANALYSIS AND CHECK-COMPLIANT INSTRUMENTS FOR THE NATIONAL AVIATION ACADEMY OF THE PHILIPPINES

Capstone Project

- Developed a web-based faculty evaluation system for the National Aviation Academy of the Philippines (NAAP).
- Uses PHP, MySQL/MariaDB, HTML, CSS, and JavaScript for system and database functions.
- Includes modules for students, faculty, HR, administrators, and other authorized users.
- Incorporates evaluation reporting and AI-enhanced feedback analysis.

FLIPIT – SPACED REPETITION FLASHCARD APPLICATION

- Developed a flashcard-based learning platform with web and Android versions.
- Implemented authentication, flashcard sets, study tools, search, and statistics features.

TRAINING AND CERTIFICATIONS

- Database Essentials: From Tables to Real-World Systems
- Ramp Handling and Safety Procedure Seminar
- OSA Onboarding Training
- Mental Health Seminar
- E-Waste Seminar

EXPERIENCE

Supreme Student Council – Central Board

Central Board Member / 3rd Year Representative | A.Y. 2026–2028

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- Represents student concerns and participates in student legislative activities.
- Participates in parliamentary meetings, resolutions, and student welfare initiatives.

AJSERB

3rd Year Representative | A.Y. 2026–2028

- Represents third-year Aviation Information Systems students.
- Coordinates with officers and assists in student activities and communication.

AJ8-AHAN Constituency Check

Co-Chairman / Secretary General | February 2028

- Assisted in event coordination and performed Secretary General duties during the activity.

ROTC | February 2023-2024

Public Information Officer

- Assisted with information dissemination, documentation, and multi-media-related activities.

Local Government Unit

Assistant – 250 Hours

- Assisted with document processing, administrative tasks, and services for community members.

~~Techcompetition~~

Chairperson

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LORENZA LANE S. VALLIDO

Noveleta, Cavite
(+63) 952 457 4435
vallidolorenzalane@gmail.com



OBJECTIVE

To develop and enhance my knowledge and skills in Information Systems, particularly in project management, system development, teamwork, communication, and problem-solving, while gaining practical experience through academic projects and activities.

EDUCATION

Bachelor of Science in Aviation Information System

National Aviation Academy of the Philippines (formerly PhilSCA)
2023 - Present

Science, Technology, Engineering, and Mathematics

Our Lady of Fatima University
2021 - 2023

2023 - 2024 Dean's Lister

2024 - 2025 Dean's Lister

SKILLS

Software/Applications:

- Visual Studio Code
- Android Studio
- Canva
- Cisco Packet Tracer
- XAMPP
- MS Office

Programming Languages:

- Java
- HTML
- CSS
- PHP
- SQL

TRAINING & CERTIFICATIONS

Duty Award - Basic ROTC Course, Philippine State College of Aeronautics, VAB

June, 2024

TRAINING ON THE GO: Inflight Servicing Procedure - Philippine State College of Aeronautics, VAB

February, 2025

AWS Cloud Practitioner Essentials - Amazon Web Services Skill Builder

March, 2026

Fundamentals of Machine Learning and Artificial Intelligence - Amazon Web Services Skill Builder

March, 2026

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EXPERIENCES

Supreme Student Council - Administrative Division

A.Y. 2023 - 2024

Student's Right and Welfare, Associate

- Assisted in addressing students' concerns, inquiries, and welfare-related issues.
- Helped promote awareness of students' rights, responsibilities, and available services.

College Red Cross Youth Council - Blood Services

A.Y. 2023 - 2024

Volunteer

- Helped promote blood donation awareness among students and the school community.
- Supported the organization's programs related to blood services and humanitarian efforts.

APPENDIX Y User Manual

NATIONAL AVIATION ACADEMY OF THE PHILIPPINES

Evaluation System — User Manual & Practical Guide

Part 1: The Main Login Portal & Account Security (mainpage.html)

1. What You Will See on the Login Screen

When you open the evaluation portal in your browser, the page is divided into two distinct sides designed to be simple, distraction-free, and easy to read:

- **The Welcome Side (Left panel):** Displays the official NAAP emblem and academy title. It confirms you are on the legitimate school server.
- **The Sign-In Box (Right panel):** Contains the two main fields you will fill out: your ID number and your password, plus buttons for assistance.

2. How to Sign In (Step-by-Step)

Follow these simple steps to log in to your dashboard:

Step 1: Enter your Student Number or Employee ID

Type your assigned NAAP ID in the top field.

- Students: Enter your student number exactly as printed on your school ID (for example, 2024-01234).
- Faculty & Staff: Enter your official Employee ID (for example, EMP-2021-089).

Step 2: Enter your Password

Carefully type your password into the password field. Remember that passwords are case-sensitive—check if your Caps Lock key is on!

Step 3: Click 'Login'

Click the blue Login button (or simply press the Enter key on your keyboard). You will see a quick loading message ('Signing in...').

Step 4: Automatic Dashboard Routing

Once verified, the system recognizes your role automatically and takes you straight to your personal dashboard. Students go to the student portal, professors go to the faculty portal, and administrative heads go to their respective management views.

3. Two-Factor Authentication (OTP) — Keeping Your Account Safe

To protect your confidential evaluation ratings and personal student records, NAAP uses a quick 6-digit email verification code (OTP). You won't have to do this every single time—only when security requires it.

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When will the system ask for an OTP code?

- Signing in from a new computer, phone, or browser you haven't used before.
- Signing in after three (3) incorrect password attempts (to make sure it's really you and not someone guessing your password).
- Your trusted device period (90 days) has expired.

How to complete the OTP verification:

1. When the pop-up window titled 'OTP Verification' appears, look at the message on screen. It shows a masked preview of your registered Gmail (such as j***@gmail.com).
2. Open your Gmail inbox in a new tab or on your phone.
3. Look for an email from NAAP Evaluation Portal containing your 6-digit code.
4. Return to the portal, type the 6 digits into the OTP Code box, and click 'Verify OTP'.
5. That's it! Your device is now verified and you'll be taken to your dashboard immediately.

4. Forgot Your Password? Here is How to Reset It

Don't worry if you forgot your password—especially during busy exam or evaluation week! You can securely reset it on your own in under two minutes without waiting for IT staff.

Step 1: Requesting a Reset Link

1. Click the 'Forgot Password?' link located right below the Login button.
2. In the window that appears, fill in two things:
 - Account Email: The official email address linked to your NAAP profile.
 - Student Number / Employee ID: Your school ID number.
3. Click 'Send Reset Link'.
4. If your information matches an active account, the system will send an email with your unique reset link.

Step 2: Choosing Your New Password

1. Open the reset email from NAAP and click the secure password recovery link. (Note: The link is active for 30 minutes for security reasons).
2. Enter your New Password (must be at least 8 characters long).
3. Type the exact same password into the Confirm New Password box.
4. Click 'Reset Password'.
5. You will see a green confirmation message. You can now log in with your brand-new password!

5. Where Do You Land After Logging In?

Because NAAP Evaluation System serves the entire institution, our login gateway automatically checks your role and brings you directly to your relevant tools:

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Your School Role	Destination File	What You Do There
Student	studentpanel.html	Fill out professor evaluation questionnaires, check deadlines, and view evaluation history.
Professor / Faculty	profesorpanel.html	Review student feedback summaries, submit peer evaluations, and generate faculty evaluation papers.
Program Coordinator	procoorpanel.html	Monitor subject evaluation completion rates and coordinate academic courses.
College Dean	daenpanel.html	Examine departmental faculty performance ratings and approve academic reports.
Human Resources (HR)	hrpanel.html	Manage evaluation credibility analytics, professor evaluation portfolios, and institutional reports.
Office of Student Affairs (OSA)	osapanel.html	Verify student evaluation compliance and manage clearance statuses.
VP for Academic Affairs (VPAA)	vpaapanel.html	Executive-level institutional overview, faculty performance metrics, and policy reports.
System Administrator	adminpanel.html	Manage user accounts, evaluation schedules, questionnaire templates, and system audit logs.

6. Important Rules to Keep in Mind

To make sure evaluation results are 100% fair and tamper-proof, the portal follows several automated safety rules:

One Device at a Time (Single Active Session): You cannot be logged into your account on two devices simultaneously. If you try logging in on your phone while your laptop session is still active, the screen will ask you to wait 5 minutes or log out of the other device first. Always click 'Logout' before closing your laptop!

5-Minute Inactivity Safety Timer: If you leave your computer unattended for more than 5 minutes without touching anything, your session will automatically freeze to prevent classmates or strangers from touching your evaluation forms.

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Temporary Security Lockout: If someone enters an incorrect password three (3) consecutive times, the system will temporarily lock attempts for a few minutes. The screen will clearly show the exact minute and second you can try again (e.g., 'Locked until 12:05 PM').

— End of Mainpage User Manual —

NATIONAL AVIATION ACADEMY OF THE PHILIPPINES

Student Evaluation Dashboard — Complete User Guide

Part 2: Navigating Your Dashboard, Rating Professors & Securing Clearance ([studentpanel.html](#))

1. Getting Oriented: Your Dashboard Layout

Once you sign in with your student account, you will land on the Student Dashboard. Here is a quick tour of what you see on your screen:

The Left Navigation Sidebar: Your main navigation bar with tabs for Home, Evaluation Forms, History, Profile, and Logout. On laptops, it stays visible on the left; on mobile phones, you can open and close it anytime using the hamburger menu button (≡).

The Top Header Bar: Shows your active page title, the Notification Bell (which lights up with a dot when evaluation schedules or announcements are posted), and your quick Profile badge displaying your name and student ID.

The Semester Hero Banner: A welcome banner at the top of Home showing your semester status with two quick-jump buttons: 'Browse Pending Forms' (takes you directly to your assigned professors) and 'View History' (opens your past evaluations).

The Three Summary Cards: Three live progress counters that automatically update as you submit forms:

- Pending Evaluations: The number of professors you still need to rate.
- Completed: The number of evaluations you have successfully submitted.
- Assigned Evaluations: Your total number of evaluation tasks for this term.

2. Your Academic Clearance Reference — Why It Matters

Right below your summary cards, you will notice the Clearance Reference panel. At NAAP, student evaluation completion is directly linked to your official semester clearance. Here is how the clearance system works:

Clearance Status	Badge Indicator	What It Means for You
Pending	Yellow / Amber Badge	You still have one or more pending professor evaluations to answer. Clearance reference is 'Not generated'.
Cleared	Green Badge	All assigned evaluations have been submitted! The system immediately generates an official

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alphanumeric reference code (e.g., NAAP-CLR-2026-88912) and timestamps it.

Not Cleared / Missed

Red Badge

The evaluation deadline has passed and you missed submitting required forms. You must file a formal proof appeal through the Office of Student Affairs (OSA).

3. Step-by-Step: How to Evaluate Your Professor

Evaluating your instructors is simple and structured. Follow this step-by-step walkthrough:

Step 1: Check Your Assigned Subjects on 'My Evaluations'

On your Home dashboard, scroll down to the 'My Evaluations' section. You will see a list of cards for each subject you are officially enrolled in, showing:

- Professor's Name (e.g., Prof. Arvin Lauron)
- Subject Code & Name (e.g., SF 5 - Student Formation 5)
- Section & Due Date
- A blue 'Start Evaluation' button.

Step 2: Click 'Start Evaluation'

Find the professor you wish to evaluate and click 'Start Evaluation'. The portal will automatically switch to the Evaluation Forms view and load the instructor's details.

Step 3: Review the Student Data Privacy Notice (Privacy Gate)

Before the questionnaire opens, you must review NAAP's Data Privacy Notice. This notice guarantees that your identity is completely protected—professors cannot see who submitted individual ratings or specific comments. Read the notice, check the box confirming 'I have read and agree...', and click 'Agree and Continue'.

Step 4: Answer the Questionnaire Sections

The evaluation is divided into structured categories (such as Teaching Effectiveness, Course Organization, Mastery of Subject Matter, and Professional Conduct).

- Use the 1 to 5 rating scale (1 = Unsatisfactory, 5 = Outstanding).
- Use the 'Next' button to move to subsequent sections and 'Back' if you need to review a previous answer.
- Watch the progress bar at the top to track your completion.

Step 5: Provide Meaningful Written Feedback

At the end of the form, you will have the opportunity to write constructive comments and suggestions. Be honest, professional, and respectful. Constructive feedback helps professors improve their teaching methods for future classes.

Step 6: Submit Your Final Evaluation

On the last section, click the green 'Submit Evaluation' button. A confirmation prompt will appear. Once confirmed, your answers are saved securely, and you are returned to your dashboard with your completed count updated!

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4. Saving Drafts: Take a Break Anytime

We know student life gets busy. If you get called to a class, need to catch a flight simulator session, or lose internet connection while answering an evaluation, you do not lose your answers!

How Draft Saving Works:

- **Manual Saving:** While answering any question, simply click the 'Save Draft' button at the bottom of the form. You will see the message update to 'Draft saved successfully at [Time]'.
- **Automatic Silent Saving:** The portal also automatically saves your answers behind the scenes whenever you click 'Next' to move between sections or after typing text.
- **Resuming Later:** When you return to the dashboard and click 'Start Evaluation' on that same professor, the portal detects your saved draft and automatically restores all your previously selected answers!

5. Reviewing Past Submissions (History View)

Want to verify which professors you evaluated last semester or review the comments you wrote? You can inspect all your historical records via the History tab in the sidebar:

- **Filter by Academic Year:** Choose a specific academic year from the dropdown (e.g., 2025–2026 or 2026–2027) or select 'All'.
- **Filter by Semester:** Select 1st Semester, 2nd Semester, or view all terms at once.
- **Keyword Search:** Type a professor's name, subject code (e.g., ENG 201), or course title in the search box and click 'Apply Filters'.
- **View Submitted Answers:** Each past evaluation has an action button that opens a clean pop-up window showing the questions, your recorded rating scale, and the written feedback you submitted.

6. Managing Your Student Profile & Account Password

Clicking 'Profile' on the left navigation opens your official student record. This view contains two primary sections:

A. Student Information (View-Only)

This section displays your official academy profile as encoded by the Registrar:

- Student ID Number & Full Name
- Academic Program (e.g., BS Aviation, BS Aircraft Maintenance, BS Avionics)
- Official Email & Enrolled Campus
- Department, Year Level, Section, and Academic Year Status

If any of this information contains a typo or does not reflect your current semester enrollment, please visit the Registrar's Office to request an official record update.

B. Change Password (Editable)

To keep your account secure, you can update your password at any time directly inside the portal:

1. Enter your Current Password in the first box.
2. Enter your New Password (must be at least 8 characters long).

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3. Type the same new password into Confirm New Password.
4. Click the small eye icon next to any box if you want to double-check what you typed.
5. Click 'Update Password'. A confirmation message will confirm your credentials have been updated.

7. What If You Missed the Deadline? (OSA Proof Appeal)

Life happens—whether due to medical confinement, family emergency, or official flight training duty, some students may occasionally miss the evaluation deadline. NAAP provides an official appeal process through the Office of Student Affairs (OSA):

How to Submit Missed Evaluation Proof:

1. If the evaluation period closes and you have incomplete forms, the 'Evaluation Proof Requirement' window will appear when you log in.
2. In the Reason field: Clearly describe why you were unable to complete your evaluations on time.
3. In the Google Drive Proof Link field: Provide a valid link to your supporting documentation (medical certificate, hospitalization discharge slip, official military/flight excuse memo).
4. Click 'Submit Proof'.
5. Your submission is immediately forwarded to the Office of Student Affairs (OSA) for verification.
6. While under review, your portal will display: 'Your submitted proof is under OSA review. Please wait for the decision.'
7. Once approved by OSA, your clearance restriction will be lifted, or a special evaluation grace period will be granted.

— End of Student Dashboard User Manual —

NATIONAL AVIATION ACADEMY OF THE PHILIPPINES

Professor Evaluation Dashboard — Complete User Guide

Part 3: Performance Reports, Peer Reviews & Faculty Paper Workflow (profesorpanel.html)

1. Professor Dashboard Overview & Quick Actions

Upon signing in, you are greeted with your personal faculty overview. The interface is structured for quick navigation across your teaching responsibilities:

Left Navigation Sidebar: Provides fast access to the 5 primary faculty views: Dashboard, Peer Evaluation, Reports, Faculty Paper, and Profile.

Top Faculty Header: Displays your active role, Announcement Bell (alerting you to evaluation deadlines and report unlock dates), and the Profile quick-card showing your Faculty ID, Department, Academic Rank, and Current Semester.

Faculty Workspace Hero Banner: Equipped with three one-click shortcut buttons:

- 'Open Peer Evaluation': Jumps straight to evaluating assigned colleagues.
- 'View Reports': Opens class performance ratings and analytics.
- 'Open Faculty Paper': Takes you directly to your Section C acknowledgement draft.

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Three Live Progress Cards: • Evaluations Received: Live ratio of submitted student reviews versus total enrolled students across your assigned sections.

- Average Score: Your overall rating score (scaled out of 5.0) for the current active semester.
- Response Rate: The percentage of enrolled students who have completed their forms so far.

Evaluation Deadline & Pending Tasks Widgets: A live countdown timer showing remaining days/hours before the evaluation window closes, alongside a list of pending peer evaluations that still require your input.

2. Completing Peer-to-Peer Evaluations

Peer evaluation allows faculty members within an academic department to provide collegial, professional feedback on teaching performance, teamwork, and institutional participation.

How to Submit a Peer Evaluation:

1. Click 'Peer Evaluation' in the sidebar or click 'Open Peer Evaluation' on your hero card.
2. In the 'Select Colleague' dropdown menu, choose the assigned faculty member you are evaluating.
3. The system dynamically loads the official Peer Questionnaire.
4. Review and rate each criterion (1 to 5 scale) covering classroom conduct, collaboration, syllabi adherence, and professionalism.
5. Provide constructive comments highlighting your colleague's strengths and collegial contributions.
6. Click 'Submit Evaluation'. Once confirmed, the task is marked completed in your Pending Tasks list.

3. Accessing Evaluation Reports & Understanding the Blackout Window

The Reports section (reportsView) provides comprehensive data visualization, criteria breakdown, and student comments. However, there is an essential institutional policy regarding report availability:

The Evaluation Blackout Window (Why Reports are Temporarily Locked)

If you visit the Reports tab while the semester evaluation window is still active, you will see the message: 'Reports not available yet. Reports will be available after the evaluation period ends on [Unlock Date].'

Why this policy exists: To maintain total academic integrity and prevent grading bias, professors cannot view student scores or comments until students have finished submitting all their forms and final semester marks are safely recorded.

What Becomes Available After the Unlock Date:

Once the evaluation window concludes, the full analytics suite automatically unlocks:

- Evaluation Type Selector: Switch seamlessly between Student Evaluation (SET), Peer Evaluation (SAF-Peer), and Supervisor Evaluation (SAF-Dean).
- Semester Filter: Access and compare data from previous academic terms.
- Faculty Breakdown Table: Displays your handled subjects, section count, total evaluations received, percentage response rate, and average rating per course.
- Criteria Breakdown: Shows your exact score in each specific category (e.g., Mastery of Subject Matter, Instructional Design, Communication Skills, Classroom Facilitation).

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4. Reading Feedback Comments & AI Summarization

Student comments offer qualitative context to your quantitative scores. The NAAP portal features an integrated feedback reader and AI summarizer:

1. In the Reports view, scroll to the 'Anonymous Feedback' panel.
2. Click the 'View' button to load all anonymized student remarks for the chosen subject and semester.
3. Click the 'AI Summarize' button (with the magic wand icon).
4. The system will analyze all submitted student remarks and generate a synthesized summary highlighting:
 - Key Student Commendations (what students appreciated most in your lectures and activities).
 - Constructive Feedback Themes (areas where students requested additional examples, slower pacing, or more consultation time).
5. Click 'Close' when you wish to collapse the feedback list.

5. 4-Semester Analytics & Visual Performance Charts

To support long-term faculty growth, promotion dossiers, and tenure applications, the portal provides an automated longitudinal track record:

- **Semestral Evaluation Trend Table & Chart:** Compares your overall performance across the last four (4) consecutive semesters. It displays a multi-line trend graph showing the trajectories of your Student, Peer, and Supervisor scores over time.
- **Student Evaluation Average Scores (Bar Chart):** A graphical bar chart displaying your exact average rating for every evaluation criterion.
- **Rating Distribution (Pie Chart):** A visual breakdown illustrating the percentage of ratings you received in each score tier (from 5-star Outstanding to 1-star Needs Improvement).
- **Peer & Supervisor Comparative Graphs:** Dedicated bar and pie charts comparing peer review patterns with supervisory assessments.

6. The Faculty Acknowledgement Paper Workflow

The Faculty Paper (facultyPaperView) is the official academic document summarizing your semester ratings. It serves as the formal bridge between faculty self-reflection and Dean consultation.

Paper Section	Editable By	Contents & Purpose
Section A	Read-Only (Auto-Generated)	Faculty Demographic Snapshot: Faculty ID, Name, Department, Academic Rank, Semester, and Load Type (Main vs Excess Load).
Section B	Read-Only (Auto-Generated)	Quantitative Evaluation Scores: Official SET Rating (Student Evaluation %), SAF Rating (Supervisor Assessment %), and composite evaluation totals.

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Section C

Jointly Editable (Professor & Dean)

Action & Development Plan: Contains three key fields:

1. Areas for Improvement
2. Proposed Learning & Development Activities
3. Action Plan

Step-by-Step Guide to Completing Your Faculty Paper:

Step 1: Choose Your Load Type and Create a Draft

Select either 'Main Load' or 'Excess Load' from the dropdown and click 'Create/Refresh Draft'. A new active paper row will appear in your Active tab.

Step 2: Select the Paper and Review Sections A & B

Click on the paper in the table. The 'Paper Detail' card opens below, displaying your confirmed SET and SAF scores.

Step 3: Fill in Section C (Or Use AI Smart Recommendations!)

Type your self-reflection into the three fields: Areas for Improvement, Proposed Activities, and Action Plan.

- Smart Shortcut: Click 'AI Smart Recommendations' to automatically generate customized development suggestions based on your semester scores! You can freely edit or expand any generated text.

Step 4: Configure Signatures & Preview PDF

Use the checkboxes to auto-fill your name and date on the final document if desired. Click 'Preview PDF' to inspect the formatted print-ready PDF.

Step 5: Click 'Save Section C' and 'Send to Dean'

Click 'Save Section C' to store your changes. When ready, click 'Send to Dean'. Your draft is transmitted to your College Dean's portal for formal review, dean endorsement, and archival.

7. Managing Your Faculty Profile, Photo & Credentials

Under the Profile tab, professors can manage personal account details and security credentials:

- Uploading a Profile Photo: Click 'Upload Photo' to select a JPG or PNG professional faculty picture. Your photo updates immediately across your dashboard.
- Updating Your Official Email: Click 'Change Mail Account' in Account Actions. Enter your new email address, confirm it, and submit. This email will receive your 2FA OTP codes and system alerts.
- Changing Your Password: Click 'Change Password'. Enter your current password, type your new 8+ character password, re-enter to confirm, and save. Use the eye icons to verify typing before submitting.

— End of Professor Dashboard User Manual —

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Dean Dashboard & Academic Oversight — Complete Operating Guide

Part 4: Department Analytics, Supervisor Reviews, Paper Endorsements & Peer Management (daenpanel.html)

1. Dean Dashboard Architecture & Department Overview

When you log in, your primary view is the Dean Dashboard (dashboardView). The interface is organized to provide high-level clarity while keeping operational tools just one click away:

Left Navigation Sidebar: Provides instant switching between the six (6) dedicated Dean modules:

- Dashboard (Executive overview & analytics)
- Evaluation (Supervisor evaluation questionnaire)
- Faculty Paper Inbox (Professor acknowledgement drafts)
- Profile (Dean credentials & contact details)
- Faculty Response Rate & Counts (Detailed department results, comments, trends & files)
- Faculty Peer Management (Automated peer evaluation assignments)

Top Executive Header: Displays your administrative title, the Announcement Bell (notifying you of evaluation calendar deadlines and policy memos), and the Profile quick-panel showing your Faculty ID, Department scope, Academic Rank, and Current Semester.

Dean Workspace Hero Banner: Equipped with three primary operational shortcut buttons:

- 'Open Evaluation' (Immediately launches a supervisor assessment)
- 'View Faculty Results' (Jumps directly to faculty response tables)
- 'Open Paper Inbox' (Opens assigned faculty acknowledgement papers)

Three Real-Time Summary Cards: • Evaluations Received: Tracks total submissions received versus total expected student submissions across all course sections under your department.

- Professors Supervised: Total count of active faculty members under your departmental jurisdiction.
- Response Rate: Live percentage participation for the selected semester.

Faculty Dashboard Summary & Criteria List: A live criteria breakdown displaying average scores across core evaluation areas (Instructional Methodology, Course Syllabi Adherence, Classroom Facilitation, Subject Mastery).

2. Merged Evaluation Analytics & Detailed Category Breakdown

Directly below your summary cards, the Dean Dashboard integrates a rich visualization suite combining both Student Evaluation (SET) and Peer Evaluation data:

- **Student Evaluation Charts:** • Average Scores (Bar Chart): Shows the departmental mean rating for each student questionnaire criterion.
- Rating Distribution (Pie Chart): Illustrates the percentage breakdown of student scores across ratings 1 to 5.
- **Peer Evaluation Charts:** • Peer Average Scores (Bar Chart): Shows departmental peer averages for collegiality, syllabus adherence, and professionalism.
- Peer Rating Distribution (Pie Chart): Displays the proportion of peer scores across performance tiers.

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The Detailed Summary Table (5-Star Distribution)

The Detailed Summary Table gives you a deep quantitative audit across every evaluation category. It tabulates total responses and sorts them into the 5 standard rating brackets:

Category Name	Average Score	Total Responses	Score Distribution
Instructional Delivery	4.65 / 5.0	342 submissions	Excellent: 280 Good: 48 Fair: 14
Subject Matter Mastery	4.82 / 5.0	342 submissions	Excellent: 302 Good: 35 Fair: 5
Classroom Management	4.41 / 5.0	342 submissions	Excellent: 210 Good: 95 Fair: 37
Assessment & Feedback	4.38 / 5.0	342 submissions	Excellent: 198 Good: 110 Fair: 34

3. Conducting Supervisor Evaluations

As Dean, your supervisory evaluation constitutes the SAF (Supervisor Assessment of Faculty) component of each professor's official semester rating. To evaluate a faculty member under your supervision:

Step 1: Navigate to 'Evaluation'

Click 'Evaluation' in the left sidebar or click 'Open Evaluation' on your hero banner.

Step 2: Select the Faculty Member

In the 'Select Employee' search box, start typing the professor's name or employee ID. An interactive suggestion datalist will filter matching faculty members under your department. Click the employee to select them.

Step 3: Complete the Supervisor Questionnaire

The system loads the official Supervisor Evaluation Form. Rate each criterion on the 1 to 5 Likert scale across administrative performance areas:

- Timely Submission of Grades and Syllabi
- Punctuality and Attendance during Lectures and Labs
- Participation in Departmental Meetings and Curriculum Committees
- Research Productivity, Professional Development, and Institutional Engagement

Step 4: Provide Qualitative Supervisory Remarks

Enter constructive comments highlighting specific commendations or institutional development goals.

Step 5: Submit the Evaluation

Click 'Submit Evaluation'. The assessment is securely saved and automatically integrated into the professor's SAF score.

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4. The Faculty Paper Inbox: Reviewing & Endorsing Acknowledgements

When professors complete their Section C self-reflection, they click 'Send to Dean'. Their papers arrive immediately in your Faculty Paper Inbox (facultyPaperInboxView):

Navigating the Paper Inbox:

1. Open the 'Faculty Paper Inbox' tab in the sidebar.
2. The Assigned Faculty Papers Table lists all transmitted drafts with Paper ID, Professor Name, Semester, Status, and Date Sent.
3. Click on any paper to open the 'Faculty Paper Detail' card below.

Reviewing & Editing Section C:

The detail card displays the professor's confirmed SET Rating (Student %) and SAF Rating (Supervisor %). In Section C, you can review what the professor typed and add your supervisory feedback directly into the three fields:

- Areas for Improvement: Highlight specific teaching practices or administrative areas needing attention.
- Proposed Learning and Development Activities: Recommend seminars, training workshops, or industry flight certifications.
- Action Plan: Jointly agree on measurable milestones for the upcoming semester.

Configuring Auto-Fill Signatures & Previewing PDF:

- Checkbox 'Auto-fill supervisor name on generated PDF': Automatically prints your full Dean title on the endorsement block.
- Checkbox 'Auto-fill supervisor current date signed on generated PDF': Automatically adds today's date stamp.
- Click 'Save Section C': Stores all joint edits into the database.
- Click 'Preview PDF': Generates the official print-ready PDF containing the complete three-section report (Demographics, Scores, and Action Plan) ready for final filing.

5. Faculty Response Tracking & Department Report Access Control

Under 'Faculty Response Rate & Counts' (facultyResponseView), you have comprehensive search, audit, and access management tools across your teaching roster:

A. Department Report Access Control (Crucial Dean Feature)

At the top of the view, locate the Department Report Access Panel:

- Current Status: Displays whether professor reports are currently 'Unlocked / Available' or 'Restricted / Locked'.
- 'Restrict Department Reports' / 'Allow Department Reports' Button: Allows the Dean to override and lock report visibility for all professors in the department if grades are pending or departmental review is ongoing. Clicking this button toggles access instantly for your entire department.

B. Searching Faculty & Switching Evaluation Views

- Search Input: Type a professor's name or employee ID and click 'Search' (or 'Reset' to clear).
- Semester Filter: Switch between current and past academic terms.
- View Toggles: Click 'Student Evaluation Results', 'Peer Evaluation Results', or 'Supervisor Evaluation Results' to update the table columns and ratings for the selected evaluation category.

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C. Faculty Breakdown Table Columns

The table displays:

- Employee ID & Faculty Name
- Institute, Employment Type (Full-Time / Part-Time), and Academic Rank
- Evaluations Received vs Expected
- Response Rate percentage
- Average Score (e.g., 4.75)
- Status & 'Comments' action button

6. Qualitative Feedback, AI Summarization & 4-Semester Trends

When you click 'Comments' next to any professor in the Faculty Response table, two powerful panels open below:

1. Professor Comments Panel:

- Loads all student or peer comments submitted for that specific instructor.
- 'AI Summarize' Button: Click the magic wand button to trigger instant AI synthesis. The system digests all qualitative remarks and produces an executive summary dividing feedback into Commendations and Areas for Improvement.
- Click 'Close' to dismiss the comments panel.

2. Semestral Evaluation Trend Section (deanSemestralTrendSection):

- Opening a professor's comments automatically triggers their 4-Semester Performance Trend!
- Trend Table: Displays four consecutive semesters with Overall Average, Student score, Peer score, and Supervisor score.
- Interactive Line Graph: Renders a multi-semester chart comparing performance trajectory over time. This is invaluable for tenure reviews, promotion deliberations, or faculty coaching sessions.

7. Professor Files & The IFER Document Directory

Located at the bottom of the Faculty Response view, the 'Professor Files' section (dean-ifer-section) manages official institutional IFER (Institutional Faculty Evaluation Report) documents:

- **The ifer.docx Document Pill:** Highlights the active standardized evaluation document template used across the academy.
- **Semester Selection & Search Bar:** Filter faculty by academic semester or search by professor name / employee ID.
- **Professor Files Directory Table:** Lists each professor with their Employee ID, Name, Institute, Position, and associated Files count.
- **Action Controls:** Allows the Dean to inspect uploaded files or download completed acknowledgement dossiers directly.

8. Faculty Peer Management & Automated Non-Reciprocal Generation

One of the Dean's most important administrative responsibilities is establishing peer evaluation assignments. The Faculty Peer Management view (peerManagementView) automates this process while strictly enforcing academic fairness rules:

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The Non-Reciprocal Algorithmic Rule (Why It Matters)

In traditional manual assignments, if Professor A evaluates Professor B, Professor B often evaluates Professor A. This reciprocal pairing creates psychological bias ('you rate me high, I rate you high').

NAAP's peer generation engine strictly prevents reciprocal pairs! If Professor A is assigned to evaluate Professor B, the system guarantees that Professor B will NEVER be assigned to evaluate Professor A. Furthermore, peer reviews are strictly confined within the same academic program.

How to Generate Peer Assignments:

1. Open 'Faculty Peer Management' in the left navigation sidebar.
2. In the 'Generate Peer Assignments' card, select the academic Program from the dropdown (e.g., BS Aviation, BS Avionics, BS AMT).
3. In 'Peers Per Professor', enter the required evaluation count (default is 5).
4. Click 'Generate Assignments'. The algorithm runs instantly and creates a balanced distribution matrix.
5. Check the 'Program Overview' table to verify the status (Professors count, Requested peers, and Actual assignments).
6. In 'Assignment Details', select the program and click 'View Assignments'.
7. The table reveals the complete peer matrix: exactly who each professor will evaluate, who will evaluate them, and their real-time Pending vs Submitted counts.

9. Managing Dean Profile, Official Email & Password

Under the Profile tab, Deans can maintain account security and visual identity:

- Profile Photo Upload: Click 'Upload Photo' to attach a professional photo (JPG/PNG). Your photo updates in the navigation header and appears on official system communications.
- Updating Official Mail Account: Click 'Change Mail Account' under Account Actions. Enter your new email, confirm, and save. This address receives your 2FA OTP codes and critical system notices.
- Updating Password: Click 'Change Password'. Enter your current password, type your new secure password (8+ characters), confirm, and click 'Update Password'. Toggle the eye icons to verify typing accuracy.

— End of Dean Dashboard Operating Manual —

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Office of Student Affairs (OSA) — Operating Manual & Clearance Guide

Part 5: Student Evaluation Monitoring, Proof Adjudication & Clearance Verification (osapanel.html)

1. OSA Dashboard Architecture & Live Monitoring Overview

Upon logging in, you land on the primary OSA Monitoring Dashboard (dashboardView). The interface is optimized for rapid tracking across the entire student population:

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Left Navigation Sidebar: Provides smooth navigation across the four (4) specialized OSA workspaces:

- Dashboard (Live descriptive analytics & completion metrics)
- Student Evaluation Status (Search directory, clearance lookups & manual clearing)
- Proof Reviews (Dedicated queue for adjudicating missed-evaluation excuse links)
- Profile (Official OSA contact information, Gmail update & password controls)

Top Monitoring Header: Displays the active page title, brief context description, and your current OSA User identity badge.

OSA Hero Card & Action Launchers: Features quick-launch operational shortcuts:

- 'Open Status List' (Instantly opens the full student directory table)
- 'Open Proof Reviews' (Jumps directly to pending student excuse submissions)

Four Real-Time Summary Cards: • Assigned Students: Total number of active enrolled students required to evaluate faculty for the active semester.

- Evaluated: Total students who have completed 100% of their assigned evaluations.
- Not Evaluated: Students who still have one or more pending professor forms.
- Completion Rate (%): Academy-wide progress percentage calculated in real time.

2. Descriptive Analytics: Campus, Department & Program Drill-Down

Directly below the summary cards lies the Descriptive Analytics section. This interactive module allows OSA officers to spot which colleges or academic programs are lagging behind before the evaluation deadline closes:

A. Campus Selector Dropdown (analyticsCampusSelect):

Filter data for 'All Campuses' or isolate a specific campus (e.g., Main Flight Campus, Pasay Extension, Clark Training Facility). All tables and completion cards instantly re-calculate for the chosen location.

B. By Department Table (departmentAnalyticsBody):

Lists each academic department (e.g., Aviation Management, Flight Operations, Aircraft Maintenance Technology) with columns:

- Assigned Students Count
- Evaluated Count
- Not Evaluated Count
- Department Completion Percentage (%)

C. Interactive Program Drill-Down (Programs under Department):

Click on any department row in the table! The row highlights in blue, and the lower table ('Programs by Department') instantly displays the specific academic degrees under that department (e.g., BS Aviation, BS Avionics, Commercial Pilot Program) showing their individual completion percentages.

3. Student Evaluation Status Directory & Search

The Student Evaluation Status view (studentStatusView) is your comprehensive student lookup and clearance audit tool. Here, OSA officers can review individual student progress and investigate clearance status:

Searching and Filtering Students:

Use the search bar at the top of the section. The search engine responds in real time as you type and matches across:

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- Student Full Name (e.g., 'Dela Cruz, Juan')
- Official Student Number (e.g., '2024-00123')
- Official Clearance Reference Code (e.g., 'CLR-2026-000001' or 'NAAP-CLR-2026-88912')

Status Badges & Progress Indicators:

The table displays real-time progress ratios (e.g., '4/4' or '1/4' completed) and visually classifies each student:

Status Badge	Visual Tone	Operational Meaning
Done	Green Checkmark	The student has evaluated 100% of their enrolled professors. Automatic clearance reference generated.
Cleared	Teal / Blue Badge	Student is officially cleared (either automatically upon completion or via approved OSA appeal/manual clearance).
Not Done	Red X-Mark	The student has pending evaluations and has not yet submitted an appeal.
Pending Review	Amber Hourglass	The student missed the deadline and submitted a formal proof reason + Google Drive link. Awaiting OSA review.
Rejected	Red Warning Triangle	The student's submitted proof was rejected by OSA. Student must provide valid documentation.

4. The Official Clearance Verification Tool

During enrollment, graduation audits, or flight training check-ins, students present clearance reference numbers. To safeguard against forged clearance slips, the OSA panel features a built-in cryptographic server validator:

1. In the 'Student Evaluation Status' view, locate the 'Verify Clearance Reference' card.
2. In the input box, enter the exact clearance code presented by the student (e.g., 'CLR-2026-000001' or 'NAAP-CLR-2026-88912').
3. Click 'Verify Reference' (or press Enter).
4. The system queries the secure server database and displays the verified record details:
 - Verification Status: 'Verified clearance reference' in green.
 - Official Student Identity: Full Name and Student Number.
 - Academic Term & Campus: Semester label and verified campus location.
 - Clearance Method: 'Automatic' (if earned via completing forms) or 'Manual' (if granted via approved appeal).
 - Generation Timestamp: Exact official date and time recorded on the institutional database.

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5. If an invalid or fabricated code is entered, the system flashes a red warning: 'No stored clearance matches this reference.'

[SECURITY AUDIT] Zero Forgery Policy: The Registrar and College Deans will only accept clearance codes that return a valid confirmation through this tool. It eliminates the need for paper clearance signatures and stamps.

5. Adjudicating Missed-Evaluation Proof Requests (Approve / Reject)

When the evaluation window officially closes, students who failed to submit their forms can no longer evaluate professors. Their student portal prompts them to file a formal appeal stating their reason and submitting a shareable Google Drive link containing official proof documents (hospital discharge slips, medical certificates, flight training duty memos).

The Evaluation Period Closed Rule:

By system policy, the 'Review' action buttons (Approve, Reject, and Mark Cleared) activate ONLY after the Student-to-Professor evaluation period has officially ended! While evaluations are still open, students must complete their forms normally rather than seeking exemptions.

How to Review and Decide a Proof Appeal:

1. Open either the 'Student Evaluation Status' table or the dedicated 'Proof Reviews' queue (proofReviewView).
2. Locate the student with a 'Pending Review' status badge.
3. Read the student's submitted explanation under the 'Reason' column.
4. Click the 'Open proof' link. This securely launches the student's Google Drive document in a new tab.
5. Inspect the documentation (verify doctor's signature, date of illness, or official academy memo).

Option A: Approving the Appeal ('Approve' Button)

- Click 'Approve'. A confirmation prompt will ask to confirm your decision.
- Once confirmed, the student is instantly marked as 'Cleared' in the system, generating an official clearance reference code.
- The decision is locked and immediately unblocks the student's enrollment status.

Option B: Rejecting the Appeal ('Reject' Button)

- Click 'Reject'.
- The system prompts you to enter a mandatory rejection note (e.g., 'Google Drive link was restricted / inaccessible', 'Medical certificate dates do not match evaluation period', 'Document unreadable').
- The rejection note is displayed in the student's portal, giving them clear instructions to resubmit valid proof.

6. Manual Clearance Modal & Permanent Audit Lock

In exceptional circumstances where a student presents physical documentation in person at the OSA office, or where an institutional exemption was granted by the Dean or VPAA, OSA officers can manually clear a student:

1. In the Student Status table, click the 'Mark Cleared' button in the Review column.
2. The 'Mark Student as Cleared' dialog modal appears on screen (manualClearModal).
3. Type a mandatory, detailed administrative reason into the Clearance Reason box (e.g., 'Verified physical

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medical certificate presented in person at Room 108; 'Official varsity athletics tournament memo from VPAA').

4. Click 'Save Clearance'.

5. The system generates an official manual clearance record and tags the student as 'Cleared (Manual)'.

Crucial Audit Rule — Permanent Clearance Lock:

Once a clearance is saved (whether automatic, appeal-approved, or manual), IT BECOMES PERMANENTLY LOCKED! The system strictly prohibits editing or revoking existing clearances. This permanent immutability protects student records against administrative tampering and accidental data loss.

7. The Dedicated Proof Review Queue (proofReviewView)

While the Student Status view displays all enrolled students, the 'Proof Reviews' view provides a filtered, streamlined queue focused solely on students who submitted appeals for the current semester:

- **Clean Adjudication Roster:** Displays only students who filed excuse requests, sorted with newest submissions at the top.
- **Quick-Access Columns:** Student Number, Full Name, Semester ID, Review Status, Stated Reason, Drive Link, OSA Review Note, and Action Buttons.
- **Status Transparency:** Highlights whether an appeal is 'Pending Review', 'Approved (Cleared)', or 'Rejected' with exact timestamps for each action.
- **Audit Historical Notes:** Allows officers to review previous rejection comments to verify if a student's resubmitted document addressed earlier deficiencies.

8. Managing OSA Profile, Official Gmail & Password

Under the Profile tab (profileView), OSA officers can maintain office contact details and system credentials:

- **Office Profile Photo:** Click 'Upload Photo' to attach an official department logo or staff photo (JPG, PNG, or WEBP). The avatar updates in the top header.
- **Official Gmail Configuration:** Click the 'Change Gmail' toggle under Account Actions. Enter your new official Gmail address, confirm it, and click 'Save Gmail'. This address receives all 2FA verification PINs and automated system notices.
- **Password Management:** Click 'Change Password'. Enter your current password, type your new password (minimum 8 characters with at least one number), confirm, and click 'Save Password'. Use the eye-toggle buttons to verify typing accuracy before saving.

— End of OSA Panel Operating Manual —

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Office of the VPAA — Institutional Evaluation Executive Guide

Part 6: Macro Academic Analytics, Tri-Perspective Insights & Institutional SASR Reporting
 (vpaapanel.html)

1. Executive Dashboard Architecture & Institutional KPI Cards

Upon entering the VPAA portal, you arrive at the Institutional Overview (dashboard-view). The dashboard presents high-level academic signals at a single glance:

Left Executive Navigation Sidebar: Provides streamlined switching between the three core executive modules:

- Overview (Macro KPIs, Tri-Perspective Charts & Category Scores)
- Reports (Descriptive analytics, AI highlights, multi-parameter filters & Overall SASR)
- Profile (VPAA office details, official Gmail & security credentials)

Top Executive Header: Displays the portal title, real-time context line, mobile menu toggle, and the VPAA Office user badge.

VPAA Hero Banner & Operational Shortcuts: Equipped with two executive action buttons:

- 'Open Reports' (Jumps directly to the reports, filters, and SASR tools)
- 'View Highlights' (Smoothly scrolls to the automated AI performance highlights)

Four Institutional KPI Metric Cards:

- **Total Students:** Total active enrolled student population across all academy campuses.

- **Completion Rate (%):** Live institutional percentage of submitted evaluations versus total expected course enrollments.

- **Pending Evaluations:** Real-time count of evaluations still awaiting student submission before the deadline.

- **Active Professors:** Total faculty roster undergoing evaluation for the active semester.

2. The Tri-Perspective Evaluation Framework (360-Degree Feedback)

NAAP's academic excellence model relies on three complementary evaluative angles. The VPAA Overview provides dedicated visual chart cards for all three evaluation streams:

Evaluation Stream	Evaluator Group	Key Institutional Focus
Student to Professor (SET)	Enrolled Students	Classroom teaching effectiveness, clarity of technical instruction, fairness in grading, and availability for academic consultation.
Professor to Professor (Peer)	Department Faculty Colleagues	Collegiality, adherence to departmental curriculum standards, syllabus pacing, and collaborative academic contributions.

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**Supervisor to
Professor (SAF)**

College Deans & Program
Chairs

Administrative punctuality, timely grade
submission, research productivity, committee
service, and institutional alignment.

Visual Chart Pairs for Each Stream:

Each of the three streams features identical dual-chart visualizations powered by Chart.js:

- **Average Scores by Category (Bar Chart):** Shows departmental average scores across core criteria (Teaching Effectiveness, Clarity, Fairness, Engagement, Professionalism).
- **Rating Distribution (Pie Chart):** Illustrates the exact proportion of ratings across tiers (from 5-star Outstanding to 1-star Poor).
- **Live Stream Footers:** Displays the stream's overall average rating (out of 5.0), total evaluations submitted, and total professors evaluated.

3. Descriptive Analytics: Campus & Degree Program Drill-Down

Under the Reports view (reports-view), the Descriptive Analytics section enables deep administrative auditing across all operational units of the academy:

1. **Campus Filtering (vpaaAnalyticsCampusSelect):**
Switch between 'All Campuses' or isolate specific flight centers and ground schools (e.g., Main Flight Campus, Clark Airport Hub, Pasay Campus).
2. **By Department Table (vpaaDepartmentAnalyticsBody):**
Summarizes completion metrics for each academic department: Assigned Students, Evaluated, Not Evaluated, and Completion %.
3. **Interactive Program Breakdown (Programs under Department):**
Click any department row in the table. The row highlights, and the lower table ('Programs by Department') instantly displays each degree program (e.g., BS Commercial Aviation, BS Aircraft Maintenance, BS Avionics, Flight Dispatch) with its specific student completion counts and percentages.

4. Key Highlights: Automated Institutional Signals

To save executive time when reviewing hundreds of course offerings, the VPAA panel runs automatic algorithmic analysis to extract the three most critical institutional signals:

Top Rating (Trophy Icon): Identifies the highest-performing faculty member or course offering for the semester, highlighting exceptional instructional delivery and student commendations. Ideal for faculty teaching awards and merit recognition.

Most Comments (Chat Bubble Icon): Highlights the instructor or subject with the highest volume of student qualitative remarks. Signals strong student engagement or intense course dialogue.

Needs Attention (Warning Triangle Icon): Flags faculty members or subject clusters experiencing notable score declines or elevated negative feedback patterns. Gives the VPAA early notice to coordinate developmental coaching with the respective College Dean.

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5. Multi-Parameter Evaluation Filters & Faculty Sorting

Below the highlights section, the VPAA panel provides an executive search and filtering console that allows you to slice institutional evaluation data precisely:

- **Search Input:** Type a professor's full name or employee ID and click 'Search' to locate an instructor instantly.
- **Semester Selector:** Switch between the active term and historical semesters to analyze longitudinal performance.
- **Campus Selector:** Filter results by specific campus or flight training center.
- **Department Selector:** Filter down to a single college or academic department.
- **Sort By Dropdown:** Sort the entire faculty grid by:
 - Rating (high to low)
 - Rating (low to high)
 - Response rate (%)
 - Professor Name (alphabetical)
- **Reset Filters Button:** Instantly restores all filter dropdowns to their default institutional overview state.

6. The Overall SASR Institutional Report (Summary Assessment & Supervisory Report)

Located at the header of the Professor Evaluation Reports section, the 'Overall SASR' button (with the green spreadsheet icon) is one of the VPAA's most powerful executive tools:

What the Overall SASR Does:

- Gathers and compiles all faculty evaluation records across every department for the active semester into a single unified report.
- Computes the official composite institutional ratings combining Student Evaluation of Teachers (SET) and Supervisory Assessment of Faculty (SAF).
- Exports formatted data suitable for official submission to the Board of Trustees, Executive Committee, and external accreditation bodies.

How to Generate the Overall SASR:

1. Set your desired Semester, Campus, and Department filters (or leave on 'All' for academy-wide compilation).
2. Click the 'Overall SASR' button (vpaa-overall-sasr-btn).
3. The system compiles the complete institutional dataset and generates the downloadable summary report dossier.

7. Professor Evaluation Reports Grid & Analytics Dossier Modal

The Professor Evaluation Reports section renders an interactive card grid (professorGrid) for all evaluated faculty:

Card Contents:

Each card displays the professor's Name, Employee ID, Department, Campus, Academic Rank, overall rating score, response rate percentage, and total submitted student reviews.

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The Professor Analytics Dossier Modal:

Clicking 'View Analytics' on any card opens the comprehensive Professor Analytics Modal (professor-analytics-modal):

- **Category Radar & Bar Breakdown:** Shows how the instructor scored across every distinct evaluation category.
- **Tri-Perspective Composite Breakdown:** Displays side-by-side comparison of Student vs Peer vs Supervisor ratings.
- **Full Anonymized Feedback Repository:** Lists all qualitative comments submitted by students and peers, complete with sentiment badges.
- Click the 'X' button or press Escape to close the modal and return to the main grid.

8. Managing VPAA Office Profile, Official Gmail & Password

Under the Profile view (profile-view), the executive office maintains institutional identity and access credentials:

- **Official Office Photo / Seal:** Click 'Upload Photo' to attach an official executive seal or portrait (JPG/PNG). The image updates in the navigation header and appears on official report previews.
- **Updating Official Gmail:** Click 'Change Gmail'. Enter the updated executive email address, confirm, and save. This address receives high-priority 2FA OTP security PINs and institutional alert memos.
- **Password Management:** Click 'Change Password'. Enter the current password, type the new 8+ character password (with letters and numbers), confirm, and click 'Update Password'. Toggle the eye icons to verify typing accuracy.

— End of VPAA Evaluation Overview Manual —

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Program Coordinator Dashboard & Curriculum Supervision Guide

Part 7: Program Analytics, Supervisor Reviews, Section C Endorsements & Peer Generation
(procoorpanel.html)

1. Role Definition & Program Scope Architecture

To ensure institutional clarity, NAAP's academic governance is divided into distinct administrative tiers.

Understanding how your role as Program Coordinator integrates into this structure ensures accurate reporting:

- **Vice President for Academic Affairs (VPAA):** Maintains macro-level, institutional oversight across all colleges, departments, and accreditation lifecycles.
- **College Deans:** Direct college-wide faculty rosters, multi-program resource allocations, and college-wide departmental performance.
- **Program Coordinators (Your Role):** Lead the curriculum, faculty teaching assignments, peer mentoring, and operational quality assurance for a specific degree program (e.g., BS Aeronautical Engineering).
- **Faculty Members (Professors):** Conduct classroom instruction, submit self-appraisals, complete peer evaluations, and sign official acknowledgement papers.

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[PROGRAM SCOPE BOUNDING] Automatic Program Scope Bounding: In procoorpanel.html, the supervisor configuration (NAAP_SUPERVISOR_PANEL) strictly enforces role='procoor' with scopeLabel='program' and scopeDescriptor='program scope'. You will only see faculty members, course sections, student evaluations, and peer assignments that belong to your assigned degree curriculum. Furthermore, student identities in qualitative comment threads are permanently masked to preserve confidentiality.

2. Program Coordinator Dashboard & Real-Time KPIs

Upon authenticating with your credentials and 2FA passcode, you will land on your primary Dashboard (dashboardView). The interface provides immediate operational awareness through curated visual sections:

Left Navigation Sidebar: Enables seamless one-click navigation across the six (6) dedicated coordinator views:

1. Dashboard (KPIs, criteria breakdown & merged analytics)
2. Evaluation (Supervisor evaluation questionnaire)
3. Faculty Paper Inbox (Assigned faculty acknowledgement papers & Section C)
4. Profile (Coordinator credentials, photo upload & security settings)
5. Faculty Response Rate & Counts (Detailed program results, comments, AI summaries, trend charts & IFERs)
6. Faculty Peer Management (Program-scoped peer generation & assignment matrix)

Top Executive Header: Displays your title ('Program Coordinator Dashboard'), the Announcement Bell icon (alerts for evaluation deadlines and academic memoranda), and your user badge with department/program details.

Workspace Hero Banner: Highlights your workspace mission ('Monitor faculty performance and program actions from one dashboard') and provides three rapid action shortcuts:

- [Open Evaluation] — Instantly launches the supervisor assessment questionnaire.
- [View Faculty Results] — Jumps straight to program-wide response tables and comments.
- [Open Paper Inbox] — Opens pending faculty acknowledgement papers requiring Section C review.

Program Real-Time KPI Cards:

- **Evaluations Received:** Tracks total completed student submissions versus expected submissions across all courses in your program.

- **Professors Supervised:** Live count of faculty members currently assigned to teach under your program curriculum.
- **Response Rate:** Current semester percentage completion across all classes within your program scope.

Average Rating per Criteria Summary: A live summary list reporting aggregate mean scores across the four foundational evaluation criteria (Instructional Methodology, Course Syllabi Adherence, Classroom Facilitation, Subject Mastery).

Integrated Evaluation Analytics (SET & Peer Charts)

Directly below your KPI summary cards, the dashboard renders comprehensive graphical analytics combining both Student Evaluation of Teaching (SET) and Peer-to-Peer Evaluation data:

- **Student Evaluation - Average Scores (Bar Chart):** Illustrates average performance scores (scale 1.00 to 5.00) across each student questionnaire parameter.
- **Student Evaluation - Rating Distribution (Pie Chart):** Displays the proportional percentage of student ratings falling into 5-Star (Excellent), 4-Star (Good), 3-Star (Fair), 2-Star (Poor), and 1-Star (Very Poor) tiers.

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- **Peer Evaluation - Average Scores (Bar Chart):** Displays program-wide peer scores evaluating collegiality, instructional rigor, and syllabus adherence.
- **Peer Evaluation - Rating Distribution (Pie Chart):** Shows the spread of peer ratings across performance brackets to ensure departmental grading calibration.

The Detailed Summary Table (5-Star Distribution)

For quantitative audits, the Detailed Summary Table breaks down every criterion with exact response tallies:

Criteria Category	Program Avg	Total Responses	Rating Spread (5 to 1)
A. Instructional Methodology & Delivery	4.68 / 5.00	842 Responses	5: 612 4: 180 3: 42 2: 6 1: 2
B. Course Syllabi & Curriculum Adherence	4.74 / 5.00	842 Responses	5: 660 4: 148 3: 28 2: 4 1: 2
C. Classroom Facilitation & Conduct	4.59 / 5.00	842 Responses	5: 580 4: 195 3: 55 2: 9 1: 3
D. Subject Mastery & Aviation Practicality	4.82 / 5.00	842 Responses	5: 710 4: 110 3: 18 2: 3 1: 1

3. Conducting Supervisor Evaluations for Program Faculty

As Program Coordinator, you provide formal administrative appraisals (Supervisor Evaluation Form) evaluating the instructional execution, workplace conduct, and institutional collaboration of professors teaching within your program scope.

Step 1: Access the Evaluation Module — Click 'Evaluation' in the left navigation sidebar, or click the [Open Evaluation] shortcut button on your dashboard hero.

Step 2: Select the Faculty Member — In the 'Search Employee / Program' input field, begin typing the professor's name or employee ID. The system queries the dynamic datalist (peerProfessorOptions) restricted strictly to faculty members within your program scope. Click the suggested professor name to populate their record.

Step 3: Complete Dynamic Evaluation Criteria — The questionnaire container loads official institutional assessment indicators across core supervisory dimensions:

- **Curriculum Alignment:** Adherence to CHED/CAAP standardized course outlines and laboratory exercises.
- **Punctuality & Syllabus Pace:** Regular class attendance, prompt submission of grades, and pacing consistency.
- **Pedagogical Innovation:** Integration of modern aviation simulators, interactive teaching tools, and industry safety protocols.
- **Professional Conduct & Collaboration:** Departmental meeting participation, peer support, and student

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consultation responsiveness.

Select the corresponding Likert rating (1 = Unsatisfactory to 5 = Outstanding) for each indicator.

Step 4: Provide Formative Supervisory Feedback — In the qualitative remarks section, record specific commendations, documented observations, and targeted areas for professional development. This feedback directly informs Section C of the Faculty Paper.

Step 5: Submit Evaluation — Click [Submit Evaluation]. The system securely commits your appraisal to the institutional database. A green confirmation banner will verify successful submission. Once submitted, the scores are incorporated into the professor's composite SAF rating.

4. Faculty Paper Inbox: Reviewing & Endorsing Section C

At the conclusion of each evaluation cycle, professors generate their official Faculty Acknowledgement Paper. These papers route to the Program Coordinator's Faculty Paper Inbox (facultyPaperInboxView) for joint review and endorsement.

Assigned Faculty Papers Table

The inbox displays all papers submitted by faculty in your program with key tracking columns:

- Paper ID: Unique tracking code (e.g., FP-2026-AE-014).
- Professor: Full name and academic rank of the faculty member.
- Semester: Academic Year and Semester of the evaluation period.
- Status: Workflow state ('Pending Supervisor Review', 'Under Joint Revision', 'Endorsed').
- Sent At: Timestamp of submission from the professor panel.
- Actions: Action button ('Review Paper') that opens the comprehensive detail card below.

Joint Section C Formulation & Sign-Off Options

Clicking 'Review Paper' loads the Faculty Paper Detail Card. This card displays verified evaluation metrics, including:

- SET Rating (%): Student Evaluation of Teaching composite score.
- SAF Rating (%): Supervisor / Self / Peer composite evaluation score.

Directly beneath the metadata is Section C (Editable by Professor and Program Coordinator/Dean). Section C constitutes the formal Professional Development Plan and contains three editable fields:

1. Areas for Improvement (deanSectionCAreas): Synthesize specific pedagogical or operational items identified during student, peer, or supervisor reviews (e.g., 'Enhance timely return of laboratory lab reports' or 'Integrate more practical simulator scenarios').

2. Proposed Learning & Development Activities (deanSectionCActivities): Recommend concrete professional training, workshops, or academic conferences (e.g., 'Attend CAAP Safety Management Systems (SMS) Workshop' or 'Enroll in Advanced Avionics Didactics').

3. Action Plan (deanSectionCActionPlan): Define measurable milestones, expected outputs, and completion timelines for the forthcoming academic term.

Endorsement & Digital PDF Generation Checkboxes:

Before finalizing the document, two automated options streamline the sign-off workflow:

- ☐ Auto-fill supervisor name on generated PDF (deanFpApprovalNamesAutoFillInput): Automatically prints

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your full name and official Program Coordinator title onto the formal endorsement block.

- [] Auto-fill supervisor current date signed on generated PDF (deanFpApprovalDatesAutoFillInput): Stamps the official system timestamp as the signature date.

Actions Available:

- [Save Section C]: Commits your edits directly into the database so the professor can view the updated action plan.
- [Preview PDF]: Generates a live on-screen PDF preview of the complete 3-page NAAP Faculty Acknowledgement Document, allowing you to verify formatting and signatures before archiving.

5. Faculty Response Rate, Results & AI Comment Synthesis

The Faculty Response Rate & Counts view (facultyResponseView) serves as your primary diagnostic tool. It provides granular insight into student, peer, and supervisor feedback across every professor in your program.

Search & 3-Way Evaluation View Toggles

The top control bar allows you to filter professors by typing their Name or Employee ID, and selecting the desired Semester. Three interactive toggle buttons let you instantly pivot the entire data table between evaluation dimensions:

- **[Student Evaluation Results] (Active by default):** Displays student participation counts, course section response rates, and composite student scores (SET) for each instructor.
- **[Peer Evaluation Results]:** Displays peer evaluation submission counts, peer response rates, and peer-awarded mean scores within the program.
- **[Supervisor Evaluation Results]:** Displays official supervisory ratings, submission status, and administrator review timestamps.

Program Faculty Results Table

The data table lists all faculty members in your program with ten (10) detailed tracking columns:

1. Employee ID | 2. Faculty Name | 3. Institute | 4. Employment Type (Regular / Contractual) | 5. Position / Academic Rank | 6. Evaluations Received | 7. Response Rate (%) | 8. Average Score | 9. Status (Completed / In Progress / Pending) | 10. Comments ([View] Button).

Qualitative Comments Drawer & The 'AI Summarize' Engine

Clicking the [View] button in the Comments column slides open the Faculty Comments Panel. In accordance with NAAP privacy regulations, all student identifying information is permanently masked. Instead of reading through hundreds of raw qualitative entries individually, you can leverage the built-in AI Summarization tool:

Step 1: Open Comments Drawer — Click [View] next to the target professor. The panel displays the chronological list of raw comments.

Step 2: Trigger AI Summarization — Click the [AI Summarize] button with the magic wand icon (facultyCommentsAiSummarize) at the top of the comments drawer.

Step 3: Review Structured AI Analysis — Within 3 to 5 seconds, the AI engine processes all comments for that professor and returns a structured brief comprising:

- Key Strengths & Commendations: Core teaching qualities praised by students (e.g., clear derivations, approachable demeanor).

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- **Constructive Areas for Growth:** Recurring pain points (e.g., pacing during complex aerodynamics equations, delayed syllabus uploads).
- **Overall Class Sentiment:** Net positive, neutral, or critical sentiment breakdown.
- **Actionable Recommendations:** Specific suggestions to discuss during faculty mentoring sessions.

Step 4: Close or Copy — Use these AI-generated insights directly when formulating Section C development goals. Click [Close] when finished.

6. 4-Semester Longitudinal Trends & Professor Files Directory

Sustainable academic excellence requires tracking faculty growth over time rather than evaluating isolated semesters. The coordinator panel equips you with two powerful multi-semester audit modules:

Semestral Evaluation Trend (4-Semester Longitudinal Tracking)

When you select a professor, the 'Semestral Evaluation Trend' section (`deanSemestralTrendSection`) automatically unhides. This section maps the professor's performance trajectory across the last four (4) consecutive academic semesters:

- **Multi-Metric Table Breakdown:** Tabulates each of the 4 semesters showing:
 - Semester Code (e.g., AY 2025-2026 Sem 1, Sem 2, AY 2026-2027 Sem 1, Sem 2)
 - Overall Average Score (composite weighted mean)
 - Student Evaluation (SET) Average
 - Peer Evaluation Average
 - Supervisor Evaluation (SAF) Average
- **Trend Delta & Performance Status:** Calculates the net score change (+0.18, -0.05) and displays an automated trajectory status ('Consistent High Performer', 'Upward Growth Trend', 'Requires Monitoring', 'Needs Urgent Intervention').
- **Visual Trend Canvas (`deanSemestralTrendChart`):** Renders an interactive line graph tracking score fluctuations over time, making it effortless to identify positive responses to previous faculty development plans.

Professor Files & IFER Directory (`deanIfferSection`)

At the bottom of the results view lies the Professor Files Directory. This tool manages official Instructional Faculty Evaluation Report (IFER) documents and institutional acknowledgement archives:

- **Semester Selector:** Switch between past and current academic terms to pull historical documentation.
- **Template Pill (`ifer.docx`):** Confirms the active institutional evaluation template being referenced.
- **Search Bar:** Rapidly locate specific instructors by entering their name or employee ID.
- **Professor Files Table:** Displays Employee ID, Professor Name, Institute, Position, Files Available (e.g., 'IFER Report.docx', 'Summary.pdf'), and direct Action buttons to open or download the certified documentation.

7. Faculty Peer Management & Non-Reciprocal Pairing Algorithm

The Faculty Peer-to-Peer Management view (`peerManagementView`) is one of the most vital administrative modules in your panel. It eliminates manual pairing spreadsheets by automating current-semester peer evaluation assignments using a certified algorithm.

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Generating Program Peer Assignments

To generate peer evaluation pairs for your academic program, follow these straightforward steps:

Step 1: Select Your Degree Program — From the 'Program' dropdown (peerMgmtProgramSelect), select your designated academic program (e.g., 'BS Aeronautical Engineering').

Step 2: Define Peer Assignment Quota — In the 'Peers Per Professor' input field (peerMgmtProfessorCount), specify how many peer evaluations each instructor must perform. The standard institutional default is 5 peers per professor (minimum 1).

Step 3: Execute Assignment Generation — Click [Generate Assignments] (peerMgmtCreateRoomBtn). The system runs the automated non-reciprocal assignment generator (generateCoordinatorProgramPeerAssignments).

Step 4: Verify Success Notification — The system validates that the program has sufficient active faculty members to fulfill the quota. A green confirmation banner will announce: 'Assignments successfully generated for [Program Name] with [N] peers per professor.'

The Non-Reciprocal Pairing Algorithm Explained

To prevent bias, collusion, or retaliatory grading, NAAP enforces a strict Non-Reciprocal Peer Pairing Rule:

Reviewing Assignment Details & Progress

Once generated, the Program Overview Table records the Program Name, Total Professors, Requested Peer Quota, Actual Assigned Quota, and Status ('Active / In Progress').

Clicking 'View Assignments' populates the Assignment Details Table (peerMgmtAssignmentDetailsTable), showing:

- Professor: The faculty member's name.
- Will Evaluate: The list of designated colleagues this professor must evaluate.
- Will Be Evaluated By: The list of colleagues evaluating this professor.
- Pending: The count of peer evaluations yet to be submitted.
- Submitted: The count of peer evaluations successfully completed.

8. Coordinator Profile & Security Settings

The Profile view (profileView) enables you to verify your institutional credentials, upload an official avatar, and maintain robust security across your academic computing accounts.

View-Only HR Sync Data: Displays verified institutional data synchronized directly from NAAP Human Resources:

- Faculty ID Number | • Full Legal Name | • Academic Department | • Assigned Program Lead | • Faculty Rank (e.g., Assistant Professor III) | • Active Mail Account | • Current AY & Semester.

Profile Photo Upload: Upload a clear professional headshot in JPG or PNG format (max 5MB). Click [Upload Photo] and select the image file. The preview circular avatar updates immediately across your dashboard header and faculty paper endorsements.

Update Mail Account (emailUpdateCard): If your official institutional email changes, click [Change Mail Account]. Enter your new mail account twice for validation and click [Update Mail Account]. A confirmation handshake ensures zero typo errors.

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Update Account Password (passwordUpdateCard): To safeguard academic integrity, coordinators should change their passwords periodically. Click [Change Password], enter your current password, and provide a new strong password (minimum 8 characters, combination of upper/lowercase letters and numerals). Use the eye icon button to toggle password visibility. Click [Update Password] to save.

— End of Program Coordinator Panel Operating Manual —

NATIONAL AVIATION ACADEMY OF THE PHILIPPINES

Human Resources Panel & System Administration Guide

Part 8: Faculty Records, Evaluation Calendars, Questionnaires & AI Auditing (hrpanel.html)

1. HR Panel Architecture & System Navigation

The HR Panel operates as the central engine of the evaluation workflow. The layout is optimized for efficiency, allowing administrative staff to transition rapidly between strategic oversight and granular data entry:

Dashboard (dashboard): The primary executive command screen featuring real-time KPI metrics, professor ranking leaderboards, system notification feeds, multi-semester trend graphs, and merged 3-way evaluation analytics.

User Management (users): Comprehensive faculty directory facilitating search, multi-campus filtering, professor profile creation, status toggles, individual analytics drilldowns, and the master 'Overall SASR' Excel export.

Activity Log (activity-log): Immutable administrative audit trail recording user logins, account alterations, evaluation submissions, and system events paired with IP addresses and timestamps.

Evaluation Settings (settings): Operational configuration suite for defining start/end submission windows per evaluation type, managing academic semesters, and customizing automated email reminder frequencies and templates.

Questionnaire (questionnaire): Dynamic questionnaire authoring studio allowing HR to design and customize survey sections, Likert-scale questions, qualitative text prompts, data privacy agreements, and previous-term template duplication.

AI Insights (ai-insights): Advanced quality-assurance laboratory featuring Human-in-the-Loop Credibility Reviews (<70 score threshold), Behavior Anomaly Scoring (fast/uniform answers), NLP Bias Detection, Cross-Source Discrepancy checks, and Printable Summaries.

Profile (profile): Personal administrative profile displaying HR employee credentials, professional headshot photo upload, and secure email and password management forms.

Top Executive Header & Global Shortcuts

The persistent top header bar provides administrative situational awareness:

- **Mobile Toggle Button (hrMenuToggle):** Expands and collapses the sidebar on compact displays or tablets.
- **Dynamic Page Title & Context Block:** Instantly reflects your active view and current operational guidance.
- **Notification Bell Icon (notification-icon):** Displays an active badge counter (e.g., '3') and reveals a dropdown feed of urgent system alerts.

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- **User Profile Badge:** Identifies the logged-in HR officer with quick status confirmation.
- **Logout Button (hrLogoutBtn):** Securely terminates session tokens and redirects to the main authentication gateway.

2. Executive Dashboard, Operational KPIs & Merged Analytics

Upon authentication, HR personnel land on the primary Dashboard (dashboard-view). This screen provides immediate macro-level visibility over the active evaluation cycle.

The HR Workspace Hero Banner

The header hero card frames your operational mission: 'Coordinate evaluation operations and current-semester content from one panel.' It features three prominent operational jump buttons:

- **[Open User Management] (heroOpenUsersBtn)** — Jumps directly to faculty records and roster controls.
- **[Open Questionnaire] (heroOpenQuestionnaireBtn)** — Opens the dynamic survey builder for immediate edits.
- **[Open AI Insights] (heroOpenAiInsightsBtn)** — Navigates straight to submission audit and credibility checks.

Five Real-Time Overview KPI Cards

Directly beneath the hero card are five real-time statistical cards providing campus-wide operational metrics:

- 1. Total Students:** Displays the total enrolled student population eligible to evaluate faculty for the active semester.
- 2. Completion Rate:** Percentage of required student evaluations that have been officially submitted to date.
- 3. Completed Evaluations:** Gross total of finished and submitted evaluation records across all course sections.
- 4. Pending Evaluations:** Count of assigned student evaluations that remain incomplete or saved as drafts.
- 5. Number of Active Professors:** Total count of active faculty teaching accounts currently loaded into the system.

Professor Ranking Leaderboard & Department Filters

The left column of the dashboard houses the Professor Ranking section (professor-ranking-list). This live leaderboard showcases the top-performing faculty members based on composite evaluation scores. Two dropdown filters enable rapid segmentation:

- **Employment Type Filter (ranking-employment-filter):** Filter by All, Permanent, COS (Contract of Service), or Temporary.
- **Department Filter (ranking-dept-filter):** Filter across All Departments, ICS (Computer Studies), ILAS (Liberal Arts & Sciences), or ENGI (Engineering).

Top Trends & Merged Tri-Perspective Evaluation Analytics

The dashboard merges high-level charting with detailed multi-perspective evaluation statistics:

- **Evaluation Overview Chart (hr-evaluation-overview-chart):** Visual doughnut chart categorizing submissions into Completed, Pending Drafts, and Not Started.
- **Semestral Performance Chart (hr-semestral-performance-chart):** Bar/line graph tracking total student submissions across the latest 4 academic terms.

Beneath the trends section are three dedicated Evaluation Type Cards providing deep statistical breakdowns:

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- **Student to Professor Evaluation:** • Average Scores by Category (Bar Chart) & Rating Distribution (Pie Chart).
 - Quick Stats: Average Rating (/5.0), Total Student Evaluations, and Number of Professors Evaluated.
- **Professor to Professor Analytics (Peer Reviews):** • Peer Average Scores by Category (Bar Chart) & Peer Rating Distribution (Pie Chart).
 - Quick Stats: Peer Mean Rating, Total Peer Evaluations, and Peer Professors Evaluated.
- **Supervisor to Professor Evaluation (Deans & Coordinators):** • Supervisor Scores by Category (Bar Chart) & Rating Distribution (Pie Chart).
 - Quick Stats: Supervisor Average Rating, Total Supervisory Reviews, and Professors Evaluated.

3. System Notifications & Targeted Announcement Composer

HR personnel hold the institutional authority to broadcast official announcements, evaluation deadlines, and administrative policy notices across the entire academy.

Viewing Notifications & System Alerts

The right column of the dashboard features the System Notifications feed (hr-system-notifications). This feed streams chronological system milestones, such as batch email reminder dispatches, period openings, and security events.

The Targeted Announcement Composer Modal

Clicking the [Create Announcement] button (hr-open-announcement-compose-btn on the dashboard or hr-settings-announcement-compose-btn in Settings) opens the Announcement Compose Modal (hr-announcement-compose-modal). This tool enables precise role-based targeting:

- **Title (hr-announcement-compose-title):** Clear, concise subject line (e.g., 'Reminder: Midterm Faculty Evaluation Closes Feb 10').
- **Message (hr-announcement-compose-message):** Detailed body text (max 3,000 characters) containing directives, instructions, or policy memos.
- **Target Role (hr-announcement-target-role):** Select the specific audience: All Users, Admin, HR Staff, VPAA, OSA, Dean, Program Coordinator, Professor, or Student.
- **Campus Filter (hr-announcement-target-campus):** Optional dropdown to isolate recipients to a specific campus (or 'All Campuses').
- **Program Filter (hr-announcement-target-program):** Optional dropdown to target students or faculty belonging to a specific degree program.
- **Student Completion Filter (hr-announcement-student-completion):** Appears automatically when 'Student' role is selected. Allows targeting:
 - All Students
 - Completed (send congratulatory or next-step clearance notes)
 - Not Completed (send urgent compliance warnings)
- **Live Audience Preview (hr-announcement-audience-preview):** Dynamically computes and displays the target recipient description before publishing.
- **Action Buttons:** [Cancel] (discards draft) or [Publish Announcement] (commits notice instantly to recipient login screens and notification bells).

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4. User & Professor Management (Rosters, Profiles & Overall SASR)

The User Management view (user-management-view) gives HR complete oversight over faculty accounts, teaching assignments, employment classifications, and individual performance dossiers.

Toolbar Controls & Multi-Parameter Filtering

At the top of the professor directory is a comprehensive filtering and action toolbar:

- Search Input (professor-search): Search by Professor Name or Employee ID (e.g., 'EMP-1001').
- Campus Filter (professor-campus-filter): Filter by institutional campus branch.
- Status Filter (professor-status-filter): Filter by Active or Inactive status.
- Program Filter (professor-program-filter): Filter by academic degree curriculum.
- [Search] Button (professor-search-btn): Executes the filtered query against the database.
- [Clear] Button (professor-clear-btn): Resets all search parameters to default.
- Department Tabs: Instant tabbed switching between All Departments, ICS, ILAS, and ENGI.

Adding and Editing Professor Accounts

Clicking the [Add Professor] button (add-professor-btn) launches the Professor Account Modal (professor-modal):

1. Full Name (professor-name): Legal name of the instructor.
2. Email (professor-email): Official institutional email address.
3. Employee ID (professor-employee-id): Unique institutional identifier.
4. Department (professor-department): Academic institute assignment (ICS, ILAS, ENGI).
5. Position / Academic Rank (professor-position): Title (e.g., Associate Professor III, Instructor I).
6. Employment Type (professor-employment-type): Classification (Permanent, COS, Temporary).
7. Active Status Checkbox (professor-active): Controls whether the account is eligible for active evaluation loading.

Click [Save Professor] to commit the record, or [Cancel] to exit.

Professor Details & Individual Analytics Dossier

Each professor card in the directory includes interactive action triggers:

- View Details: Opens the Professor Details Modal (professor-details-modal), displaying assigned course subjects, student headcounts, and contact info.
- View Analytics: Opens the Professor Analytics Modal (professor-analytics-modal), presenting a complete dossier of the instructor's SET, Peer, and Supervisor scores, historical trends, and qualitative feedback summaries.

The 'Overall SASR' Master Export

The [Overall SASR] button (overall-sasr-btn) generates the master Summary Assessment & Supervisory Report. This function aggregates every evaluated faculty member across the entire institution, calculating composite scores (SET 50%, Peer 20%, Supervisor 30%), qualitative ratings, and accreditation classifications into a certified institutional Excel spreadsheet for submission to the VPAA and Academic Council.

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5. Evaluation Settings: Period Windows, Semesters & Email Alerts

The Evaluation Settings view (settings-view) controls the temporal rules and automated communication schedules governing institutional evaluation cycles.

Configuring Evaluation Period Windows

NAAP operates under a staggered evaluation lifecycle. HR sets distinct Start and End dates for each evaluation category:

- **Student to Professor (student-professor-start to student-professor-end):** The window during which students and cadets can access studentpanel.html to submit course ratings.
- **Professor to Professor (professor-professor-start to professor-professor-end):** The collegial evaluation window during which faculty complete peer reviews generated by Program Coordinators.
- **Supervisor to Professor (supervisor-professor-start to supervisor-professor-end):** The administrative window for Deans and Program Coordinators to submit formal supervisory assessments.
- **[Save Evaluation Periods] (save-eval-periods-btn):** Commits the calendar dates. Evaluations attempted outside these dates are automatically locked by system gates.

Active Academic Term & New Semester Provisioning

The Current Semester card manages the institutional academic calendar:

- **Active Semester Selector (current-semester):** Sets the active term referenced across all student, professor, and administrative panels.
- **Provisioning New Terms:** Select Academic Year Start, Academic Year End, and Semester Type (1st, 2nd, or 3rd/Summer Semester).
- **Term Preview & [Add Term] (btn-add-term):** Formats and appends the new term to institutional databases.
- **[Save Current Semester] (save-current-semester-btn):** Activates the term campus-wide.

Automated Email Reminder Schedules & Templating

To ensure high student completion rates, HR configures the background automated notification engine:

- **Email Reminders Toggle (student-eval-reminder-enabled):** Master switch enabling or pausing automated reminder dispatches.
- **Reminder Frequency (reminder-freq):** Defines minimum interval between reminders per student (default: 7 days).
- **Email Subject (student-eval-reminder-subject):** Customizable subject line (max 200 characters).
- **Email Message Body (student-eval-reminder-body):** Multi-line template supporting dynamic system placeholders:
 - `{{student_name}}` : Replaced with cadet's registered full name.
 - `{{evaluation_end_date}}` : Injects the official deadline date.
 - `{{academic_year}}` : Injects the active academic year (e.g., '2026-2027').
 - `{{semester}}` : Injects the active term (e.g., '2nd Semester').
- **Actions:** [Reset to Default] (restores standard notice) and [Save Reminder Settings] (commits configuration).

6. Dynamic Questionnaire Builder, Privacy Consent & Template Reuse

The Questionnaire view (questionnaire-view) is an interactive design studio empowering HR to maintain and evolve institutional evaluation instruments without writing code.

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Selecting Questionnaire Type & Semester

Use the top selectors to load the desired instrument:

- Semester Selector (semester-select): Choose the target academic term.
- Questionnaire Type (questionnaire-type-select): Select between 'Student to Professor', 'Professor to Professor', or 'Supervisor to Professor'.

Action Buttons Available:

- [Use Previous Set] (copy-questionnaire-btn): Opens the template copy modal.
- [Add Section] (add-section-btn): Creates a new thematic evaluation section.
- [Add Question] (add-question-btn): Appends an assessment question.

Data Privacy Agreement Configuration

To comply with the Philippine Data Privacy Act of 2012 (RA 10173), every questionnaire includes configurable consent controls:

- [x] Require agreement checkbox (questionnaire-privacy-required): Enforces mandatory acceptance before a student can begin rating.
- Consent Version & Text Identifier: Tracks policy versioning (e.g., 'DPA-V2.1-2026').
- Notice Title & Notice Text: Full legal privacy disclosure explaining data confidentiality and anonymous aggregation.
- Agreement Checkbox Text: The exact confirmation statement the user must tick.

Sections, Questions & Likert/Qualitative Configuration

Questionnaires are organized into structured sections and individual questions:

- Add/Edit Section Modal (section-modal): Set Section Letter (A, B, C), Section Title (e.g., 'Instructional Methodology'), and Section Description.
 - Add/Edit Question Modal (question-modal):
 - Section Assignment: Links question to a specific section.
 - Question Text: The clear prompt presented to evaluators.
 - Question Type: Choose between 'Rating (Numeric Scale)' or 'Qualitative (Text Response)'.
 - Rating Scale Options: Set Maximum Rating Number (scale 2 to 10; default: 5 for 5-star Likert).
 - Qualitative Options: Set Maximum Character Length (50 to 2,000 characters; default: 500).
 - Checkboxes: 'Required Question' (mandatory answer) and 'Exception Reporting' (special anomaly tracking).
- Click [Save Questionnaire] (save-questionnaire-btn) to commit all additions and structural changes.

Reusing Previous Questionnaire Sets

To eliminate repetitive survey re-entry, the Copy Previous Questionnaire Modal (copy-questionnaire-modal) allows HR to select a source previous semester, choose the scope checkboxes (Student, Peer, and/or Supervisor), preview the destination semester, and click [Copy to Current Semester] to duplicate verified question sets in one click.

7. AI Insights: Credibility Review, Behavior Analysis & Bias Audits

The AI Insights view (ai-insights-view) is NAAP's premier data-integrity laboratory. It employs machine learning and statistical models to protect faculty evaluations against fraudulent, careless, or biased submissions.

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1. Human-in-the-Loop Credibility Review (<70 Threshold)

Evaluations that score below the institutional threshold of 70 Reliability Points are automatically sequestered and require HR adjudication before they can affect a professor's official grade:

- Filtering Controls: Filter by Review Status (Pending review, Accepted by HR, Rejected by HR, Auto accepted), Evaluation Type, Semester, or Faculty Name.
- Batch Actions: Check [Check All displayed reviewable evaluations], then click [Accept Selected] or [Reject Selected].
- Individual Review Dialog (cred-review-dialog): Review full response details, attach an optional internal HR Note, and click [Accept] or [Reject].
- Audit Principle: Accepting changes analytical eligibility; the original evaluation record remains permanently stored for legal auditability.

2. Evaluation Behavior Score Analysis (Fast/Uniform Pattern Detection)

This engine analyzes raw submission metadata to detect unreflective responses:

- Select Semester and click [Run Behavior Analysis] (hr-run-behavior-analysis-btn).
- Three Anomaly Flags:
 - Fast Flag: Identifies forms completed in under 45 seconds (speed-clicking).
 - Uniform Flag: Identifies straight-lining (e.g., selecting '5' across all 25 questions without variation).
 - Repetitive Flag: Identifies identical patterned selections repeated across multiple instructors.
- Tabulated Outputs: Student Aggregate Scores table (Student Number, Forms Analyzed, Avg Behavior Score, Assessment Level) and Submission-Level Analysis table.

3. NLP Bias Detection in Student Comments

An integrated Natural Language Processing classifier examines qualitative student remarks:

- Select Semester and click [Run Bias Detection] (hr-run-bias-detection-btn).
- Classification Labels: Categorizes comments as 'Constructive', 'Neutral', or 'Biased'.
- Bias Reason & Source: Identifies comments rooted in personal grievances rather than instructional delivery.
- Tabulated Outputs: Displays Comment, Label, Reason, Source, and Timestamp.

4. Cross-Source Evaluation Discrepancy Checks

Flags statistical divergence where student ratings diverge significantly from supervisor reviews:

- Select Semester and click [Run Discrepancy Check] (hr-run-discrepancy-analysis-btn).
- Threshold Rule: Flags cases where Student Average and Supervisor Average differ by 2.0 or more on a 5.0 scale.
- Outputs: Table listing Professor, Student Avg, Supervisor Avg, Score Difference, Flag Reason, and Severity.

5. Evaluation Credibility Score (Composite Reliability)

Combines behavior, bias, and cross-source checks into an overall composite reliability score:

- Click [Run Credibility Analysis] (hr-run-credibility-analysis-btn).
- Summary Metrics: Total forms, Highly Reliable, Acceptable, Needs Review, and Mean Credibility Score.
- Printable Dossier: Click [Print AI Insights Summary] (hr-print-ai-insights-btn) to generate an official print/PDF audit sheet.

8. Activity Log: Security Auditing & IP Tracking

The Activity Log view (activity-log-view) provides an immutable, transparent record of all operations conducted across the evaluation platform.

Session Snapshot & Detailed Audit Controls

The Activity Summary card presents an immediate snapshot of current session metrics. Below the summary, HR officers can perform forensic searches using multi-parameter controls:

- Date From (hr-activity-from) & Date To (hr-activity-to): Restricts audit query to specific date ranges.
- Search Type (hr-activity-type): Filter by All Types, User Login, User Update, Evaluation, or System.
- Search Query (hr-activity-search): Free-text search matching user IDs, IP addresses, or action keywords.
- [Search] Button (hr-activity-search-btn): Executes the filtered audit log query.

Audit Table Column Structure

The Activity Log Table tabulates seven (7) critical audit dimensions:

1. IP Address: Originating client IP address (detects unauthorized off-campus network access).
2. Timestamp: Exact system date and time of the event.
3. Description: Detailed narrative explaining the action performed.
4. Action: Standardized operation code (e.g., 'LOGIN', 'EVAL_SUBMIT', 'UPDATE_PROFILE').
5. Role: User tier executing the action ('student', 'professor', 'hr', 'dean').
6. User ID: Unique account identifier.
7. Log ID: Primary key database sequence identifier for forensic verification.

9. Profile & Account Security Management

The Profile view (profile-view) enables HR officers to maintain their credentials, upload an official avatar, and manage login security.

Synced HR Credentials: Displays verified institutional data: Faculty ID, Full Legal Name, Department (Human Resources), Rank (e.g., HR Manager), Active Gmail Account, and Current AY/Semester.

Profile Photo Upload: Upload an official headshot in JPG or PNG format. Click [Upload Photo] (profilePhotoInput) to refresh your circular avatar across all system interfaces.

Update Gmail Account (emailUpdateCard): Click [Change Gmail]. Enter your new institutional Gmail address twice to validate, and click [Update Gmail].

Update Account Password (passwordUpdateCard): Click [Change Password]. Enter your current password, input a strong new password (minimum 8 characters with numbers and uppercase letters), confirm the password, and toggle visibility with the eye icon. Click [Update Password] to save.

— End of HR Panel Operating Manual —

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System Administrator Portal & Infrastructure Operations Guide

Part 9: System Governance, Infrastructure, Bulk Onboarding & Operational Controls
(adminpanel.html)

1. Administrator Authority & Control Center Architecture

The Admin Panel is engineered for institutional oversight, multi-campus scalability, and disaster resilience. The left navigation sidebar (adminSidebar) provides instant access to eight primary administrative suites:

Dashboard (dashboard): The master operational launchpad featuring high-level KPIs, 7 rapid action shortcuts, an live session activity ticker, overview trend graphs, and merged tri-perspective evaluation analytics.

User Management (users): Campus-wide user directory supporting granular multi-parameter searches, hierarchical organizational breakdowns (HR, VPAA, OSA, Admin, and Academic Departments), individual CRUD modals, campus/program managers, and bulk Excel registration.

Profile (profile): Administrator profile management interface displaying synchronized HR credentials, headshot photo upload, and secure email and password modification forms.

Professor Management (hr-professors): Dedicated faculty registry interface enabling department-filtered roster reviews, individual teaching profile edits, faculty analytics drilldowns, and the master 'Overall SASR' institutional export.

Subject Management (subject-management): Curriculum management workstation containing the Subject Catalog (course creation, editing, and Excel import) and Offering Assignment (pairing subjects and sections with professors, excess load management, and student enrollments).

Evaluation Settings (hr-evaluation-settings): Academic cycle management hub for defining exact start/end submission windows across evaluation types, provisioning terms, and configuring automated email reminder intervals and dynamic templates.

Questionnaire (hr-questionnaire): Interactive survey creation studio supporting Likert-scale and qualitative question authoring, section grouping, Philippine Data Privacy Act (RA 10173) compliance consent controls, and template reuse.

System Settings (system): Core infrastructure control room for institution-wide SMTP server configuration, bulk credential distribution, OpenAI API key management with per-panel feature toggles, encrypted database backups, OTP security rules, and system health checks.

Top Header & Session Navigation

The persistent top executive header bar features:

- Mobile Navigation Toggle (adminMenuToggle): Expands/collapses navigation on portable administrator tablets or laptops.
- Page Title & Context Block: Reflects the active module and provides operational guidance.
- Administrator Identity Badge (admin-username): Displays the logged-in root administrator identity.
- Secure Logout (logout-btn): Immediately invalidates the active JSON Web Token (JWT) session and returns to mainpage.html.

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2. Executive Dashboard, Quick Actions & Real-Time Metrics

Upon authentication, administrators enter the primary Dashboard view (dashboard-view). The interface is organized to provide rapid access to the most frequent maintenance tasks:

The Control Center Hero Banner & Shortcuts

The hero card ('Keep users, settings, and evaluations aligned') provides two direct navigation jump buttons:

- [Open User Management] (heroOpenUsersBtn) — Jumps directly to user accounts, campus rosters, and bulk imports.
- [Review Settings] (heroOpenSettingsBtn) — Navigates straight to system infrastructure and evaluation period configurations.

Four Global Operational KPI Cards

Real-time counters provide instant academy-wide status:

1. Total Professors (total-professors): Active teaching faculty currently loaded into the system.
2. Total Students (total-students): Total enrolled student body tracked for evaluation clearance.
3. Completion Rate (completion-rate-overview): Overall percentage of required evaluations submitted to date.
4. Pending Evaluations (pending-evaluations): Count of assigned evaluations awaiting completion across all roles.

Seven Quick Action Shortcuts Grid

The Quick Actions Grid gives administrators immediate one-click execution of critical operational routines:

- **[Add User] (data-action='add-user')**: Instantly launches the Add User Modal to provision a single account.
- **[Manage Users] (data-action='manage-users')**: Navigates directly to the comprehensive User Management view.
- **[Manage Subjects] (data-action='manage-subjects')**: Opens the Subject Catalog and Offering Assignment workstation.
- **[Manage Campus] (data-action='manage-campus')**: Pops up the Manage Campuses and Departments modal.
- **[Announcement] (data-action='send-announcement')**: Launches the Targeted Announcement Compose modal.
- **[System Backup] (data-action='system-backup')**: Navigates directly to the Encrypted Backup Management console.
- **[Settings] (data-action='system-settings')**: Opens the System Settings and Infrastructure configuration screen.

Live Activity Summary & Tri-Perspective Reports

The dashboard integrates a live Activity Summary feed (activity-list) showing recent logins and submissions, paired with a [View Activity Log] button (js-open-activity-log) that jumps straight to forensic logs.

Beneath the activity feed are two high-level visual charts:

- Evaluation Overview (evaluation-chart): Doughnut chart displaying completion percentages.
- Semestral Performance (performance-chart): Bar graph tracking multi-semester submission volume.

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Furthermore, complete statistical cards for Student to Professor, Professor to Professor, and Supervisor to Professor evaluations are embedded directly in the dashboard, displaying category average bar charts, rating distribution pie charts, and total evaluation counts.

3. User & Campus Management (Multi-Role Accounts & Hierarchy)

The User Management view (users-view) is the administrative core for managing NAAP personnel across all campuses, departments, and governance levels.

Campus Hierarchy & Management Modals

NAAP operates multiple physical airbases and instructional campuses (Villamor, Basa, Medelin, Mactan, Fernando). The campus bar provides vital structural controls:

- Campus Selector (campus-filter-select): Filters user lists, departments, and course offerings by campus.
- [Manage Campuses] Button (manage-campuses-btn): Opens the Manage Campuses Modal (manage-campuses-modal) to create new campus locations and assign departmental units (e.g., ICS, ILAS, Aero Eng).
- [Manage Programs] Button (manage-programs-btn): Opens the Manage Programs Modal (manage-programs-modal) to define degree programs (e.g., BS Aeronautical Engineering, Associate in Information Systems) per campus and department.

Granular User Search & Filtering Toolbar

Administrators can locate any user among thousands of records using the multi-field filter toolbar:

- Search Input (user-search): Search by Full Name or ID Number (Student or Employee Number).
- Role Filter (user-role-filter): Student, Professor, Dean, Program Coordinator, OSA, VPAA, HR Staff, or Administrator.
- Department Filter (user-department-filter): Filter by institute/department code.
- Status Filter (user-status-filter): Active or Inactive.
- Program Filter (user-program-filter): Filter by enrolled/assigned degree program.
- [Search] Button (user-search-btn): Executes the query and displays results in the User Results Table (user-results-table).

Collapsible Organization Structure Tree

For hierarchical browsing, the view features collapsible accordion sections displaying live user counts:

- HR Staff Section (hr-section): Lists all Human Resources officers.
- VPAA Section (vpaa-section): Lists Vice Presidential and executive academic accounts.
- OSA Section (osa-section): Lists Office of Student Affairs officers.
- Administrators Section (administrators-section): Lists certified system administrators.
- Academic Departments (departments-dynamic): Dynamically renders faculty and students grouped by institute and program.

Adding and Editing User Accounts

- Floating [Add New User] Button (add-user-btn): Opens the Add User Modal (add-user-modal). The form dynamically alters its input fields based on the selected Role. If 'Student' is selected, Student Number and Year & Section fields appear. If employee roles are selected, Employee ID and Employment Type (Regular, Temporary, COS) appear.

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• **Edit User Modal (edit-user-modal):** Allows administrators to update user credentials, reassign campuses or programs, reset passwords, or toggle account status between Active and Inactive.

4. Bulk Onboarding & Registration (Excel Import Engine)

To accommodate hundreds of new cadets and faculty members at the beginning of each academic year, the Admin Panel integrates a high-performance Excel bulk onboarding engine.

Executing Bulk Registration

1. Click the [Bulk Register] button (bulk-register-btn) on the User Management toolbar.
2. Select a standardized Excel (.xlsx or .xls) file from your local workstation.
3. The system immediately displays the Bulk Register Loading modal (bulk-register-loading) while parsing rows in batches.
4. Valid accounts are inserted into the database, role-specific profiles are generated, and a success summary is displayed.

5. Subject Management & Offering Allocations (Catalog & Rosters)

The Subject Management view (subject-management-view) manages the academic curriculum and connects courses to professors and enrolled students.

Subject Catalog Management

The Subject Catalog section (subject-catalog-section) houses the master repository of institutional courses:

- **Campus & Department Filters:** Narrow down course lists by airbase campus and academic department.
- **[Add Subject] Button (add-subject-btn):** Opens the Subject Modal (subject-modal) to input Campus, Department, Subject Code (e.g., 'AERO-312'), and Subject Name.
- **[Import Subjects] Button (bulk-subject-import-btn):** Uploads an Excel roster of curriculum courses in bulk.
- **Catalog Table:** Displays Campus, Department, Subject Code, Subject Name, and Action buttons (Edit / Delete).

Offering Assignments & Teaching Loads

The Offering Assignment section (subject-offering-section) allocates specific course sections to teaching professors:

- **Offering Form Grid (offering-form):** Select Semester, Campus, Department, Program, Subject, Professor, and specify Section (e.g., '3-1'). Click [Save Assignment] (offering-save-btn) to create the teaching load.
- **[Import Offerings] (bulk-offering-import-btn):** Bulk imports semester course offerings via Excel.
- **[Import Excess Load] (bulk-excess-load-import-btn):** Imports overload teaching assignments for part-time or adjunct instructors.
- **Offering Table (offering-list-body):** Lists Subject, Section, Professor, Employee ID, Program Code, Campus/Dept, Load, Status, and Actions.

Assigning Students to Course Offerings

Clicking 'Assign Students' on any course offering row opens the Offering Students Modal (offering-students-modal). Administrators can filter by program, search students by name or student number, tick enrolled cadets, and click [Save Students] (save-offering-students-btn). This dynamically binds students to that professor's evaluation roster.

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6. Professor Records & Master Overall SASR Generation

The Professor Management view (hr-professors-view) enables specialized faculty oversight:

- Search & Filter Toolbar: Filter faculty by Name/ID, Campus, Status (Active/Inactive), and Program.
- Department Tabs: Quick filtering across All Departments, ICS, ILAS, and ENGI.
- [Add Professor] (add-professor-btn): Opens the Professor Modal to register a faculty member with rank and employment type.
- Profile Action Triggers: View Details (assigned subjects, course loads) and View Analytics (evaluation scores, trends, comments).
- Master SASR Export: The [Overall SASR] button (overall-sasr-btn) compiles academy-wide evaluation results (SET 50%, Peer 20%, Supervisor 30%) into the official accreditation Excel format.

7. Evaluation Settings: Period Windows, Semesters & Email Alerts

The Evaluation Settings view (hr-evaluation-settings-view) manages operational timelines:

- Evaluation Periods: Configure Start and End dates for Student to Professor, Professor to Professor, and Supervisor to Professor windows. Click [Save Evaluation Periods] (save-eval-periods-btn) to lock dates into system gates.
- Current Semester Management: Select active semester or provision new terms (Academic Year Start, End, Semester Type, and [Add Term]). Click [Save Current Semester] (save-current-semester-btn) to activate term campus-wide.
- Automated Email Reminders: Toggle switch (student-eval-reminder-enabled), reminder frequency in days (reminder-freq), custom email subject, and dynamic email body with placeholders ({{student_name}}, {{evaluation_end_date}}, {{academic_year}}, {{semester}}). Click [Save Reminder Settings] (student-eval-reminder-save-btn) to commit.

8. Dynamic Questionnaire Builder & Data Privacy Consent

The Questionnaire view (hr-questionnaire-view) empowers administrators to maintain evaluation instruments:

- Instrument Selectors: Choose Semester and Questionnaire Type (Student to Professor, Professor to Professor, Supervisor to Professor).
- Section & Question Modals: Add Sections (letter, title, description) and Add Questions (rating scale 2-10, qualitative text 50-2000 chars, required toggle, exception reporting toggle).
- Data Privacy Agreement (questionnaire-privacy-settings): Complies with RA 10173 with mandatory agreement checkbox, consent versioning, notice title, legal notice text, and checkbox confirmation text.
- [Save Questionnaire] Button (save-questionnaire-btn): Commits instrument to live student and peer evaluation forms.

9. System Infrastructure: SMTP, Credential Distributor & OpenAI

The System Settings view (system-view) provides comprehensive governance over technical infrastructure, integrations, and communications.

General Platform Configuration

The General Settings group allows administrators to customize the platform's institutional identity:

- Institution Name (general-institution-name): Official academy title.
- System Email (general-system-email): Automatically synchronized with the active SMTP From Email.

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- Main Campus Selector (general-main-campus): Sets primary administrative headquarters (Villamor, Basa, Medelin, Mactan, Fernando).
- [Save General Settings] Button (general-settings-save-btn): Commits platform identity variables.

SMTP Server & Bulk Credential Distributor

The Bulk Credential Distributor group (credential-distributor-group) manages all outgoing email infrastructure:

- SMTP Parameters: SMTP Host, Port (e.g., 587), Encryption (TLS, SSL, None), Authentication toggle, SMTP Username, Timeout (seconds), From Email, From Name, and SMTP Password.
- Action Buttons: [Save SMTP Config] (credential-save-config-btn) and [Clear Saved Password] (credential-clear-password-btn).
- Bulk Credential Dispatch: Upload an Excel file with user credentials, click [Select XLSX File], then click [Distribute Credentials] (credential-distributor-send-btn). The system dispatches personalized onboarding emails in batches. If any addresses bounce, click [Download Failed Rows CSV] (credential-download-failures-btn) for remediation.
- Single SMTP Test Email Box: Input recipient email, subject, and test message, and click [Send Test Email] (credential-test-smtp-send-btn) to verify mailer connectivity safely before mass dispatches.

OpenAI API Configuration & Per-Panel Access Toggles

The OpenAI Configuration section controls AI-assisted features across the platform:

- OpenAI Model (gemini-model): Configurable model endpoint (e.g., gpt-5.6-luna).
- Timeout (gemini-timeout-ms): Request duration threshold in milliseconds (default: 30,000 ms).
- OpenAI API Key (gemini-api-key): Master API token stored with AES-256 database encryption.
- Action Buttons: [Save OpenAI Config] (gemini-save-config-btn) and [Clear Saved API Key] (gemini-clear-key-btn).

Granular Per-Panel AI Access Switches:

Administrators can selectively enable or disable AI capabilities for individual user panels without altering the master API key:

- Professor Panel Toggle: Feedback summaries & Section C recommendations.
- Dean Panel Toggle: Professor qualitative comment summaries.
- Program Coordinator Panel Toggle: Program-level comment summaries.
- HR Panel Toggle: AI credibility insights and anomaly detection.
- Admin Panel Toggle: AI reporting tools inside administrative analytics.
- VPAA Panel Toggle: Institution-wide AI evaluation insights.

10. Security Controls, Encrypted Backups & System Health

Security and disaster preparedness are strictly enforced through dedicated administrative modules in System Settings:

Authentication & OTP Security Settings

- Require OTP on First Login & New Devices (trusted-device-otp-enabled): Master toggle enforcing two-factor authentication via email OTP codes for untrusted browser sessions. Multiple wrong password attempts trigger mandatory OTP challenges regardless of this toggle.
- Session Timeout (30 minutes): Enforces automatic session expiration after 30 minutes of administrator

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inactivity.

- [Save OTP Setting] Button (otp-security-save-btn): Commits authentication security policies.

Encrypted Backup & Restoration Testing

The Encrypted Backup Management group (backup-management-group) safeguards institutional records against data loss:

- Automated Schedule: Configured for daily execution at 02:00 Asia/Manila.
- Retention Policy: Retains the newest 30 encrypted backup artifacts automatically.
- Status & Readiness Metrics: Last Backup timestamp, Archive Size, Integrity SHA-256 checksum status, and Restoration Test result.
- Manual Backup Execution: Click [Run Encrypted Backup] (backup-last-24-btn) to create an immediate snapshot of the MySQL database and persistent faculty documents.
- Backup History Table (backup-history-body): Lists Backup ID, Creation Date, Source, Status, Size, Integrity, and CLI Restore test logs.

System Health & Update Verification

The System Health group (system-health-group) provides automated diagnostics to certify system integrity following updates:

- [Run Verification] Button (system-health-run-btn): Executes read-only integrity probes across database schemas, table indexes, file storage permissions, and authentication endpoints.
- Diagnostic Badges & Metrics: Displays Last Run date, Administrator ID, and Summary (e.g., '14 passed, 0 warning, 0 failed').
- Check Results Table: Lists specific test items, technical details, and timestamps.
- Verification History Table: Maintains an immutable log of historical verification runs for institutional IT audit evidence.

11. Forensic Activity Logging & Audit Trail Investigation

The Activity Log view (activity-log-view) serves as the academy's official forensic record. Every critical system operation, authentication event, profile modification, and evaluation submission is recorded immutably.

Audit Search Controls:

- Date Range: Select 'From' (activity-from) and 'To' (activity-to) calendar boundaries.
- Search Type (activity-type): Filter by All Types, User Login, User Update, Evaluation, or System events.
- Keyword Search (activity-search): Search by specific user ID, IP address, or description keywords.
- [Search] Button (activity-search-btn): Executes the filtered audit log query.

Log Table Dimensions (activity-log-body):

1. IP Address: Originating client IP address (enables tracing of unauthorized off-campus network access).
2. Timestamp: Exact date and time formatted in Asia/Manila standard time.
3. Description: Plain-language operational narrative (e.g., 'User Alex M. Reyes updated SMTP configuration').
4. Action: Standardized operation verb ('LOGIN', 'UPDATE', 'CREATE', 'EVAL_SUBMIT').
5. Role: User privilege tier initiating the action.
6. User ID: Unique account identifier.
7. Log ID: Primary key database sequence number for tamper-evident verification.

12. Admin Profile & Credential Security Management

The Profile view (profile-view) enables administrators to review personal records and maintain root account security:

- Synced Profile Attributes: Displays Faculty ID (ADM-0001), Full Name, Department (Administration), Rank (System Administrator), Gmail address, active AY/Semester, and Assigned Institutes.
- Photo Upload: Upload a professional headshot in JPG or PNG format via [Upload Photo] (profilePhotoInput).
- Change Email (adminEmailUpdateCard): Input current email, new email, and confirmation email, then click [Update Email].
- Change Password (adminPasswordUpdateCard): Input current password, enter a strong new password (minimum 8 characters, mixed case, numerals, and special symbols), confirm password, toggle visibility with the eye icon, and click [Update Password].